

SingleRAN

Interoperability Between E-UTRAN and NG-RAN in Connected Mode Feature Parameter Description

Issue	Draft B
Date	2026-01-30



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1

Change History

1

2

About This Document

10

2.1

General Statements

10

2.2

Applicable RAT

10

2.3

Features in This Document

11

2.4

Differences

12

3

Overview

16

4

Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode

20

4.1

Principles

20

4.1.1

Overall Procedure

20

4.1.2

Function Initiation Determination

21

4.1.3

Processing Mode Selection

21

4.1.4

Measurement Configuration

29

4.1.4.1

Measurement Objects

29

4.1.4.1.1

Measurement Object Selection

30

4.1.4.1.2

NR Frequency Selection by the eNodeB

30

4.1.4.1.3

LTE Frequency Selection by the gNodeB

34

4.1.4.1.4

Blacklisted Cells for Cell Measurement

37

4.1.4.2

Reporting Configurations

38

4.1.4.2.1

Measurement Events

38

4.1.4.2.2

Triggering Quantity

49

4.1.4.2.3

QCI-specific Measurement Parameter Configurations

52

4.1.5

Other Configurations

53

4.1.5.1

Measurement Gap Configuration

53

4.1.5.2

Non-Gap-assisted B1 Measurement Configuration by the eNodeB

57

4.1.5.3

Measurement Object Preemption

57

4.1.5.4

Multi-Antenna-Port Configuration Indication

58

4.1.6

Measurement Reporting

58

4.1.7

RSRP Prediction Based on Virtual Grids

58

4.1.7.1

LN RSRP Model Building and Maintenance

61

4.1.7.2

NL RSRP Model Building and Maintenance

65

4.1.7.3 Application to Functions in Interoperability Between E-UTRAN and NG-RAN.....	68
4.1.8 Target Cell or Frequency Evaluation.....	69
4.1.8.1 General Rules.....	70
4.1.8.2 NR-specific Rules.....	71
4.1.8.3 LTE-specific Rules.....	72
4.1.9 Mobility Policy Execution.....	74
4.1.10 Roaming UEs Preferentially Back to HPLMN Cells.....	79
4.2 Network Analysis.....	80
4.2.1 Benefits.....	80
4.2.2 Impacts.....	82
4.3 Requirements.....	84
4.3.1 Licenses.....	85
4.3.2 Software.....	85
4.3.3 Hardware.....	85
4.3.4 Others.....	85
4.4 Operation and Maintenance (NG-RAN).....	85
4.4.1 Data Configuration.....	85
4.4.1.1 Data Preparation.....	85
4.4.1.2 Using MML Commands.....	96
4.4.1.3 Using the MAE-Deployment.....	98
4.4.2 Activation Verification.....	98
4.4.3 Network Monitoring.....	99
4.5 Operation and Maintenance (E-UTRAN).....	99
4.5.1 Data Configuration.....	99
4.5.1.1 Data Preparation.....	99
4.5.1.2 Using MML Commands.....	116
4.5.1.3 Using the MAE-Deployment.....	119
4.5.2 Activation Verification.....	119
4.5.3 Network Monitoring.....	120
5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode.....	121
5.1 Overview of Interoperability in Connected Mode.....	121
5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN.....	122
5.2.1 Principles.....	122
5.2.2 Network Analysis.....	128
5.2.2.1 Benefits.....	128
5.2.2.2 Impacts.....	128
5.2.3 Requirements.....	128
5.2.3.1 Licenses.....	129
5.2.3.2 Software.....	129
5.2.3.2.1 Prerequisite Functions.....	129
5.2.3.2.2 Mutually Exclusive Functions.....	129
5.2.3.2.3 Function Impacts.....	130

5.2.3.3 Hardware.....	133
5.2.3.4 Others.....	134
5.2.4 Operation and Maintenance.....	134
5.2.4.1 Data Configuration.....	134
5.2.4.1.1 Data Preparation.....	134
5.2.4.1.2 Using MML Commands.....	135
5.2.4.1.3 Using the MAE-Deployment.....	136
5.2.4.2 Activation Verification.....	136
5.2.4.3 Network Monitoring.....	137
5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN.....	139
5.3.1 Principles.....	139
5.3.2 Network Analysis.....	144
5.3.2.1 Benefits.....	144
5.3.2.2 Impacts.....	145
5.3.3 Requirements.....	145
5.3.3.1 Licenses.....	146
5.3.3.2 Software.....	146
5.3.3.2.1 Prerequisite Functions.....	146
5.3.3.2.2 Mutually Exclusive Functions.....	146
5.3.3.2.3 Function Impacts.....	147
5.3.3.3 Hardware.....	149
5.3.3.4 Others.....	150
5.3.4 Operation and Maintenance.....	150
5.3.4.1 Data Configuration.....	150
5.3.4.1.1 Data Preparation.....	150
5.3.4.1.2 Using MML Commands.....	152
5.3.4.1.3 Using the MAE-Deployment.....	153
5.3.4.2 Activation Verification.....	153
5.3.4.3 Network Monitoring.....	154
5.4 Load-based Inter-RAT Mobility from NG-RAN to E-UTRAN.....	155
5.4.1 Principles.....	155
5.4.2 Network Analysis.....	158
5.4.2.1 Benefits.....	158
5.4.2.2 Impacts.....	158
5.4.3 Requirements.....	158
5.4.3.1 Licenses.....	158
5.4.3.2 Software.....	159
5.4.3.2.1 Prerequisite Functions.....	159
5.4.3.2.2 Mutually Exclusive Functions.....	159
5.4.3.2.3 Function Impacts.....	160
5.4.3.3 Hardware.....	160
5.4.3.4 Others.....	161

5.4.4 Operation and Maintenance.....	161
5.4.4.1 Data Configuration.....	161
5.4.4.1.1 Data Preparation.....	161
5.4.4.1.2 Using MML Commands.....	162
5.4.4.1.3 Using the MAE-Deployment.....	163
5.4.4.2 Activation Verification.....	163
5.4.4.3 Network Monitoring.....	163
5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN.....	163
5.5.1 Principles.....	163
5.5.2 Network Analysis.....	170
5.5.2.1 Benefits.....	170
5.5.2.2 Impacts.....	170
5.5.3 Requirements.....	171
5.5.3.1 Licenses.....	171
5.5.3.2 Software.....	171
5.5.3.2.1 Prerequisite Functions.....	171
5.5.3.2.2 Mutually Exclusive Functions.....	171
5.5.3.2.3 Function Impacts.....	172
5.5.3.3 Hardware.....	172
5.5.3.4 Others.....	173
5.5.4 Operation and Maintenance.....	173
5.5.4.1 Data Configuration.....	173
5.5.4.1.1 Data Preparation.....	173
5.5.4.1.2 Using MML Commands.....	174
5.5.4.1.3 Using the MAE-Deployment.....	175
5.5.4.2 Activation Verification.....	175
5.5.4.3 Network Monitoring.....	176
5.6 E-UTRAN to NG-RAN Traffic Steering.....	178
5.6.1 Principles.....	178
5.6.2 Network Analysis.....	187
5.6.2.1 Benefits.....	187
5.6.2.2 Impacts.....	188
5.6.3 Requirements.....	188
5.6.3.1 Licenses.....	188
5.6.3.2 Software.....	188
5.6.3.2.1 Prerequisite Functions.....	188
5.6.3.2.2 Mutually Exclusive Functions.....	189
5.6.3.2.3 Function Impacts.....	189
5.6.3.3 Hardware.....	189
5.6.3.4 Others.....	190
5.6.4 Operation and Maintenance.....	191
5.6.4.1 Data Configuration.....	191

5.6.4.1.1 Data Preparation.....	191
5.6.4.1.2 Using MML Commands.....	194
5.6.4.1.3 Using the MAE-Deployment.....	195
5.6.4.2 Activation Verification.....	195
5.6.4.3 Network Monitoring.....	195
5.7 Voice Fallback from NG-RAN to E-UTRAN.....	197
5.7.1 Principles.....	197
5.7.1.1 Normal Voice Fallback.....	198
5.7.1.2 Emergency Voice Fallback.....	205
5.7.1.3 Fast Voice Fallback.....	205
5.7.1.4 Collaboration with Other Functions.....	208
5.7.1.5 Protocol Update.....	210
5.7.2 Network Analysis.....	211
5.7.2.1 Benefits.....	211
5.7.2.2 Impacts.....	211
5.7.3 Requirements.....	213
5.7.3.1 Licenses.....	213
5.7.3.2 Software.....	213
5.7.3.2.1 Prerequisite Functions.....	213
5.7.3.2.2 Mutually Exclusive Functions.....	214
5.7.3.2.3 Function Impacts.....	214
5.7.3.3 Hardware.....	214
5.7.3.4 Others.....	215
5.7.4 Operation and Maintenance.....	215
5.7.4.1 Data Configuration.....	215
5.7.4.1.1 Data Preparation.....	215
5.7.4.1.2 Using MML Commands.....	220
5.7.4.1.3 Using the MAE-Deployment.....	222
5.7.4.2 Activation Verification.....	223
5.7.4.3 Network Monitoring.....	224
5.8 Fast Return from E-UTRAN to NG-RAN.....	226
5.8.1 Principles.....	226
5.8.2 Network Analysis.....	234
5.8.2.1 Benefits.....	234
5.8.2.2 Impacts.....	234
5.8.3 Requirements.....	234
5.8.3.1 Licenses.....	235
5.8.3.2 Software.....	235
5.8.3.2.1 Prerequisite Functions.....	235
5.8.3.2.2 Mutually Exclusive Functions.....	235
5.8.3.2.3 Function Impacts.....	236
5.8.3.3 Hardware.....	236

5.8.3.4 Others.....	237
5.8.4 Operation and Maintenance.....	238
5.8.4.1 Data Configuration.....	238
5.8.4.1.1 Data Preparation.....	238
5.8.4.1.2 Using MML Commands.....	241
5.8.4.1.3 Using the MAE-Deployment.....	243
5.8.4.2 Activation Verification.....	243
5.8.4.3 Network Monitoring.....	244
5.9 VoNR-to-VoLTE Handover Based on Voice Quality.....	244
5.9.1 Principles.....	244
5.9.2 Network Analysis.....	245
5.9.2.1 Benefits.....	246
5.9.2.2 Impacts.....	246
5.9.3 Requirements.....	246
5.9.3.1 Licenses.....	246
5.9.3.2 Software.....	246
5.9.3.2.1 Prerequisite Functions.....	247
5.9.3.2.2 Mutually Exclusive Functions.....	247
5.9.3.2.3 Function Impacts.....	248
5.9.3.3 Hardware.....	248
5.9.3.4 Others.....	249
5.9.4 Operation and Maintenance.....	249
5.9.4.1 Data Configuration.....	249
5.9.4.1.1 Data Preparation.....	249
5.9.4.1.2 Using MML Commands.....	250
5.9.4.1.3 Using the MAE-Deployment.....	251
5.9.4.2 Activation Verification.....	251
5.9.4.3 Network Monitoring.....	252
5.10 Voice-Capability-based Redirection Back to the HPLMN.....	252
5.10.1 Inter-RAT Redirection Back to the HPLMN Based on Roaming UE Identification.....	252
5.10.1.1 Principles.....	253
5.10.1.2 Network Analysis.....	255
5.10.1.2.1 Benefits.....	255
5.10.1.2.2 Impacts.....	255
5.10.1.3 Requirements.....	255
5.10.1.3.1 Licenses.....	255
5.10.1.3.2 Software.....	255
5.10.1.3.3 Hardware.....	261
5.10.1.3.4 Others.....	261
5.10.1.4 Operation and Maintenance.....	262
5.10.1.4.1 Data Configuration.....	262
5.10.1.4.2 Activation Verification.....	263

5.10.1.4.3 Network Monitoring.....	264
5.10.2 Dedicated-Priority-based Inter-RAT Cell Reselection Back to the HPLMN.....	264
5.10.2.1 Principles.....	264
5.10.2.2 Network Analysis.....	265
5.10.2.2.1 Benefits.....	265
5.10.2.2.2 Impacts.....	265
5.10.2.3 Requirements.....	265
5.10.2.3.1 Licenses.....	265
5.10.2.3.2 Software.....	265
5.10.2.3.3 Hardware.....	265
5.10.2.3.4 Others.....	266
5.10.2.4 Operation and Maintenance.....	266
5.10.2.4.1 Data Configuration.....	266
5.10.2.4.2 Activation Verification.....	268
5.10.2.4.3 Network Monitoring.....	268
6 Multi-Operator Sharing Scenarios.....	269
6.1 Principles.....	269
6.1.1 Function Description.....	269
6.1.1.1 Operator-specific Measurement Variables.....	269
6.1.1.2 Operator-specific Mobility Policy.....	271
6.1.1.3 Operator-specific NR Frequency Policy.....	274
6.1.1.4 Operator-specific LTE Frequency Configuration.....	275
6.1.2 Typical Multi-Operator Sharing Scenarios.....	276
6.2 Network Analysis.....	281
6.2.1 Benefits.....	281
6.2.2 Impacts.....	281
6.3 Requirements.....	281
6.3.1 Licenses.....	281
6.3.2 Software.....	282
6.3.2.1 Prerequisite Functions.....	282
6.3.2.2 Mutually Exclusive Functions.....	282
6.3.2.3 Function Impacts.....	282
6.3.3 Hardware.....	282
6.3.4 Others.....	283
6.4 Operation and Maintenance.....	283
6.4.1 Data Configuration.....	284
6.4.1.1 Data Preparation.....	284
6.4.1.2 Using MML Commands.....	287
6.4.1.3 Using the MAE-Deployment.....	294
6.4.2 Activation Verification.....	294
6.4.2.1 Operator-specific Dedicated Reselection Priority.....	295
6.4.2.2 Operator-specific Measurement Variables.....	295

6.4.2.3 Operator-specific Mobility Policy.....	296
6.4.2.4 Operator-specific NR Frequency Delivery.....	297
6.4.3 Network Monitoring.....	297
7 Appendix.....	300
7.1 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN.....	300
7.2 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN.....	303
7.3 E-UTRAN to NG-RAN Traffic Steering.....	308
7.4 Voice Fallback from NG-RAN to E-UTRAN.....	311
7.5 Fast Return from E-UTRAN to NG-RAN.....	320
7.6 VoNR-to-VoLTE Handover Based on Voice Quality.....	322
8 Parameters.....	325
9 Counters.....	327
10 Glossary.....	328
11 Reference Documents.....	329

1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for counting the number of coverage-based handover execution attempts and successful coverage-based handovers from E-UTRAN to NG-RAN. For details, see 5.5.4.2 Activation Verification .	None	LTE FDD LTE TDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Added support for flexible configuration of handover measurement parameters for voice and data service UEs based on RFSPs or SPIDs. For details, see 4.1.4.2.1 Measurement Events .	None	LTE FDD LTE TDD NR FDD Low-frequency NR TDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for separate time-to-trigger configuration for events A1 and A2 in inter-RAT mobility from E-UTRAN to NG-RAN. For details, see: • 4.1.4.2.1 Measurement Events • 4.5.1.1 Data Preparation • 4.5.1.2 Using MML Commands	Added the InterRatHoCommGroup.InterRatHoNrA1A2TimeToTrig parameter on the LTE side.	LTE FDD LTE TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
 - Added a frequency filtering rule for the eNodeB in NR frequency selection. For details, see [4.1.4.1.2 NR Frequency Selection by the eNodeB](#).
 - Revised descriptions of non-overlapping between gap and NR resources. For details, see [4.1.5.1 Measurement Gap Configuration](#).
 - Added a command example for deactivating the frequency-specific LTE-NR interoperability prohibition function. For details, see [4.5.1.2 Using MML Commands](#).
 - Revised descriptions of the impacts of optimized non-overlapping between gap and NR resources. For details, see [4.2.2 Impacts](#).
 - Revised descriptions of the mobility policy when both the **EUTRAN_HO_SWITCH** and **EUTRAN_REDIRECT_SWITCH** options of the **NRInterRatHoParam.HoModeSwitch** parameter are selected. For details, see [5.2.1 Principles](#).
 - Revised descriptions of principles and data preparation related to load-based inter-RAT mobility from NG-RAN to E-UTRAN. For details, see [5.4.1 Principles](#) and [5.4.4.1.1 Data Preparation](#).
 - Revised descriptions of signaling tracing for activation verification of inter-RAT mobility from NG-RAN to E-UTRAN. For details, see [5.2.4.2 Activation Verification](#), [5.3.4.2 Activation Verification](#), [5.7.4.2 Activation Verification](#), and [5.9.4.2 Activation Verification](#).
 - Revised descriptions of signaling tracing for activation verification of inter-RAT mobility from E-UTRAN to NG-RAN. For details, see [5.5.4.2 Activation Verification](#), [5.6.4.2 Activation Verification](#), and [5.8.4.2 Activation Verification](#).
 - Revised descriptions of the **NRCeUtranNFreq.VolteBlindPriority** parameter. For details, see [5.7.1.2 Emergency Voice Fallback](#), [5.7.4.1.2 Using MML Commands](#), and [6.4.1.2 Using MML Commands](#).

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 08 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added frequency-specific LTE-NR interoperability prohibition. For details, see: • 4.1.4.1.2 NR Frequency Selection by the eNodeB • 4.2.2 Impacts • 4.5.1.1 Data Preparation • 4.5.1.2 Using MML Commands	Modified parameter on the LTE side: Added the L2NR_MOB_FORBID_FL AG option to the NrNFreq.AggregationAttribute parameter.	LTE FDD LTE TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 Lamp Site and DBS5900 Lamp Site

Change Description	Parameter Change	RAT	Base Station Model
Added the "optimized non-overlapping between gap and NR resources" function. For details, see: 4.1.5.1 Measurement Gap Configuration 4.2.2 Impacts 4.4.1.1 Data Preparation 4.4.1.2 Using MML Commands	Modified parameter on the NR side: Added the GAP_AVOID_NR_RES_OPT_SW option to the NRDUCellMeas.MeasAlgorithmSwitch parameter.	NR FDD Low-frequency NR TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 Lamp Site and DBS5900 Lamp Site

Change Description	Parameter Change	RAT	Base Station Model
Added support for separate priority configuration for voice services in blind mode. For details, see: • 4.1.4.1.3 LTE Frequency Selection by the gNodeB • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands • 5.2.1 Principles • 5.2.4.1.1 Data Preparation • 5.2.4.1.2 Using MML Commands • 5.7.1.1 Normal Voice Fallback • 5.7.1.2 Emergency Voice Fallback • 5.7.4.1.1 Data Preparation • 5.7.4.1.2 Using MML Commands • 6.1.1.4 Operator-specific LTE Frequency Configuration • 6.4.1.1 Data Preparation • 6.4.1.2 Using MML Commands	Added parameters on the NR side: • NRCeUtrNfFreq.VolteBlindPriority • gNBFreqPriorityGroup.VolteBlindPriority	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 Lamp Site and DBS5900 Lamp Site

Change Description	Parameter Change	RAT	Base Station Model
Added incompatibility of BTS5929R with inter-RAT mobility from E-UTRAN to NG-RAN. For details, see 5.5.3.3 Hardware .	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929 R
Added incompatibility of BTS5929R with E-UTRAN to NG-RAN traffic steering. For details, see 5.6.3.3 Hardware .	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929 R
Added incompatibility of BTS5929R with fast return from E-UTRAN to NG-RAN. For details, see 5.8.3.3 Hardware .	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929 R
Added the impact relationship between "inter-RAT redirection back to the HPLMN based on roaming UE identification" and "RF module fast deep dormancy". For details, see Function Impacts .	None	NR FDD Low-frequency NR TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Optimized downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN. For details, see: • 5.2.1 Principles • 5.2.4.1.1 Data Preparation • 5.2.4.1.2 Using MML Commands	Modified parameters on the NR side: • Added the DL_COV_N2L_MOB_OFF_SWITCH option to the NRInterRatHoParam.HoModeSwitch parameter. • Added the DL_COV_N2L_MOB_OFF_SWITCH option to the NRCellOpPolicy.InterRatHoPolicySwitch parameter.	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 Lamp Site and DBS5900 Lamp Site
Added penalty optimization for E-UTRAN-to-NG-RAN handover preparation failures due to resource-related causes. For details, see: • 4.1.9 Mobility Policy Execution • 4.2.2 Impacts • 4.5.1.1 Data Preparation • 4.5.1.2 Using MML Commands	Modified parameter on the LTE side: Added the LNR_RES_HO_FAIL_PENALTY_OPT_SW option to the ENodeBAlgoExtSwitch.NrHandoverAlgoSwitch parameter.	LTE FDD LTE TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 Lamp Site and DBS5900 Lamp Site

Change Description	Parameter Change	RAT	Base Station Model
Added downlink-SINR-based mobility to E-UTRAN. For details, see: • 5.2.1 Principles • 5.2.2.2 Impacts • 5.2.4.1.1 Data Preparation • 5.2.4.1.2 Using MML Commands	Added parameters on the NR side: • NRCellMobilityConfig.InterRatHoExtSwitch • NRCellInterRHoMeaGrp.DlSinrHoToEutranA1Thld • NRCellInterRHoMeaGrp.DlSinrHoToEutranA2Thld	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 Lamp Site and DBS5900 Lamp Site
Optimized the activation verification of RSRP prediction based on virtual grids by adding the counter measuring the maximum number of LTE-NR inter-RAT grid models available for a cell. For details, see 4.4.2 Activation Verification.	None	NR FDD Low-frequency NR TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 Lamp Site and DBS5900 Lamp Site

Change Description	Parameter Change	RAT	Base Station Model
Optimized the activation verification of virtual grid model building by introducing new counters that indicate the numbers of good-RSRP and poor-RSRP virtual grid models. For details, see 4.5.2 Activation Verification .	None	LTE FDD LTE TDD	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 Lamp Site and DBS5900 Lamp Site

- Editorial changes
 - Added the reference information for ANR functions. For details, see [3 Overview](#) and [11 Reference Documents](#).
- Revised descriptions in the document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
NR	FOFD-021209	Inter-RAT Mobility from NG-RAN to E-UTRAN	4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode 5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN 5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN 5.4 Load-based Inter-RAT Mobility from NG-RAN to E-UTRAN 5.9 VoNR-to-VoLTE Handover Based on Voice Quality
LTE FDD	LNOFD-15133 6	Inter-RAT Mobility from E-UTRAN to NG-RAN	4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode 5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN
LTE TDD	TDLNOFD-151501	Inter-RAT Mobility from E-UTRAN to NG-RAN	4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode 5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN
LTE FDD	LNOFD-15133 7	E-UTRAN to NG-RAN Traffic Steering	4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode 5.6 E-UTRAN to NG-RAN Traffic Steering

RAT	Feature ID	Feature Name	Chapter/Section
LTE TDD	TDLNOFD-15 1502	E-UTRAN to NG-RAN Traffic Steering	4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode 5.6 E-UTRAN to NG-RAN Traffic Steering
NR	FOFD-021210	Voice Fallback	4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode 5.7 Voice Fallback from NG-RAN to E-UTRAN
LTE FDD	LNOFD-15133 5	Fast Return from E-UTRAN to NG-RAN	4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode 5.8 Fast Return from E-UTRAN to NG-RAN
LTE TDD	TDLNOFD-15 1503	Fast Return from E-UTRAN to NG-RAN	4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode 5.8 Fast Return from E-UTRAN to NG-RAN

2.4 Differences

Differences Between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
Downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN	None	5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN
E-UTRAN to NG-RAN traffic steering	None	5.6 E-UTRAN to NG-RAN Traffic Steering
Fast return from E-UTRAN to NG-RAN	None	5.8 Fast Return from E-UTRAN to NG-RAN

Differences Between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	None	5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN
Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	None	5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN
Load-based inter-RAT mobility from NG-RAN to E-UTRAN	None	5.4 Load-based Inter-RAT Mobility from NG-RAN to E-UTRAN
Voice fallback from NG-RAN to E-UTRAN	None	5.7 Voice Fallback from NG-RAN to E-UTRAN
VoNR-to-VoLTE handover based on voice quality	None	5.9 VoNR-to-VoLTE Handover Based on Voice Quality

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	Supported in SA networking as well as NSA and SA hybrid networking, but not in NSA networking	5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN
Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	Supported in SA networking as well as NSA and SA hybrid networking, but not in NSA networking	5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN

Function Name	Difference	Chapter/Section
Load-based inter-RAT mobility from NG-RAN to E-UTRAN	Supported in SA networking as well as NSA and SA hybrid networking, but not in NSA networking	5.4 Load-based Inter-RAT Mobility from NG-RAN to E-UTRAN
Voice fallback from NG-RAN to E-UTRAN	Supported in SA networking as well as NSA and SA hybrid networking, but not in NSA networking	5.7 Voice Fallback from NG-RAN to E-UTRAN
VoNR-to-VoLTE handover based on voice quality	Supported in SA networking as well as NSA and SA hybrid networking, but not in NSA networking	5.9 VoNR-to-VoLTE Handover Based on Voice Quality

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	Supported only in low frequency bands	5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN
Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	Supported only in low frequency bands	5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN

Function Name	Difference	Chapter/Section
Load-based inter-RAT mobility from NG-RAN to E-UTRAN	Supported only in low frequency bands	5.4 Load-based Inter-RAT Mobility from NG-RAN to E-UTRAN
Voice fallback from NG-RAN to E-UTRAN	Supported only in low frequency bands	5.7 Voice Fallback from NG-RAN to E-UTRAN
VoNR-to-VoLTE handover based on voice quality	Supported only in low frequency bands	5.9 VoNR-to-VoLTE Handover Based on Voice Quality

3 Overview

Standalone (SA) networking is gaining in popularity for New Radio (NR) deployment. NR operates in high frequency bands (the C-band or higher), causing limited cell coverage. During initial NR network construction, it is difficult to provide continuous coverage. As a result, the coverage continuity of NR networks is worse than that of LTE networks. **Interoperability between evolved universal terrestrial radio access network (E-UTRAN) and Next Generation Radio Access Network (NG-RAN) uses the LTE network with continuous coverage as the basic coverage to ensure service continuity and selects the appropriate bearer network (NR/LTE) according to services to achieve better user experience.**

Interoperability between E-UTRAN and NG-RAN mainly involves RRC_CONNECTED UEs and RRC_IDLE UEs, which is therefore split into interoperability for UEs in idle mode and UEs in connected mode. For details about interoperability between E-UTRAN and NG-RAN in idle mode, see *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*. This document describes interoperability between E-UTRAN and NG-RAN in connected mode, including multiple functions.

- The basic procedures for these functions, including measurement object selection, measurement event configuration, measurement reporting, target cell or frequency evaluation, and basic configuration of mobility management, are the same. For details, see [4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).
- For details about the functions and application scenarios of interoperability between E-UTRAN and NG-RAN in connected mode, see [Table 3-1](#).
- In multi-operator sharing scenarios, operator-specific parameters can be configured for inter-RAT interoperability in connected mode. For details, see [6 Multi-Operator Sharing Scenarios](#).

NOTE

A UE in inactive mode can be transferred from NG-RAN to E-UTRAN through cell reselection, the procedure of which is the same as the reselection from NG-RAN to E-UTRAN for a UE in idle mode. After the reselection to an LTE cell, the UE is in idle mode. For details about cell reselection from NG-RAN to E-UTRAN for UEs in idle mode, see *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 3-1 Functions and application scenarios of interoperability between E-UTRAN and NG-RAN in connected mode

Function Name	Application Scenario	Reference
Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	When a UE moves from an NR cell to an LTE cell, the signal quality of the serving NR cell is poor while that of the neighboring LTE cell is good. To ensure service continuity or better user experience, downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN is triggered for the UE.	5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN
Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	When a UE moves from an NR cell to an LTE cell, the uplink coverage of the serving NR cell signals is poor while that of the neighboring LTE cell signals is good. To ensure service continuity or better user experience, uplink coverage-based inter-RAT mobility from NG-RAN to E-UTRAN is triggered for the UE.	5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN
Downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN	When a UE moves from an LTE cell to an NR cell, the signal quality of the serving LTE cell is poor while that of the neighboring NR cell is good. To ensure service continuity or better user experience, downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN is triggered for the UE.	5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN
E-UTRAN to NG-RAN traffic steering	The traffic steering is triggered when the following conditions are met: (1) One of the following conditions is met: A. a UE accesses an LTE cell and sets up services; B. services are added or modified after a UE accesses an LTE cell and sets up services; C. a UE is handed over to an LTE cell and sets up services. (2) eNodeB determines that the UE capability supports NR. (3) No handover-prohibited service is running on the UE.	5.6 E-UTRAN to NG-RAN Traffic Steering
Voice fallback from NG-RAN to E-UTRAN	Voice services initiated by UEs in an NR cell that does not support VoNR can fall back to an LTE cell by means of voice fallback, which is also known as EPS fallback. Voice fallback is classified into normal voice fallback, emergency voice fallback, and fast voice fallback.	5.7 Voice Fallback from NG-RAN to E-UTRAN

Function Name	Application Scenario	Reference
Fast return from E-UTRAN to NG-RAN	When a UE completes its voice or data service in an LTE network and has the voice or data bearer released, a fast return procedure is initiated for the UE to return to an NR cell, if no handover-prohibited service is running on the UE.	5.8 Fast Return from E-UTRAN to NG-RAN
VoNR-to-VoLTE handover based on voice quality	If the gNodeB detects that the voice quality of a VoNR UE is poor, it initiates a voice-quality-based VoNR-to-VoLTE handover to enable the UE to use an LTE network with better signal quality to make a voice call. This reduces the packet loss rate and improves user experience.	5.9 VoNR-to-VoLTE Handover Based on Voice Quality
Voice-capability-based redirection back to the HPLMN	Inter-RAT redirection/reselection back to the home public land mobile network (HPLMN) based on roaming UE identification applies when the NR coverage area of the serving PLMN of a roaming UE overlaps the LTE coverage area of the HPLMN of the UE. This function can quickly transfer a VoNR-incapable UE back to an LTE cell on a frequency that belongs to the UE's HPLMN using redirection or reselection.	5.10 Voice-Capability-based Redirection Back to the HPLMN
NSA/SA selection based on downlink traffic volume	In NSA and SA hybrid networking of NR cells, UEs in connected mode that support both NSA and SA can access NR cells in SA or NSA mode. When SA coverage is poor, the gNodeB initiates SA-to-NSA switching based on downlink traffic volume after identifying a UE with high downlink traffic, in order to provide better service experience for the UE.	<i>NSA-SA Selection Based on User Experience</i>
NSA/SA selection based on uplink coverage	In NSA and SA hybrid networking of NR cells, UEs in connected mode that support both NSA and SA can access NR cells in SA or NSA mode. Based on the coverage conditions of the two networking modes, the gNodeB initiates SA-to-NSA switching based on weak uplink coverage or NSA-to-SA switching based on strong uplink coverage.	<i>NSA-SA Selection Based on User Experience</i>

 NOTE

Emergency call UEs can be treated specially on the eNodeB side. If the **EMERG_CALL_NR_RESTRICTION_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter is selected and the eNodeB identifies a UE as an emergency call UE, various LTE-to-NR mobility functions do not take effect for the UE. These functions include downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN, E-UTRAN to NG-RAN traffic steering, and fast return from E-UTRAN to NG-RAN. These functions take effect for the UE again after the eNodeB detects that the UE has exited the emergency call state. This treatment requires that the emergency call UE identification function be enabled in LTE cells. For details, see Emergency Call in *eRAN Feature Documentation*.

NR cells support interoperability with LTE cells in NSA and SA hybrid networking. The interoperability process is the same as that in SA networking.

In this document, any automatic neighbor relation (ANR) function on the LTE side refers to the inter-RAT event-triggered ANR function (controlled by the **NR_EVENT_ANR_SW** option of the **CellAlgoSwitch.AnrFunctionSwitch** parameter). For details about the inter-RAT event-triggered ANR function and the ANR's requirements for UEs, see ANR Management in *eRAN Feature Documentation*.

In this document, any ANR function on the NR side refers to the inter-RAT ANR function (controlled by the **NR_EUTRAN_ANR_SW** option of the **NRCCellAlgoSwitch.AnrSwitch** parameter). For details about the inter-RAT ANR function and the ANR's requirements for UEs, see *ANR* in *5G RAN Feature Documentation*.

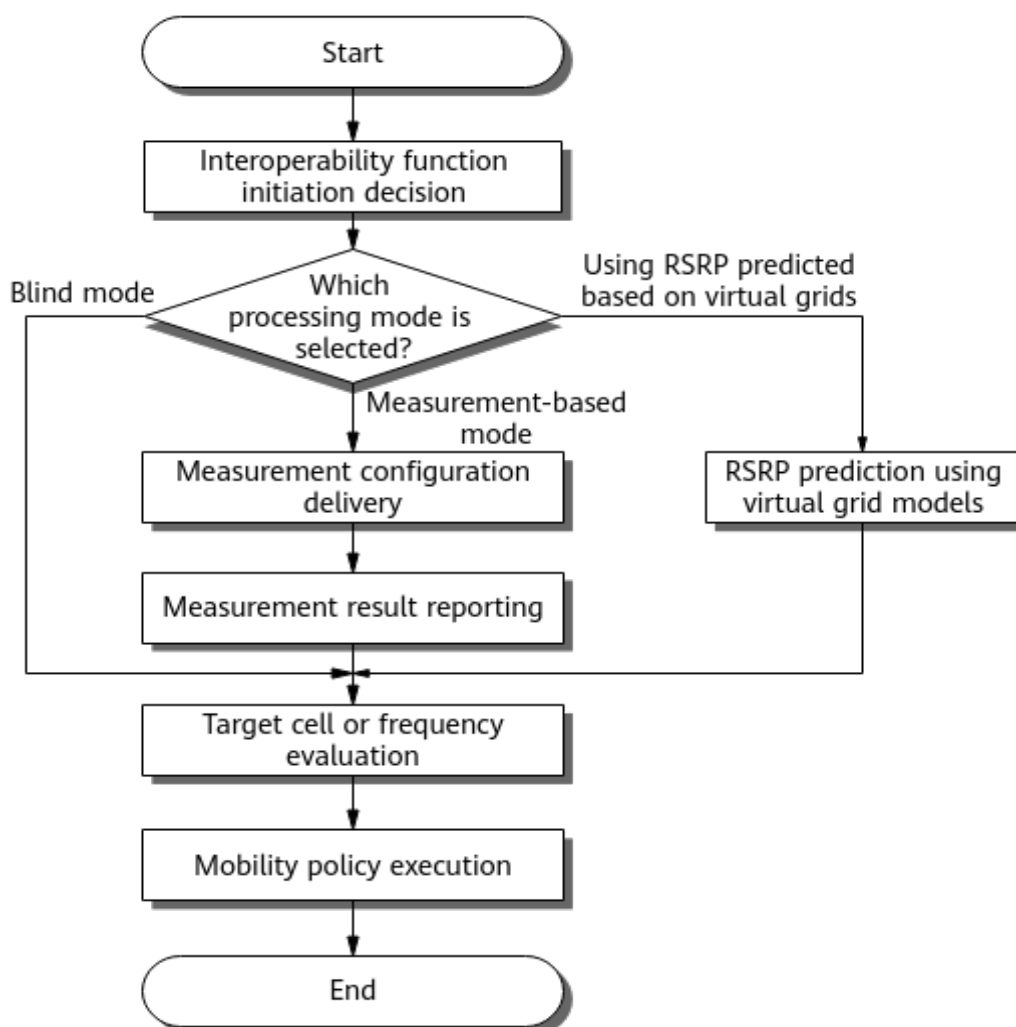
4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode

This chapter describes the basic procedure for interoperability between E-UTRAN and NG-RAN. Unless otherwise specified, this procedure applies to all functions of interoperability between E-UTRAN and NG-RAN.

4.1 Principles

4.1.1 Overall Procedure

Figure 4-1 shows the basic procedure for interoperability between E-UTRAN and NG-RAN.

Figure 4-1 Basic procedure for interoperability between E-UTRAN and NG-RAN

4.1.2 Function Initiation Determination

Handover function initiation determination refers to the process of determining whether to initiate a handover function based on the function-specific initiation conditions. For details about those specific initiation conditions for each function, see [Table 4-1](#).

4.1.3 Processing Mode Selection

After a handover function is started, the base station selects a processing mode based on the signal quality of the serving cell and related events. **Depending on whether a UE measures the signal quality of candidate target cells before a handover, the processing mode consists of measurement-based mode, blind mode, and the mode using RSRP prediction based on virtual grids.**

- Measurement-based mode

A UE measures the signal quality of candidate cells and the base station generates a target cell list based on measurement reports. For details, see [4.1.4 Measurement Configuration](#) and [4.1.6 Measurement Reporting](#).

- Blind mode
A UE does not measure the signal quality of candidate cells, and the base station generates a target cell or frequency list based on priority parameter settings. The UE is likely to experience an access failure in the target cell in this mode. Therefore, this mode is used only when an interoperability procedure needs to be performed immediately.
- RSRP prediction based on virtual grids
The virtual grid models are used to predict the RSRP values of candidate target cells. For details, see [4.1.7 RSRP Prediction Based on Virtual Grids](#).

For details about the processing modes supported by each function, see [Table 4-1](#).

Table 4-1 Initiation conditions and supported processing modes for each function

Function	Initiation Condition	Processing Mode	Handover Policy	Reference
Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	• The MOBILITY_TO_E_UTRAN_SW option of the NRCellAlgoSwitch.InterRatServiceMobilitySwitch parameter is selected. • The DL_COV_N2L_MOB_OFF_SWITCH option of the NRInterRatHoParam.HoModeSwitch parameter is deselected. • The DL_COV_N2L_MOB_OFF_SWITCH option of the NRCellOpPolicy.InterRatHoPolicySwitch parameter is deselected. • An event A2 measurement report is received.	• Measurement-based mode • Blind mode	• Handover • Redirection • Blind redirection	5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN

Function	Initiation Condition	Processing Mode	Handover Policy	Reference
Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	- The UL_SINR_MOBILITY_TO_EUTRAN_SW and MOBILITY_TO_EUTRAN_SW options of the NRCellAlgoSwitch. <i>InterRatServiceMobilitySw</i> parameter are selected. - The UE is identified as a UE under weak uplink coverage. - Any of the following conditions is met: - The VOICE_UL_COV_BASED_HO_SW option of the NRCellAlgoSwitch. <i>VoiceStrategySwitch</i> parameter is deselected. - Assume that the VOICE_UL_COV_BASED_HO_SW option of the NRCellAlgoSwitch. <i>VoiceStrategySwitch</i> parameter is selected. When the UE has no 5QI 1 bearer, the VOLUME_IDENTIFIER_SW option of the	Measurement-based mode	- Handover - Redirection	5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN

Function	Initiation Condition	Processing Mode	Handover Policy	Reference
	NRCellAlgoSwitch.ServiceFunctionSwitch parameter is selected and the UE traffic volume meets the threshold. The VOICE_UL_COV_BASED_HO_SWITCH option of the NRCellAlgoSwitch.VoiceStrategySwitch parameter is selected and the UE has a 5QI 1 bearer.			
Downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN	- The INTER_RAT_MOBILITY_TO_NRSW option of the CellAlgoExtSwitch.HoAllowedSwitch parameter is selected. - The NrHoSwitch or NrRedirectSwitch option of the CellHoParaCfg.HoModeSwitch parameter is selected. - An event A2 measurement report is received.	Measurement-based mode	- Handover - Redirection	5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN

Function	Initiation Condition	Processing Mode	Handover Policy	Reference
E-UTRAN to NG-RAN traffic steering	• The ServBasedNrHoSwitch option of the CellAlgoSwitch.HoAllowedSwitch parameter is selected. • The NrHoSwitch or NrRedirectSwitch option of the CellHoParaCfg.HoModeSwitch parameter is selected. If both options are selected, whether handover can be performed for the UE is preferentially evaluated. If handovers fail, the procedure ends.	• Measurement-based mode • RSRP prediction based on virtual grids	• Handover • Redirection	5.6 E-UTRAN to NG-RAN Traffic Steering

Function	Initiation Condition	Processing Mode	Handover Policy	Reference
Voice fallback from NG-RAN to E-UTRAN	Normal voice fallback and fast voice fallback: - The EPS_FB_SWITCH option of the NRCellAlgoSwitch.VoiceStrategySwitch parameter is selected. - Fast voice fallback also requires that the EPS_FB_IN_ADVANCE_MEAS_SWITCH option of the NRCellAlgoSwitch.VoiceStrategySwitch parameter be selected. Emergency voice fallback: This function is enabled by default and is not controlled by parameters.	- Measurement-based mode - RSRP prediction based on virtual grids - Blind mode	- Handover - Redirection - Blind handover - Blind redirection	5.7 Voice Fallback from NG-RAN to E-UTRAN

Function	Initiation Condition	Processing Mode	Handover Policy	Reference
Fast return from E-UTRAN to NG-RAN	• The FAST_RETURN_TO_NR_SWITCH option of the CellAlgoExtSwitch.HoAllowedSwitch parameter is selected. • The NrHoSwitch or NrRedirectSwitch option of the CellHoParaCfg.HoModeSwitch parameter is selected. Alternatively, the FastReturnBlindRedirectSwitch and NrRedirectSwitch options of the CellHoParaCfg.HoModeSwitch parameter are both selected. If all the above options are selected, the mobility policy evaluation for the UE proceeds in sequence: blind redirection first, then measurement-based handover, and finally measurement-based redirection.	• Measurement-based mode • RSRP prediction based on virtual grids • Blind mode	• Handover • Redirection • Blind redirection	5.8 Fast Return from E-UTRAN to NG-RAN

Function	Initiation Condition	Processing Mode	Handover Policy	Reference
VoNR-to-VoLTE handover based on voice quality	- The MOBILITY_TO_EUTRAN_SW option of the NRCellAlgoSwitch.InterRatServiceMobilitySwitch parameter is selected. - The QLTY_BASED_EUTRAN_HO_SW option of the gNBQciAlgoParamGrp.QciAlgoSwitch parameter is selected.	Measurement-based mode	Handover	5.9 VoNR-to-VoLTE Handover Based on Voice Quality
Voice-capability-based redirection back to the HPLMN	- The UE is a roaming VoNR-incapable UE. - The VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter is selected. - In the frequency list group bound to the roaming UE, there is a frequency with gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag set to TRUE, and this frequency is configured as a neighboring frequency of the serving frequency.	Measurement-based mode	- Redirection - Reselection	5.10 Voice-Capability-based Redirection Back to the HPLMN

4.1.4 Measurement Configuration

After a radio bearer (SRB3) is set up for a UE, the eNodeB or gNodeB delivers measurement configurations to the UE based on the mobility trigger and policy settings. If the measurement configurations are updated, the eNodeB or gNodeB sends an RRC Connection Reconfiguration or RRC Reconfiguration message to deliver the updated measurement configurations.

Measurement configurations include:

- Measurement objects
Measurement objects include RATs and frequencies. They specify which frequencies for the UE to measure.
- Measurement events
Measurement events indicate the signal quality of the serving cell and neighboring cells. Measurement event configurations specify the conditions of and criteria for measurement reporting.
- Other configurations
Other configurations involve measurement gaps and measurement object preemption.

4.1.4.1 Measurement Objects

The eNodeB or gNodeB selects a target RAT for measurement and then obtains the frequencies or cells to be measured from the neighboring cell list of the selected RAT. [Table 4-2](#) lists the key attributes of measurement objects for different RATs.

Table 4-2 Measurement object configurations

RAT	Configuration
NR	Set the following key attributes of E-UTRAN: • E-UTRA absolute radio frequency channel number (EARFCN), which is specified by the NRCellEutranNFreq.DIEarfcn parameter • Measurement bandwidth, which is specified by the NRCellEutranNFreq.MeasurementBandwidth parameter • Frequency-specific offset for UEs in connected mode, which is specified by the NRCellEutranNFreq.FreqSpecificOffset parameter
LTE	Set the following key attributes of NG-RAN: • NR-ARFCN, which is specified by the NrNFreq.DlArfcn parameter • Measurement bandwidth, which is specified by the NrNFreq.NrDlBandwidth parameter • Frequency-specific offset for UEs in connected mode, which is specified by the NrNFreq.FreqSpecificOffset parameter

In addition, in multi-operator sharing scenarios, NR/LTE frequency policy parameters can be configured on a per operator basis for inter-RAT interoperability in connected mode. For details, see [6.1.1.3 Operator-specific NR Frequency Policy](#) and [6.1.1.4 Operator-specific LTE Frequency Configuration](#).

4.1.4.1.1 Measurement Object Selection

The eNodeB or gNodeB selects a RAT based on measurement events.

- The eNodeB selects LTE to measure for events A1 and A2 and selects NR to measure for events B1 and B2.
- The gNodeB selects NR to measure for events A1 and A2 and selects LTE to measure for events B1 and B2.

4.1.4.1.2 NR Frequency Selection by the eNodeB

The eNodeB filters NR frequencies based on the rules described in [Frequency Filtering](#), and then filters neighboring cells on the remaining NR frequencies based on the rules described in [Neighboring Cell Filtering](#). The eNodeB selects the NR frequencies with at least one NR neighboring cell from the remaining ones and delivers measurement configurations of them to a UE. The total number of frequencies contained in the measurement configurations cannot exceed the maximum number described in [Frequency Delivery Specifications](#).

Frequency Filtering

The eNodeB filters NR frequencies based on the following rules:

- The eNodeB filters out NR frequencies not supported by the UE. For details about the fields related to UE capabilities, see "UE requirements" in the "Others" sections for the mobility procedures from E-UTRAN to NG-RAN in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).
- In roaming scenarios, if no N26 interface is configured between the home LTE core network and the visited NR core network, the eNodeB does not deliver measurement configurations of a neighboring NR frequency that meets the following conditions to VoNR-incapable UEs: (1) The frequency belongs to the visited network. (2) The **VOICE_NR_HO_POLICY_INDICATION** option of the **NrNFreq.AggregationAttribute** parameter is selected for the frequency.
- If ANR has been enabled on the LTE network, the eNodeB obtains the frequencies to be measured from the neighboring frequencies maintained by ANR (namely the frequencies for which the **ANR_INDICATION** option of the **NrNFreq.AggregationAttribute** parameter is selected), filters out the neighboring NR frequencies not supported by the UE, and delivers the remaining frequencies in the measurement configuration to the UE (without filtering the frequencies using the neighboring cell filtering rules).
- If ANR is not enabled on the LTE side, the eNodeB filters out certain neighboring cells from the neighboring relation table (NRT). (For details, see [Neighboring Cell Filtering](#).) The eNodeB selects the NR frequencies with at least one NR neighboring cell from the remaining ones and delivers measurement configurations of them to a UE.
- If a neighboring NR frequency belongs to multiple frequency bands, the eNodeB does not include the neighboring NR frequency in the measurement configuration if the frequency meets any of the following conditions:

- The **NrMfbiFreq** MO is configured on the eNodeB, the **NR_MULTIBAND_SUPPORT_SW** option of the **EnodebAlgoExtSwitch.NrMfbiAlgoSwitch** parameter is selected, and the UE does not support the frequency bands specified by the **NrMfbiFreq.FrequencyBand** and **NrMfbiFreq.AdditionalFrequencyBand** parameters.
- The **NrMfbiFreq** MO is not configured on the eNodeB or the **NR_MULTIBAND_SUPPORT_SW** option of the **EnodebAlgoExtSwitch.NrMfbiAlgoSwitch** parameter is deselected.
- For a RedCap UE (the value of the **supportOfRedCap-r17** field in the *UE-NR-Capability* IE is **supported**), if the policy for handovers of RedCap UEs to NG-RAN (controlled by the **REDCAP_UE_NR_HO_POLICY_SW** option of the **EnodebAlgoExtSwitch.NrHandoverAlgoSwitch** parameter) is enabled, and the RedCap UE supports 1RX or 2RX but the corresponding external NR cell does not support 1RX or 2RX (the **NrExternalCell.CellBarredRedCap** parameter is set to a value other than **NOT_BARRED**), the eNodeB filters out the cell. If all cells on an NR frequency are filtered out, the eNodeB filters out the NR frequency. For details about RedCap UEs, see "RedCap Introduction" in *RedCap in 5G RAN Feature Documentation*.

If the *UE-NR-Capability* IE of a RedCap UE does not contain the

supportOfRedCap-r17 field, it is recommended that the

REDCAP_LTE_TO_NR_MOB_SW option of the

UeCompat.WhitelistCtrlExtSwitch1 parameter be selected for the RedCap

UE. In this way, LTE-to-NR mobility for RedCap UEs and the NR handover

policy for RedCap UEs can take effect for whitelisted RedCap UEs. For details about LTE-to-NR mobility for RedCap UEs, see *Terminal Awareness Differentiation* in *eRAN Feature Documentation*.

- The rules for selecting frequencies to deliver vary depending on whether flexible user steering is enabled:
 - If flexible user steering is enabled, the eNodeB selects frequencies based on the SPID of the UE.
The **SPID_L2N_TARGET_FREQ_CTRL_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter must be selected, and frequencies are configured using the **RatFreqPriorityGroup.DLEarfcn** parameter for the SPID. The eNodeB selects frequencies for the UE based on these configurations.
 - If flexible user steering is not enabled, the eNodeB selects frequencies for measurements in descending order of priorities specified by the **NrNFreq.ConnFreqPriority** parameter. A larger value indicates a higher priority. The value **0** indicates the lowest priority. If multiple NR frequencies are assigned the same value of the **NrNFreq.ConnFreqPriority** parameter, frequencies to deliver are randomly selected. The total number of frequencies contained in the measurement configurations cannot exceed the maximum number described in [Frequency Delivery Specifications](#).
- If the **FORBID_NEC_HO_SW** option of the **RatFreqPriorityGroup.AggregationAttribute** parameter is selected, necessary handovers or redirections to frequencies specified by the **RatFreqPriorityGroup.DLEarfcn** parameter are prohibited for UEs with SPIDs or bearers of QCI associated with the **RatFreqPriorityGroup** MO. In addition, the base station does not deliver measurement configurations related to the

corresponding frequencies to these UEs. However, this function does not affect ANR, MR, and MDT measurements.

- If the **FORBID_UNNEC_HO_SW** option of the **RatFreqPriorityGroup.AggregationAttribute** parameter is selected, unnecessary handovers or redirections to frequencies specified by the **RatFreqPriorityGroup.DLEarfcn** parameter are prohibited for UEs with SPIDs or bearers of QCI associated with the **RatFreqPriorityGroup** MO. In addition, the base station does not deliver measurement configurations related to the corresponding frequencies to these UEs. However, this function does not affect ANR, MR, and MDT measurements.
- When the **NR_TARGET_FREQ_CTRL_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter is selected, for a UE for which a bearer with a QCI (specified by the **CellQciPara.Qci** parameter) has been set up, the eNodeB only allows the UE to be handed over to cells on the NR frequency whose downlink frequency number is specified by the **RatFreqPriorityGroup.DLEarfcn** parameter in the **RatFreqPriorityGroup** MO (identified by the **RatFreqPriorityGroup.RatFreqPriorityGroupId** parameter) associated with the **CellQciPara** MO through the **CellQciPara.RatFreqPriorityGroupId** parameter.
- If the **FREQ_MEAS_FLAG** option of the **NrNFreq.AggregationAttribute** parameter is deselected, the eNodeB does not deliver measurement configurations related to the neighboring NR frequency.
- If the **DSS_FREQ_INDICATION** option of the **NrNFreq.AggregationAttribute** parameter is selected but the UE does not support CRS rate matching, the eNodeB does not deliver measurement configurations related to the neighboring NR frequency. For details about CRS rate matching, see *LTE FDD and NR Dynamic Spectrum Sharing*.
- If the **NR_DL_SCC_FREQ_ONLY_IND** option of the **NrNFreq.AggregationAttribute** parameter is selected, the eNodeB does not deliver measurement configurations related to the neighboring NR frequency.
- After frequency-specific LTE-NR interoperability prohibition is enabled, the eNodeB can block (i) the delivery of measurement configurations for event B1 related to specific frequencies, and (ii) interoperability procedures between LTE and NR—including handovers, redirections, and reselections—toward those frequencies. This function is controlled by the **L2NR_MOB_FORBID_FLAG** option of the **NrNFreq.AggregationAttribute** parameter.

Neighboring Cell Filtering

Neighboring cells that meet any of the following conditions are filtered out:

- Neighboring cells that belong to PLMNs (including the PLMNs of primary and secondary operators) not on the PLMN list of the serving cell. This list is composed of the following PLMNs:
 - PLMN that the serving cell belongs to
 - PLMNs configured in the **InterPlmnHoList** MO when the **InterPlmnHoSwitch** option of the **ENodeBAlgoSwitch.HoAlgoSwitch** parameter is selected
 - Equivalent PLMNs obtained by the eNodeB from the *Handover Restriction List* IE sent from the MME when both the **InterPlmnHoSwitch**

and **EPlmnSwitch** options of the **ENodeBAlgoSwitch.HoAlgoSwitch** parameter are selected

- Blacklisted NR cells
- NSA-only NR cells, that is, cells with the **NrExternalCell.NrNetworkingOption** parameter set to **NSA**
- Neighboring NR cells with Extended Cell Range enabled, for which the **OVER_DISTANCE_FLAG** option of the **NrNRRelationship.AggregationAttribute** parameter is selected
- (Applicable only when the NR side is working in RAN sharing with dedicated carrier mode) Neighboring cells of secondary operators, when the handover is a handover to the primary PLMN of the neighboring cells (A neighboring NR cell is a cell of secondary operators if the **NrExternalCell.MasterPlmnReservedFlag** parameter is set to **TRUE** for that cell.)
- (Applicability dependent on the **NrBwCheckSwitch** setting of the **ENodeBAlgoSwitch.CompatibilityCtrlSwitch** parameter) Any cell whose NR cell bandwidth and minimum NR carrier bandwidth are both different from any cell bandwidth supported by the UE. The downlink and uplink NR cell bandwidth are specified by the **NrNFreq.NrDLBandwidth** and **NrNFreq.NrULBandwidth** parameters, respectively. The downlink and uplink minimum NR carrier bandwidth is specified by the **NrNFreq.NrDLMinCarrierBw** and **NrNFreq.NrULMinCarrierBw** parameters, respectively. If both the **NrNFreq.NrDLBandwidth** and **NrNFreq.NrULBandwidth** parameters are set to **NOT_CONFIG**, these NR cells are not filtered out.
- For a RedCap UE (the value of the **supportOfRedCap-r17** field in the **UE-NR-Capability** IE is **supported**), if the **REDCAP_UE_NR_HO_POLICY_SW** option of the **ENodeBAlgoExtSwitch.NrHandoverAlgoSwitch** parameter is selected, and the RedCap UE supports 1RX or 2RX but the corresponding external NR cell does not support 1RX or 2RX (the **NrExternalCell.CellBarredRedCap** parameter is set to a value other than **NOT_BARRED**), the eNodeB filters out the NR cell.
- If (1) a neighboring NR cell experiences a PCI conflict or missing configuration, and (2) the ANR function is disabled, then the eNodeB filters out that neighboring NR cell.

Frequency Delivery Specifications

The maximum number of NR frequencies that can be contained in the measurement configurations delivered by the eNodeB is specified by the **Max NR Measure Freq Number** parameter. The number of NR frequencies that can be contained in the measurement configurations delivered by the eNodeB must also be less than or equal to the difference between **NSA Max LTE Inter-Freq Measurement Object Num** and the number of non-NR frequencies that have been delivered. That is, the maximum number of NR frequencies that can be contained in the measurement configurations delivered by the eNodeB is $\min(\text{Max NR Measure Freq Number}, \text{NSA Max LTE Inter-Freq Measurement Object Num} - \text{Number of non-NR frequencies that have been delivered})$. For example, if **Max NR Measure Freq Number** is set to eight and the difference between **NSA Max LTE Inter-Freq Measurement Object Num** and the number of non-NR frequencies contained in the measurement configurations is six, the measurement

configurations contain six NR frequencies at most. [Table 4-3](#) lists the parameter names and corresponding parameter IDs.

Table 4-3 Parameter IDs corresponding to the parameter names

Parameter Name	Parameter ID
Max NR Measure Freq Number	CellUeMeasControlCfg.MaxNrMeasFreqNum
NSA Max LTE Inter-Freq Measurement Object Num	CellUeMeasControlCfg.NsaMaxLteInterFMeasObjNum

4.1.4.1.3 LTE Frequency Selection by the gNodeB

The gNodeB filters frequencies and then delivers measurement configurations containing the filtered frequencies based on frequency delivery specifications.

Frequency Filtering

[Table 4-4](#) describes the common filtering rules.

Table 4-4 Common filtering rules used by the gNodeB for LTE frequencies

No.	Rule
1	The bands and frequencies that are not supported by the UE are filtered out based on the UE band capability.
2	All inter-RAT frequencies are filtered out if the UE does not support inter-RAT measurements.
3	All inter-RAT frequencies are filtered out if the UE supports inter-RAT measurements but does not support inter-RAT redirections nor inter-RAT handovers.

No.	Rule
4	When the MOBILITY_RESTRICTION_SW option of the gNBMMobilityCommParam.MobilityAlgoSwitch parameter is selected, the gNodeB restricts UE handovers to E-UTRAN based on the <i>Mobility Restriction List</i> IE sent from the AMF. If the UE is not performing an emergency call (emergency voice fallback or emergency VoNR call), the gNodeB filters out all LTE frequencies in an available PLMN of the UE if the PLMN meets any of the following conditions: • In the <i>Mobility Restriction List</i> IE received by the gNodeB, the <i>RAT Restrictions</i> IE is present and contains the <i>RAT Restriction Information</i> IE that indicates "e-UTRA (0)" for the PLMN. • In the <i>Mobility Restriction List</i> IE received by the gNodeB, the <i>Core Network Type Restriction for Serving PLMN</i> IE and the <i>Core Network Type Restriction for Serving PLMN</i> IE that indicates "EPCForbidden" for the serving PLMN of the UE are present. • In the <i>Mobility Restriction List</i> IE received by the gNodeB, the <i>Core Network Type Restriction for Equivalent PLMNs</i> IE and the <i>Core Network Type Restriction</i> IE that indicates "EPCForbidden" for the available PLMN of the UE are present.
5	In blind processing mode, inter-RAT frequencies for which NRCellEutranNFreq.VolteBlindPriority is set to 0 or 255 are filtered out for EPS fallback and VoNR-to-VoLTE redirection; inter-RAT frequencies for which NRCellEutranNFreq.VolteBlindPriority is set to 255 are filtered out for emergency voice fallback. In other processing modes, inter-RAT frequencies for which NRCellEutranNFreq.VoltePriority is set to 255 are filtered out for EPS fallback, emergency voice fallback, VoNR-to-VoLTE handover, and VoNR-to-VoLTE redirection.
6	Frequencies for which NRCellEutranNFreq.MeasActvFlag is set to INACTIVE are filtered out.
7	Frequencies for which the HIGH_SPEED_UE_REDIRECT_ONLY_SW option of the NRCellEutranNFreq.FrequencyCtrlSwitch is selected are filtered out. However, this filtering rule does not take effect if the HIGH_SPEED_UE_REDIRECT_SW option of the NRCellAlgoSwitch.HighSpeedDedPolicySw parameter is also selected.
8	Frequencies for which NRCellEutranNFreq.FreqIntrffFlag is set to INTRF are filtered out for unnecessary handovers, but not for necessary handovers. Necessary handovers include downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, voice fallback from NG-RAN to E-UTRAN, and VoNR-to-VoLTE handover based on voice quality.

No.	Rule
9	A frequency for which gNBFreqPriorityGroup.RoamingUeHplmnFreq-Flag is set to FALSE in the frequency list group associated with a roaming UE is filtered out from the measurement configurations for the local operator's UEs that do not support VoNR on the frequency, when the VOICE_INTER_FREQ_HO_POLICY_SW option of the NRCellFreqRelation.AggregationAttribute parameter is selected.
10	The frequency selection rules vary depending on whether flexible user steering is enabled: • If flexible user steering is enabled, the gNodeB selects frequencies based on the RFSP index of the UE. The NR_TO_EUTRAN_FREQ_LIST_CTRL_SW option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter must be selected, and frequencies must be configured using the gNBFreqPriorityGroup.DLEarfcn parameter for the RFSP. The gNodeB selects frequencies for measurements in descending order of priorities specified by gNBFreqPriorityGroup.ConnFreqPriority or gNBFreqPriorityGroup.VoltePriority. Note that, if gNBFreqPriorityGroup.ConnFreqPriority (for data services) is set to 65535, the gNodeB selects frequencies for measurements in descending order of priorities specified by NRCellEutranNFreq.ConnFreqPriority; if gNBFreqPriorityGroup.VoltePriority (for voice services) is set to 65535, the gNodeB does so by NRCellEutranNFreq.VoltePriority. During blind handovers or blind redirections for voice services, the gNodeB selects frequencies for measurements in descending order of priorities specified by gNBFreqPriorityGroup.VolteBlindPriority. Note that, if gNBFreqPriorityGroup.VolteBlindPriority is set to 65535, the gNodeB selects frequencies for measurements in descending order of priorities specified by NRCellEutranNFreq.VolteBlindPriority. • If flexible user steering is not enabled and the value of gNBRfspConfig.RfspIndex is invalid, the gNodeB selects frequencies for measurements in descending order of priorities specified by NRCellEutranNFreq.ConnFreqPriority or NRCellEutranNFreq.VoltePriority. Note that frequencies for which NRCellEutranNFreq.VoltePriority is set to 255 are filtered out. If multiple LTE frequencies have the same priority, the frequencies with smaller downlink EARFCNs are preferentially selected. During blind handovers or blind redirections for voice services, the gNodeB selects frequencies for measurements in descending order of priorities specified by NRCellEutranNFreq.VolteBlindPriority.
11	In load-based inter-RAT mobility from NG-RAN to E-UTRAN, if the MOB_INTERRAT_FREQ_FILTER_SW option of the NRCellEutranNFreq.FrequencyCtrlSwitch parameter is selected, the gNodeB does not include the corresponding frequency in the measurement configurations related to inter-RAT event B1.

Frequency Delivery Specifications

The gNodeB can deliver a maximum of 13 LTE frequencies for measurement, including a maximum of 7 LTE FDD frequencies and a maximum of 7 LTE TDD frequencies. For LTE measurements by a UE in NG-RAN, the gNodeB delivers LTE cell information in addition to frequency information when the cell individual offset (CIO) for the LTE cells (specified by the **NRCellEutraNRelation.EutraCellIndividualOffset** parameter) has a non-zero value.

4.1.4.1.4 Blacklisted Cells for Cell Measurement

When the **NRCellMobilityConfig.MeasExcludedCellListSwitch** parameter is set to **ON** in the serving cell, the gNodeB includes the blacklisted cells in the *excludedCellsToAddModList* IE in *measObjectNR* during measurement configuration delivery. If a UE receives the measurement configuration and finds that *measObjectNR* includes the *excludedCellsToAddModList* IE, the UE does not measure those blacklisted cells specified in that IE during neighboring cell measurement. Some UEs may still send measurement reports of blacklisted cells to the gNodeB. If this is the case, the gNodeB filters out these cells during target cell determination, and does not deliver reportCGI measurement configurations related to these cells to UEs.

Specifically:

- Cell blacklist configuration:

Blacklisted cells can be configured for a cell in two steps: configuring multiple cell blacklists on the base station, and associating a cell blacklist with the cell.

- a. Use the **gNBCellBlacklist** MO on the gNodeB to configure cell blacklists.
 - i. Set the **gNBCellBlacklist.CellBlacklistId** parameter to specify a cell blacklist. Note that the **gNBCellBlacklist.RatType** parameter must be set to **EUTRAN** for the cell to which the **gNBCellBlacklist.CellBlacklistId** parameter is mapped.
 - ii. Set the start physical cell identifier (PCI) and PCI range to specify a blacklisted cell range. Note that a maximum of 16 blacklisted cell ranges can be configured for each cell blacklist. *N* consecutive PCI-specific cells starting from the one with the specified start PCI are blacklisted.

Specifically:

- The start physical cell ID is specified by the **gNBCellBlackList.StartPhysicalCellId** parameter.
 - The physical cell ID range is specified by the **gNBCellBlackList.PhysicalCellIdRange** parameter.
 - *N* is a positive integer to which the value of the **gNBCellBlackList.PhysicalCellIdRange** parameter is mapped.
- b. Associate a cell blacklist to a cell on the gNodeB.
Associate a cell blacklist (specified by the **gNBCellBlacklist.CellBlacklistId** parameter) with a cell using the **NRCellEutraNFreq.EutraExcludedCellListId** parameter. This way, the E-UTRAN cell blacklist associated with the cell determines the blacklisted E-UTRAN cells for the local cell.

 NOTE

- If the **NRCeUtranNFreq.EutranExcludedCellListId** parameter is not set to **65535**, its value must be one of the values of the **gNBCellBlacklist.CellBlacklistId** parameter, and the corresponding **gNBCellBlacklist.RatType** parameter must be set to **EUTRAN**.
- The values of **gNBCellBlacklist.RatType** in all records with the same **gNBCellBlacklist.CellBlacklistId** value must be the same.
- Cell blacklist delivery specifications:
A frequency can be associated with only one cell blacklist, and each cell blacklist can be configured with a maximum of 16 blacklisted cell ranges.

4.1.4.2 Reporting Configurations

Reporting configurations include:

- **Measurement event, which can be A1, A2, A3, A4, B1, or B2.** For details about the definition, entering conditions, and leaving conditions for each event, see [4.1.4.2.1 Measurement Events](#).
- **Triggering quantity, which specifies the policy used to trigger event reporting and can be the reference signal received power (RSRP), reference signal received quality (RSRQ), or signal to interference plus noise ratio (SINR).** For details, see [4.1.4.2.2 Triggering Quantity](#).

4.1.4.2.1 Measurement Events

In measurement-based mode, UEs report measurement events and the base station acquires cell signal quality from the measurement reports. The measurement events involved in this document are defined as follows:

- Event A1 indicates that the signal quality of the serving cell becomes higher than a specific threshold.
- Event A2 indicates that the signal quality of the serving cell becomes lower than a specific threshold.
- Event A3 indicates that the signal quality of a neighboring cell is higher than that of the serving cell by a certain offset.
- Event B1 indicates that the signal quality of an inter-RAT neighboring cell becomes higher than a specific threshold. Event B1 used for NG-RAN measurements in E-UTRAN and event B1 used for E-UTRAN measurements in NG-RAN differ in their entering and leaving conditions.
- Event B2 indicates that the signal quality of the serving cell drops below threshold 1 and the signal quality of an inter-RAT neighboring cell becomes higher than threshold 2. Event B2 used for NG-RAN measurements in E-UTRAN and event B2 used for E-UTRAN measurements in NG-RAN differ in their entering and leaving conditions.

Events A1 and A2 are used to evaluate the signal quality of the serving cell in the interoperability function initiation decision phase. Events B1 and B2 are used to evaluate the signal quality of neighboring cells in the target cell or frequency decision phase.

[Table 4-5](#) presents the entering and leaving conditions for these events.

Table 4-5 Entering and leaving conditions for events

Event Type	Entering Condition	Leaving Condition	Setting Notes
A1	$(Ms - Hys > Thresh)$ is true throughout <i>TimeToTrig</i> .	$(Ms + Hys < Thresh)$ is true throughout <i>TimeToTrig</i> .	For details, see Table 4-6 and Table 4-7 .
A2	$(Ms + Hys < Thresh)$ is true throughout <i>TimeToTrig</i> .	$(Ms - Hys > Thresh)$ is true throughout <i>TimeToTrig</i> .	For details, see Table 4-8 and Table 4-9 .
A3	$(Mn + Ofn + Ocn - Hys > Ms + Ofn + Ocs + Off)$ is true throughout <i>TimeToTrig</i> .	$(Mn + Ofn + Ocn + Hys < Ms + Ofn + Ocs + Off)$ is true throughout <i>TimeToTrig</i> .	For details, see Event A3 .
E-UTRAN side: event B1 related to neighboring NR cells	$(Mn + Ofn - Hys > Thresh)$ is true throughout <i>TimeToTrig</i> .	$(Mn + Ofn + Hys < Thresh)$ is true throughout <i>TimeToTrig</i> .	For details, see Table 4-13 .
E-UTRAN side: event B2 related to neighboring NR cells	$(Ms + Hys < Thresh1)$ and $(Mn + Ofn - Hys > Thresh2)$ are true throughout <i>TimeToTrig</i> .	$(Ms - Hys > Thresh1)$ or $(Mn + Ofn + Hys < Thresh2)$ is true throughout <i>TimeToTrig</i> .	For details, see Table 4-15 .
NG-RAN side: event B1 related to neighboring LTE cells	$(Mn + Ofn + Ocn - Hys > Thresh)$ is true throughout <i>TimeToTrig</i> .	$(Mn + Ofn + Ocn + Hys < Thresh)$ is true throughout <i>TimeToTrig</i> .	For details, see Table 4-12 .

Event Type	Entering Condition	Leaving Condition	Setting Notes
NG-RAN side: event B2 related to neighboring LTE cells	$(Ms + Hys < Thresh1)$ and $(Mn + Ofn + Ocn - Hys > Thresh2)$ are true throughout <i>TimeToTrig</i> .	$(Ms - Hys > Thresh1)$ or $(Mn + Ofn + Ocn + Hys < Thresh2)$ is true throughout <i>TimeToTrig</i> .	For details, see Table 4-14 .

The variables in the preceding table are explained as follows:

- *Ms* is the measurement result of the serving cell.
- *Mn* is the measurement result of a neighboring cell.
- *Hys* is the hysteresis for an event.
- *Thresh*, *Thresh1*, and *Thresh2* are the thresholds for the event.
- *TimeToTrig* is the time-to-trigger, throughout which a condition is met before the event can be reported.
- *Ofn* is the frequency-specific offset of the neighboring cell.
- *Ocn* is the cell-specific offset of the neighboring cell.
- *Ofs* is the frequency-specific offset of the serving cell.
- *Ocs* is the cell-specific offset of the serving cell.
- *Off* is the offset for the event.

In addition, in multi-operator sharing scenarios, measurement event parameters can be configured on a per operator basis for inter-RAT interoperability in connected mode. For details, see [6.1.1.1 Operator-specific Measurement Variables](#).

Event A1

Entering condition for event A1: $(Ms - Hys > Thresh)$ is true throughout the duration specified by *TimeToTrig*. [Figure 4-2](#) shows how event A1 is triggered. [Table 4-6](#) and [Table 4-7](#) describe the parameters for the variables.

Figure 4-2 Event A1 triggering

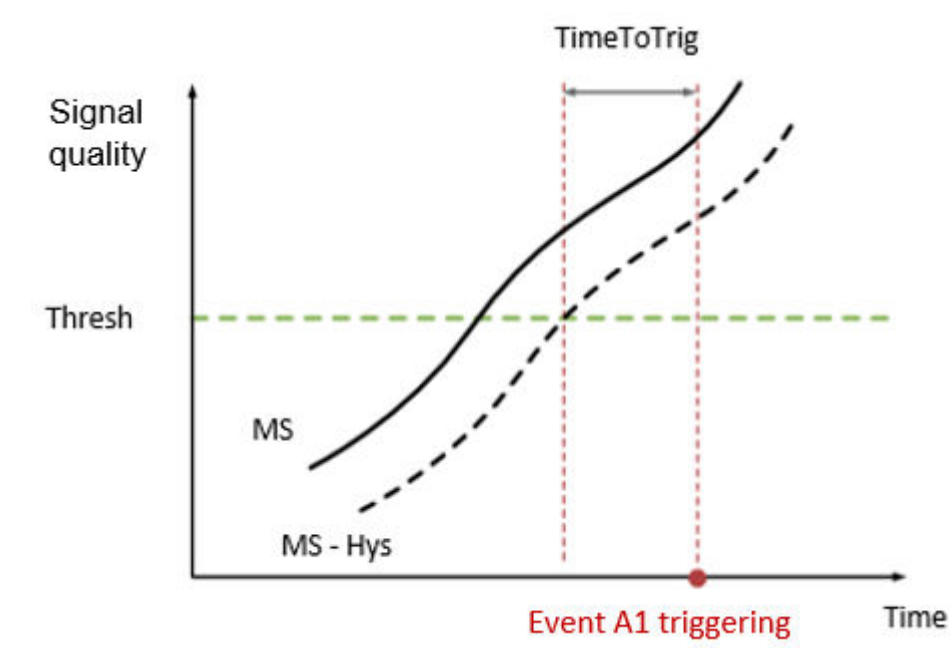


Table 4-6 Parameter settings for event A1 on the gNodeB side

Variable	Parameter
Hys	<i>NRCellInterRHoMeaGrp.InterRatHoA1A2Hyst</i>
Thresh	- RSRP threshold: <i>NRCellInterRHoMeaGrp.InterRatHoA1RsrpThld</i> - RSRQ threshold: <i>NRCellInterRHoMeaGrp.InterRatHoA1RsrqThld</i>
TimeToTrig	<i>NRCellInterRHoMeaGrp.InterRatHoA1A2TimeToTrig</i>

Table 4-7 Parameter settings for event A1 on the eNodeB side

Variable	Parameter
Hys	<i>InterRatHoCommGroup.InterRatHoA1A2Hyst</i>
Thresh	<i>InterRatHoCommGroup.InterRatHoNrA1ThldRsrp</i>
TimeToTrig	<i>InterRatHoCommGroup.InterRatHoA1A2TimeToTrig</i>, when <i>InterRatHoCommGroup.InterRatHoNrA1A2TimeToTrig</i> is set to NOT_CONFIG <i>InterRatHoCommGroup.InterRatHoNrA1A2TimeToTrig</i>, when <i>InterRatHoCommGroup.InterRatHoNrA1A2TimeToTrig</i> is set to a value other than NOT_CONFIG

Event A2

Entering condition for event A2: $(Ms + Hys < Thresh)$ is true throughout the duration specified by *TimeToTrig*. Figure 4-3 shows how event A2 is triggered. Table 4-8 and Table 4-9 describe the parameters for the variables.

Figure 4-3 Event A2 triggering

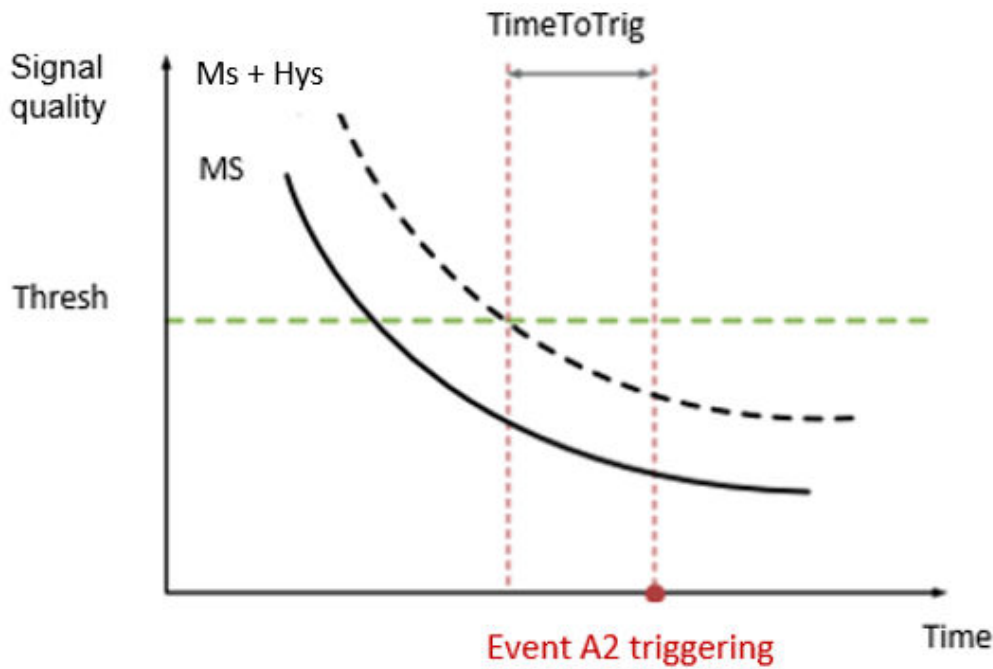


Table 4-8 Parameter settings for event A2 on the gNodeB side

Variable	Parameter
Hys	NRCellInterRHoMeaGrp.InterRatHoA1A2Hyst
Thresh	For inter-RAT event A2: - RSRP threshold: NRCellInterRHoMeaGrp.InterRatHoA2RsrpThld - RSRQ threshold: NRCellInterRHoMeaGrp.InterRatHoA2RsrqThld
	For blind event A2: - RSRP threshold: NRCellInterRHoMeaGrp.CovHoToEutranBlindA2Thld - RSRQ threshold: NRCellInterRHoMeaGrp.CovHoToLteBlindA2RsrqThld - SINR threshold: NRCellInterFHoMeaGrp.InterFBlindRedirA2SinrThld (This threshold is used only by blind redirection in downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN.)

Variable	Parameter
TimeToTrig	NRCellInterRHoMeaGrp.<i>InterRatHoA1A2TimeToTrig</i>

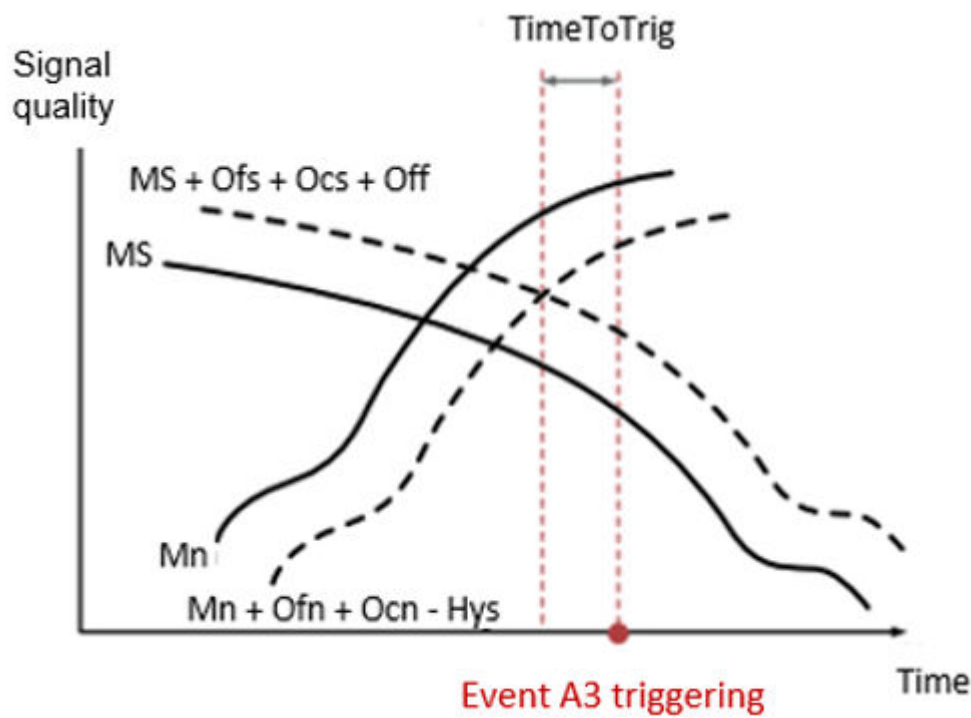
Table 4-9 Parameter settings for event A2 on the eNodeB side

Variable	Parameter
Hys	InterRatHoCommGroup.<i>InterRatHoA1A2Hyst</i>
Thresh	InterRatHoCommGroup.<i>InterRatHoNrA2ThldRsrp</i>
TimeToTrig	• InterRatHoCommGroup.<i>InterRatHoA1A2TimeToTrig</i>, when InterRatHoCommGroup.<i>InterRatHoNrA1A2TimeToTrig</i> is set to NOT_CONFIG • InterRatHoCommGroup.<i>InterRatHoNrA1A2TimeToTrig</i>, when InterRatHoCommGroup.<i>InterRatHoNrA1A2TimeToTrig</i> is set to a value other than NOT_CONFIG

Event A3

Entering condition for event A3: $(Mn + Ofn + Ocn - Hys > Ms + Ofs + Ocs + Off)$ is true throughout the time specified by *TimeToTrig*. [Figure 4-4](#) shows how event A3 is triggered.

Figure 4-4 Event A3 triggering



On the gNodeB side, event A3 is used for RSRP prediction based on virtual grids. [Table 4-10](#) describes the parameters for the variables.

Table 4-10 Parameter settings for event A3 on the gNodeB side

Variable	Parameter
Hys	This variable is fixed at 1 dB.
Off	This variable is fixed at -15 dB.
TimeToTrig	This variable is fixed at 160 ms.
Ofn	This variable is fixed at 0.
Ocn	This variable is fixed at 0.
Ofs	This variable is fixed at 0.
Ocs	This variable is fixed at 0.

On the eNodeB side, event A3 is used for RSRP prediction based on virtual grids. [Table 4-11](#) describes the parameters for the variables.

Table 4-11 Parameter settings for event A3 on the eNodeB side

Variable	Parameter
Hys	This variable is fixed at 2 dB.
Off	This variable is fixed at -30 dB.
TimeToTrig	LnrCarrierSelection.IntraFreqMeasA3TimeToTrig
Ofn	This variable is fixed at 0.
Ocn	EutranIntraFreqNCell.CellIndividualOffset
Ofs	Cell.QoffsetFreq
Ocs	Cell.CellSpecificOffset

Event B1

Entering condition for event B1 used for NG-RAN measurements in E-UTRAN: $(Mn + Ofn - Hys > Thresh)$ is true throughout the time specified by *TimeToTrig*. Entering condition for event B1 used for E-UTRAN measurements in NG-RAN: $(Mn + Ofn + Ocn - Hys > Thresh)$ is true throughout the time specified by *TimeToTrig*. Using E-UTRAN measurements in NG-RAN as an example, [Figure 4-5](#) shows how event B1 here is triggered. [Table 4-12](#) and [Table 4-13](#) describe the parameters for the variables.

Figure 4-5 Event B1 triggering (using event B1 for E-UTRAN measurements in NG-RAN as an example)

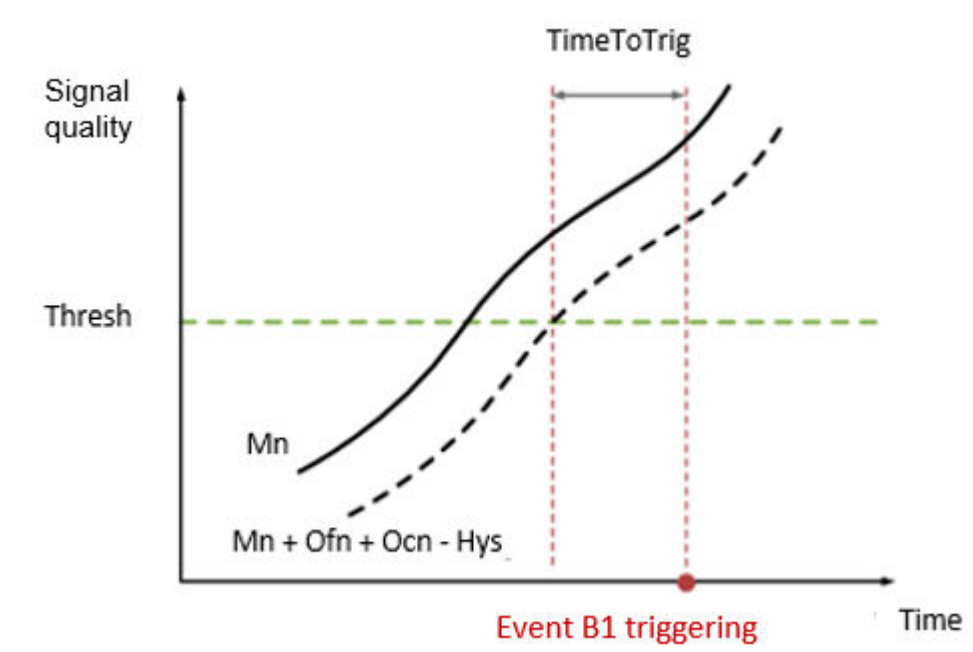


Table 4-12 Parameter settings for event B1 on the gNodeB side

Variable	Parameter
Hys	For downlink-coverage-based inter-RAT event B1: NRCellHoEutranMeaGrp.CovBasedHoB1B2Hyst For event B1 related to voice fallback from NG-RAN to E-UTRAN: NRCellHoEutranMeaGrp.EpsFbB1Hyst A fixed value is used for other functions.
Thresh	For downlink-coverage-based inter-RAT event B1: - RSRP threshold: NRCellHoEutranMeaGrp.CovBasedHoB1RsrpThld - RSRQ threshold for filtering: NRCellHoEutranMeaGrp.CovBasedHoB1RsrqThld
	For inter-RAT event B1 related to EPS fallback: - RSRP threshold: NRCellHoEutranMeaGrp.EpsFbB1RsrpThld - RSRQ threshold for filtering: NRCellHoEutranMeaGrp.EpsFbB1RsrqThld

Variable	Parameter
	For event B1 related to VoNR-to-VoLTE handover based on voice quality: • RSRP threshold: NRCeIISevExp.VonrQltyEutranB1RsrpThld • RSRQ threshold for filtering: NRCeIISevExp.VonrQltyEutranB1RsrqThld
	For event B1 related to voice-capability-based redirection back to the HPLMN: NRCeIIHoEutranMeaGrp.CovBasedHoB1RsrpThld
	RSRP threshold for event B1 related to load-based handover to E-UTRAN: NRCeIIHoEutranMeaGrp.LoadMobToEutranB1RsrpThld
TimeToTrig	For downlink-coverage-based inter-RAT event B1: NRCeIIHoEutranMeaGrp.CovBasedHoB1B2TimeToTrig For event B1 related to voice fallback from NG-RAN to E-UTRAN: NRCeIIHoEutranMeaGrp.EpsFbB1TimeToTrig A fixed value is used for other functions.
Ofn	NRCeIIeEutranNFreq.FreqSpecificOffset
Ocn	NRCeIIeEutraNRelation.EutranCellIndividualOffset

Table 4-13 Parameter settings for event B1 on the eNodeB side

Variable	Parameter
Hys	InterRatHoNrParamGrp.NrB1B2Hysteresis
Thresh	For coverage-based inter-RAT event B1: • RSRP threshold: InterRatHoNrParamGrp.CovBasedNrB1RsrpThld • RSRQ threshold for filtering: InterRatHoNrParamGrp.CovBasedNrB1RsrqThld
	For service-based inter-RAT event B1: • RSRP threshold: InterRatHoNrParamGrp.ServBasedNrB1RsrpThld • RSRQ threshold for filtering: InterRatHoNrParamGrp.ServBasedNrB1RsrqThld
TimeToTrig	InterRatHoNrParamGrp.NrB1B2TimeToTrigger For event B1 related to fast return: InterRatHoNrParamGrp.FastReturnB1TimeToTrigger
Ofn	NrNFreq.FreqSpecificOffset

There is a high probability that a UE supporting both NSA and SA network access capabilities (referred to as an NSA/SA dual-mode UE for short) reports NSA DC event B1 measurement results first, as the **NrScgFreqConfig.NrB1TimeToTrigger** parameter of NSA DC event B1 has a smaller default value than the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter of NR event B1. In this situation, the time-to-trigger adjustment for NR event B1 function can be enabled to prioritize NR SA network measurement. With this function enabled, the eNodeB aligns the time-to-trigger for NSA DC event B1 with the maximum time-to-trigger for NR event B1 among the UE's bearers if the following conditions apply: (1) The cell supports the delivery of both NR event B1 and NSA DC event B1 measurement configurations. (2) The maximum time-to-trigger for NR event B1 among the UE's bearers is greater than the time-to-trigger for the first delivered NSA DC event B1. The eNodeB does not make such an adjustment if the maximum time-to-trigger for NR event B1 among the UE's bearers is smaller than the time-to-trigger for NSA DC event B1. The time-to-trigger adjustment for NR event B1 function is controlled by the **NrB1TimeToTrigAdjSw** option of the **CellHoParaCfg.CellHoAlgoSwitch** parameter.

Event B2

Entering condition for event B2 used for NG-RAN measurements in E-UTRAN: ($M_s + Hys < Thresh1$ and $M_n + Ofn - Hys > Thresh2$) is true throughout the time specified by *TimeToTrig*. **Entering condition for event B2 used for E-UTRAN measurements in NG-RAN:** ($M_s + Hys < Thresh1$ and $M_n + Ofn + Ocn - Hys > Thresh2$) is true throughout the time specified by *TimeToTrig*. Using E-UTRAN measurements in NG-RAN as an example, [Figure 4-6](#) shows how event B2 here is triggered. [Table 4-14](#) and [Table 4-15](#) describe the parameters for the variables.

Figure 4-6 Event B2 triggering (using event B2 for E-UTRAN measurements in NG-RAN as an example)

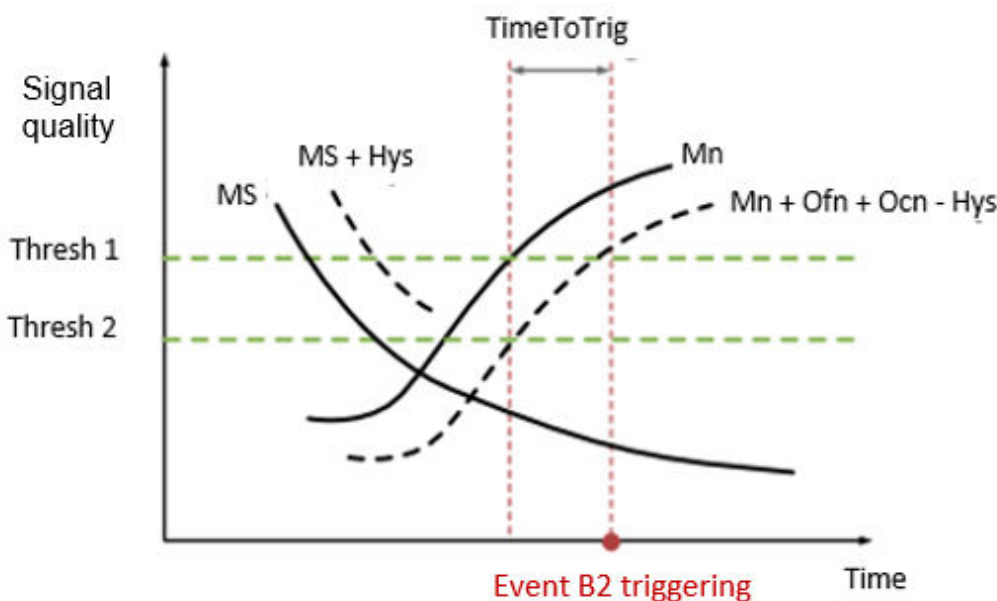


Table 4-14 Parameter settings for event B2 on the gNodeB side

Variable	Parameter
Hys	For downlink-coverage-based inter-RAT event B2: NRCellHoEutranMeaGrp.CovBasedHoB1B2Hyst
Thresh	For downlink-coverage-based inter-RAT event B2: • RSRP <i>Thresh1</i>: NRCellInterRHoMeaGrp.CovHoToEutranB2RsrpThld1 If the RSRP_OR_RSRQ option of the NRCellMobilityConfig.MeasTrigQuantity parameter is deselected, RSRP threshold 1 for downlink-coverage-based inter-RAT event B2 is specified by this parameter. If this option is selected, this threshold is always -43 dBm. • RSRP <i>Thresh2</i>: NRCellHoEutranMeaGrp.CovBasedHoB1RsrpThld • RSRQ <i>Thresh2</i>: NRCellHoEutranMeaGrp.CovBasedHoB1RsrqThld For event B2 related to uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN: • RSRP <i>Thresh1</i>: NRCellInterRHoMeaGrp.UlSinrHoEutranB2RsrpThld1 • RSRP <i>Thresh2</i>: NRCellHoEutranMeaGrp.NetworkingOptionOptB1Thld Other functions are not involved.
TimeToTrig	For downlink-coverage-based inter-RAT event B2: NRCellHoEutranMeaGrp.CovBasedHoB1B2TimeToTrig Other functions are not involved.
Ofn	NRCellEutranNFreq.FreqSpecificOffset
Ocn	NRCellEutraNRelation.EutranCellIndividualOffset

Table 4-15 Parameter settings for event B2 on the eNodeB side

Variable	Parameter
Hys	InterRatHoNrParamGrp.NrB1B2Hysteresis
Thresh	• RSRP <i>Thresh1</i>: InterRatHoCommGroup.NrB2Thld1Rsrp • RSRP <i>Thresh2</i>: InterRatHoNrParamGrp.CovBasedNrB1RsrpThld • RSRQ threshold for filtering: InterRatHoNrParamGrp.CovBasedNrB1RsrqThld

Variable	Parameter
TimeToTrig	InterRatHoNrParamGrp.NrB1B2TimeToTrigger
Ofn	NrNFreq.FreqSpecificOffset

NOTE

- Event B2 used for NG-RAN measurements in E-UTRAN involves only downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN.
- Assume that flexible user steering is enabled by selecting the **NR_TO_EUTRAN_FREQ_LIST_CTRL_SW** option of the **NRCCellAlgoSwitch.RfspAlgoSwitch** parameter. On the gNodeB side, the RSRP thresholds for inter-RAT handover events A1 and A2 as well as RSRP threshold 1 for downlink-coverage-based inter-RAT event B2 are all affected by the **gNBRfspConfig.InterRatA1A2B2RsrpThldOfs** parameter. For details, see *Flexible User Steering* in *5G RAN Feature Documentation*.
- During handovers from NG-RAN to E-UTRAN, the gNodeB supports flexible configuration of handover measurement parameters for voice and data service UEs based on RFSPs. For details, see *Flexible User Steering* in *5G RAN Feature Documentation*.
- During handovers from E-UTRAN to NG-RAN, the eNodeB supports flexible configuration of handover measurement parameters for voice and data service UEs based on SPIDs. For details, see *Flexible User Steering* in *eRAN Feature Documentation*.
- For super uplink UEs, the RSRP thresholds for inter-RAT handover events A1 and A2 on the gNodeB side are affected by the value of the **NRCCellMeasConfig.SulCovHoRsrpThldOffset** parameter. For details, see *Super Uplink* in *5G RAN Feature Documentation*.

4.1.4.2.2 Triggering Quantity

A triggering quantity refers to the policy that triggers event reporting. This policy involves RSRP, RSRQ, and SINR.

- **RSRP:** Indicates the strength of the received reference signals. RSRP refers to the linear average power of signals received by a UE on all resource elements (REs) that carry reference signals within the duration of an orthogonal frequency division multiplexing (OFDM) symbol across a measurement bandwidth. RSRP reflects the strength rather than the quality of downlink reference signals in a cell, excluding noise and interference.
- **RSRQ:** Indicates the quality of received reference signals and is similar to the result of RSRP/RSSI.
- **SINR:** Indicates a ratio of the strength of received wanted signals to the strength of received unwanted signals (noise and interference).

The triggering quantities of measurement events include:

- **RSRP-based triggering:** A UE evaluates measurement reporting based on RSRP. The base station selects the target cell or frequency based on the UE-reported RSRP values of frequencies.
- **RSRQ-based triggering:** A UE evaluates measurement reporting based on RSRQ. The base station selects the target cell or frequency based on the UE-reported RSRQ values of frequencies.

- RSRP/RSRQ-based triggering: A UE evaluates measurement reporting based on RSRP or RSRQ. The UE reports a measurement event when either the RSRP or RSRQ value meets the entering condition for the event. The base station selects the target cell or frequency based on the UE-reported RSRP or RSRQ values of frequencies.
- RSRP-based triggering combined with RSRQ-based filtering: A UE evaluates measurement reporting based on RSRP and reports both the RSRP and RSRQ values of frequencies. The base station selects the target cell or frequency based on the UE-reported RSRP values of the frequencies, and evaluates whether the leaving condition for a measurement event is met based on the UE-reported RSRQ values. If the condition is met, the base station filters out the relevant cell.
- RSRP/SINR-based triggering: A UE evaluates measurement reporting based on RSRP or SINR. The UE reports a measurement event when either the RSRP or SINR value meets the entering condition for the event. The base station selects the target cell or frequency based on the UE-reported RSRP or SINR values of frequencies.

Table 4-16 lists the triggering quantities of measurement events supported by the eNodeB.

Table 4-16 Triggering quantities supported by the eNodeB

Measurement Event	Supported Triggering Quantity
A1	RSRP-based triggering
A2	RSRP-based triggering
A3	RSRP-based triggering
B1	RSRP-based triggering RSRP-based triggering combined with RSRQ-based filtering
B2	RSRP-based triggering RSRP-based triggering combined with RSRQ-based filtering

The triggering quantities for events B1 and B2 are controlled by the **NR_B1_B2_PLUS_RSRQ_INDICATION** option of the **NrNFreq.AggregationAttribute** parameter.

- When this option is deselected, RSRP-based triggering is used.
- When this option is selected, RSRP-based triggering combined with RSRQ-based filtering is used.

Table 4-17 lists the triggering quantities of measurement events supported by the gNodeB.

Table 4-17 Triggering quantities supported by the gNodeB

Measurement Event	Supported Triggering Quantity
A1	RSRP-based triggering RSRP/RSRQ-based triggering
A2	RSRP-based triggering RSRP/RSRQ-based triggering RSRP/SINR-based triggering (supported only by blind redirection in downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN)
A3	RSRP-based triggering
B1	RSRP-based triggering RSRP-based triggering combined with RSRQ-based filtering
B2	RSRP-based triggering RSRP-based triggering combined with RSRQ-based filtering

Note that:

- The triggering quantities for events A1 and A2 are specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter.
 - When the parameter is set to **RSRP**, RSRP-based triggering is used.
 - When the parameter is set to **RSRP_OR_RSRQ**, RSRP- or RSRQ-based triggering is used. The gNodeB delivers RSRP- or RSRQ-based A2 measurement configurations. When an A2 measurement report with either of the quantities is received, a handover or redirection will take place.
 - When the parameter is set to **RSRP_OR_SINR**, RSRP- or SINR-based triggering is only used for event A2 related to blind redirection in downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN. For event A2 in other scenarios and event A1, RSRP-based triggering is used. In blind redirection scenarios, the gNodeB delivers both RSRP- and SINR-based A2 measurement configurations. When the gNodeB receives an A2 measurement report with either of the quantities, a blind redirection will take place.

NOTE

In downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, the triggering quantities for events A1 and A2 related to handover and redirection are also subject to downlink-SINR-based mobility to E-UTRAN. For details about downlink-SINR-based mobility to E-UTRAN, see [5.2.1 Principles](#).

- The triggering quantities for events B1 and B2 are specified by the **NRCellEutranNFreq.FreqIntrfFlag** parameter.
 - When this parameter is set to **NO_INTRF**, RSRP-based triggering is used.

- When this parameter is set to **INTRF**, RSRP-based triggering combined with RSRQ-based filtering is used.

Table 4-18 lists the measurement triggering quantities supported by different functions. For details, see the descriptions of the corresponding functions.

Table 4-18 Measurement event triggering quantities supported by different functions

Function Name	Supported Measurement Event Triggering Quantity
Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	RSRP-based triggering RSRP/RSRQ-based triggering RSRP/SINR-based triggering (supported only by event A2 related to blind redirection) RSRP-based triggering combined with RSRQ-based filtering
Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	RSRP-based triggering RSRP/RSRQ-based triggering
Downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN	RSRP-based triggering RSRP-based triggering combined with RSRQ-based filtering
E-UTRAN to NG-RAN traffic steering	RSRP-based triggering RSRP-based triggering combined with RSRQ-based filtering
Voice fallback from NG-RAN to E-UTRAN	RSRP-based triggering RSRP-based triggering combined with RSRQ-based filtering
Fast return from E-UTRAN to NG-RAN	RSRP-based triggering RSRP-based triggering combined with RSRQ-based filtering
VoNR-to-VoLTE handover based on voice quality	RSRP-based triggering RSRP-based triggering combined with RSRQ-based filtering
Voice-capability-based redirection back to the HPLMN	RSRP-based triggering

4.1.4.2.3 QCI-specific Measurement Parameter Configurations

The settings of parameters described in the previous sections can vary depending on QoS class identifiers (QCIs), to flexibly control mobility of QCI-specific services. If measurements are triggered for a UE running services with

different QCIs, the gNodeB or eNodeB delivers the group of measurement configurations mapped to RATs and services.

- On the gNodeB side:

The gNodeB delivers the group of measurement parameters mapped to the service with the highest-priority QCI among these services to the UE. If different QCIs are mapped to the same priority, a smaller QCI indicates a higher priority.

 - If the **gNBQciBearer.QciPriorityForHo** parameter is set to a value other than **255**, the priorities specified by this parameter are used for sorting. A smaller value of this parameter indicates a higher QCI priority.
 - If the **gNBQciBearer.QciPriorityForHo** parameter is set to **255**, then:
 - QCI priorities on the NR side are determined by the *priorityLevelQos* IE carried in the *Non Dynamic 5QI Descriptor* IE of the latest INITIAL CONTEXT SETUP REQUEST, PDU SESSION RESOURCE SETUP REQUEST, or PDU SESSION RESOURCE MODIFY REQUEST message from the core network. A smaller value of the IE indicates a higher QCI priority.
 - If the message does not carry the *priorityLevelQos* IE, QCI priorities are determined by the **gNBQciBearer.PriorityLevel** parameter. A smaller value of this parameter indicates a higher QCI priority.
- On the eNodeB side:
 - For downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN, the eNodeB delivers the group of measurement parameters mapped to the service with the highest-priority QCI among these services to the UE.

On the LTE side, QCI priorities are determined by the **CellQciPara.QciPriorityForHo** parameter. A smaller value indicates a higher priority.

 - For E-UTRAN to NG-RAN traffic steering and fast return from E-UTRAN to NG-RAN, the eNodeB selects the group of measurement parameters with the largest measurement thresholds for services.

Measurement parameters are mapped to QCIs by using the **NRCellQciBearer** MO on the NR side and the **CnOperatorQciPara** MO on the LTE side.

The preceding statement applies to functions other than E-UTRAN to NG-RAN traffic steering. For details about the measurement parameter selection rules for E-UTRAN to NG-RAN traffic steering, see [5.6 E-UTRAN to NG-RAN Traffic Steering](#).

4.1.5 Other Configurations

Other configurations include measurement gap configuration, non-gap-assisted B1 measurement configuration by the eNodeB, and measurement object preemption.

4.1.5.1 Measurement Gap Configuration

Measurement gaps are time periods during which UEs stop services on their serving frequencies and perform measurements on non-serving frequencies. The eNodeB derives the signal quality of NR cells from the measurement results of

synchronization signal and PBCH block (SSB) signals from the gNodeB. To this end, SSB signals from NR cells must be transmitted within measurement gaps on the E-UTRAN side. Otherwise, the eNodeB fails to initiate handovers. The start position of a gap for NR measurement calculated by the eNodeB is relevant to the SSB period and SSB offset. The recommended settings of the SSB period and SSB offset vary depending on the synchronization mode used by the eNodeB and gNodeB. It is recommended that time synchronization be used. When the eNodeB initiates gap-assisted measurements of NR frequencies, it does not deliver B1- or B2-related measurement gap configurations to the UE if the UE is capable of non-gap-assisted measurements of NR frequencies and the **NR_B1_NO_GAP_SW** option of the **EnodeBAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter is selected. The gNodeB supports gap pattern adaptation based on UE capabilities. Measurement gap configurations are delivered to UEs through measurement configurations and do not need to be configured using parameters.

SSB Period

- For time synchronization, the configuration requirements are as follows:

The eNodeB parameter **NrNFreq.SsbPeriod** must be set to the same value as the gNodeB parameter **NRDUCell.SsbPeriod**. If multiple NR cells operate on the same frequency on the NR network and have different SSB period settings, it is recommended that, on the E-UTRAN side, the SSB period for that frequency be set to the maximum value among these SSB period settings.

- The following describes the configuration requirements for frequency synchronization.

It is recommended that the SSB period (specified by the eNodeB parameter **NrNFreq.SsbPeriod** and the gNodeB parameter **NRDUCell.SsbPeriod**) be set to 5MS.

SSB Offset

- For time synchronization, the configuration requirements are as follows:

On the NG-RAN side, the frame offset is specified by the **gNodeBParam.FrameOffset** or **gNBFreqBandConfig.FrameOffset** parameter. The value ranges of the two parameters are the same. The former applies to a gNodeB, and the latter applies to a frequency band. If both parameters are configured, the latter takes effect. The following uses the **gNBFreqBandConfig.FrameOffset** parameter as an example.

- The time difference between the start time of the subframe for the NR SSB and the global navigation satellite system (GNSS) standard time, also known as the offset of SSB to the clock source, is calculated based on the NG-RAN frame offset.
 - If the value of the **gNBFreqBandConfig.FrameOffset** parameter is less than or equal to 261120, the offset of SSB to the clock source is calculated as follows: $(\text{gNBFreqBandConfig.FrameOffset}/30720) \text{ MOD } \text{NrNFreq.SsbPeriod}$.
 - If the value of the **gNBFreqBandConfig.FrameOffset** parameter is greater than or equal to 275943, the offset of SSB to the clock source is calculated as follows: $(\text{gNBFreqBandConfig.FrameOffset}/30720 - 10 + \text{NrNFreq.SsbPeriod}) \text{ MOD } \text{NrNFreq.SsbPeriod}$.

If the NR cells operate on different frequencies, the **gNBFreqBandConfig.FrameOffset** parameter must be set to an identical value across the frequencies.

When the frame offset requirements of a frequency band between gNodeBs are different, the **gNBFreqBandConfig.AdditionalFrameOffset** parameter can be used to implement compatibility between different frame offset measurements on the same frequency band. In this case, the offset of SSB to the clock source can be calculated by choosing the smallest valid value from **gNBFreqBandConfig.AdditionalFrameOffset**, **gNodeBParam.FrameOffset**, and **gNBFreqBandConfig.FrameOffset**, and then using it in one of the preceding formulas.

- b. The SSB offset is corrected based on the eNodeB frame offset.

The offset of SSB to the clock source calculated in the previous step is used as the SSB offset before correction. Perform the following steps to correct the SSB offset:

- i. The NG-RAN frame offset remainder and E-UTRAN frame offset remainder are calculated as follows:
 - NG-RAN frame offset remainder = $(\text{gNBFreqBandConfig.FrameOffset} \text{ MOD } 30720)/30720$
 - E-UTRAN frame offset remainder = $(\text{CellFrameOffset.FrameOffset} \text{ MOD } 30720)/30720$
- ii. If the E-UTRAN frame offset remainder is greater than the NG-RAN frame offset remainder and is less than 0.5 ms, the corrected SSB offset is calculated as follows: $(\text{SSB offset before correction} - 0.5 + \text{NrNFreq.SsbPeriod}) \text{ MOD } \text{NrNFreq.SsbPeriod}$.
- iii. If the preceding condition is not met, the corrected SSB offset is equal to the SSB offset before correction.
- c. The corrected SSB offset is rounded down to its nearest integer, and the integer is used as the value of the **NrNFreq.SsbOffset** parameter.

NOTE

The unit of the frame offset is Ts, which equals 1/30720 ms. Frame offset can be expressed in the unit of ms as long as the value in Ts is divided by 30720.

- The following describes the configuration requirements for frequency synchronization.

Set the eNodeB parameter **NrNFreq.SsbOffset** to 0.

Non-Overlapping Between Gap and NR Resources

The uplink and downlink throughput of UEs decrease in the measurement gaps for inter-frequency or inter-RAT measurements because UEs cannot transmit uplink signals or receive downlink signals on the serving frequency throughout the measurement gaps. In addition, if the time-domain positions of NR channel resources conflict with those of the measurement gaps, user experience is affected. Examples are as follows:

- If the PUCCH SR period is longer than or equal to the gap period, PUCCH SRs and gap-assisted measurements will conflict in the time domain, resulting in service drops.

- If the SRS period is shorter than the gap period, SRSs and gap-assisted measurements will conflict at certain time-domain positions, increasing uplink IBLER and uplink RBLER and causing significant signal fluctuation. This negatively affects the timeliness of measurement reporting, resulting in deteriorated service experience for moving UEs.

In this context, the following functions are introduced:

- For the scenarios where the NR resource allocation period is longer than or equal to the gap period:
 - **Non-overlapping between gap and NR resources:** When the target cells are NR cells only, this function staggers the channel resources of specific signals (including SRS, PUCCH SR, CSI-RS for CM, and CSI-RS for CM measurement reporting in low frequency bands) from measurement gaps in the time domain. The non-overlapping between gap and NR resources function is controlled by the **GAP_AVOID_NR_RES_SW** option of the **gNBMobilityCommParam.MeasAlgoSwitch** parameter. For more details about non-overlapping between gap and NR resources, see *SA Mobility Management in Connected Mode* in 5G RAN feature documentation.
 - **Inter-frequency and inter-RAT gap offset modification:** When the target cells are LTE and NR cells, this function implements non-overlapping between channel resources and inter-frequency and inter-RAT measurement gaps in the time domain. The gap offset is calculated based on the SMTC-indicated position of the measured frequency when both inter-frequency and inter-RAT measurement configurations are delivered. This function is controlled by the **INTERRAT_GAP_OFFSET_MOD_SW** option of the **gNBMobilityCommParam.MeasAlgoSwitch** parameter and depends on non-overlapping between gap and NR resources.
 - **Non-overlapping between inter-RAT measurement gaps and NR resources:** When the target cells are LTE cells only, this function staggers inter-RAT measurement gaps and NR resources in the time domain. This function is controlled by the **INTERRAT_ONLY_GAP_OFFSET_SW** option of the **gNBMobilityCommParam.MeasAlgoSwitch** parameter and depends on non-overlapping between gap and NR resources.
- For the scenarios where the NR resource allocation period is shorter than the gap period:
 - **Optimized non-overlapping between gap and NR resources:** When the target cells are LTE cells, this function staggers the channel resources of specific signals (including SRS, PUCCH SR, CSI-RS for CM, and CSI-RS for CM measurement reporting in low frequency bands) from measurement gaps in the time domain. Optimized non-overlapping between gap and NR resources is controlled by the **GAP_AVOID_NR_RES_OPT_SW** option of the **NRDUCellMeas.MeasAlgoSwitch** parameter and is supported only in low frequency bands. It is recommended that this function be enabled in the case of significant signal fluctuations.
 - If this option is deselected, the channel resources for the aforesaid signals are not staggered from measurement gaps in the time domain. In this case, measurement gaps preferentially take effect on UEs, which do not transmit or receive the aforesaid signals during measurement gaps.
 - If this option is selected, the channel resources for the aforesaid signals are staggered from measurement gaps in the time domain. This reduces

uplink IBLER and uplink RBLER and decreases the number of delayed measurement reports, but the number of reconfiguration signaling messages may increase.

4.1.5.2 Non-Gap-assisted B1 Measurement Configuration by the eNodeB

In SA networking, a UE initiates gap-assisted measurements when measuring NR frequencies, causing the user-perceived rate to decrease. Non-gap-assisted B1 measurement configuration by the eNodeB can be enabled to address this issue.

The impact of gap-assisted NR frequency measurement on data rates can be reduced for certain UEs by the non-gap-assisted B1 measurement function, which is controlled by the **NO_GAP_B1_MEAS_SW_ON** option of the **UeCompat.WhiteLstCtrlSwitch** parameter.

- If this option is selected, non-gap-assisted B1 measurement takes effect for UEs specified by the **UeCompat** MO. This type of measurement does not apply to frequencies in blacklisted combinations.
- If this option is deselected, non-gap-assisted B1 measurement does not take effect.

Combination *xx*, identified by the **LnrFreqComb.FreqCombId** parameter, can be blacklisted by selecting the **B1_NO_GAP_FREQ_COMB_ID_xx** option of the **UeCompat.LnrNoGapFreqCombBlacklist** parameter. When this option is selected and a UE is served by the LTE frequency identified by the **LnrFreqComb.DLEarfcn** parameter in combination *xx*, non-gap-assisted B1 measurement does not apply to the NR frequency in this combination, which is identified by the **LnrFreqComb.DINrArfcn** parameter.

Assume that a UE does not support non-gap-assisted measurement on the NR frequency with an NR-ARFCN of 636666 when it is working on the LTE frequency with an EARFCN of 1500. This frequency combination needs to be blacklisted. Specifically, select the **B1_NO_GAP_FREQ_COMB_ID_0** option of the **UeCompat.LnrNoGapFreqCombBlacklist** parameter, set the **LnrFreqComb.FreqCombId** parameter to **0**, and set the corresponding **LnrFreqComb.DLEarfcn** and **LnrFreqComb.DINrArfcn** parameters to **1500** and **636666**, respectively.

4.1.5.3 Measurement Object Preemption

UEs have limited capabilities of gap-assisted measurements on inter-RAT frequencies. Measurement object preemption allows a UE with such a limited capability to measure higher-priority frequencies when possible for user experience consideration. It takes effect only on the LTE side.

This function is controlled by the **NSA_SA_MEAS_OBJ_PREEMPTION_SW** option of the **CellUeMeasControlCfg.MeasOptAlgoSwitch** parameter. If this option is selected and the UE's frequency measurement capability is limited, the eNodeB checks whether the priority of the algorithm to be started is higher than the highest priority of earlier started algorithms. If it is, the eNodeB enables this algorithm to preempt measurement resources, which means that the UE will measure the frequencies involved in the algorithm to be started.

Among the mobility management algorithms for UEs from E-UTRAN to NG-RAN, the coverage-based mobility management algorithm takes priority over the service-based mobility management and fast return algorithms.

4.1.5.4 Multi-Antenna-Port Configuration Indication

Neighboring LTE cells can be configured with a different number of antenna ports: some with one, and others with two or more. When delivering measurement configurations of LTE frequencies, NR needs to include the number of antenna ports of the neighboring LTE cells in the LTE measurement control message for better UE measurement. The multi-antenna-port configuration indication function is introduced for this purpose. This function applies only to the NR side.

This function is recommended when all neighboring cells on the LTE frequencies in MeasObjectEUTRA are configured with two or more antenna ports. With this function, the value of the *eutra-PresenceAntennaPort1* IE is **TRUE**. If at least one neighboring cell on an LTE frequency in MeasObjectEUTRA is configured with a single antenna port, this function needs to be disabled. In this case, the value of the *eutra-PresenceAntennaPort1* IE is **FALSE**.

This function is controlled by the **PRESENCE_ANTENNA_PORT1_SW** option of the **NRCellEutranNFreq.FrequencyCtrlSwitch** parameter.

4.1.6 Measurement Reporting

A UE performs measurements based on the measurement configurations received from an eNodeB or gNodeB. When determining that the reporting criteria are met, the UE sends measurement reports to the eNodeB or gNodeB. The eNodeB or gNodeB generates a candidate cell list based on the measurement reports.

4.1.7 RSRP Prediction Based on Virtual Grids

UEs with the same radio signal characteristics can be classified into a category based on multi-dimensional measurements. The eNodeB or gNodeB (collectively called base station) considers the UEs with the same RSRP measurement result on a frequency to be located in one virtual radio grid, referred to as a virtual grid, for the frequency. For example, if UE 1's measurement result on a frequency is [(Cell 1, RSRP 1), (Cell 2, RSRP 2)] and UE 2's measurement result is the same, the base station considers the two UEs to be in the same virtual grid. The base station uses machine learning technology to explore the mapping in signal characteristics from all virtual grids in a cell to a frequency. This mapping is called a virtual grid model.

Interoperability between E-UTRAN and NG-RAN involves two types of virtual grid models:

- Virtual grid models used by the eNodeB to predict RSRP of NR frequencies (also known as LN RSRP models): These models are built on LTE cells. After obtaining the virtual grid information of a UE (based on A3 measurement reports) in an LTE cell, the eNodeB can query the virtual grid model for a specific neighboring NR frequency to quickly predict the RSRP of the UE on the NR frequency.
- Virtual grid models used by the gNodeB to predict RSRP of LTE frequencies (also known as NL RSRP models): These models are built on NR cells. After obtaining the virtual grid information of a UE (based on A3 measurement

reports) in an NR cell, the gNodeB can query the virtual grid model for a specific neighboring LTE frequency to quickly predict the RSRP of the UE on the LTE frequency.

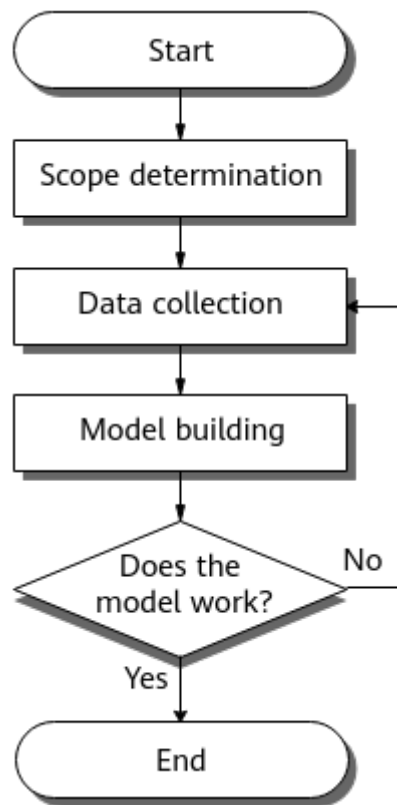
If the RSRP predicted using a virtual grid model meets the entering condition for inter-RAT event B1, this prediction result is a good-RSRP prediction. If the RSRP predicted using a virtual grid model meets the leaving condition for inter-RAT event B1, this prediction result is a poor-RSRP prediction.

RSRP prediction based on virtual grids is a trial function. Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

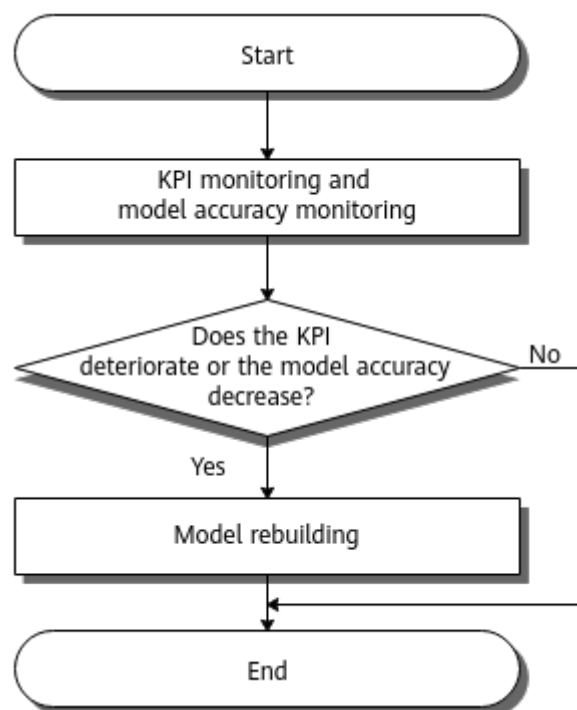
Virtual Grid Model Building

The building procedure for LN RSRP models is similar to that for NL RSRP models, as shown in [Figure 4-7](#).

Figure 4-7 Virtual grid model building

Virtual Grid Model Maintenance

The maintenance procedure for LN RSRP models is similar to that for NL RSRP models, as shown in [Figure 4-8](#).

Figure 4-8 Virtual grid model maintenance

This section provides details on the building and maintenance procedures for LN and NL RSRP models.

4.1.7.1 LN RSRP Model Building and Maintenance

LN RSRP model building is controlled by the **LNR_VIRTUAL_GRID_SW** option of the eNodeB parameter **LnrCarrierSelection.LnrCarrierSelectionSw**.

Model Building

The following is the procedure for building LN RSRP models:

1. The eNodeB determines the scope of LN RSRP models.

[*LTE cell, Neighboring NR frequency*] serves as the index for LN RSRP models. The scope of LN RSRP models is indicated by the LTE cell and neighboring NR frequency attributes.

LN RSRP models are not built for the following LTE cells:

- Cells with multiple physical cells each with a different ID, for example, a single frequency network (SFN) cell or a cell with the **Cell.MultiRruCellFlag** parameter set to **BOOLEAN_TRUE**
- High-speed-mobility cells, for example, cells with high speed mobility enabled and cells with ultra high speed mobility enabled
- Cells in which UEs are stationary and therefore handovers are not involved, such as eMTC-only cells
- (FDD) NB-IoT cells
- Cells that can only serve as SCells, where UE admission is not supported and therefore samples cannot be collected, for example, cells with the

Cell.WorkMode parameter set to **DL_ONLY** or **LAA** and cells with the **ForbidCellSiSwitch** option of the **CellSiMap.SiSwitch** parameter selected

- Cells with a cell radius greater than 100 km
- Cells for which the **PATTERN_ADJ_SWITCH** option of the **SectorSplitGroup.SectorSplitSwitch** parameter is selected
- Cells that are on the same frequencies as the cells for which the **BEAM_NUM_ADJ_SWITCH** option of the **SectorSplitGroup.SectorSplitSwitch** parameter is selected

The following neighboring NR frequencies will not be selected for virtual grid model building:

- Frequencies for which the **VG_MODEL_FORBID_BUILD_FLAG** option of the **NrNFreq.AggregationAttribute** parameter is selected
- Frequencies for which the **FREQ_MEAS_FLAG** option of the **NrNFreq.AggregationAttribute** parameter is deselected

One LN RSRP model provides RSRP data of one NR frequency for one LTE cell. If an eNodeB serves S cells and each cell has N neighboring NR frequencies, a total of $S \times N$ LN RSRP models are required. The LN RSRP model capacities are described in [Table 4-19](#).

Table 4-19 Model capacities of each main control board

Main Control Board	Related Parameter Configuration	LN RSRP Model Capacity per eNodeB
UMPTe	• Option 1: AI_ENHANCEMENT_SWITCH option of the eNodeBResModeAlgo.ServiceMode parameter • Option 2: AI_ENHANCEMENT_OPT_SWITCH option of the eNodeBResModeAlgo.ServiceMode parameter	• When option 1 is selected and option 2 is deselected: 90 • When both option 1 and option 2 are deselected: 36 • When option 2 is selected: – If the UMPTe series board works in LN/GULN mode: 18 – If the UMPTe series board works in another mode: 36
UMPTga		• When either option is selected: 90 • When neither option is selected: 36
UMPTHa		When this option is selected: 168 When this option is deselected: 90

Main Control Board	Related Parameter Configuration	LN RSRP Model Capacity per eNodeB
UMPTg	UMPTG_ENHANCEMENT_SWITCH option of the eNodeBResModeAlgo.ServiceMode parameter	When this option is selected: 168 When this option is deselected: 90

NOTE

Each of the preceding virtual grid model capacities is the total allowed number of both LN RSRP models and LTE-only RSRP prediction models. The capacity consumption rule of "first-come, first-built" applies to the two types of models as a whole.

If the number of LN RSRP models to be built exceeds the limit, these models are prioritized according to the following rules, and then higher-priority models are selected for building as long as the limit allows:

- For LTE cells, a larger cell bandwidth indicates a higher LN RSRP model building priority.
 - For NR frequencies, a higher frequency priority for connected-mode UEs indicates a higher LN RSRP model building priority.
2. The eNodeB randomly selects a maximum of 15 UEs (capable of EN-DC or SA) every 40s within 24 hours in each LTE cell for which models are to be built, and collects their measurement reports about each frequency as the sample data for LN RSRP models.

On an LTE network, measurement configurations are generally delivered by the eNodeB to a UE through an RRC Connection Reconfiguration message. The UE performs measurement based on the measurement configurations and reports measurement results to the eNodeB through a measurement report. Measurements are triggered periodically or based on events. By default, the eNodeB periodically triggers measurement. If the **VG_MDL_NCELL_MR_OPT_SW** option of the **MultiCarrUnifiedSch.MultiCarrierUnifiedSchEnSw** parameter is selected, inter-RAT neighboring cell measurements are triggered by event B1 during model training. Event B1 indicates that the signal quality of an inter-RAT neighboring cell becomes higher than a specific threshold. The threshold has a fixed value of -156 dBm.

In LN RSRP model training, if B1-related measurement configuration is delivered, UEs without inter-frequency coverage do not report inter-frequency measurement reports. If the **SAM_EXPAND_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected, such samples will be used for LN RSRP model training. If this option is deselected, such samples will not be used for LN RSRP model training.

Samples without intra-frequency neighboring cell measurement results impair LN RSRP model training. If the **NO_INTRA_FREQ_SAM_DEL_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected, such samples will not be used for LN RSRP model training. If this option is deselected, such samples will be used for the training.

3. After completing data collection, the eNodeB starts LN RSRP model training and calculates the accuracy of each model.

When the **SAM_BALANCING_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected, the learning weight is increased for the samples with a minor proportion (the samples with a minor proportion are samples with good or poor RSRP prediction results, whichever accounts for a smaller proportion of all samples) and decreased for the samples with a major proportion. This improves the prediction accuracy of the model through sample balancing.

4. The eNodeB checks the LN RSRP model accuracy to evaluate whether the conditions for using the models are met. If the accuracy of a model meets conditions, the model has been built successfully and can be put into use. Otherwise, the model has failed to be built and the procedure goes back to 2. If the building fails seven consecutive times, it cannot be tried again until 28 days later.

NOTE

The accuracy of virtual grid models upon their launch is independent of the accuracy requirements for using virtual grid models in mobility functions. A higher accuracy requirement of a mobility function for virtual grid models may result in a lower probability that this function can use virtual grid models to acquire RSRP information. In extreme cases, this function fails to use virtual grid models to acquire RSRP information.

5. After completing model building, the eNodeB randomly selects UEs involved in sample collection (the number of such UEs is specified by the **LnrCarrierSelection.SampleUeNum** parameter) every 40s, collects 7x24-hour data, and builds new models. If a new model is more accurate than the old one, the new one is put into use. If a new model is less accurate than the old one, the old model is still used.

After a virtual grid model is launched, its good-RSRP prediction and poor-RSRP prediction are performed separately and good- and poor-RSRP prediction results are used separately.

- For a launched virtual grid model whose good-RSRP predictions are usable but poor-RSRP predictions are unusable, if the prediction result is positive (good RSRP), a handover can be performed; if the prediction result is negative (poor RSRP), the corresponding B1 measurement configurations need to be delivered.
- For a virtual grid model whose good-RSRP predictions are unusable but poor-RSRP predictions are usable, if the prediction result is negative, the corresponding B1 measurement configurations are not delivered; if the prediction result is positive, the corresponding B1 measurement configurations need to be delivered.

Model Maintenance

After models are put into use, the eNodeB starts monitoring and updating the models. The following are possible actions:

- Model update triggered by KPI monitoring
After a model is put into use, the eNodeB monitors the success rate of handovers from E-UTRAN to NG-RAN (at the NR frequency level) in real time.

If the success rate of handovers from E-UTRAN to NG-RAN using the virtual grid model is lower than the value of the

LnrCarrierSelection.VgMdlHoSuccRateThld parameter for two consecutive hours, the eNodeB suspends the use of the model and initiates model rebuilding. For the rebuilding, the eNodeB randomly selects UEs involved in sample collection (the number of such UEs is specified by the **LnrCarrierSelection.SampleUeNum** parameter) every 40s and collects 7x24-hour data. Otherwise, the model continues its use.

- Model update triggered by model accuracy monitoring

Every other two days, the eNodeB selects one UE randomly and delivers intra-frequency and inter-RAT measurement configurations to the UE at an interval of 40s, collects one-day measurement reports, and performs a model accuracy test. If the model accuracy is lower than expected, the eNodeB considers that the model is aged. It then suspends the use of the model and initiates model rebuilding. For the rebuilding, the eNodeB randomly selects UEs involved in sample collection (the number of such UEs is specified by the **LnrCarrierSelection.SampleUeNum** parameter) every 40s and collects 7x24-hour data. For details, see [Model Building](#). Otherwise, the model continues its use.

Model Building/Use Stop and Model Removal

Models can be stopped from being built, stopped from being used, or removed.

- Model building/use stop

When the **LNR_VIRTUAL_GRID_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is deselected, the eNodeB stops the building and use of virtual grid models. The models that have been successfully built can be stored for a maximum of seven days. They can be used again if this option is selected within the upcoming seven days.

- Model removal

If the **VG_MODEL_FORBID_BUILD_FLAG** option of the eNodeB parameter **NrNFreq.AggregationAttribute** is selected, the eNodeB removes the successfully built virtual grid model for the specified frequency.

4.1.7.2 NL RSRP Model Building and Maintenance

NL RSRP model building is controlled by the **LNR_HO_MEAS_QTY_PRED_FUN_SW** option of the gNodeB parameter

NRCellSmartMultiCarr.LnrMultiCarrierAlgoSwitch.

Model Building

The following is the procedure for building NL RSRP models:

1. The gNodeB determines the scope of NL RSRP models.

[*NR cell, Neighboring LTE frequency*] serves as the index for models. The scope of models is indicated by the NR cell and neighboring LTE frequency attributes.

NL RSRP models are not built for the following NR cells:

- Cells for which **NRDUCell.NrDuCellNetworkingMode** is set to **HYPER_CELL** or **HYPER_CELL_COMBINE_MODE**

- Cells for which **NRDUCell.DuplexMode** is set to **CELL_SDL**
- Cells with the **NRDUCell.HighSpeedFlag** parameter set to **HIGH_SPEED**

NL RSRP models are not built for neighboring LTE frequencies with the **VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellEutranNFreq.AggregationAttribute** parameter deselected.

One NL RSRP model provides RSRP data of one LTE frequency for one NR cell. If a gNodeB serves *S* NR cells and each NR cell has *N* neighboring LTE frequencies, a total of *S* x *N* NL RSRP models are required. [Table 4-20](#) describes the model capacities.

Table 4-20 Model capacities of each main control board

Main Control Board	Working Mode	NL RSRP Model Capacity
UMPTg/UMPTga series	NR-only	108
	Co-MPT multiple modes	108
UMPTga series	NR-only	54
	Co-MPT multiple modes	36
UMPTe series	NR-only	54
	Co-MPT multiple modes ^a	0
a: UMPTe series boards working in co-MPT multimode scenarios do not support NL RSRP models.		

 NOTE

Part of the model capacities described in the preceding table is shared between NL RSRP models and NR-only RSRP models. The capacity consumption rule of "first-come, first-built" applies to the shared models. The shared model capacity is equal to the NR-only RSRP models. For details, see *Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation*. For example, if a UMPTg series board working in NR-only mode supports a maximum of 36 NR-only RSRP models, a total number of 36 NL RSRP models and NR-only RSRP models is supported.

2. The gNodeB randomly selects a maximum of 15 SA UEs every 40s within 24 hours in each cell for which models are to be built, and collects their measurement reports about each frequency as the sample data for NL RSRP models.

Generally, NR measurement configurations are delivered by a gNodeB to a UE through an RRC Reconfiguration message. The UE performs measurement based on the measurement configurations and reports measurement results to the gNodeB through a measurement report. Measurements are triggered periodically or based on events. By default, the gNodeB periodically triggers measurements. If the **NR_VG_MDL_MR_IMP_SW** option of the **NRCellSmartMultiCarr.NrMultiCarrierAlgoSwitch** parameter is selected,

inter-RAT neighboring cell measurements are triggered by event B2 during model training. Event B2 indicates that the signal quality of the serving cell drops below threshold 1 and that of an inter-RAT neighboring cell exceeds threshold 2. Threshold 1 and threshold 2 are fixed at -43 dBm and -156 dBm, respectively.

3. After completing data collection, the gNodeB starts NL RSRP model training and calculates the accuracy of each model.
4. The gNodeB evaluates whether individual models can be used based on their model accuracy.
 - If the accuracy of a model meets conditions, the model has been built successfully and can be put into use.
 - Otherwise, the model has failed to be built and the procedure goes back to 2. If the building fails seven consecutive times, it cannot be tried again until 28 days later.

NOTE

Mobility functions have their own accuracy requirements on virtual grid models when they use the models. The accuracy requirements for launching virtual grid models are independent of those for using virtual grid models in mobility functions. A higher accuracy requirement of a mobility function for virtual grid models may result in a lower probability that this function can use virtual grid models to acquire RSRP information. In extreme cases, this function fails to use virtual grid models to acquire RSRP information.

5. After completing model building based on 24-hour data, the gNodeB randomly selects at most three UEs every 40s, collects 7x24-hour data, and builds new models. If a new model is more accurate than the old one, the new one is put into use.

After a virtual grid model is launched, its good-RSRP prediction and poor-RSRP prediction are performed separately and good- and poor-RSRP prediction results are used separately.

- When virtual grids are used for NSA/SA selection based on user experience:
 - For a virtual grid model whose good-RSRP predictions are usable but poor-RSRP predictions are unusable, if the prediction result is positive (good RSRP), a blind handover can be performed; if the prediction result is negative (poor RSRP), B1 measurement configurations are delivered.
 - For a virtual grid model whose good-RSRP predictions are unusable but poor-RSRP predictions are usable, if the prediction result is negative, B1 measurement configurations are not delivered; if the prediction result is positive, B1 measurement configurations are delivered.
- When virtual grids are used for voice fallback:
 - For a virtual grid model whose good-RSRP predictions are usable but poor-RSRP predictions are unusable, if the prediction result is positive (good RSRP), a blind handover can be performed; if the prediction result is negative (poor RSRP), B1 measurement configurations are delivered.
 - For a virtual grid model whose good-RSRP predictions are unusable but poor-RSRP predictions are usable, B1 measurement configurations are delivered.

Model Maintenance

After models are put into use, the gNodeB starts monitoring and updating the models. The following are possible actions:

- Model update triggered by KPI monitoring
After a model is put into use, the gNodeB monitors the success rate of handovers from NG-RAN to E-UTRAN (at the LTE frequency level) in real time. If the success rate of handovers from NG-RAN to E-UTRAN using the virtual grid model is lower than the value of the **NRCellSmartMultiCarr.LnrHoSuccRateProtectThld** parameter for two consecutive hours, the gNodeB suspends the use of the model and initiates model rebuilding. (For the rebuilding, the gNodeB randomly selects three UEs every 40s and collects 7x24-hour data.) Otherwise, the model continues its use.
- Model update triggered by model accuracy monitoring
Every other two days, the gNodeB selects one UE randomly and delivers intra-frequency and inter-RAT measurement configurations to the UE at an interval of 40s, collects one-day measurement reports, and performs a model accuracy test. If the model accuracy is lower than expected, the gNodeB considers that the model is aged. It then suspends the use of the model and initiates model rebuilding. For the rebuilding, the gNodeB randomly selects three UEs every 40s and collects 7x24-hour data. For details, see [Model Building](#). Otherwise, the model continues its use.

Model Building/Use Stop and Model Removal

Models can be stopped from being built, stopped from being used, or removed.

- Model building/use stop
When the **LNR_HO_MEAS_QTY_PRED_FUN_SW** option of the **NRCellSmartMultiCarr.LnrMultiCarrierAlgoSwitch** parameter is deselected, the gNodeB stops the building and use of virtual grid models, and stores the models that have been successfully built. These models can be stored for a maximum of seven days. They can be used again if this option is selected within the seven days.
- Model removal
If the **VG_MODEL_ALLOW_BUILD_FLAG** option of the gNodeB parameter **NRCellEutranNFreq.AggregationAttribute** is deselected, the gNodeB removes the successfully built virtual grid model for the specified frequency.

4.1.7.3 Application to Functions in Interoperability Between E-UTRAN and NG-RAN

LN and NL RSRP models can be used for the following functions in interoperability between E-UTRAN and NG-RAN:

- LN RSRP models
 - E-UTRAN to NG-RAN traffic steering. For details, see [5.6 E-UTRAN to NG-RAN Traffic Steering](#).
 - Fast return from E-UTRAN to NG-RAN. For details, see [5.8 Fast Return from E-UTRAN to NG-RAN](#).

- NL RSRP models
 - NSA/SA selection based on user experience. For details, see the "NSA/SA Selection Based on Downlink Traffic Volume" and "NSA/SA Selection Based on Uplink Coverage" sections in *NSA-SA Selection Based on User Experience*.
 - Voice fallback from NG-RAN to E-UTRAN. For details, see [5.7 Voice Fallback from NG-RAN to E-UTRAN](#).

4.1.8 Target Cell or Frequency Evaluation

The process includes:

- Measurement report processing (needed only in measurement-based mode)
- Target cell or frequency generation
- Handover policy determination

Measurement Report Processing

The eNodeB or gNodeB processes measurement reports in a first-come-first-handled manner. Based on the results, the eNodeB or gNodeB generates candidate cells or frequencies.

Target Cell or Frequency Generation

The target cell or frequency is generated as follows, depending on the **processing modes and handover policies**:

The eNodeB filters NR frequencies specified by the **NrNFreq.DlArfcn** parameter according to [4.1.8.1 General Rules](#) and [4.1.8.3 LTE-specific Rules](#). Then, the eNodeB generates a candidate frequency list based on the frequency priorities. After the candidate frequency list is generated, the gNodeB instructs the UE to perform further processing based on the handover policy. In the case of a handover, the eNodeB instructs the UE to be handed over to the cell that is the first to report to have met the measurement conditions. In the case of a redirection, the eNodeB selects the frequency with the highest priority as the target frequency for redirection.

- If the handover policy is handover, the eNodeB generates candidate cells based on measurement reports and then filters the candidate cells according to [4.1.8.1 General Rules](#) and [4.1.8.3 LTE-specific Rules](#) to generate the target cell.
- If the handover policy is redirection, the eNodeB generates a candidate frequency based on a measurement report. This candidate frequency is the frequency of cells in the measurement report. Then, the eNodeB releases the RRC connection of the UE and instructs the UE to access a cell working on this frequency.

The gNodeB filters the LTE frequencies specified by the **NRCeUtranNFreq.DlEarfcn** parameter according to [4.1.8.1 General Rules](#) and [Frequency Filtering Rules](#). Then, the gNodeB generates a candidate frequency list based on the frequency priorities. After the candidate frequency list is generated, the gNodeB instructs the UE to perform further processing based on the measurement mode and handover policy. In the case of a handover, the gNodeB

instructs the UE to be handed over to the cell that is the first to report to have met the measurement conditions. In the case of a redirection, the gNodeB selects the frequency with the highest priority as the target frequency for redirection.

- Measurement-based mode
 - If the handover policy is handover, the gNodeB generates candidate cells based on measurement reports and then filters the candidate cells according to [4.1.8.1 General Rules](#) and [Frequency Filtering Rules](#) to generate the target cell.
 - If the handover policy is redirection, the gNodeB generates a candidate frequency based on a measurement report. This candidate frequency is the frequency of cells in the measurement report. Then, the gNodeB releases the RRC connection of the UE and instructs the UE to access a cell working on this frequency.
- Blind mode
 - If the handover policy is redirection, the gNodeB releases the RRC connection of the UE and instructs the UE to access a cell working on the target frequency.

Handover Policy Determination

The eNodeB or gNodeB selects a handover policy for a UE based on information about candidate cells or frequencies, the UE's current service type, and UE capabilities. Handover policies include:

- Handover: When referred to as a handover policy, it means that the base station, while retaining the RRC connections of a UE, transfers the UE from a source cell to a target cell to ensure service continuity.
- Redirection: The base station releases the RRC connections of a UE and instructs the UE to access a cell working on a certain frequency.

NOTE

- Unless otherwise stated, handover is used in this chapter as a generic term that covers procedures related to mobility management in connected mode.
- When referred to as a handover policy, handover is a specific term in contrast to redirection and describes a procedure in which the serving cell of a UE changes.

4.1.8.1 General Rules

If ANR is disabled and physical cell identifier (PCI) conflicts between candidate cells are detected, the eNodeB or gNodeB filters out these cells or the operating frequencies of these cells.

If ANR is enabled, the eNodeB or gNodeB does not filter neighboring cells. It derives candidate cells or frequencies from an ANR-maintained neighboring cell list.

If the mobility policy is handover, the eNodeB or gNodeB selects the cell with the best signal quality from the candidate cell list.

If the mobility policy is redirection, the eNodeB or gNodeB selects the operating frequency of the cell with the best signal quality on the candidate cell list.

4.1.8.2 NR-specific Rules

Frequency Filtering Rules

- If the **NRCeUtranNFreq.FreqIntrfFlag** parameter is set to **INTRF** for the operating frequency of a cell in the neighboring cell list, the gNodeB filters out this frequency when the cell meets the RSRQ-based leaving condition. This rule is applicable for downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, EPS fallback, and voice-quality-based handover from VoNR to VoLTE.
- When the PCI of the neighboring LTE cell with the best signal quality is not included in the neighboring cell list configured on the gNodeB or is involved in PCI confusion, the UE can be redirected to the operating frequency of this neighboring LTE cell. This applies to the following mobility functions:
 - Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN: The **EUTRAN_STRONGEST_NCELL_RED_SW** option of the **NRInterRatHoParam.HoModeSwitch** parameter is selected.
 - Inter-RAT ANR between E-UTRAN and NG-RAN is not enabled, or this ANR function is enabled but the UE does not support this ANR function.
 - ANR is enabled, but the **NR_EUTRAN_COV_REDIRECT_ANR_SW** option of the **NRCeUtranNFreq.AnrSwitch** parameter is deselected.
 - Inter-RAT ANR between E-UTRAN and NG-RAN and the ANR for coverage-based redirection from NG-RAN to E-UTRAN function are enabled, and the UE supports this ANR function. However, the UE does not send CGI measurement reports.
 - Voice fallback from NG-RAN to E-UTRAN: The **EPS_FB_STRONGEST_NCELL_RED_SW** option of the **NRCeUtranNFreq.VoiceStrategySwitch** parameter is selected.
 - Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN: The **UL_SINR_RED_TO_STRONGEST_NCELL_SW** option of the **NRInterRatHoParam.HoModeSwitch** parameter is selected.
- If redirection is set as the mobility policy and no neighbor relationship is configured for a candidate frequency:
 - During EPS fallback, this frequency can be selected for redirections.
 - In other scenarios, this frequency cannot be selected for redirections. It can be selected for redirections only when neighbor relationships are configured for it.

Neighboring Cell Filtering Rules

- Neighboring cells that belong to PLMNs (including the PLMNs of primary and secondary operators) outside the PLMN list of the serving cell are filtered out. PLMNs are checked against the following components of this list in descending order of priority:
 - PLMN that the serving cell belongs to
 - Equivalent PLMNs obtained by the gNodeB from the AMF-delivered *Mobility Restriction List* IE and the **gNBReqvPlmn** settings (AMF-delivered

equivalent PLMNs take precedence), when the **INTER_RAT_HO_WITH_GNB_EPLMN_SW** option of the **gNodeBParam.EqvPlmnAlgoSwitch** parameter is selected.

- When the LTE side is working in RAN sharing with special dedicated carrier mode, neighboring cells of secondary operators are filtered out if the handover is to the primary PLMN of the neighboring cells. A neighboring LTE cell is a cell of secondary operators if the **MASTER_PLMN_RESERVED_FLAG** option of the **gNBEutraExternalCell.AggregationAttribute** parameter is selected for that cell.
- The gNodeB cannot hand over UEs to the E-UTRAN cells served by the eNodeBs that have been added to the blacklist using the **gNBEutranBlacklist** MO. However, this restriction does not apply to emergency voice fallback.

4.1.8.3 LTE-specific Rules

Neighboring Cell Filtering Rules

After a UE sends a measurement report, the eNodeB filters neighboring cells based on the cell IDs corresponding to the PCIs in the measurement report. The eNodeB does not deliver mobility commands related to neighboring NR cells or neighboring frequencies that meet any of the following conditions:

- Neighboring cells that belong to PLMNs (including the PLMNs of primary and secondary operators) not on the PLMN list of the serving cell. This list is composed of the following PLMNs:
 - PLMN that the serving cell belongs to
 - PLMNs configured in the **InterPlmnHoList** MO when the **InterPlmnHoSwitch** option of the **ENodeBAlgoSwitch.HoAlgoSwitch** parameter is selected
 - Equivalent PLMNs obtained by the eNodeB from the *Handover Restriction List* IE sent from the MME when both the **InterPlmnHoSwitch** and **EPlmnSwitch** options of the **ENodeBAlgoSwitch.HoAlgoSwitch** parameter are selected
- Blacklisted NR cells
- NSA-only NR cells, that is, cells with the **NrExternalCell.NrNetworkingOption** parameter set to **NSA**
- If the operating frequency of a neighboring NR cell belongs to multiple frequency bands, the **NrExternalCell.FrequencyBand** and **NrExternalCell.AdditionalFrequencyBand** parameters are configured, and the **NR_MULTIBAND_SUPPORT_SW** option of the **EnodebAlgoExtSwitch.NrMfbiAlgoSwitch** parameter is selected, the eNodeB filters out the cells that are included in NR measurement reports and operate in the UE-incapable frequency bands specified by the **NrExternalCell.FrequencyBand** and **NrExternalCell.AdditionalFrequencyBand** parameters.
- If the PCI of the strongest neighboring cell in a B1 measurement report received by the eNodeB from the UE is unknown or the cell experiences a PCI conflict, a redirection to the operating NR frequency of the strongest neighboring cell is triggered. This function requires that the **NrRedirectEnhSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter be selected and the following conditions be met:

- The redirection switch (controlled by the **NrRedirectSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter) is turned on.
- Inter-RAT ANR between E-UTRAN and NG-RAN is not enabled, or this ANR function is enabled but the UE does not support this ANR function.
- If the **NR_B1_B2_PLUS_RSRQ_INDICATION** option of the eNodeB parameter **NrNFreq.AggregationAttribute** is selected for an NR frequency, the eNodeB filters out cells that work on this NR frequency and meet the RSRQ-based leaving condition from the candidate cell list.
- Neighboring NR cells with Extended Cell Range enabled, for which the **OVER_DISTANCE_FLAG** option of the **NrNRelationship.AggregationAttribute** parameter is selected
- (Applicable only when the NR side is working in RAN sharing with dedicated carrier mode) Neighboring cells of secondary operators, when the handover is a handover to the primary PLMN of the neighboring cells (A neighboring NR cell is a cell of secondary operators if the **NrExternalCell.MasterPlmnReservedFlag** parameter is set to **TRUE** for that cell.)
- (Applicability dependent on the setting of the **NrBwCheckSwitch** option of the **ENodeBAlgoSwitch.CompatibilityCtrlSwitch** parameter) Any cell whose NR cell bandwidth and minimum NR carrier bandwidth are both different from any cell bandwidth supported by the UE. The downlink and uplink NR cell bandwidths are specified by the **NrNFreq.NrDLBandwidth** and **NrNFreq.NrULBandwidth** parameters, respectively. The downlink and uplink minimum NR carrier bandwidths are specified by the **NrNFreq.NrDLMinCarrierBw** and **NrNFreq.NrULMinCarrierBw** parameters, respectively. If both the **NrNFreq.NrDLBandwidth** and **NrNFreq.NrULBandwidth** parameters are set to **NOT_CONFIG**, these NR cells are not filtered out.
- For a RedCap UE (the value of the **supportOfRedCap-r17** field in the *UE-NR-Capability* IE is **supported**), if the **REDCAP_UE_NR_HO_POLICY_SW** option of the **ENodeBAlgoExtSwitch.NrHandoverAlgoSwitch** parameter is selected, and the RedCap UE supports 1RX or 2RX but an external NR cell does not support 1RX or 2RX (the **NrExternalCell.CellBarredRedCap** parameter is set to a value other than **NOT_BARRED**), the eNodeB filters out the corresponding NR cell.

Measurement Report Processing Mode on the eNodeB

After receiving measurement reports on event B1 or B2 from the UE, the eNodeB processes the measurement reports based on the mode indicated by the **CellHoParaCfg.NrMrProcessMode** parameter.

- If this parameter is set to **NORMAL**, new processing modes do not take effect. In the case of a handover, the eNodeB makes a handover decision immediately after receiving a measurement report. In the case of a redirection, the eNodeB selects a target frequency for redirection based on the latest measurement report before receiving a UE context release command from the core network.
- If this parameter is set to **HANDOVER_IMMEDIATELY**, whether to perform a handover or redirection is determined immediately after a measurement report is received.

- If this parameter is set to **BASED_ON_SIGNAL_STRENGTH**, a handover to the cell with the strongest signal strength or a redirection to the operating frequency of this cell indicated by measurement reports is initiated after a waiting period specified by **CellHoParaCfg.NrMrWaitingTimer** elapses.
- If this parameter is set to **BASED_ON_FREQ_PRIORITY**, and a measurement report of a cell on the highest-priority frequency is received, a handover to this cell or redirection to this frequency is immediately initiated. Otherwise, a handover to the cell on the highest-priority frequency or redirection to this frequency indicated in the received reports on event B1 or B2 is initiated after a waiting period specified by the **CellHoParaCfg.NrMrWaitingTimer** parameter elapses.

4.1.9 Mobility Policy Execution

The eNodeB or gNodeB delivers the target cell or frequency to the UE, instructing the UE to perform a handover or redirection.

During a handover, handover preparation precedes handover execution. Handover preparation and handover execution may fail. Upon failures in these phases, penalties will be imposed to reduce the number of unnecessary handover retries on target cells. This section describes penalty upon failures of handovers from E-UTRAN to NG-RAN. For details about penalty upon failures of handovers from NG-RAN to E-UTRAN, see *SA Mobility Management in Connected Mode in 5G RAN Feature Documentation*.

NOTE

During a redirection to E-UTRAN, the gNodeB discards the data that it received from the core network but has not delivered to the UE. The core network will use the retransmission mechanism of upper-layer services to send the undelivered data to the UE through the eNodeB.

To increase the preparation success rate of handovers between a Huawei eNodeB and a non-Huawei gNodeB, if the non-Huawei gNodeB requires that the PLMN ID in the Target Cell ID IE in the LTE-to-NR handover request message be the **Primary PLMN** (PLMN ID of an external NR cell), select the **LteToNrHoTargetCellIdPlmnSw** option of the **ENodeBAlgoSwitch.HoSignalingOptSwitch** parameter; if the non-Huawei gNodeB allows that the PLMN ID in the Target Cell ID IE in the LTE-to-NR handover request message be the **Selected PLMN**, deselect the **LteToNrHoTargetCellIdPlmnSw** option.

Penalties Upon Failures in the Handover Preparation Phase

A penalty is imposed in the following scenarios in the handover preparation phase:

- The base station receives a Handover Preparation Failure message. The preparation failures may be either resource-related failures or non-resource-related failures.
- After sending a handover request (Handover Required), the base station does not receive a handover command (Handover Command). This triggers handover cancellation.

If the handover preparation fails, the eNodeB initiates a handover attempt to the next best cell in the candidate cell list. If the eNodeB has tried all cells in the candidate cell list but handover preparation still fails, the eNodeB waits for a second measurement report from the UE.

Handover preparation may fail in different scenarios. The penalty and retry policies vary depending on the scenario.

- If the handover preparation in a cell fails for a resource reason, a penalty whose duration is determined by a penalty timer is imposed on the cell. No retry can be initiated toward that cell until the penalty timer expires. The penalty timer on the E-UTRAN side is specified by the **HoMeasComm.NrHoResPrepFailPenaltyTmr** parameter. If the **HoMeasComm.NrHoResPrepFailPenaltyTmr** parameter is set to a value other than 0, it is recommended that the **LNR_RES_HO_FAIL_PENALTY_OPT_SW** option of the **EnodebAlgoExtSwitch.NrHandoverAlgoSwitch** parameter be selected to enable penalty optimization for E-UTRAN-to-NGRAN handover preparation failures due to resource-related causes. With the optimization, the penalty for handover preparation failures becomes more accurate, and the eNodeB imposes the penalty on more UEs.

NOTE

- Only the following reasons are the resource reasons for handover preparation failures:
 - No radio resources available in target cell
 - Transport resource unavailable
 - Not enough User Plane Processing Resources
 - Radio resources not available
 - Control Processing Overload
- If the handover preparation in a cell fails for a non-resource reason or a handover is canceled, a penalty whose duration is defined by a maximum number of penalty times is imposed on the cell. This number is specified by the **HoMeasComm.NonResHoPreFailPunishTimes** parameter. No retry can be initiated toward that cell until the number of penalty times reaches the maximum number. When the cell is in the penalty state, the number of penalty times is incremented by 1 if the measurement result indicates that the cell meets handover conditions. After the penalty on the cell is complete, retries may occur. A handover request toward the cell can be initiated next time the cell meets handover conditions. If all the retries on the cell fail when the number of retries reaches the maximum, no more handover attempts for the UE can be initiated toward the cell. The maximum number of retries is specified by the **HoMeasComm.NonResHoPreFailRetryTimes** parameter.

Penalties Upon Failures in the Handover Execution Phase

If the RRC connection of a UE is reestablished with the source cell upon the execution failure of a handover to a cell (for example, cell A), the source eNodeB still initiates a handover retry toward cell A next time a handover attempt is initiated and the handover preparation is successful for the retry. If the retries on cell A still fail after a number of handover attempts, the source eNodeB no longer sends a handover request to cell A for the UE. The number of attempts is specified by the **CellHoParaCfg.InterRatUuHoFailRetryTimes** parameter. This prevents the service drop rate from rising due to exceptions.

Handling of Oversized Handover Request Messages

How the base station handles oversized NG-RAN-to-E-UTRAN handover request messages:

During a handover from NG-RAN to E-UTRAN, if the size of the handover request message is excessively large, the handover request message may exceed the specifications of the core network, and thus be truncated or rejected by the core network. As a result, the handover fails. In this case, the value of the **gNBMobilityCommParam.InterRatHoReqUeCapbSize** parameter needs to be properly configured on the base station, which should be based on the capability of the core network in terms of handling the handover request messages. Before sending a handover request, the gNodeB checks whether the reported UE capability field size (E-UTRA/EN-DC/NR) exceeds the value of **gNBMobilityCommParam.InterRatHoReqUeCapbSize**.

- If **gNBMobilityCommParam.InterRatHoReqUeCapbSize** is set to a non-zero value:
 - If the UE capability field size exceeds the value of **gNBMobilityCommParam.InterRatHoReqUeCapbSize**, the NG-RAN-to-E-UTRAN handover request message carries only the E-UTRA capability.
 - If the UE capability field size does not exceed the value of **gNBMobilityCommParam.InterRatHoReqUeCapbSize**, the NG-RAN-to-E-UTRAN handover request message normally carries the reported UE capability.
- When **gNBMobilityCommParam.InterRatHoReqUeCapbSize** is set to 0:
 - In this case, the NG-RAN-to-E-UTRAN handover request message carries the reported UE capability.

How the base station handles oversized E-UTRAN-to-NG-RAN handover request messages:

During a handover from E-UTRAN to NG-RAN, if the size of the handover request message is excessively large, the handover request message may exceed the specifications of the core network, and thus be truncated or rejected by the core network. As a result, the handover fails. In this case, the value of the **EnodebAlgoExtSwitch.NrHoReqUeCapbSize** parameter needs to be properly configured on the base station, which should be based on the capability of the core network in terms of handling the handover request messages. Before sending a handover request, the eNodeB checks whether the reported UE capability field size (E-UTRA/EN-DC/NR) exceeds the value of **EnodebAlgoExtSwitch.NrHoReqUeCapbSize**.

- If **EnodebAlgoExtSwitch.NrHoReqUeCapbSize** is set to a non-zero value:
 - If the UE capability field size exceeds the value of **EnodebAlgoExtSwitch.NrHoReqUeCapbSize**, the E-UTRAN-to-NG-RAN handover request message carries only the NR capability.
 - If the UE capability field size does not exceed the value of **EnodebAlgoExtSwitch.NrHoReqUeCapbSize**, the E-UTRAN-to-NG-RAN handover request message carries the complete reported UE capability.
- If **EnodebAlgoExtSwitch.NrHoReqUeCapbSize** is set to 0.
 - In this case, the E-UTRAN-to-NG-RAN handover request message carries the complete reported UE capability.

Handling of Conflicts Between Handover and QoS Procedures

In the case of downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN or NSA/SA selection based on user experience, a conflict occurs between

handover and QoS procedures if the gNodeB receives a PDU session setup or modification message in the outgoing handover preparation phase and the message indicates QoS flow addition or modification. The **QOS_FLOW_FIRST_SW** option of the **NRCellQciBearer.HoQosFlowConflProcStrat** parameter determines how to handle this conflict.

- If the **QOS_FLOW_FIRST_SW** option is selected, the gNodeB preferentially handles the QoS procedure and initiates a handover cancellation request. It is recommended that this option be selected when the core network does not support PDU session setup or modification in the target cell during a handover procedure.
- If the **QOS_FLOW_FIRST_SW** option is deselected, the gNodeB preferentially handles the handover procedure and notifies the core network that the gNodeB is executing a handover. When the execution is complete, the core network re-initiates a QoS procedure, as specified in 3GPP TS 23.501. This QoS procedure will fail if the core network does not support the preceding procedure.

In the case of voice fallback from NG-RAN to E-UTRAN, if the gNodeB receives a PDU session setup or modification message in the outgoing handover preparation phase and the message indicates QoS flow addition or modification, the gNodeB continues the outgoing handover procedure for voice fallback, irrespective of the **QOS_FLOW_FIRST_SW** option setting of the **NRCellQciBearer.HoQosFlowConflProcStrat** parameter.

Handling of Conflicts Between Handover and PDU Session-related Procedures

This function allows for proper handling of conflicts between a handover procedure and a PDU session-related procedure. Such conflicts occur when the delay between NEs on the core network is abnormal, as in such scenarios the signaling procedure on the core network will be abnormal. For example, if the gNodeB receives a PDU session release message in the preparation phase of an NG-based inter-RAT outgoing handover, the NG-based handover procedure conflicts with the PDU session procedure. As a result, the handover fails, and the relevant UE cannot be handed over or released normally.

The conflict handling function is applicable to conflicts between a PDU session release procedure and an NG-based outgoing handover procedure other than those triggered by voice fallback from NG-RAN to E-UTRAN. This function allows the selected procedure to be normally processed, and is controlled by the **PDU_SESS_REL_NG_HO_REQ_SEL_SW** option of the **gNBMobilityCommParam.HoPduSessConflHandlingSw** parameter.

- If this option is selected, the gNodeB preferentially handles the PDU session release procedure and initiates a handover cancellation request.
- If this option is deselected, the gNodeB preferentially handles an NG-based handover procedure and sends a NAS NON DELIVERY INDICATION message to the core network, notifying the core network that it is in the middle of a handover procedure.

In the case of voice fallback from NG-RAN to E-UTRAN, if the gNodeB receives a PDU session release message in the preparation phase of an outgoing handover procedure, the gNodeB continues the outgoing handover procedure for voice

fallback, irrespective of the **PDU_SESS_REL_NG_HO_REQ_SEL_SW** option setting of the **gNBMobilityCommParam.HoPduSessConflHandlingSw** parameter.

This function is supported only in SA networking and NSA and SA hybrid networking. It is recommended that this function only be enabled if the signaling procedure on the core network is abnormal and is not recommended otherwise.

Prohibiting Large-Packet UEs in LTE Cells from Returning to NR Cells

The eNodeB can identify the service status of UEs that have been handed over from NR cells to LTE cells for resource optimization reasons. This is to ensure that UEs running large-packet services camp on LTE cells as much as possible. If a UE is identified as a large-packet UE (that is, the average amount of remaining data in the uplink buffer per TTI is greater than or equal to the

CellMultiCarrUniSch.ULTrafficVolumeThld parameter value), the eNodeB does not periodically deliver measurement configurations to the UE and prohibits the UE from returning to an NR cell. After the eNodeB identifies that the UE is not a large-packet UE, the eNodeB periodically delivers measurement configurations to the UE so that the UE can return to an NR cell.

For the UE that has been identified as a large-packet UE, the eNodeB no longer identifies it as one if one of the following conditions is met:

- The **FAST_TRAFFIC_AWARENESS_SCS_SW** option of the **MultiCarrUnifiedSch.MultiCarrierUnifiedSchEnSw** parameter is deselected and the UE does not meet the conditions of a large-packet UE for three consecutive seconds.
- The **FAST_TRAFFIC_AWARENESS_SCS_SW** option of the **MultiCarrUnifiedSch.MultiCarrierUnifiedSchEnSw** parameter is selected and the UE does not meet the conditions of a large-packet UE for a consecutive period specified by the value of **CellMultiCarrUniSch.TrafficModeExitPeriodNum** multiplied by 100 ms.

This function is controlled by the **UL_TFC_NR_LTE_MOBILITY_ENH_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter.

 NOTE

- The interoperability functions related to resource optimization are as follows:
 - Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN
 - NSA/SA selection based on downlink traffic volume
 - NSA/SA selection based on uplink coverage
- The gNodeB identifies large packets in a way different from that of eNodeB.
 - The gNodeB identifies large packets based on the volume of uplink traffic to be transmitted by a UE.
 - The eNodeB identifies large packets based on the average amount of remaining data in the uplink buffer per TTI.

Assume that the gNodeB identifies a UE as a large-packet UE and then the UE is handed over to an LTE cell. If the air interface capability of the LTE cell is strong and the average amount of remaining data in the uplink buffer per TTI does not meet the threshold requirement, the UE will not be identified as a large-packet UE and will return to the NR cell.

Increasing the time required for identifying a UE as a non-large-packet UE requires more time for the UE to return from the E-UTRAN to the NG-RAN, but the accuracy of identifying that a UE is no longer a large-packet UE in the uplink is also higher. Decreasing the time required for identifying a UE as a non-large-packet UE requires less time for the UE to return from the E-UTRAN to the NG-RAN, but the accuracy of identifying that a UE is no longer a large-packet UE in the uplink is also lower.

4.1.10 Roaming UEs Preferentially Back to HPLMN Cells

When a UE is roaming in an NR cell not belonging to the home public land mobile network (HPLMN), the UE can preferentially return to an LTE cell belonging to the HPLMN. This function is applicable to [5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN](#), [5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN](#), and [5.7 Voice Fallback from NG-RAN to E-UTRAN](#).

This function is controlled by the **INTER_RAT_HO_WITH_HPLMN_SW** option of the **gNodeBAlgo.HplmnAlgoSwitch** parameter. It requires that the RFSPs of roaming UEs be configured as follows on the gNodeB:

- The **gNBRfspConfig.RfspIndex** parameter is set to a non-zero value.
- The frequency belongs to the HPLMN of roaming UEs. That is, in the **gNBRfspConfig.gNBRfspPriorityGroup** MO identified by the **gNBRfspConfig.gNBRfspPriorityGroupPld** parameter corresponding to the **gNBRfspConfig.RfspIndex** parameter, the **gNBRfspPriorityGroup.RoamingUeHplmnFreqFlag** parameter is set to **TRUE**.

The gNodeB performs different mobility procedures for roaming UEs and non-roaming UEs as follows:

- When the gNodeB delivers inter-RAT event B1 or B2 measurement configurations (handover-based) or selects a frequency (redirection-based):
 - If the **NRCeUtranNFreq.MeasureType** parameter is set to **ROAMING_MEAS**, only roaming UEs can measure or select this frequency.
 - If the **NRCeUtranNFreq.MeasureType** parameter is set to **NORMAL**, all UEs can measure or select this frequency.

The preceding restrictions do not apply to emergency call UEs. These UEs can measure all frequencies. For details about the measurement configuration delivery procedure, see [4.1.4 Measurement Configuration](#).

- After receiving measurement reports on event B1 or B2 from the UE, the gNodeB selects a target cell as follows:
 - Assume that the RFSP and HPLMN are both configured for a roaming UE on the gNodeB through the **gNBRfspHplmn** MO and the PLMN of the target cell intersects with the serving PLMN or HPLMN, the handover is allowed. If target cells belonging to the serving PLMN and HPLMN both exist, the UE is preferentially handed over to a target cell belonging to the HPLMN.
 - Assume that the RFSP is configured for a roaming UE on the gNodeB but not the HPLMN corresponding to the RFSP, or the RFSP is not configured on the gNodeB. The gNodeB cannot obtain the HPLMN of the roaming UE. Under this setting, the gNodeB selects an LTE cell belonging to the serving PLMN.

For details about the target cell evaluation procedure, see [4.1.8 Target Cell or Frequency Evaluation](#).

4.2 Network Analysis

4.2.1 Benefits

The benefits are described in the "Benefits" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

RSRP Prediction Based on Virtual Grids

RSRP prediction using virtual grid models removes the need for UE measurements of neighboring frequencies. It shortens the time required for acquiring the coverage data of neighboring frequencies.

LTE-based NR RSRP prediction offers the following benefits to LTE networks:

- In scenarios without NR coverage, it reduces the number of invalid measurements, which are present in periodic UE measurements.
- In scenarios with NR coverage, it can quickly obtain the RSRP values of neighboring NR frequencies, reduce the overall durations of various interoperability procedures, and accelerate UE handovers to target cells.

NR-based LTE RSRP prediction offers the following benefits to NR networks:

- For voice fallback, it shortens the end-to-end call delay in normal voice fallback.
- For NSA/SA selection based on downlink traffic volume and NSA/SA selection based on uplink coverage:
 - In scenarios without LTE coverage, it reduces the number of invalid measurements, which are present in periodic UE measurements.

- In scenarios with LTE coverage, it can quickly obtain the RSRP values of neighboring LTE frequencies and transfer UEs to LTE cells, reducing the throughput loss caused by gap-assisted measurements.

Whether virtual-grid-based prediction results can be used for inter-RAT mobility between E-UTRAN and NG-RAN depends on factors such as the validity of A3 measurement reports from UEs as input for grid-based query and the grid-based prediction accuracy. Therefore, not all procedures for inter-RAT mobility between E-UTRAN and NG-RAN can benefit from virtual grids. The overall benefits of RSRP prediction based on virtual grids are affected by UE location distributions and service changes.

Rules for Configuring the PLMN ID in the Target Cell ID IE Carried in an LTE-to-NR Handover Request Message

If the **LteToNrHoTargetCellIdPlmnSw** option of the **ENodeBAlgoSwitch.HoSignalingOptSwitch** parameter is selected and the non-Huawei gNodeB requires that the PLMN ID in the Target Cell ID IE in the LTE-to-NR handover request message be the **Primary PLMN**, the success rate of handover preparations from LTE to NR increases. If the non-Huawei gNodeB requires that the PLMN ID in the Target Cell ID IE in the LTE-to-NR handover request message be the **Selected PLMN**, the success rate of handover preparations from LTE to NR decreases.

Handling of Oversized Handover Request Messages

- The values of the **N.HO.InterRAT.N2E.PrepAttOut** and **N.HO.InterRAT.N2E.PrepFailOut.PrepFailure** counters, indicating the number of outgoing handover preparation attempts from NG-RAN to E-UTRAN and number of outgoing handover preparation failures from NG-RAN to E-UTRAN, respectively, both decrease.
- The values of the **L.IRATHO.E2N.PrepAttOut** and **L.IRATHO.E2N.PrepFailOut.PrepFailure** counters, indicating the number of outgoing handover preparation attempts from E-UTRAN to NG-RAN and number of outgoing handover preparation failures from E-UTRAN to NG-RAN, respectively, both decrease.
- Assume that the **gNBMobilityCommParam.InterRatHoReqUeCapbSize** parameter is set to a non-zero value. A smaller value of this parameter results in a higher probability that an inter-RAT handover request message contains only the E-UTRA capability of a UE, a lower probability that the handover request is rejected by the core network, a higher handover success rate, but a higher probability that the target base station needs to query for the UE's capability of other RATs. A larger value of this parameter results in a lower probability that an inter-RAT handover request message contains only the E-UTRA capability of a UE, a higher probability that the handover request is rejected by the core network, and a lower handover success rate.
- Assume that the **EnodeBAlgoExtSwitch.NrHoReqUeCapbSize** parameter is set to a non-zero value. A smaller value of this parameter results in a higher probability that an LTE-to-NR handover request message contains only the NR capability of a UE, a lower probability that the handover request is rejected by the core network, a higher handover success rate, but a higher probability that the target base station needs to query for the UE's capability of other RATs. A larger value of this parameter results in a lower probability that an

LTE-to-NR handover request message contains only the NR capability of a UE, a higher probability that the handover request is rejected by the core network, and a lower handover success rate.

4.2.2 Impacts

The impacts between this function and other functions are described in the "Software" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

The "optimized non-overlapping between gap and NR resources" function lowers the uplink IBLER and uplink RBLER, reduces the loss in uplink and downlink UE throughput caused by gap-assisted measurement, reduces the number of delayed measurement reports, but may increase the number of reconfiguration signaling messages. In addition, this function may cause the values of the following cell-level indicators to change: **N.ThpTime.UL.Cell** (increase); **N.ThpTime.DL.Cell** (increase); **Cell Uplink Average Throughput (DU)** (decrease); **Cell Downlink Average Throughput (DU)** (decrease).

When RSRP-based triggering combined with RSRQ-based filtering is used for events B1 and B2, the possible impacts are as follows:

- Impacts on the NG-RAN: LTE cells must additionally meet RSRQ requirements before they can be selected by the gNodeB. If no candidate LTE cells are available after filtering, the value of the **N.HO.InterRAT.N2E.ExecSuccOut** counter decreases, indicating fewer successful handovers from NG-RAN to E-UTRAN.
- Impacts on the E-UTRAN: NR cells must additionally meet RSRQ requirements before they can be selected by the eNodeB. If no candidate NR cells are available after filtering, the value of the **L.IRATHO.E2N.ExecSuccOut** counter decreases, indicating fewer successful handovers from E-UTRAN to NG-RAN.

After frequency-specific LTE-NR interoperability prohibition is enabled, the values of the following counters decrease if there are no other candidate NR cells:

- **L.IRATHO.E2N.NoNRT** (number of times handovers from E-UTRAN to NG-RAN cannot be initiated because of no corresponding neighbor relationships)
- **L.IRATHO.E2N.Coverage.PrepAttOut** (number of coverage-based handover preparation attempts from E-UTRAN to NG-RAN)
- **L.IRATHO.E2N.Service.PrepAttOut** (number of service-based handover preparation attempts from E-UTRAN to NG-RAN)
- **L.IRATHO.E2N.PrepAttOut** (number of EUTRAN-to-NG-RAN outgoing handover attempts)
- **L.RRCRedirection.E2N** (number of redirections from E-UTRAN to NG-RAN)

RSRP prediction using virtual grid models has the following impacts:

- Measurement data needs to be collected for building virtual grid models. This increases the amount of air interface signaling and the number of SRBs. The average MCS index may decrease because small-index MCSs are used for SRBs. In addition, RRC connection reconfiguration messages may fail to be sent and therefore the service drop rate may increase.
- The CPU usage of main control boards increases when the eNodeB and gNodeB build virtual grid models and frequently query models.

CPU flow control works at the base stations:

- During data collection, a base station monitors the CPU usage, which is indicated by **Average CPU Load (LTE eNodeB, 5G gNodeB)**. If the value exceeds a threshold (60% by default), the base station stops building virtual grid models.
- After a virtual grid model is launched, its use is stopped when the CPU usage exceeds 60%, and its use is resumed when the CPU usage decreases to less than 45%.

When this function is enabled, more resources of the main control board are consumed. If a UMPTe, UMPTga, or UMPTHa series board is used as the main control board and it is working in LN or GULN mode, once the number of UEs in busy hours is too large and congestion occurs earlier than expected, the RRC connection setup success rate decreases, and the service drop rate increases.

- For LTE: The RRC connection setup success rate is measured using **RRC Setup Success Rate**. The service drop rate is measured using **Call Drop Rate (VoIP)** and **Service Drop Rate**.
- For NR: The RRC connection setup success rate is measured using **RRC Setup Success Rate (CU)**. The service drop rate is measured using **Service Call Drop Rate (CU)** and **Call Drop Rate (CU, VoNR)**.
- When the required accuracy of LN RSRP models is set to a value other than 100%, the results of prediction using virtual grid models can be used for handovers. Indicators related to the handover success rate may deteriorate for the following reasons:
 - The blind handover performance of certain UEs is poor.
 - When the network structure changes (for example, due to RF optimization), the predicted RSRP values in certain grids are inaccurate.
 - Target NR cells selected for handovers based on virtual grids are unavailable due to exceptions, and therefore handover preparations fail.
- When NL RSRP models are used for voice fallback or NSA/SA selection based on user experience, the results of prediction using virtual grid models can be used for handovers if the **MEAS_QTY_PRED_HO_FORBID_SW** option of the **NRCellSmartMultiCarr.HoProtectAlgoSwitch** parameter is deselected. Indicators related to the handover success rate may deteriorate for the following reasons:
 - The blind handover performance of certain UEs is poor.
 - When the network structure changes (for example, due to RF optimization), the predicted RSRP values in certain grids are inaccurate.
 - Target LTE cells selected for handovers based on virtual grids are unavailable due to exceptions, and therefore handover preparations fail.
- The eNodeB does not build virtual grid models within 14 days after an eNodeB software upgrade. However, models built before the upgrade can still be used during this period. After the 14 days, the eNodeB restarts the model building process. Therefore, the upgrade may have the following impacts:
 - The number of signaling messages over the air interface decreases.
 - The number of RRC connection reconfiguration messages decreases.
 - The service drop rate decreases.

Therefore, RSRP prediction based on virtual grids is not recommended in the following scenarios:

- High-speed mobility scenarios such as high-speed railways and highways: requiring the prevention of network performance deterioration caused by inaccurate NR coverage evaluation due to fast UE movement
- Big event assurance scenarios: requiring the prevention of network performance deterioration caused by CPU usage increases
- Network reconstruction scenarios: requiring the prevention of network performance deterioration caused by decreases in the accuracies of virtual grid models
- Scenarios with a small number of UEs in a cell: A large number of UE measurement reports must be collected for training of virtual grid models used for RSRP prediction. In cells serving a small number of UEs, it takes a long time to collect data and build virtual grid models. Therefore, RSRP prediction based on virtual grids is not recommended for networks serving a small number of UEs.

If the eNodeB uses signal-strength-based or frequency-priority-based handover mode, the time spent on waiting for inter-RAT measurement reports will be prolonged in certain scenarios, and the service drop rate increases.

Assume that coverage-based redirection to the strongest neighboring LTE cell and NR ANR with E-UTRAN are both enabled. If the **NR_EUTRAN_COV_REDIRECT_ANR_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected, indicators such as the inter-RAT handover failure rate and service drop rate may fluctuate.

When the LTE-to-NR handover policy for RedCap UEs is enabled (by selecting the **REDCAP_UE_NR_HO_POLICY_SW** option of the **EnodebAlgoExtSwitch.NrHandoverAlgoSwitch** parameter), and not all NR cells support RedCap, the number of LTE-to-NR outgoing handover preparation failures decrease and the number of redirections from LTE to NR decreases.

When penalty optimization for E-UTRAN-to-NG-RAN handover preparation failures due to resource-related causes is enabled (by selecting the **LNR_RES_HO_FAIL_PENALTY_OPT_SW** option of the **EnodebAlgoExtSwitch.NrHandoverAlgoSwitch** parameter), the impacts are as follows:

- The number of EUTRAN-to-NG-RAN outgoing handover attempts (indicated by the **L.IRATHO.E2N.PrepAttOut** counter) decreases.
- The number of EUTRAN-to-NG-RAN outgoing handover executions (indicated by the **L.IRATHO.E2N.ExecAttOut** counter) decreases.
- The success rate of handovers from E-UTRAN to NG-RAN (**L.IRATHO.E2N.ExecSuccOut/L.IRATHO.E2N.PrepAttOut**) increases.

For other impacts, see the "Impacts" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

4.3 Requirements

4.3.1 Licenses

For details, see the "Licenses" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

4.3.2 Software

For details, see the "Software" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

RSRP prediction using virtual grid models has the following hardware requirements:

- LN RSRP models: The main control board of the eNodeB must be a UMPTe, UMPTga, UMPTga, or UMPTg series board.
- NL RSRP models: The main control board of the gNodeB must be a UMPTe (NR only), UMPTga, UMPTga, or UMPTg series board.

For other hardware requirements, see the "Hardware" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

4.3.4 Others

RSRP prediction based on virtual grids:

- Building of LN RSRP models requires more than 15 NR-capable UEs in an LTE cell. This number of UEs is indicated by the **L.NsaDc.Capable.User.RRC.Avg** counter.
- Building of NL RSRP models requires more than 15 SA UEs in an NR cell. This number of UEs is equal to **(N.User.RRCCConn.Active.Avg – N.User.NsaDc.PSCell.Avg)**.

For other requirements, see the "Others" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

4.4 Operation and Maintenance (NG-RAN)

4.4.1 Data Configuration

4.4.1.1 Data Preparation

This section describes the basic MOs that need to be configured on the NG-RAN side for mobility management in connected mode. You are advised to configure the MOs based on the network plan. The MOs are as follows:

- **NRCellEutranNFreq**, which represents neighboring LTE frequencies of NR cells
A maximum of 32 LTE frequencies can be configured on a gNodeB.
- **gNBEutraExternalCell**, which represents external LTE cells
The maximum number of external LTE cells that can be configured on a gNodeB is equal to the maximum number of cells supported by the gNodeB multiplied by 32. For details about the maximum number of cells supported by a gNodeB, see "Capacity Specifications" in the corresponding BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- **NRCellEutraNRelation**, which represents the neighbor relationships of NR cells with LTE cells
A maximum of 384 neighbor relationships with LTE cells can be configured on an NR cell. The gNodeB-supported maximum number of neighbor relationships with LTE cells is equal to the maximum number of cells supported by a gNodeB multiplied by 256.

Setting notes are available for certain parameters, as described in the following tables.

Table 4-21 Neighboring LTE frequency of an NR cell

Parameter Name	Parameter ID	Setting Notes
Frequency-specific Offset	NRCellEutranNFreq.FreqSpecificOffset	The default value is recommended for function activation. During reconfiguration: • A smaller value of this parameter results in a lower probability of inter-RAT handovers. • A larger value results in a higher probability.
Frequency Priority for Connected Mode	NRCellEutranNFreq.ConnFreqPriority	The default value is recommended for function activation. During reconfiguration: • Set this parameter to 1 to 16 for LTE frequencies providing continuous coverage to specify their priorities. A larger value is recommended for a frequency in a lower frequency band. • Set this parameter to 0 for an LTE frequency providing discontinuous coverage. This frequency will not be selected as the target frequency for a blind redirection.
VoLTE Priority	NRCellEutranNFreq.VoltePriority	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
VOLTE Blind Priority	NRCellEutranNFr eq. <i>VolteBlindPriority</i>	Set this parameter based on the network plan.
Measurement Type	NRCellEutranNFr eq. <i>MeasureType</i>	Set this parameter based on the network plan.
Measurement Excluded Cell List Switch	NRCellMobilityC onfig. <i>MeasExcludedCellListSwitch</i>	Set this parameter based on the network plan.
RAT Type	gNBCellBlacklist. <i>RatType</i>	Set this parameter based on the network plan.
E-UTRAN Excluded Cell List ID	NRCellEutranNFr eq. <i>EutranExcludedCellListId</i>	Set this parameter based on the network plan.

Table 4-22 Handover measurement parameters related to events A1 and A2 for an NR cell

Parameter Name	Parameter ID	Setting Notes
Measurement Trigger Quantity	NRCellMobilityC onfig. <i>MeasTriggerQuantity</i>	- To enable RSRP-based triggering for events A1 and A2, set this parameter to RSRP. - To enable RSRP- or RSRQ-based triggering for events A1 and A2, set this parameter to RSRP_OR_RSRQ. - To enable RSRP- or SINR-based triggering for event A2 related to blind redirection in downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, set this parameter to RSRP_OR_SINR. For event A2 in other scenarios and event A1, RSRP-based triggering is used.
Inter-RAT Handover A1A2 Hysteresis	NRCellInterRHo MeaGrp. <i>InterRatHoA1A2Hyst</i>	The default value is recommended for function activation. The greater the value of this parameter, the more likely it is to prevent incorrect handover decisions and ping-pong handovers.

Parameter Name	Parameter ID	Setting Notes
Inter-RAT Handover A1 RSRP Threshold	NRCellInterRHoMeaGrp.InterRatHoA1RsrpThld	It is recommended that these parameters be set to values 4 dB greater than the respective thresholds for event A2. Assume that flexible user steering is enabled, that is, the NR_TO_EUTRAN_FREQ_LIST_CTRL_SW option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter is selected. On the gNodeB side, the RSRP threshold for inter-RAT handover event A1 is affected by the value of the gNBRfspConfig.InterRatA1A2B2RsrpThldOfs parameter. For details, see <i>Flexible User Steering</i> in <i>5G RAN Feature Documentation</i> . For super uplink UEs, the RSRP threshold for the inter-RAT handover event A1 on the gNodeB side is affected by the value of NRCellMeasConfig.SulCovHoRsrpThldOffset . For details, see <i>Super Uplink</i> in <i>5G RAN Feature Documentation</i> .
Inter-RAT Handover A1 RSRQ Threshold	NRCellInterRHoMeaGrp.InterRatHoA1RsrqThld	

Parameter Name	Parameter ID	Setting Notes
Inter-RAT Handover A2 RSRP Threshold	NRCeIIInterRHoMeaGrp.InterRatHoA2RsrpThld	Set the parameters based on site conditions. Usually, set them to the RSRP in the area where the UE throughput in the serving cell and that in the neighboring cell are similar. The RSRP or RSRQ threshold for inter-RAT event A2 must be less than that for inter-RAT event A1. Assume that flexible user steering is enabled, that is, the NR_TO_EUTRAN_FREQ_LIST_CTRL_SW option of the NRCeIIAlgoSwitch.RfspAlgoSwitch parameter is selected. On the gNodeB side, the RSRP threshold for inter-RAT handover events A2 is affected by the value of the gNB RfspConfig.InterRatA1A2B2RsrpThldOfs parameter. For details, see <i>Flexible User Steering in 5G RAN Feature Documentation</i>. For super uplink UEs, the RSRP threshold for the inter-RAT handover event A2 on the gNodeB side is affected by the value of NRCeII MeasConfig.SulCovHoRsrpThldOffset. For details, see <i>Super Uplink in 5G RAN Feature Documentation</i>. The SINR threshold for event A2 related to inter-frequency blind redirection is used as the SINR threshold for event A2 related to blind redirection in downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN.
Inter-RAT Handover A2 RSRQ Threshold	NRCeIIInterRHoMeaGrp.InterRatHoA2RsrqThld	
Cov HO to E-UTRAN Blind A2 RSRP Thld	NRCeIIInterRHoMeaGrp.CovHoToEutranBlindA2Thld	
Cov HO to E-UTRAN Blind A2 RSRQ Thld	NRCeIIInterRHoMeaGrp.CovHoToLteBlindA2RsrqThld	
Inter-Freq Blind Redirect A2 SINR Thld	NRCeIIInterFHoMeaGrp.InterFBli ndRedirA2SinrThld	

Parameter Name	Parameter ID	Setting Notes
Cov HO to E-UTRAN B2 RSRP Thld1	NRCeIIInterRHoMeaGrp.CovHoToEutranB2RsrpThld1	It is recommended that you adjust the parameter value based on site conditions. Increase the parameter value if delayed inter-RAT handovers occur. <i>Thresh2</i> is specified by a parameter that is also used for event B1. Assume that flexible user steering is enabled, that is, the NR_TO_EUTRAN_FREQ_LIST_CTRL_SW option of the NRCeIIAlgoSwitch.RfspAlgoSwitch parameter is selected. On the gNodeB side, the RSRP Thresh1 for downlink-coverage-based inter-RAT event B2 is affected by the value of the gNB RfspConfig.InterRatA1A2B2RsrpThldOfs parameter. For details, see <i>Flexible User Steering in 5G RAN Feature Documentation</i>.
Inter-RAT Handover A1A2 Time To Trigger	NRCeIIInterRHoMeaGrp.InterRatHoA1A2TimeToTrigger	The default value is recommended for function activation. During reconfiguration: • A larger value of this parameter results in a lower probability of event reporting triggered by instantaneous signal changes, a smaller average number of handovers, and a smaller number of unexpected handovers, preventing unnecessary handovers. • A smaller value results in the opposite effects.
Inter-RAT Handover Measurement Group ID	NRCeIIQciBearer.InterRatHoMeasGroupId	Set this parameter based on the network plan.

Table 4-23 Handover measurement parameters related to events B1 and B2 for an NR cell

Parameter Name	Parameter ID	Setting Notes
Frequency Interference Flag	NRCellEutranNFr eq.FreqIntrfFlag	If a frequency is exposed to interference, set this parameter to INTRF for the frequency. In this case, the method of RSRP-based triggering combined with RSRQ-based filtering is used.
Coverage-based HO B1B2 Hysteresis	NRCellHoEutran MeaGrp.CovBase dHoB1B2Hyst	The default value is recommended for function activation. The greater the value of this parameter, the more likely it is to prevent incorrect handover decisions and ping-pong handovers.
Coverage-based HO B1 RSRP Threshold	NRCellHoEutran MeaGrp.CovBase dHoB1RsrpThld	It is recommended that you adjust the parameter values based on site conditions. Decrease the parameter values if delayed inter-RAT handovers occur.
Coverage-based HO B1 RSRQ Threshold	NRCellHoEutran MeaGrp.CovBase dHoB1RsrqThld	
Coverage-based HO B1B2 Time to Trigger	NRCellHoEutran MeaGrp.CovBase dHoB1B2TimeTo Trig	The default value is recommended for function activation. During reconfiguration: • A larger value of this parameter results in a lower probability of event reporting triggered by instantaneous signal changes, a smaller average number of handovers, and a smaller number of unexpected handovers, preventing unnecessary handovers. • A smaller value results in the opposite effects.
Frequency-specific Offset	NRCellEutranNFr eq.FreqSpecificOf fset	The default value is recommended for function activation. During reconfiguration: • A smaller value of this parameter results in a lower probability of inter-RAT handovers. • A larger value results in a higher probability.

Parameter Name	Parameter ID	Setting Notes
E-UTRAN Cell Individual Offset	NRCellEutraNRel ation.EutranCell/ ndividualOffset	The default value is recommended for function activation. During reconfiguration: • A smaller value of this parameter leads to a lower probability of triggering B1 or B2 measurement reporting and a smaller number of handovers. • A larger value results in the opposite effects.
Inter-RAT HO to E-UTRAN Meas Group ID	NRCellQciBearer. InterRHoToEutra nMeasGrpld	Set this parameter based on the network plan.
VoNR Quality E-UTRAN HO B1 RSRP Thld	NRCellServExp.V onrQltyEutranB1 RsrpThld	Set this parameter based on the network plan.
VoNR Quality E-UTRAN HO B1 RSRQ Thld	NRCellServExp.V onrQltyEutranB1 RsrqThld	Set this parameter based on the network plan.

Table 4-24 (Optional) Parameters related to equivalent PLMNs

Parameter Name	Parameter ID	Setting Notes
Equivalent PLMN Algorithm Switch	gNodeBParam.Eq vPlmnAlgoSwitc h	When the INTER_RAT_HO_WITH_GNB_EPLMN_SW option is selected, the gNodeB can acquire equivalent PLMNs from the AMF or the gNBEqvPlmn MO configuration.
Operator ID	gNBEqvPlmn.Op eratorId	This parameter needs to be set only when the INTER_RAT_HO_WITH_GNB_EPLMN_SW option is selected. Set this parameter based on the network plan.
Equivalent Mobile Country Code	gNBEqvPlmn.Equ ivalentMcc	This parameter needs to be set only when the INTER_RAT_HO_WITH_GNB_EPLMN_SW option is selected. Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Equivalent Mobile Network Code	gNBEqvPlmn.EquivalentMnc	This parameter needs to be set only when the INTER_RAT_HO_WITH_GNB_EPLMN_SW option is selected. Set this parameter based on the network plan.

Table 4-25 (Optional) Parameters related to flexible user steering

Parameter Name	Parameter ID	Setting Notes
RFSP Based Algorithm Switch	NRCellAlgoSwitch.RfspAlgoSwitch	Select the NR_TO_EUTRAN_FREQ_LIST_CTRL_SW option.
Downlink EARFCN	gNBFreqPriorityGroup.DLEarfcn	Set this parameter based on the network plan.
Frequency Priority for Connected Mode	gNBFreqPriorityGroup.ConnFreqPriority	Set this parameter based on the network plan.
Cell Reselection Priority	gNBFreqPriorityGroup.CellReselPri	Set this parameter based on the network plan.
Cell Reselection Sub-Priority	gNBFreqPriorityGroup.CellReselSubPri	Set this parameter based on the network plan.
VoLTE Priority	gNBFreqPriorityGroup.VoltePriority	Set this parameter based on the network plan.
VOLTE Blind Priority	gNBFreqPriorityGroup.VolteBlindPriority	Set this parameter based on the network plan.
HPLMN Algo Switch	gNodeBAlgo.HplmnAlgoSwitch	To enable the "roaming UEs preferentially back to HPLMN cells" function, select the INTER_RAT_HO_WITH_HPLMN_SW option.
Roaming UE HPLMN Frequency Flag	gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag	To enable the "roaming UEs preferentially back to HPLMN cells" function, set this parameter to TRUE .

Table 4-26 (Optional) Parameters related to RSRP prediction based on virtual grids

Parameter Name	Parameter ID	Setting Notes
LNR Multi-Carrier Algorithm Switch	NRCellSmartMultiCarr.LnrMultiCarrierAlgoSwitch	Select the LNR_HO_MEAS_QTY_PRED_FUN_SW option.
LNR Handover Success Rate Protect Thld	NRCellSmartMultiCarr.LnrHoSuccessRateProtectThld	The default value is recommended.
Prediction Accuracy for Voice Thld	NRCellSmartMultiCarr.PredAccuracyForVoiceThld	The default value is recommended.
Prediction Accuracy for Data Thld	NRCellSmartMultiCarr.PredAccuracyForDataThld	The value ACCURACY95 is recommended.
Handover Protect Algorithm Switch	NRCellSmartMultiCarr.HoProtectAlgoSwitch	Deselect the MEAS_QTY_PRED_HO_FORBID_SW option.
Aggregation Attribute	NRCellEutranNfrEq.AggregationAttribute	- Select the VG_MODEL_ALLOW_BUILD_FLAG option for LTE frequencies for which virtual grid models are allowed to be built. - Deselect the VG_MODEL_ALLOW_BUILD_FLAG option for LTE frequencies for which virtual grid models are not allowed to be built.
NR Multi-Carrier Algorithm Switch	NRCellSmartMultiCarr.NrMultiCarrierAlgoSwitch	Select the NR_VG_MDL_MR_IMP_SW option if inter-RAT neighboring cell measurements need to be triggered by event B2 during model training.

Table 4-27 (Optional) Compatibility between coverage-based redirection to the strongest neighboring LTE cell and NR ANR with E-UTRAN

Parameter Name	Parameter ID	Setting Notes
ANR Switch	NRCellAlgoSwitch.h.AnrSwitch	To enable compatibility between coverage-based redirection to the strongest neighboring LTE cell and NR ANR with E-UTRAN, select the NR_EUTRAN_ANR_SW and NR_EUTRAN_COV_REDIRECT_ANR_SW options.

Table 4-28 (Optional) UE capability field size in an NG-RAN-to-E-UTRAN handover request message

Parameter Name	Parameter ID	Setting Notes
Inter-RAT Handover Request UE Capability Size	gNBMobilityCom mParam.<i>InterRat HoReqUeCapbSiz e</i>	If optimization for handover from NG-RAN to E-UTRAN is required, set this parameter to the planned value. It is recommended that this function be enabled when the size of the NG-RAN-to-E-UTRAN handover request message exceeds the specifications of the core network.

Table 4-29 (Optional) Parameters related to non-overlapping between gap and NR resources

Parameter Name	Parameter ID	Setting Notes
Measurement Algorithm Switch	gNBMobilityCom mParam.<i>MeasAlg oSwitch</i>	If the target cells are NR cells only, select the GAP_AVOID_NR_RES_SW option if non-overlapping between gap and NR resources in the time domain is required.
Measurement Algorithm Switch	gNBMobilityCom mParam.<i>MeasAlg oSwitch</i>	If the target cells are LTE and NR cells, select the GAP_AVOID_NR_RES_SW and INTERRAT_GAP_OFFSET_MOD_SW options if non-overlapping between measurement gaps and NR resources in the time domain is required.
Measurement Algorithm Switch	gNBMobilityCom mParam.<i>MeasAlg oSwitch</i>	If the target cells are LTE cells only, select the GAP_AVOID_NR_RES_SW and INTERRAT_ONLY_GAP_OFFSET_SW options if non-overlapping between measurement gaps and NR resources in the time domain is required.
Measurement Algorithm Switch	NRDUCellMeas. <i>MeasAlgoSwitch</i>	If the target cells are LTE cells only, select the GAP_AVOID_NR_RES_OPT_SW option if optimized non-overlapping between gap and NR resources is required.

Table 4-30 (Optional) Multi-antenna-port configuration indication

Parameter Name	Parameter ID	Setting Notes
Frequency Control Switch	NRCellEutranNFr eq.FrequencyCtrl Switch	If all neighboring cells on the LTE frequencies in MeasObjectEUTRA are configured with two or more antenna ports, the LTE measurement control message needs to include the number of antenna ports of the neighboring LTE cells to improve the measurement accuracy. In this case, select the PRESENCE_ANTENNA_PORT1_SW option of this parameter.

4.4.1.2 Using MML Commands

Activation Command Examples

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Adding a neighboring LTE frequency
ADD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1300, MeasurementBandwidth=MBW15,
ConnFreqPriority=1, VoltePriority=2, VolteBlindPriority=1, MeasureType=NORMAL, FreqSpecificOffset=DB0;
//Adding an external LTE cell whose MCC and MNC are 460 and 20, respectively
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="20", EnodebId=122, CellId=1, DLEarfcn=1300,
PhysicalCellId=3, Tac=1;
//Optional) Enabling inter-PLMN handovers
MOD GNODEBPARAM: EqvPlmnAlgoSwitch=INTER_RAT_HO_WITH_GNB_EPLMN_SW-1;
//Optional; required for inter-PLMN handovers) Configuring the equivalent PLMN list based on the
network plan, with the MCC and MNC of the operator set to 460 and 01, respectively
ADD GNBEQVPLMN: OperatorId=1, EquivalentMcc="460", EquivalentMnc="01";
//Adding a neighbor relationship with the LTE cell
ADD NRCELLEUTRANRELATION: NrCellId=0, Mcc="460", Mnc="20", EnodebId=122, CellId=1;
//Optional) Setting the measurement triggering quantity to RSRP_OR_RSRQ if RSRP- or RSRQ-based
triggering is used
MOD NRCELLMOBILITYCONFIG: NrCellId=0, MeasTrigQuantity=RSRP_OR_RSRQ;
//Optional) Setting the measurement triggering quantity to RSRP_OR_SINR if RSRP- or SINR-based
triggering is required for event A2 related to blind redirection in downlink-coverage-based inter-RAT
mobility from NG-RAN to E-UTRAN
MOD NRCELLMOBILITYCONFIG: NrCellId=0, MeasTrigQuantity=RSRP_OR_SINR;
//Optional) Setting the frequency interference flag to enable RSRP-based triggering combined with RSRQ-
based filtering
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1300, FreqIntrfFlag=INTRF;
//Optional) Adjusting the RSRP and RSRQ thresholds for VoNR-to-VoLTE handover based on voice quality
MOD NRCELLSERVEXP: NrCellId=0, VonrQtyEutranB1RsrpThld=-120, VonrQtyEutranB1RsrqThld=-17;
//Adding a parameter group related to measurements for inter-RAT handovers from the NR cell
ADD NRCELLINTERRHOMEAGRP: NrCellId=0, InterRatHoMeasGroupId=10, InterRatHoA1RsrpThld=-116,
InterRatHoA2RsrpThld=-118, InterRatHoA1A2Hyst=2, InterRatHoA1A2TimeToTrig=320MS,
CovHoToEutranBlindA2Thld=-126, CovHoToEutranB2RsrpThld=-122, CovHoToLteBlindA2RsrqThld=-38,
```

```

InterRatHoA1RsrqThld=-30, InterRatHoA2RsrqThld=-34;
//(Optional) Setting the SINR threshold for event A2 related to inter-frequency blind redirection, which is
also used as the SINR threshold for event A2 related to blind redirection in downlink-coverage-based inter-
RAT mobility from NG-RAN to E-UTRAN in the case that RSRP- or SINR-based triggering is used for event
A2
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, InterFBlindRedirA2SinrThld=-16;
//Adding a parameter group related to measurements for handovers from the NR cell to E-UTRAN
ADD NRCELLHOEUTRANMEAGRP: NrCellId=0, InterRHoToEutranMeasGrpId=10,
CovBasedHoB1RsrpThld=-104, CovBasedHoB1B2Hyst=2, CovBasedHoB1B2TimeToTrig=320MS,
CovBasedHoB1RsrqThld=-20;
//Binding the parameter groups to QCI 5 and 9 in the NR cell
MOD NRCELLQCIBEARER: NrCellId=0, Qci=5, InterRatHoMeasGroupId=10, InterRHoToEutranMeasGrpId=10;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, InterRatHoMeasGroupId=10, InterRHoToEutranMeasGrpId=10;
//Turning on the measurement excluded cell list switch
MOD NRCELLMOBILITYCONFIG: NrCellId=0, MeasExcludedCellListSwitch=ON;
//Adding a cell blacklist for the gNodeB
ADD GNBCELLBLACKLIST: CellBlacklistId=1, StartPhysicalCellId=231, PhysicalCellIdRange=N4,
RatType=EUTRAN;
//Modifying the E-UTRAN excluded cell list ID
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1300, EutranExcludedCellListId=1;
//(Optional; required only when RFSP-specific dedicated priorities are used) Adding a frequency priority
group on the gNodeB side
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0, RatType=EUTRAN,
DLEarfcn=1300, CellReselPri=1, CellReselSubPri=0DOT2, ConnFreqPriority=2, VoltePriority=2,
VolteBlindPriority=1;
//(Optional; required only when RFSP-specific dedicated priorities are used) Adding gNodeB RFSP
configurations
ADD GNBRFSPCONFIG: OperatorId=1, RfspIndex=1, gNBFreqPriorityGroupId=0;
//(Optional; required only when RFSP-specific dedicated priorities are used) Turning on
NR_TO_EUTRAN_FREQ_LIST_CTRL_SW
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=NR_TO_EUTRAN_FREQ_LIST_CTRL_SW-1;
//(Optional; required for RSRP prediction based on virtual grids) Running the following commands to start
building virtual grid models
MOD NRCELLEUTRANNFREQ: NrCellId=0,
DLEarfcn=1300, AggregationAttribute=VG_MODEL_ALLOW_BUILD_FLAG-1;
MOD NRCELLSMARTMULTICARR: NrCellId=0,
LnrMultiCarrierAlgoSwitch=LNR_HO_MEAS_QTY_PRED_FUN_SW-1, LnrHoSuccRateProtectThld=950,
PredAccuForDataThld=ACCURACY95, PredAccuForVoiceThld=ACCURACY95,
HoProtectAlgoSwitch=MEAS_QTY_PRED_HO_FORBID_SW-0, NrMultiCarrierAlgoSwitch=NR_VG_MDL_MR_IMP
_SW-1;
//(Optional) Enabling RFSP-based handover back to the HPLMN
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1300, MeasureType=NORMAL;
MOD GNODEBALGO: HplmnAlgoSwitch=INTER_RAT_HO_WITH_HPLMN_SW-1;
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0, RatType=EUTRAN,
DLEarfcn=1300, RoamingUeHplmnFreqFlag=TRUE;
//(Optional) Enabling compatibility between coverage-based redirection to the strongest neighboring LTE
cell and NR ANR with E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0,
AnrSwitch=NR_EUTRAN_ANR_SW-1&NR_EUTRAN_COV_REDIRECT_ANR_SW-1;
//Configuring the UE capability field size carried in an NG-RAN-to-E-UTRAN handover request message
MOD GNBMOBILITYCOMMPARAM: InterRatHoReqUeCapbSize=5;
//(Optional) Selecting the GAP_AVOID_NR_RES_SW and INTERRAT_GAP_OFFSET_MOD_SW options if the
target cells are LTE and NR cells and non-overlapping between gap and NR resources in the time domain is
required
MOD GNBMOBILITYCOMMPARAM:
MeasAlgoSwitch=GAP_AVOID_NR_RES_SW-1&INTERRAT_GAP_OFFSET_MOD_SW-1;
//(Optional) Selecting the GAP_AVOID_NR_RES_SW and INTERRAT_ONLY_GAP_OFFSET_SW options if the
target cells are LTE cells and non-overlapping between gap and NR resources in the time domain is required
MOD GNBMOBILITYCOMMPARAM:
MeasAlgoSwitch=GAP_AVOID_NR_RES_SW-1&INTERRAT_ONLY_GAP_OFFSET_SW-1;
//(Optional) Selecting the following option if the target cells are LTE cells and optimized non-overlapping
between gap and NR resources is required
MOD NRDUCELLMEAS: NrDuCellId=0, MeasAlgoSwitch=GAP_AVOID_NR_RES_OPT_SW-1;
//(Optional) Selecting the PRESENCE_ANTENNA_PORT1_SW option if each of the neighboring cells on the
LTE frequencies in MeasObjectEUTRA is configured with two or more antenna ports and therefore the LTE
measurement configurations need to include the number of antenna ports of the neighboring LTE cells to
improve the measurement accuracy (when no neighboring LTE frequency for the NR cell has been added)
ADD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1000, MeasurementBandwidth=MBW6,
FrequencyCtrlSwitch=PRESENCE_ANTENNA_PORT1_SW-1;

```

```

//(Optional) Selecting the PRESENCE_ANTENNA_PORT1_SW option if each of the neighboring cells on the
LTE frequencies in MeasObjectEUTRA is configured with two or more antenna ports and the LTE
measurement configurations need to include the number of antenna ports of the neighboring LTE cells to
improve the measurement accuracy (when a neighboring LTE frequency for the NR cell has been added)
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1000, MeasurementBandwidth=MBW6,
FrequencyCtrlSwitch=PRESENCE_ANTENNA_PORT1_SW-1;

```

Deactivation Command Examples

The data configurations described in this section are fundamental to the functions related to mobility from NG-RAN to E-UTRAN. Deactivation is not recommended.

```

//(Optional) Deselecting PRESENCE_ANTENNA_PORT1_SW
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1000, MeasurementBandwidth=MBW6,
FrequencyCtrlSwitch=PRESENCE_ANTENNA_PORT1_SW-0;
//(Optional) Disabling non-overlapping between measurement gaps and NR resources in the time domain
if the target cells are LTE and NR cells
MOD GNBMOBILITYCOMMPARAM:
MeasAlgoSwitch=GAP_AVOID_NR_RES_SW-0&INTERRAT_GAP_OFFSET_MOD_SW-0;
//(Optional) Disabling non-overlapping between measurement gaps and NR resources in the time domain
if the target cells are LTE cells
MOD GNBMOBILITYCOMMPARAM:
MeasAlgoSwitch=GAP_AVOID_NR_RES_SW-0&INTERRAT_ONLY_GAP_OFFSET_SW-0;
//(Optional) Disabling optimized non-overlapping between gap and NR resources if the target cells are LTE
cells
MOD NRDUCELLMEAS: NrDuCellId=0, MeasAlgoSwitch=GAP_AVOID_NR_RES_OPT_SW-0;
//(Optional) Running the following command to disable compatibility between coverage-based redirection
to the strongest neighboring LTE cell and NR ANR with E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0,
AnrSwitch=NR_EUTRAN_ANR_SW-1&NR_EUTRAN_COV_REDIRECT_ANR_SW-0;
//(Optional) Running the following commands to disable RFSP-based handover back to the HPLMN
MOD GNODEBALGO: HplmnAlgoSwitch=INTER_RAT_HO_WITH_HPLMN_SW-0;
//(Optional) Running the following command to disable RSRP prediction based on virtual grids
MOD NRCELLSMARTMULTICARR: NrCellId=0,
LnrMultiCarrierAlgoSwitch=LNR_HO_MEAS_QTY_PRED_FUN_SW-0;
//(Optional) Turning off the NR-to-E-UTRAN frequency list control switch
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=NR_TO_EUTRAN_FREQ_LIST_CTRL_SW-0;
//(Optional) Disabling inter-PLMN handovers
MOD GNODEBPARAM: EqvPlmnAlgoSwitch=INTER_RAT_HO_WITH_GNB_EPLMN_SW-0;

```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

When gNodeB data configuration is complete, you are advised to use the consistency check function on the MAE-Deployment to check data consistency. For details, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Consistency Check**. You are also advised to select the check items in the category "External Cell" to check whether the parameter settings of external LTE cells on the gNodeB side are consistent with those of the LTE cells on the eNodeB side.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

This section describes how to verify the activation of basic functions related to mobility management from NG-RAN to E-UTRAN in connected mode. For details,

see the "Activation Verification" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

RSRP Prediction Based on Virtual Grids

For RSRP prediction based on virtual grids, virtual grid model building can be completed only after sufficient UE measurement reports are collected, and it takes at least 24 hours to successfully build a virtual grid model. Therefore, it is recommended that the following operations be performed 24 hours after virtual grid model building is started:

- Using MML commands
Run the **DSP NRCELLAIGRIDMODEL** command to query the status of virtual grid models in a cell. If **RSRP Forecast Status** is **Available**, RSRP prediction based on virtual grids has been successfully activated.
- Using counters
LTE RSRP prediction based on virtual grids in an NR cell can be observed using the **N.LnrVirtualGrid.Model.Max** and **N.LnrVirtualGrid.Model.Avg** counters. If the value of either counter is not 0, NL RSRP models have been successfully built.

Blacklisted Cells for Cell Measurement

After related configurations are completed on the gNodeB, observe SIB5 of the current serving cell according to the following steps:

1. Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
2. If SIB5 delivered by the gNodeB carries the *EUTRA-FreqExcludedCellList* IE, and cells specified by this IE are consistent with the blacklisted cells specified by the **NRCeUtranNFreq.EutranExcludedCellListId** parameter, the blacklisted cells for cell measurement function has taken effect.

4.4.3 Network Monitoring

For details, see the "Network Monitoring" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

4.5 Operation and Maintenance (E-UTRAN)

4.5.1 Data Configuration

4.5.1.1 Data Preparation

This section describes the basic MOs that need to be configured on the E-UTRAN side for mobility management in connected mode. You are advised to configure the MOs based on the network plan. The MOs are as follows:

- **NrNFreq**, which represents neighboring NR frequencies
A maximum of 16 neighboring NR frequencies can be configured for each cell. The maximum number of neighboring NR frequencies that can be configured on an eNodeB is equal to the maximum number of cells supported by the eNodeB multiplied by 16. For details about the maximum number of cells supported by an eNodeB, see "Capacity Specifications" in the corresponding BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
The **NrNFreq.SubcarrierSpacing** parameter must be set to **15KHZ** for NR FDD and **30KHZ** for NR TDD.
- **NrMfbiFreq**, which represents the MFBI information about NR frequencies
According to 3GPP TS 38.104, if a frequency belongs to multiple NR frequency bands, the frequency band and additional frequency band must be configured for this frequency on the eNodeB side. Take a neighboring NR frequency whose NR-ARFCN is in the range of 422000–434000 as an example. This NR frequency belongs to bands n1 and n66. The **NrMfbiFreq** MO on the eNodeB side must be set based on the NR network plan, with this NR frequency configured in the frequency band and additional frequency band (n1 and n66). Neighboring NR frequencies whose NR-ARFCNs are outside the preceding ranges each belong to a single frequency band and do not require MFBI parameters to be set. The support for additional frequency bands is controlled by the **NR_MULTIBAND_SUPPORT_SW** option of the **EnodebAlgoExtSwitch.NrMfbiAlgoSwitch** parameter.
- **NrExternalCell**, which represents external NR cells
 - For the UMPTb and UMPTe series boards, a maximum of 672 external NR cells can be configured.
 - For the UMPTga series boards, a maximum of 1344 external NR cells can be configured.
 - For the UMPTg and UMPTga series boards: In LTE-only scenarios, a maximum of 2304 external NR cells can be configured. In multimode scenarios, a maximum of 1344 external NR cells can be configured.
- **NCellPlmnList**, which contains the parameters related to the PLMNs to which NR cells belong
The **NCellPlmnList.RatType** parameter must be set to **NR**, and the **NCellPlmnList.gNBIdLength** parameter must be set to the value the same as the **gNodeBFunction.gNBIdLength** parameter for the gNodeB with the same PLMN ID on the 5G network.
- **NrNRRelationship**, which represents the neighbor relationships with NR cells
 - For the UMPTb and UMPTe series boards, a maximum of 256 neighbor relationships with NR cells can be configured for each cell. For the UMPTb series boards, a maximum of 2304 neighbor relationships with NR cells can be configured on an eNodeB. For the UMPTe series boards, a maximum of 4608 neighbor relationships with NR cells can be configured on an eNodeB.
 - For the UMPTg, UMPTga, and UMPTga series boards, a maximum of 384 neighbor relationships with NR cells can be configured for each cell. The maximum number of neighbor relationships with NR cells that can be configured on an eNodeB is equal to the maximum number of cells supported by the eNodeB multiplied by 256.

Setting notes are available for certain parameters, as described in the following tables.

Table 4-31 Parameters related to basic mobility functions of an LTE cell

Parameter Name	Parameter ID	Setting Notes
Handover Mode switch	CellHoParaCfg.HoModeSwitch	- To enable handover, select the NrHoSwitch option. - To enable redirection, select the NrRedirectSwitch option. - To enable redirection enhancement, select the NrRedirectEnhSwitch option. When the PCI of the strongest neighboring cell in a B1 measurement report received by the eNodeB from a UE is unknown or is involved in a PCI conflict, a redirection to the operating NR frequency of the strongest neighboring cell is initiated. It is not recommended that this option be selected in the following scenarios: - When the strongest neighboring cell with an unknown or conflicting PCI is an NR cell in NSA networking, the UE may be redirected to an NSA frequency. - In RAN sharing with special dedicated carrier mode, if the MASTER_PLMN_RESERVED_FLAG option of the gNBEutraExternal-Cell.AggregationAttribute parameter is selected for the strongest neighboring cell with an unknown or conflicting PCI, a UE served by the primary PLMN cannot be redirected to the operating frequency of the cell. As a result, the redirection fails.
Handover Algo Switch	CellHoParaCfg.CellHoAlgoSwitch	Select the NrB1TimeToTrigAdjSw option. In this case, the eNodeB adjusts the time-to-trigger for NSA DC event B1 to be the same as that for NR event B1 for NSA/SA dual-mode UEs.

Parameter Name	Parameter ID	Setting Notes
Handover Allowed Switch	CellAlgoExtSwitch.HoAllowedSwitch	To prohibit LTE-to-NR mobility functions for emergency call UEs, select the EMERG_CALL_NR_RESTRICTION_SW option.
Max NR Measure Freq Number	CellUeMeasControlCfg.MaxNrMeasFreqNum	Set this parameter based on the network plan. If this parameter is set to a value greater than the number of NR frequencies that can be measured by a UE, UE exceptions may occur because of compatibility issues.
Measurement Optimization Algorithm Switch	CellUeMeasControlCfg.MeasOptAlgoSwitch	If the NSA_SA_MEAS_OBJ_PREEMPTION_SW option is selected and the number of measurement objects for a UE reaches the maximum, the eNodeB checks whether the priority of the algorithm to be started is higher than the highest priority of previously started algorithms. If it is, the eNodeB enables this algorithm to preempt measurement resources, which means that the UE will measure the frequencies involved in the algorithm to be started. If it is not, the preemption is not performed.
HO to NR Res Prep Failure Penalty Timer	HoMeasComm.NrHoResPrepFailurePenaltyTmr	A smaller value of this parameter leads to a larger number of invalid handover requests initiated after handover preparation fails for resource reasons. A larger value of this parameter leads to a smaller number of such invalid handover requests. However, if this parameter is set to an excessively large value, UEs will experience service drops in certain scenarios due to delayed handovers.
Handover to NR Algo Switch	EnodebAlgoExtSwitch.NrHandoverAlgoSwitch	It is recommended that the LNR_RES_HO_FAIL_PENALTY_OPT_SW option be selected when the HoMeasComm.NrHoResPrepFailurePenaltyTmr parameter is set to a value other than 0.
Non Resource Handover Prepare Fail Punish Times	HoMeasComm.NrNonResHoPrepFailurePunishTimes	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
Non Resource Handover Prepare Fail Retry Times	HoMeasComm.N onResHoPreFailR etryTimes	The default value is recommended.
Inter-RAT Uu Handover Failure Retry Times	CellHoParaCfg.In terRatUuHoFailR etryTimes	The default value is recommended.
Multi Networking Option Opt Switch	ENodeBALgoExtS witch.MultiNetw orkingOptionOpt Sw	To enable non-gap-assisted NR measurement for the eNodeB, select the NR_B1_NO_GAP_SW option.
Compatibility Control Switch	ENodeBALgoSwit ch.Compatibility CtrlSwitch	To enable the base station to filter out each NR cell whose NR cell bandwidth and minimum NR carrier bandwidth are both different from any cell bandwidth supported by the UE involved, select the NrBwCheckSwitch option.
NR MFBI Algorithm Switch	ENodebAlgoExtS witch.NrMfbiAlg oSwitch	If the NR frequency belongs to multiple frequency bands, select the NR_MULTIBAND_SUPPORT_SW option.
NR Handover Policy Configuration	InterRatPolicyCfg Group.NrHoCfg	• If VoNR is not supported on the NR side, it is recommended that the PS_HO and REDIRECTION options be deselected for QCI 1 to prevent voice call failures after handovers of VoLTE services to NR cells. • If VoNR is supported on the NR side, it is recommended that the PS_HO option be selected for QCI 1 and the REDIRECTION option be deselected for QCI 1. • It is recommended that the PS_HO and REDIRECTION options be selected for QCIs corresponding to non-voice services.
Frequency Priority for Connected Mode	NrNFreq.ConnFre qPriority	Set this parameter based on the network plan.

Table 4-32 Handover measurement parameters related to events A1 and A2 for an LTE cell

Parameter Name	Parameter ID	Setting Notes
InterRat A1A2 measurement trigger quantity	InterRatHoCom m.InterRatHoA1 A2TrigQuan	The value RSRP is recommended. If this parameter is set to RSRQ or BOTH , RSRP-based measurement configurations are delivered.
InterRAT A1A2 hysteresis	InterRatHoCom mGroup.InterRat HoA1A2Hyst	The default value is recommended for function activation. The greater the value of this parameter, the more likely it is to prevent incorrect handover decisions and ping-pong handovers.
Inter-RAT Handover to NR A1 RSRP Thld	InterRatHoCom mGroup.InterRat HoNrA1ThldRsrp	It is recommended that this parameter be set to a value 4 dB greater than the threshold for event A2.
Inter-RAT Handover to NR A2 RSRP Thld	InterRatHoCom mGroup.InterRat HoNrA2ThldRsrp	Set this parameter based on site conditions. Usually, set it to the RSRP in the area where the UE throughput in the serving cell and that in the neighboring cell are similar.
NR B2 RSRP Threshold 1	InterRatHoCom mGroup.NrB2Thl d1Rsrp	It is recommended that you adjust the parameter value based on site conditions. Increase the parameter value if delayed inter-RAT handovers occur.
InterRAT A1A2 time to trigger	InterRatHoCom mGroup.InterRat HoA1A2TimeToTr ig	• A larger value of this parameter results in a lower probability of event reporting triggered by instantaneous signal changes, a smaller average number of handovers, and a smaller number of unexpected handovers, preventing unnecessary handovers. • A smaller value results in the opposite effects.

Parameter Name	Parameter ID	Setting Notes
Inter-RAT NR A1 A2 Time To Trigger	InterRatHoCommGroup.InterRatHoNrA1A2TimeToTrig	It is recommended that this parameter value be adjusted based on site conditions. • A larger value of this parameter results in a lower probability of event reporting triggered by instantaneous signal changes, a smaller average number of handovers, and a smaller number of unexpected handovers, thereby preventing unnecessary handovers. • A smaller value results in the opposite effects. The value of this parameter does not take effect when the UMPTb series main control board works with an end-of-service-and-support (EOS) baseband processing unit.
InterRAT Handover Common Group ID	CellQciPara.InterRatHoCommGroupId	This parameter takes effect only for SA-capable UEs (SA UEs for short).
NSA DC Inter-RAT HO Common Group ID	CellQciPara.NsaDcInterRatHoCommGroupId	This parameter takes effect only for NSA/SA dual-mode UEs. If an identical inter-RAT mobility policy is used for NSA/SA UEs and SA UEs, it is recommended that the CellQciPara.NsaDcInterRatHoCommGroupId and CellQciPara.InterRatHoCommGroupId parameters be set to the same value.

Table 4-33 Handover measurement parameters related to events B1 and B2 for an LTE cell

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	NrNFreq.AggregationAttribute	RSRP-based triggering and RSRP-based triggering combined with RSRQ-based filtering are supported. - To use RSRP-based triggering, deselect the NR_B1_B2_PLUS_RSRQ_INDICATION option of this parameter. - To use RSRP-based triggering combined with RSRQ-based filtering, select the NR_B1_B2_PLUS_RSRQ_INDICATION option of this parameter.
Inter-RAT HO NR Triggering Event Type	InterRatHoComm.InterRatHoNrEventType	The default value is recommended.
NR B1B2 Hysteresis	InterRatHoNrParamGrp.NrB1B2Hysteresis	The default value is recommended for function activation. The greater the value of this parameter, the more likely it is to prevent incorrect handover decisions and ping-pong handovers.
Coverage-based NR B1 RSRP Threshold	InterRatHoNrParamGrp.CovBasedNrB1RsrpThld	It is recommended that you adjust the parameter values based on site conditions. Decrease the parameter values if delayed inter-RAT handovers occur.
Coverage-based NR B1 RSRQ Threshold	InterRatHoNrParamGrp.CovBasedNrB1RsrqThld	
Service-based NR B1 RSRP Threshold	InterRatHoNrParamGrp.ServBasedNrB1RsrpThld	
Service-based NR B1 RSRQ Threshold	InterRatHoNrParamGrp.ServBasedNrB1RsrqThld	

Parameter Name	Parameter ID	Setting Notes
NR B1B2 Time-to-Trigger	InterRatHoNrParamGrp.NrB1B2TimeToTrigger	• A larger value of this parameter results in a lower probability of event reporting triggered by instantaneous signal changes, a smaller average number of handovers, and a smaller number of unexpected handovers, preventing unnecessary handovers. • A smaller value results in the opposite effects.
Fast Return B1 Time-to-Trigger	InterRatHoNrParamGrp.FastReturnB1TimeToTrigger	• A larger value of this parameter results in a lower probability of event reporting triggered by instantaneous signal changes, a smaller average number of handovers, and a smaller number of unexpected handovers, preventing unnecessary handovers. • A smaller value results in the opposite effects.
Frequency-specific Offset	NrNFreq.FreqSpecificOffset	• A smaller value of this parameter results in a lower probability of inter-RAT handovers. • A larger value results in a higher probability.

Parameter Name	Parameter ID	Setting Notes
NR Measurement Report Processing Mode	CellHoParaCfg.NrMrProcessMode	This parameter specifies the measurement report processing mode for LTE-to-NR mobility. • If this parameter is set to NORMAL, new processing modes do not take effect, and a target frequency is selected based on the latest measurement report. In the case of a handover, the eNodeB makes a handover decision immediately after receiving a measurement report. In the case of a redirection, the eNodeB selects a target frequency for redirection based on the latest measurement report before receiving a UE context release command from the core network. • If this parameter is set to HANDOVER_IMMEDIATELY, the immediate handover mode is used. Whether to perform a handover is determined immediately after a measurement report is received. • If this parameter is set to BASED_ON_SIGNAL_STRENGTH, the signal-strength-based handover mode is used. A handover to the cell with the strongest signal strength indicated by measurement reports is initiated after a waiting period. • If this parameter is set to BASED_ON_FREQ_PRIORITY, the frequency-priority-based handover mode is used. Handovers to high-priority frequencies are preferentially initiated.

Parameter Name	Parameter ID	Setting Notes
NR Measurement Report Waiting Timer	CellHoParaCfg.NrMrWaitingTimer	This parameter specifies the NR measurement report waiting timer. - If this parameter is set to 0, the eNodeB processes the received measurement report without waiting for other measurement reports. The handover delay is short. - The larger value of this parameter, the longer the eNodeB waits for other measurement reports, and the longer the handover delay. - It is recommended that this parameter be set to a value less than CellHoParaCfg.ServBasedNrB1RptWaitingTmr.

Table 4-34 (Optional) Parameters related to equivalent PLMNs

Parameter Name	Parameter ID	Setting Notes
Handover Algo switch	ENodeBAlgoSwitch.HoAlgoSwitch	- To enable inter-PLMN handovers and obtain the PLMN IDs configured in the InterPlmnHoList MOs, select the InterPlmnHoSwitch option of this parameter. - To enable inter-PLMN handovers and obtain equivalent PLMN IDs from the MME, select both the InterPlmnHoSwitch and EPlmnSwitch options.
CN Operator ID	InterPlmnHoList.CnOperatorId	Set this parameter based on the network plan.
Target Mobile country code	InterPlmnHoList.TarMcc	Set this parameter based on the network plan.
Target Mobile network code	InterPlmnHoList.TarMnc	Set this parameter based on the network plan.

Table 4-35 (Optional) Parameters related to flexible user steering

Parameter Name	Parameter ID	Setting Notes
Handover Allowed Switch	CellAlgoExtSwitch.HoAllowedSwitch	Select the SPID_L2N_TARGET_FREQ_CTRL_SW option of this parameter.
DL Earfcn or Bandclass	RatFreqPriorityGroup.DLEarfcn	Set this parameter based on the network plan.

Table 4-36 (Optional) Parameters related to non-gap-assisted B1 measurement configuration by the eNodeB

Parameter Name	Parameter ID	Setting Notes
White List Control Switch	UeCompat.WhiteLstCtrlSwitch	Select the NO_GAP_B1_MEAS_SW_ON option.
LTE-NR No Gap Freq Comb Blacklist	UeCompat.LnrNoGapFreqCombBlacklist	Select the option that indicates the ID of the frequency combination identified by the LnrFreqComb.FreqCombId parameter. For example, if the LnrFreqComb.FreqCombId parameter is set to 0, the B1_NO_GAP_FREQ_COMB_ID_0 option must be selected.
Frequency Combination ID	LnrFreqComb.FreqCombId	This parameter specifies the ID of a blacklisted frequency combination.
Downlink EARFCN	LnrFreqComb.DLEarfcn	Set this parameter based on the network plan.
Downlink NR-ARFCN	LnrFreqComb.DLNrArfcn	Set this parameter based on the network plan.

Table 4-37 (Optional) Parameters related to RSRP prediction based on virtual grids

Parameter Name	Parameter ID	Setting Notes
LNR Carrier Selection Switch	LnrCarrierSelection.LnrCarrierSelectionSw	Select the LNR_VIRTUAL_GRID_SW option.
Virtual Grid Mdl HO Success Rate Thld	LnrCarrierSelection.VgMdlHoSuccessRateThld	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	NrNFreq.AggregationAttribute	• Deselect the VG_MODEL_FORBID_BUILD_FLAG option for NR frequencies for which virtual grid models are allowed to be built. • Select the VG_MODEL_FORBID_BUILD_FLAG option for NR frequencies for which virtual grid models are not allowed to be built.

Parameter Name	Parameter ID	Setting Notes
Service Mode	eNodeBResMode Algo.ServiceMode	• If the main control board is a UMPTg, you are advised to select the UMPTG_ENHANCEMENT_SWITCH option only for base stations where the number of RRC_CONNECTED UEs during busy hours and the number of BHCA are less than 80% of the respective maximum numbers supported by the main control board. Otherwise, the RRC connection setup success rate and handover success rate may decrease. • If the main control board is a UMPTha series board, you are advised to select the AI_ENHANCEMENT_OPT_SWITCH option only for base stations where the number of RRC_CONNECTED UEs during busy hours and the number of BHCA are less than 80% of their respective maximum numbers supported by the main control board. Otherwise, the RRC connection setup success rate and handover success rate may decrease. • If the main control board is a UMPTe or UMPTga series board, you are advised to select the AI_ENHANCEMENT_OPT_SWITCH option only for base stations where the number of RRC_CONNECTED UEs during busy hours and the number of BHCA are less than 65% of their respective maximum numbers supported by the main control board. Otherwise, the RRC connection setup success rate and handover success rate may decrease. For details about the maximum number of RRC_CONNECTED UEs and the maximum number of BHCA supported by a main control board, see related descriptions of the main control board in the "LTE Technical Specifications of Main Control Boards"

Parameter Name	Parameter ID	Setting Notes
		section in <i>3900 & 5900 Series Base Station Product Documentation</i> .
Multi-Carrier Unified Scheduling Enh Switch	MultiCarrUnifiedSch.MultiCarrierUnifiedSchEnSw	Select the VG_MDL_NCELL_MR_OPT_SW option if inter-RAT neighboring cell measurements need to be triggered by event B1 during model training.
LNR Carrier Selection Switch	LnrCarrierSelection.LnrCarrierSelectionSw	- If UEs cannot send inter-frequency measurement reports due to lack of inter-frequency coverage and samples without inter-frequency neighboring cell measurement results are expected to be included in LN RSRP model training, select the SAM_EXPAND_SW option of this parameter. - To exclude samples without intra-frequency neighboring cell measurement results from LN RSRP model training, select the NO_INTRA_FREQ_SAM_DEL_SW option. - To improve the accuracy of virtual grid models, select the SAM_BALANCING_SW option.
Sample UE Number	LnrCarrierSelection.SampleUeNum	It is recommended that this parameter be set to LOW, which indicates that three UEs are involved in sample collection. Set this parameter based on the network plan.

Table 4-38 (Optional) UE capability field size in an E-UTRAN-to-NG-RAN handover request message

Parameter Name	Parameter ID	Setting Notes
NR Handover Request UE Capability Size	EnodebAlgoExtSwitch.NrHoReqUeCapbSize	If optimization for handover from E-UTRAN to NG-RAN is required, set this parameter to the planned value. It is recommended that this function be enabled when the size of E-UTRAN to NG-RAN handover request messages exceeds the specifications of the core network.

Table 4-39 (Optional) Parameters related to the function of prohibiting large-packet UEs in LTE cells from returning to NR cells

Parameter Name	Parameter ID	Setting Notes
Handover Allowed Switch	CellAlgoExtSwitch.HoAllowedSwitch	Select the UL_TFC_NR_LTE_MOBILITY_ENH_SW option when the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is required.
Uplink Traffic Volume Threshold	CellMultiCarrUniSch.ULTrafficVolumeThld	The value 100 is recommended.

Table 4-40 (Optional) Parameters related to prohibition of VoNR-incapable UE roaming from the home network to the visited 5G network in inter-operator roaming scenarios

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	NrNFreq.AggregationAttribute	Select the VOICE_NR_HO_POLICY_INDICATION option when prohibition of VoNR-incapable UE roaming from the home LTE network to the visited 5G network is required.

Table 4-41 (Optional) Parameters related to prohibition of necessary or unnecessary handovers or redirections of UEs with specific SPIDs or bearers of specific QCLs to specified NR frequencies

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	RatFreqPriorityGroup.AggregationAttribute	Select the FORBID_NEC_HO_SW option to prohibit necessary handovers or redirections of UEs with specific SPIDs or bearers of specific QCLs to frequencies specified by the RatFreqPriorityGroup.DLEarfcn parameter.
Aggregation Attribute	RatFreqPriorityGroup.AggregationAttribute	To prohibit unnecessary handovers or redirections of UEs with specific SPIDs or bearers of specific QCLs to frequencies specified by the RatFreqPriorityGroup.DLEarfcn parameter, select the FORBID_UNNEC_HO_SW option.

Parameter Name	Parameter ID	Setting Notes
Handover Allowed Switch	CellAlgoExtSwitch.HoAllowedSwitch	To restrict the range of target NR frequencies for handovers of UEs with a QCI, select the NR_TARGET_FREQ_CTRL_SW option.

Table 4-42 (Optional) Parameters related to frequency-specific LTE-NR interoperability prohibition

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	NrNFreq.AggregationAttribute	Select the L2NR_MOB_FORBID_FLAG option to prohibit handovers, redirections, and reselections toward a specific frequency.

Table 4-43 (Optional) Rules for configuring the PLMN ID in the Target Cell ID IE in an LTE-to-NR handover request when the NR gNodeB is not provided by Huawei

Parameter Name	Parameter ID	Setting Notes
Handover Signaling Optimized Switch	ENodeBAlgoSwitch.HoSignalingOptSwitch	To increase the preparation success rate of handovers between a Huawei eNodeB and a non-Huawei gNodeB, if the non-Huawei gNodeB requires that the PLMN ID in the <i>target cell ID</i> IE in the LTE-to-NR handover request message be the primary PLMN (PLMN of the external NR cell), select the LteToNrHoTargetCellIdPlmnSw option.

Table 4-44 (Optional) Parameters related to LTE-to-NR handover policy for RedCap UEs

Parameter Name	Parameter ID	Setting Notes
Handover to NR Algo Switch	EnodebAlgoExtSwitch.NrHandoverAlgoSwitch	When not all NR cells support RedCap, it is recommended that the REDCAP_UE_NR_HO_POLICY_SW option be selected to increase the preparation success rate for outgoing inter-RAT handovers to NR.

Parameter Name	Parameter ID	Setting Notes
Cell Barred for RedCap	NrExternalCell.Ce llBarredRedCap	- When the REDCAP_SW option of the NR parameter NRDUCellFeatureSw.UeCapbBase dFunSw is selected, it is recommended that the LTE parameter NrExternalCell.CellBarredRedCap be set to the same value as that of the NR parameter NRDUCellRedCap.CellBarredRedCap. - When the REDCAP_SW option of the NR parameter NRDUCellFeatureSw.UeCapbBase dFunSw is deselected, it is recommended that the LTE parameter NrExternalCell.CellBarredRedCap be set to 1RX_2RX_BARRED.

4.5.1.2 Using MML Commands

Activation Command Examples

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTE

Assume that RSRP prediction based on virtual grids is enabled. To enable or disable the main control board enhancement mode:

- If the main control board model has been determined, you are advised to select or deselect the corresponding option.
- If the main control board model is unknown, you are advised to select or deselect both the **AI_ENHANCEMENT_OPT_SWITCH** and **UMPTG_ENHANCEMENT_SWITCH** options. The base station automatically identifies the main control board model and activates or deactivates only the enhancement mode for the corresponding main control board.

//The following uses NR FDD cells as an example.

//(Optional) selecting the **EMERG_CALL_NR_RESTRICTION_SW** option to prohibit LTE-to-NR mobility functions for emergency call UEs

MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=EMERG_CALL_NR_RESTRICTION_SW-1;

//Adding a neighboring NR frequency (For NR TDD, set SubcarrierSpacing to 30KHZ.)

```

ADD NRNFREQ: LocalCellId=0, DlArfcn=428000, ConnFreqPriority=1, VnrPriority=2,
SubcarrierSpacing=15KHZ,
AggregationAttribute=NR_B1_B2_PLUS_RSRQ_INDICATION-1&VG_MODEL_FORBID_BUILD_FLAG-0,
NrDLBandwidth=CELL_BW_20M, NrULBandwidth=CELL_BW_20M, NrDLMinCarrierBw=NOT_CONFIG,
NrULMinCarrierBw=NOT_CONFIG;
//(Optional; required when the NR frequency belongs to multiple frequency bands) Turning on
NR_MULTIBAND_SUPPORT_SW
MOD ENODEBALGOEXTSWITCH: NrMfbiAlgoSwitch=NR_MULTIBAND_SUPPORT_SW-1;
//Setting the frequency band to which the neighboring NR frequency belongs
ADD NRMFBIFREQ: DlArfcn=428000, FrequencyBand=N1, AdditionalFrequencyBand=NULL;
//Adding a neighboring-cell PLMN list, with the MCC and MNC of neighboring NR cells set to 460 and 20,
respectively
ADD NCELLPLMNLIST: Mcc="460", Mnc="20", RatType=NR, gNBIdLength=22;
//Adding an external NR cell
ADD NREXTERNALCELL: Mcc="460", Mnc="20", GnodebId=123, CellId=1, DlArfcn=428000, PhyCellId=1,
Tac=1, FrequencyBand=N1, AdditionalFrequencyBand=NULL;
//(Optional) Enabling inter-PLMN handovers
MOD ENODEBALGOSWITCH: HoAlgoSwitch=InterPlmnHoSwitch-1&EPlmnSwitch-1;
//(Optional) To enable the base station to filter out each NR cell whose NR cell bandwidth and minimum
NR carrier bandwidth are both different from any cell bandwidth supported by the UE involved, run the
following command:
MOD ENODEBALGOSWITCH: CompatibilityCtrlSwitch=NrBwCheckSwitch-1;
//(Optional; required for inter-PLMN handovers) Configuring the equivalent PLMN list based on the
network plan, with the MCC and MNC of the operator set to 460 and 03, respectively
ADD INTERPLMNHOLIST: CnOperatorId=0, TarMcc="460", TarMnc="03";
//Adding a neighbor relationship with the NR cell
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="20", GnodebId=123, CellId=1;
//Setting a cell-level UE measurement configuration
MOD CELLUEMEASCONTROLCFG: LocalCellId=0, MaxNrMeasFreqNum=5;
//Setting the number of inter-RAT handover retries upon failures due to air interface problems
MOD CELLHOPARACFG: LocalCellId=0, InterRatUuHoFailRetryTimes=10;
//(Optional) To enable non-gap-assisted NR measurement for the eNodeB, run the following command:
MOD ENODEBALGOEXTSWITCH: MultiNetworkingOptionOptSw=NR_B1_NO_GAP_SW-1;
//Modifying the penalty timer for failures of resource preparations for handovers to NG-RAN, non-resource-
caused handover preparation failure punish times, and the number of penalty times in the case of non-
resource-caused handover preparation failures
MOD HOMEASCOMM: NrHoResPrepFailPenaltyTmr=10, NonResHoPreFailRetryTimes=1,
NonResHoPreFailPunishTimes=10;
//(Optional, and recommended when the penalty timer used upon failures of resource preparations for
handovers to NG-RAN is set to a value other than 0) Selecting the LNR_RES_HO_FAIL_PENALTY_OPT_SW
option
MOD ENODEBALGOEXTSWITCH: NrHandoverAlgoSwitch=LNR_RES_HO_FAIL_PENALTY_OPT_SW-1;
//Setting a common parameter group related to inter-RAT handovers
ADD INTERRATHOCOMMGROUP: LocalCellId=0, InterRatHoCommGroupId=1, InterRatHoA1A2Hyst=2,
InterRatHoA1A2TimeToTrig=640ms, InterRatHoNrA1ThldRsrp=-111, InterRatHoNrA2ThldRsrp=-115,
NrB2Thld1Rsrp=-115;
//(Optional) Setting the time-to-trigger for event A1/A2 related to inter-RAT handover to NR
MOD INTERRATHOCOMMGROUP: LocalCellId=0, InterRatHoCommGroupId=1,
InterRatHoNrA1A2TimeToTrig=320MS;
//Setting a parameter group related to inter-RAT handovers to NR
ADD INTERRATHONRPARAMGRP: LocalCellId=0, NrHoParamGroupId=1, NrB1B2Hysteresis=2,
NrB1B2TimeToTrigger=320ms, CovBasedNrB1RsrpThld=-116, ServBasedNrB1RsrpThld=-109,
CovBasedNrB1RsrqThld=-24, ServBasedNrB1RsrqThld=-24, FastReturnB1TimeToTrigger=320ms;
//Binding the parameter groups to a QCI in the cell
MOD CELLQCIPARA: LocalCellId=0, Qci=9, InterRatHoCommGroupId=1, NsaDcInterRatHoCommGroupId=1,
NrHoParamGroupId=1;
//Setting QCI-specific parameters
MOD QCIPARA: Qci=1, InterRatPolicyCfgGroupId=1;
MOD QCIPARA: Qci=5, InterRatPolicyCfgGroupId=3;
MOD QCIPARA: Qci=9, InterRatPolicyCfgGroupId=3;
//Setting an inter-RAT policy group
//If VoNR is not supported on the NR side, it is recommended that the handover and redirection switches be
turned off for QCI 1 to prevent voice call failures after handovers of VoLTE services to NR cells.
MOD INTERRATPOLICYCFGGROUP: InterRatPolicyCfgGroupId=1, NrHoCfg=PS_HO-0&REDIRECTION-0;
//If VoNR is supported on the NR side, it is recommended that the handover switch be turned on for QCI 1
and the redirection switch be turned off for QCI 1.
MOD INTERRATPOLICYCFGGROUP: InterRatPolicyCfgGroupId=1, NrHoCfg=PS_HO-1&REDIRECTION-0;
//It is recommended that the handover and redirection switches be turned on for QCIs corresponding to
non-voice services.

```



```

MOD INTERRATPOLICYCFGGROUP: InterRatPolicyCfgGroupId=3, NrHoCfg=PS_HO-1&REDIRECTION-1;
//(Optional) Enabling non-gap-assisted B1 measurement configuration by the eNodeB to reduce the impact
of gap-assisted NR frequency measurement on data rates
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
WhiteLstCtrlSwitch=NO_GAP_B1_MEAS_SW_ON-1;
//(Optional) Adding a blacklisted LTE-NR frequency combination (for example, EARFCN 1500 and NR-
ARFCN 636666) for which the specified UEs do not support non-gap-assisted B1 measurement
ADD LNRFFREQCOMB: DLEarfcn=1500, DINrArfcn=636666, FreqCombId=0;
//(Optional) Turning on the blacklist switch for the LTE-NR frequency combination when a blacklisted LTE-
NR frequency combination needs to be configured for specified UEs for non-gap-assisted B1 measurement
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,
LnrNoGapFreqCombBlacklist=B1_NO_GAP_FREQ_COMB_ID_0-1;
//(Optional; required for RSRP prediction based on virtual grids) Starting building virtual grid models
ADD NRNFREQ: LocalCellId=0, DLEarfcn=428000, AggregationAttribute=VG_MODEL_FORBID_BUILD_FLAG-0;
MOD LNRCARRIERSELECTION: PredUnnecNrHoMeasRsrpThld=ACCURACY100_ENH,
VgMdlHoSuccRateThld=90;
MOD MULTICARRUNIFIEDSCH: MultiCarrierUnifiedSchEnSw=VG_MDL_NCELL_MR_OPT_SW-1;
//(Optional; required when RSRP prediction based on virtual grids is enabled) Enabling the main control
board enhancement mode. In the following example, the corresponding option is selected for the main
control board.
MOD ENODEBRESMODEALGO:
ServiceMode=AI_ENHANCEMENT_OPT_SWITCH-1&UMPTG_ENHANCEMENT_SWITCH-1;
//(Optional) Turning on SAM_EXPAND_SW if UEs cannot send inter-frequency measurement reports due to
lack of inter-frequency coverage and samples without inter-frequency neighboring cell measurement results
are expected to be used for LN RSRP model training
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=SAM_EXPAND_SW-1;
//(Optional) Turning on NO_INTRA_FREQ_SAM_DEL_SW to not use samples without intra-frequency
neighboring cell measurement results for LN RSRP model training
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=NO_INTRA_FREQ_SAM_DEL_SW-1;
//(Optional) Turning on SAM_BALANCING_SW to improve the accuracy of virtual grid models
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=SAM_BALANCING_SW-1;
//(Optional) Setting SampleUeNum when the number of UEs involved in sample collection during LNR
virtual grid model training needs to be adjusted
MOD LNRCARRIERSELECTION: SampleUeNum=LOW;
//(Optional) Running the following commands if the function of prohibiting large-packet UEs in LTE cells
from returning to NR cells is required
MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=UL_TFC_NR_LTE_MOBILITY_ENH_SW-1;
MOD CELLMULTICARRUNISCH: LocalCellId=0, ULTrafficVolumeThld=100;
//(Optional) Running the following command if the measurement report processing mode for LTE-to-NR
mobility needs to be set
MOD CELLHOPARACFG: LocalCellId=0, NrMrWaitingTimer=30,
NrMrProcessMode=HANDOVER_IMMEDIATELY;
//(Optional) Running the following command if prohibition of VoNR-incapable UE roaming from the home
LTE network to the visited 5G network is required
MOD NRNFREQ: LocalCellId=0, DLEarfcn=428000,
AggregationAttribute=VOICE_NR_HO_POLICY_INDICATION-1;
//(Optional) Running the following command if prohibition of necessary or unnecessary handovers or
redirections to specified NR frequencies for UEs with specific SPIDs or bearers of specific QCI is required
MOD RATFREQPRIORITYGROUP: RatFreqPriorityGroupId=0, RatType=NR, DLEarfcn=428000,
AggregationAttribute=FORBID_NEC_HO_SW-1&FORBID_UNNEC_HO_SW-1;
//(Optional) Running the following command to restrict the range of target NR frequencies for handovers
of UEs with a QCI
MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=NR_TARGET_FREQ_CTRL_SW-1;
//(Optional) Prohibiting handovers, redirections, and reselections toward a specific frequency
MOD NRNFREQ: LocalCellId=0, DLEarfcn=428000, AggregationAttribute=L2NR_MOB_FORBID_FLAG-1;
//Configuring the size of the UE capability field carried in an E-UTRAN to NG-RAN handover request
message
MOD ENODEBALGOEXTSWITCH: NrHoReqUeCapbSize=5;
//(Optional) Running the following command to increase the preparation success rate of handovers
between a Huawei eNodeB and a non-Huawei gNodeB, if the non-Huawei gNodeB requires that the PLMN
ID in the target cell ID IE in the LTE-to-NR handover request message be the primary PLMN (PLMN of the
external NR cell)
MOD ENODEBALGOSWITCH: HoSignalingOptSwitch=LteToNrHoTargetCellIdPlmnSw-1;
//(Optional) Running the following commands for the NR cells that do not support RedCap when not all
NR cells support RedCap
MOD ENODEBALGOEXTSWITCH: NrHandoverAlgoSwitch=REDCAP_UE_NR_HO_POLICY_SW-1;
ADD NREXTERNALCELL: Mcc="460", Mnc="20", GnodebId=123, CellId=1, DLEarfcn=428000,
ULArfcnConfigInd=NOT_CFG, PhyCellId=1, Tac=1;

```

```
MOD NREXTERNALCELL: Mcc="460", Mnc="20", GnodebId=123, CellId=1, DlArfcn=428000, PhyCellId=1,  
Tac=1, CellBarredRedCap=1RX_2RX_BARRED;
```

Deactivation Command Examples

The data configurations described in this section are fundamental to the functions related to mobility from E-UTRAN to NG-RAN. Deactivation is not recommended.

Optional functions can be deactivated. The following is an example.

```
//(Optional) Disabling non-gap-assisted B1 measurement configuration by the eNodeB  
MOD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,  
WhiteLstCtrlSwitch=NO_GAP_B1_MEAS_SW_ON-0;  
//(Optional) Running the following command to disable RSRP prediction based on virtual grids  
MOD LNRARRIERSELECTION: LnrCarrierSelectionSw=LNR_VIRTUAL_GRID_SW-0;  
//(Optional) Disabling the prohibition of handovers, redirections, and reselections to a specific frequency  
MOD NRNFREQ: LocalCellId=0, DlArfcn=428000, AggregationAttribute=L2NR_MOB_FORBID_FLAG-0;
```

4.5.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

When eNodeB data configuration is complete, you are advised to use the consistency check function on the MAE-Deployment to check data consistency. For details, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance > MAE-Deployment Guidelines > Enhanced Feature Management > Consistency Check**. You are also advised to select the NR-related check items in the categories "External Cell", "Neighboring Relationship", and "Conflict Check" to check whether the parameter settings of external NR cells and neighbor relationships with NR cells on the eNodeB side are consistent with those of the NR cells on the gNodeB side.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.5.2 Activation Verification

This section describes how to verify the activation of basic functions related to mobility management from E-UTRAN to NG-RAN in connected mode. For details, see the "Activation Verification" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

RSRP Prediction Based on Virtual Grids

For RSRP prediction based on virtual grids, virtual grid model building can be completed only after sufficient UE measurement reports are collected, and it takes at least 24 hours to successfully build a virtual grid model. Therefore, it is recommended that the following operations be performed 24 hours after virtual grid model building is started:

- Using MML commands
Run the **DSP AIGRIDMOD** command to query the status of virtual grid models in a cell. If at least one of **Good RSRP Forecast Status** and **Poor RSRP Forecast Status** is **AVAILABLE**, RSRP prediction based on virtual grids has been successfully activated.

- Using counters
NR RSRP prediction based on virtual grids in an LTE cell can be observed using the **L.AiGrid.LnrRsrpPred.Model.AverNum** counter. If the value of this counter is not 0, LN RSRP models have been successfully built. You can further check the numbers of good-RSRP and poor-RSRP virtual grid models through the **L.VirtualGrid.LnrGoodRsrp.Model.Avg**, **L.VirtualGrid.LnrGoodRsrp.Model.Max**, **L.VirtualGrid.LnrPoorRsrp.Model.Avg**, and **L.VirtualGrid.LnrPoorRsrp.Model.Max** counters.

4.5.3 Network Monitoring

For details, see the "Network Monitoring" section of each interoperability function described in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#).

5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode

5.1 Overview of Interoperability in Connected Mode

Interoperability in connected mode allows UEs in connected mode to continuously access services or enjoy better service experience when they are moving between LTE and NR cells. The services are classified into data services and voice services.

Mobility management functions for connected-mode UEs with data services include:

- Coverage-based mobility management
 - From NG-RAN to E-UTRAN:
 - Handovers or redirections from NR cells to LTE cells or frequencies are triggered based on the downlink coverage of UEs. For details, see [5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN](#).
 - Handovers or redirections from NR cells to LTE cells or frequencies are triggered based on the uplink coverage of UEs. For details, see [5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN](#).
 - From E-UTRAN to NG-RAN: Handovers or redirections from LTE cells to NR cells are triggered based on the downlink coverage of UEs. For details, see [5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN](#).
- Service-based mobility management

Handovers or redirections from LTE cells to NR cells are triggered based on the UE service status. For details, see [5.6 E-UTRAN to NG-RAN Traffic Steering](#).

Mobility management functions for connected-mode UEs with voice services are classified into different types based on whether VoNR is enabled for NR cells.

- If VoNR is enabled for an NR cell (that is, the **VONR_SW** option of the **NRCCellAlgoSwitch.VonrSwitch** parameter is selected), the following mobility

functions work for a UE with a voice service when it moves between the NR cell and an LTE cell:

- Downlink-coverage-based VoNR-to-VoLTE handover. For details, see [5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN](#).
- Downlink-coverage-based VoLTE-to-VoNR handover. For details, see [5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN](#).
- Voice-quality-based VoNR-to-VoLTE handover. For details, see [5.9 VoNR-to-VoLTE Handover Based on Voice Quality](#).
- If VoNR is disabled for an NR cell (that is, the **VONR_SW** option of the **NRCCellAlgoSwitch.VonrSwitch** parameter is deselected), the following mobility functions work for a UE that initiates a voice service in the NR cell:
 - Voice fallback from NG-RAN to E-UTRAN, For details, see [5.7 Voice Fallback from NG-RAN to E-UTRAN](#).
 - Fast return from E-UTRAN to NG-RAN. For details, see [5.8 Fast Return from E-UTRAN to NG-RAN](#).

In addition, operator-specific mobility policies can be configured for functions involved in interoperability between E-UTRAN and NG-RAN. For details, see [6.1.1.2 Operator-specific Mobility Policy](#).

5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN

5.2.1 Principles

When a UE moves from a serving NR cell to a neighboring LTE cell, the signal quality of the NR cell is poor while that of the LTE cell is good. To ensure service continuity or better user experience, downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN is triggered for the UE.

Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN is enabled if the parameters are set as follows:

- **MOBILITY_TO_EUTRAN_SW** option of the **NRCCellAlgoSwitch.InterRatServiceMobilitySw** parameter: selected
- **DL_COV_N2L_MOB_OFF_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter: deselected
- **DL_COV_N2L_MOB_OFF_SWITCH** option of the **NRCCellOpPolicy.InterRatHoPolicySwitch** parameter: deselected

 NOTE

The **MOBILITY_TO_EUTRAN_SW** option of the **NRCellAlgoSwitch.InterRatServiceMobilitySw** parameter applies to both downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN and NSA/SA selection based on downlink traffic volume. To disable only downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, select the **DL_COV_N2L_MOB_OFF_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter or the **DL_COV_N2L_MOB_OFF_SWITCH** option of the **NRCellOpPolicy.InterRatHoPolicySwitch** parameter. For details about NSA/SA selection based on downlink traffic volume, see *NSA-SA Selection Based on User Experience*.

Downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN supports measurement-based handover, measurement-based redirection, and blind redirection, as described in [Table 5-1](#).

Table 5-1 Mobility policies supported by downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN

Scenario Classification	Mobility Policy	Parameter
Common scenarios ^a	Handover is required for both data and VoNR services.	EUTRAN_HO_SWITCH option of the NRInterRatHoParam.HoModeSwitch parameter
	Redirection is required for both data and VoNR services.	EUTRAN_REDIRECT_SWITCH option of the NRInterRatHoParam.HoModeSwitch parameter
	Redirection is required for data services, and handover is required for VoNR services.	EUTRAN_REDIRECT_SWITCH and VONR_COV_BASED_HO_PREFER_SW options of the NRInterRatHoParam.HoModeSwitch parameter
NR working in RAN sharing with common carrier mode ^b	Handover is required for an operator.	EUTRAN_HO_SWITCH option of the NRInterRatHoParam.HoModeSwitch parameter, and EUTRAN_HO_SWITCH option of the NRCellOpPolicy.InterRatHoPolicySwitch parameter
	Redirection is required for an operator.	EUTRAN_REDIRECT_SWITCH option of the NRInterRatHoParam.HoModeSwitch parameter, and EUTRAN_REDIRECT_SWITCH option of the NRCellOpPolicy.InterRatHoPolicySwitch parameter

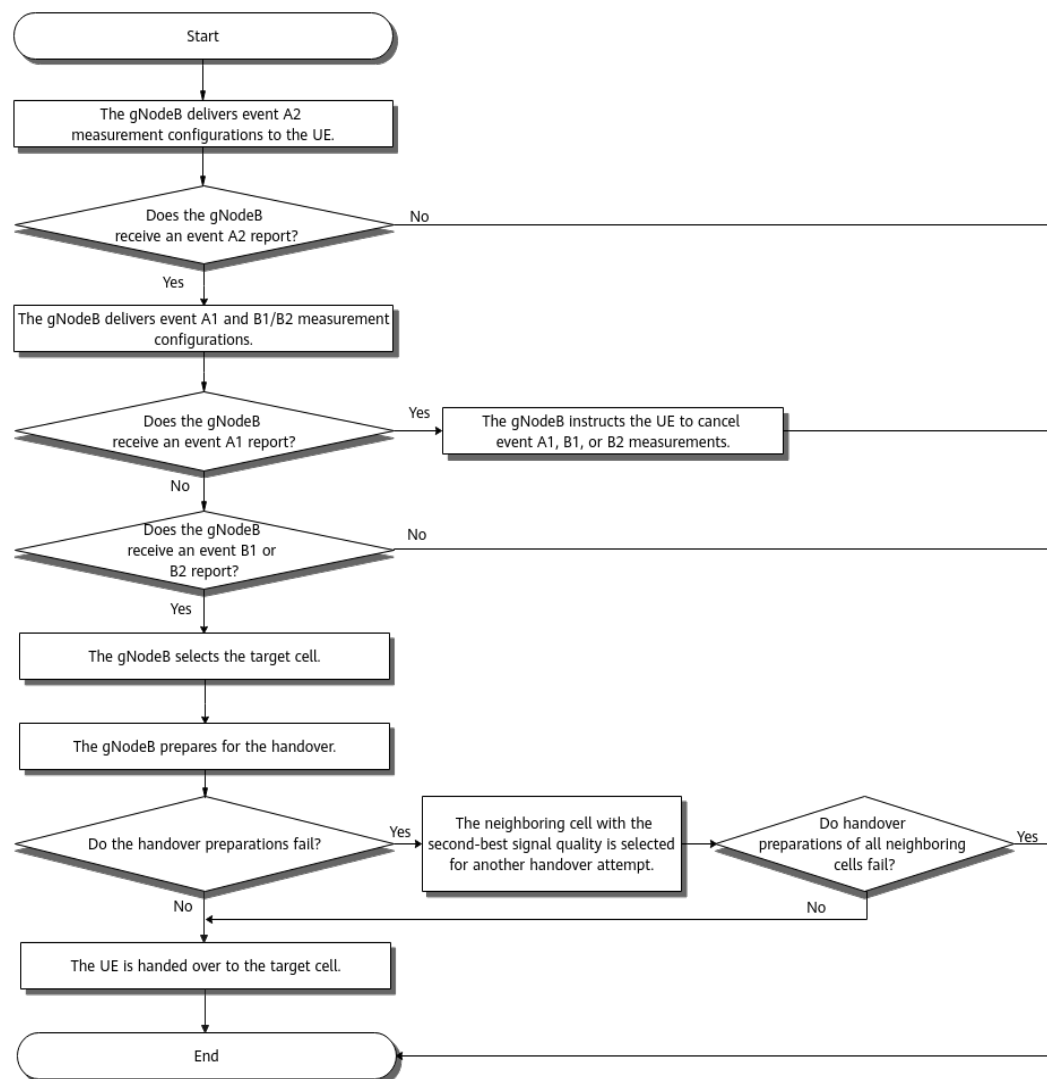
Scenario Classification	Mobility Policy	Parameter
	Prioritization of handover for VoNR services in coverage-based mobility is required for an operator.	VONR_COV_BASED_HO_PREFER_SW option of the NRInterRatHoParam.HoModeSwitch parameter, and VONR_COV_BASED_HO_PREFER_SW option of the NRCellOpPolicy.InterRatHoPolicy-Switch parameter
Note: a: • If both the EUTRAN_HO_SWITCH and EUTRAN_REDIRECT_SWITCH options of the NRInterRatHoParam.HoModeSwitch parameter are selected, handover is preferentially performed, followed by redirection. If both handover and redirection fail, blind redirection is performed. • If the EUTRAN_STRONGEST_NCELL_REDIRECTION_SW option of the NRInterRatHoParam.HoModeSwitch parameter is selected, coverage-based redirection to the strongest neighboring LTE cell takes effect. With this function in effect in coverage-based mobility from NG-RAN to E-UTRAN, the operating frequency of the cell with the strongest signal strength indicated in an E-UTRAN measurement report is selected for redirection when this cell is not configured as a neighboring cell or encounters PCI confusion. b: If the target LTE cell or frequency in the handover or redirection is shared by operators, the handover or redirection procedure is the same as that when the LTE cell or frequency is not shared.		

The downlink RSRP measurement results are not an accurate indication of actual coverage in interference scenarios, and therefore, downlink-SINR-based mobility to E-UTRAN is introduced. With this function, in addition to the triggering quantities specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter, the base station delivers SSB-SINR-based measurement configurations related to inter-RAT events A1 and A2 to a UE and, based on the downlink SINR measurement results, implements inter-RAT mobility from NG-RAN to E-UTRAN. Downlink-SINR-based mobility to E-UTRAN supports only handover and redirection.

This function is controlled by the **DL_SINR_MOBILITY_TO_EUTRAN_SW** option of the **NRCellMobilityConfig.InterRatHoExtSwitch** parameter. The thresholds for events A1 and A2 are specified by the **NRCellInterRHoMeaGrp.DlSnrHoToEutranA1Thld** and **NRCellInterRHoMeaGrp.DlSnrHoToEutranA2Thld** parameters, respectively.

Data Service Handover

Figure 5-1 shows the procedure for a downlink-coverage-based data service handover from NG-RAN to E-UTRAN.

Figure 5-1 Data service handover procedure

1. The gNodeB delivers the measurement configurations related to inter-RAT event A2 to the UE.
2. The gNodeB determines the target LTE cell.

After receiving inter-RAT event A2 measurement reports, the gNodeB delivers the measurement configurations related to inter-RAT event A1. At the same time, it delivers the measurement configurations related to inter-RAT event B1 or B2, depending on the inter-RAT handover triggering event type specified by the **NRInterRatHoParam.InterRatHoTriggerEventType** parameter (value: **EVENT_B1** or **EVENT_B2**). For details about the frequency filtering rules, see [4.1.4.1.3 LTE Frequency Selection by the gNodeB](#). For details about measurement event B1 or B2, see [4.1.4.2.1 Measurement Events](#). For details about the triggering quantities of measurement events supported by the gNodeB, see [4.1.4.2.2 Triggering Quantity](#).

- If the gNodeB receives inter-RAT event A1 measurement reports, it instructs the UE to release inter-RAT event A1/B1/B2 measurements. The procedure ends.

- If the gNodeB receives inter-RAT event B1 or B2 measurement reports, it generates a candidate cell list based on cell signal quality and selects the cell with the best signal quality as the target LTE cell. Then, the gNodeB goes to the next step.
- 3. The gNodeB performs a handover to the target LTE cell.
 - If the preparation for a handover to the target LTE cell fails, the UE will not try handovers to that cell within a certain period of time. The gNodeB selects the cell with the next best signal quality from the candidate cell list for a handover. If the preparations for handovers to all of the cells in the list fail, the procedure ends.
 - If the handover execution fails, the procedure ends.

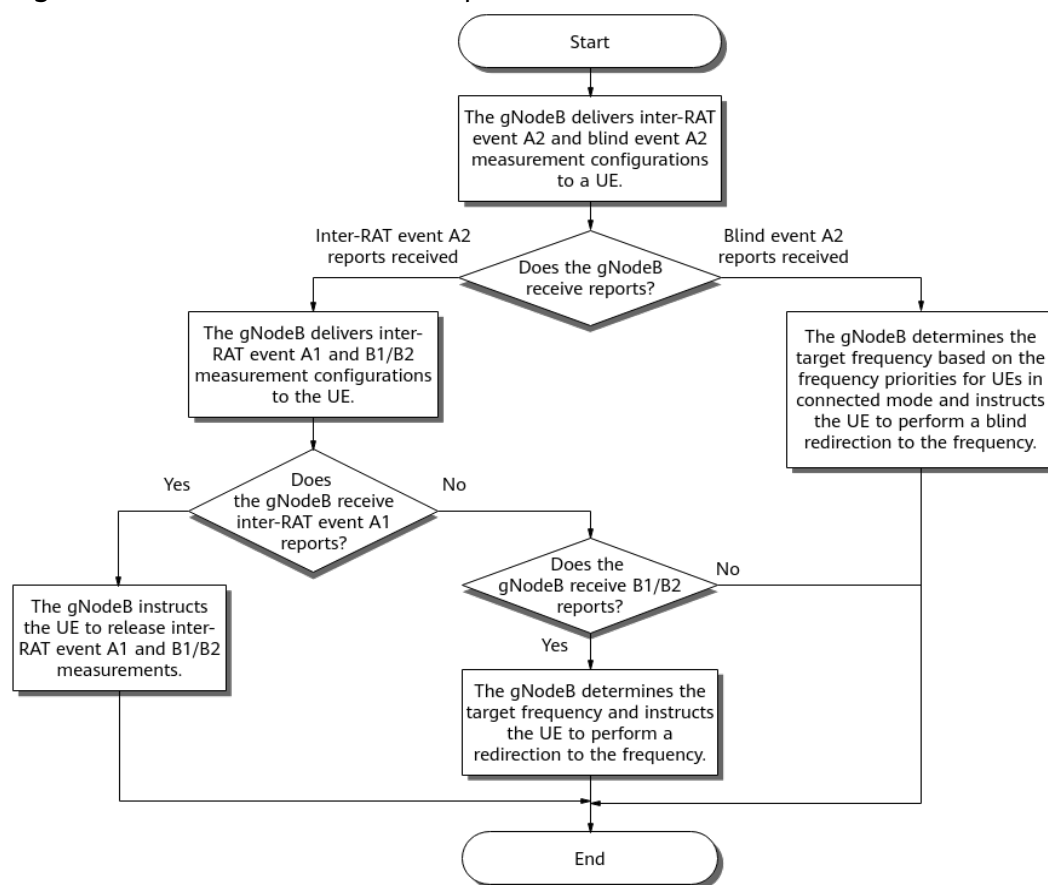
VoNR-to-VoLTE Handover

The procedure for a downlink-coverage-based VoNR-to-VoLTE handover is similar to the procedure for a downlink-coverage-based data service handover from NG-RAN to E-UTRAN. The difference is that the gNodeB determines whether to deliver the measurement configurations about an LTE frequency based on the setting of the **NRCellEutranNFreq.VoltePriority** parameter.

- If this parameter is set to **255**, the LTE frequency does not support VoLTE. The gNodeB does not deliver the measurement configurations about this frequency.
- If this parameter is set to a value other than **255**, the LTE frequency supports VoLTE. The gNodeB delivers the measurement configurations about this frequency.

Data Service Redirection

Figure 5-2 shows the procedure for a downlink-coverage-based data service redirection from NG-RAN to E-UTRAN.

Figure 5-2 Data service redirection procedure

The gNodeB delivers the measurement configurations related to inter-RAT event A2 and blind event A2 to the UE.

1. If the gNodeB receives inter-RAT event A2 measurement reports:
The gNodeB delivers the measurement configurations related to inter-RAT event A1. At the same time, it delivers the measurement configurations related to inter-RAT event B1 or B2, depending on the inter-RAT handover triggering event type specified by the **NRInterRatHoParam.InterRatHoTriggerEventType** parameter (value: **EVENT_B1** or **EVENT_B2**). For details about the frequency filtering rules, see [4.1.4.1.3 LTE Frequency Selection by the gNodeB](#). For details about measurement event B1 or B2, see [4.1.4.2.1 Measurement Events](#). For details about the triggering quantities of measurement events supported by the gNodeB, see [4.1.4.2.2 Triggering Quantity](#).
 - If the gNodeB receives inter-RAT event A1 measurement reports, it instructs the UE to release inter-RAT event A1/B1/B2 measurements. The procedure ends.
 - If the gNodeB receives inter-RAT event B1 or B2 measurement reports, it generates a candidate cell list based on cell signal quality and selects the frequency serving the cell with the best signal quality as the target LTE frequency. The gNodeB instructs the UE to perform a redirection to the target LTE frequency.
2. If the gNodeB receives measurement reports on blind event A2, it determines the target frequency for a blind redirection based on the

NRCellEutranNFreq.ConnFreqPriority parameter setting and instructs the UE to perform the blind redirection to LTE, regardless of the inter-RAT event A2 and B1/B2 reporting status.

VoNR-to-VoLTE Redirection

The procedure for a downlink-coverage-based VoNR-to-VoLTE redirection is similar to that for a downlink-coverage-based data service redirection from NG-RAN to E-UTRAN. The only difference is that the gNodeB determines whether to deliver measurement configurations for an LTE frequency based on the **NRCellEutranNFreq.VoltePriority** parameter setting for the LTE frequency, or determines a target LTE frequency for a blind redirection based on the **NRCellEutranNFreq.VolteBlindPriority** parameter setting for the LTE frequency.

5.2.2 Network Analysis

5.2.2.1 Benefits

With downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, the coverage in areas with weak NR coverage or coverage holes can be supplemented by inter-RAT neighboring cells, decreasing the service drops.

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

When downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN takes effect, the number of handovers, redirections, or blind redirections to E-UTRAN increases. For details about the related counters, see [5.2.4.3 Network Monitoring](#).

When downlink-SINR-based mobility to E-UTRAN takes effect, the values of the following counters also increase:

- **N.HO.InterRAT.N2E.DLSinr.PreAttOut**
- **N.HO.InterRAT.N2E.DLSinr.ExecAttOut**
- **N.HO.InterRAT.N2E.DLSinr.ExecSuccOut**

5.2.3 Requirements

5.2.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
NR	FOFD-021209	Inter-RAT Mobility from NG-RAN to E-UTRAN	NR0SRATMNE00	Inter-RAT Mobility from NG-RAN to E-UTRAN (NR)	gNodeB	per cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

5.2.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

None

5.2.3.2.2 Mutually Exclusive Functions

None

5.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Inter-RAT ANR	NR_EUTRAN_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>	Inter-RAT ANR requires that the measurement-based inter-RAT handover and measurement-based redirection take effect, and downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN be enabled.
NR FDD Low-frequency NR TDD	Inter-RAT ANR	NR_EUTRAN_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>	If downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN is enabled and the NRCCellEutranNFreq.FreqIntrrfFlag parameter is set to INTRF , CGI-based measurement configuration is delivered only when the RSRQ in the inter-frequency measurement report is greater than or equal to the following calculated result: $[\text{NRCCellHoEutranMeaGrp.CovBasedHoB1RsrqThld} - \text{NRCCellHoEutranMeaGrp.CovBasedHoB1B2Hyst} - \text{NRCCellEutranNFreq.FreqSpecificOffset} - (\text{NRCCellEutraNRelation.EutranCellIndividualOffset} + \text{NRCCellOpPolicy.EutraNcIoOffset})]$

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Inter-RAT ANR	NR_EUTRAN_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>	Assume that the gNodeB detects a missing neighboring cell through inter-RAT ANR. If coverage-based redirection to the strongest neighboring LTE cell is enabled, the gNodeB does not send measurement configurations of the report type <i>reportCGI</i> to UEs. As a result, the missing neighboring cell cannot be added to the neighboring cell list (NCL) and the neighbor relationship with this cell cannot be added to the neighboring relation table (NRT). In this case, you can select the NR_EUTRAN_COV_REDIRECT_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter so that the gNodeB can deliver measurement configurations with the report type of <i>reportCGI</i> to UEs and add the missing neighboring cell to the NRT/NCL.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	User-experience-based coverage expansion	COV_THLD_ADAPT_SW option of the gNB MobilityCommParam.MobilityAlgorithmSwitch parameter	<i>Multi-Frequency Smart Selection in 5G RAN Feature Documentation</i>	When both downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN and user-experience-based coverage expansion (COV_THLD_ADAPT_SW option of the gNB MobilityCommParam.MobilityAlgorithmSwitch parameter) are enabled, the following requirement must be met to prevent an increase in the number of executions of outgoing handovers from NG-RAN to E-UTRAN (N.HO.InterRAT.N2E.ExecAttOut): The RSRP threshold (gNB MobilityCommParam.AdaptCovRsrpThld) used in coverage threshold adaptation must be greater than or equal to the inter-RAT event A2 threshold (NRCellInterRHoMeaGrp.InterRatHoA2RsrpThld + gNB OperatorQciParam.EutranA2B2RsrpThldOffset + NRCellMeasConfig.SulCovHoRsrpThldOffset + gNB RfspConfig.InterRatA1A2B2RsrpThldOfs + NRCellOpPolicy.EutranA1A2B2RsrpThldOfs + NRCellInterRHoMeaGrp.InterRatHoA1A2Hyst).

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Inter-RAT Redirection Back to the HPLMN Based on Roaming UE Identification	VOICE_CAPB_MOB_TO_HP_LMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, only redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.
NR FDD Low-frequency NR TDD	Blind inter-frequency redirection	REDIRECTIO N option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode in 5G RAN Feature Documentation</i>	When intra-RAT blind inter-frequency redirection is enabled together with blind redirection for downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, the former preferentially takes effect.

5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all LTE-capable boards support this function.

- For NR, all NR-capable boards support this function.

RF Modules

No requirements

5.2.3.4 Others

- UE requirements
UEs must comply with 3GPP Release 15 or later.
- Core network requirements
 - The core networks must comply with 3GPP Release 15 or later.
 - Handover requires the N26 interface to be configured between the mobility management entity (MME) and access and mobility management function (AMF).

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

Table 5-2 describes the parameters related to downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN. For details about basic parameters for interoperability from NG-RAN to E-UTRAN, see **4.4.1.1 Data Preparation** and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-2 Parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
Inter-RAT Service Mobility Switch	NRCellAlgoSwitch.InterRatServiceMobilitySw	Select the MOBILITY_TO_EUTRAN_SW option.
Handover Mode Switch	NRInterRatHoParam.HoModeSwitch	Deselect the DL_COV_N2L_MOB_OFF_SWITCH option.
Inter-RAT Handover Policy Switch	NRCellOpPolicy.InterRatHoPolicy-Switch	Deselect the DL_COV_N2L_MOB_OFF_SWITCH option.
Handover Mode Switch	NRInterRatHoParam.HoModeSwitch	Set this parameter based on the mobility policies required for data services and VoNR services. For details, see Table 5-1 .

Parameter Name	Parameter ID	Setting Notes
Inter-RAT Handover Extend Switch	NRCellMobilityConfig . <i>InterRatHoExtSwitch</i>	To enable downlink-SINR-based mobility to E-UTRAN in interference scenarios, select the DL_SINR_MOBILITY_TO_EUTRAN_SW option.
DL SINR HO to E-UTRAN A1 Threshold	NRCellInterRHoMeaGrp . <i>DLSinrHoToEutranA1Thld</i>	Set these parameters based on the network plan. The value of NRCellInterRHoMeaGrp . <i>DLSinrHoToEutranA1Thld</i> must be greater than that of NRCellInterRHoMeaGrp . <i>DLSinrHoToEutranA2Thld</i> .
DL SINR HO to E-UTRAN A2 Threshold	NRCellInterRHoMeaGrp . <i>DLSinrHoToEutranA2Thld</i>	
Inter-RAT Handover Trigger Event Type	NRInterRatHoParam . <i>InterRatHoTriggerEventType</i>	The default value is recommended.
VoLTE Priority	NRCellEutranNFr .eq. <i>VoltePriority</i>	Set this parameter based on the network plan.
VOLTE Blind Priority	NRCellEutranNFr .eq. <i>VolteBlindPriority</i>	Set this parameter based on the network plan.

5.2.4.1.2 Using MML Commands

The basic configurations for interoperability from NG-RAN to E-UTRAN must have been completed before activating this function. For details, see [4.4.1.2 Using MML Commands](#) and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples

Before using MML commands, refer to [5.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Running the following command if handover and redirection are both required for data and VoNR services
MOD NRINTERRATHOPARAM: NrCellId=0,
HoModeSwitch=EUTRAN_HO_SWITCH-1&EUTRAN_REDIRECT_SWITCH-1,
InterRatHoTriggerEventType=EVENT_B2;
//Running the following command if redirection is required for data services and handover is preferred for voice services
MOD NRINTERRATHOPARAM: NrCellId=0,
HoModeSwitch=EUTRAN_HO_SWITCH-0&EUTRAN_REDIRECT_SWITCH-1&VONR_COV_BASED_HO_PREFER_S
```

```

W-1;
//Enabling downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=MOBILITY_TO_EUTRAN_SW-1;
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=DL_COV_N2L_MOB_OFF_SWITCH-0;
MOD NRCELLOPPOLICY: NrCellId=0, OperatorId=0,
InterRatHoPolicySwitch=DL_COV_N2L_MOB_OFF_SWITCH-0;
//Selecting DL_SINR_MOBILITY_TO_EUTRAN_SW
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterRatHoExtSwitch=DL_SINR_MOBILITY_TO_EUTRAN_SW-1;
//Setting DLSinrHoToEutranA1Thld and DLSinrHoToEutranA2Thld
MOD NRCELLINTERRHOMEAGRP: NrCellId=0, InterRatHoMeasGroupId=10, DLSinrHoToEutranA1Thld=16,
DLSinrHoToEutranA2Thld=10;
//Setting the priorities of an EARFCN (with an assumption that the EARFCN of 1300 has been configured in
an NRCELLEUTRANNFREQ MO)
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1300, ConnFreqPriority=1, VoltePriority=2,
VolteBlindPriority=1;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```

//Deselecting DL_SINR_MOBILITY_TO_EUTRAN_SW
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterRatHoExtSwitch=DL_SINR_MOBILITY_TO_EUTRAN_SW-0;

```

Scenario 1: Turn off the switch of mobility to E-UTRAN. This switch applies to both downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN and NSA/SA selection based on downlink traffic volume. Once this switch is turned off, these functions are both disabled.

```

//Turning off the switch of mobility to E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=MOBILITY_TO_EUTRAN_SW-0;

```

Scenario 2: Turn on the DL Coverage-based N2L Mobility Off Sw. You can turn on the switch in the **NRINTERRATHOPARAM** or **NRCELLOPPOLICY** MO based on the network plan, to disable only downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN.

```

//Turning on DL Coverage-based N2L Mobility Off Sw
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=DL_COV_N2L_MOB_OFF_SWITCH-1;
MOD NRCELLOPPOLICY: NrCellId=0, OperatorId=0,
InterRatHoPolicySwitch=DL_COV_N2L_MOB_OFF_SWITCH-1;

```

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

Use either of the methods described in this section to verify function activation.

Observing Performance Counter Values

Observe the values of the following counters. If either of the counters produces a non-zero value, this function has taken effect.

- **N.HO.InterRAT.N2E.Coverage.ExecSuccOut**
- **N.RRCRedirection.N2E.Coverage**

Tracing Signaling

To verify a handover from NG-RAN to E-UTRAN, perform the following steps:

- Step 1** Start Uu interface tracing and NG interface tracing for the gNodeB.
- Step 2** Observe the MobilityFromNRCommand message sent by the gNodeB to the UE over the Uu interface. If the value of the targetRAT-Type IE in the message is "eutra", the preparation for the handover from NG-RAN to E-UTRAN has succeeded.
- Step 3** Observe the signaling messages over the NG interface. If either (i) the gNodeB receives a UE CONTEXT RELEASE COMMAND message with the cause value "Successful Handover" or "Normal Release" from the AMF, or (ii) the gNodeB receives an RRC Reestablishment Complete message from the UE and the RRC connection reestablishment is successful, the handover from NG-RAN to E-UTRAN has been successfully executed.

----End

To verify a redirection or blind redirection from NG-RAN to E-UTRAN, perform the following steps:

- Step 1** Start Uu interface tracing for the gNodeB.
- Step 2** Observe the RRCRelease message sent by the gNodeB to the UE over the Uu interface. If the redirectedCarrierInfo IE in the message contains the RedirectedCarrierInfo-EUTRA IE, the redirection or blind redirection from NG-RAN to E-UTRAN has succeeded.

----End

5.2.4.3 Network Monitoring

Data Service Mobility

To evaluate the performance of data service mobility from NG-RAN to E-UTRAN, monitor the counters listed in [Table 5-3](#) and [Table 5-4](#).

Table 5-3 NR counters related to data service mobility

Counter ID	Counter Name
1911817063	N.HO.InterRAT.N2E.ExecSuccOut
1911817061	N.HO.InterRAT.N2E.ExecAttOut
1911817056	N.HO.InterRAT.N2E.PrepAttOut
1911817057	N.HO.InterRAT.N2E.AMFAbnormRsp
1911817064	N.HO.InterRAT.N2E.Prep.FailOut.AMF

Counter ID	Counter Name
1911817065	N.HO.InterRAT.N2E.Prep.FailOut.PrepFailure
1911817059	N.HO.InterRAT.N2E.Prep.FailOut.NoReply
1911817060	N.HO.InterRAT.N2E.Coverage.ExecSuccOut
1911817062	N.HO.InterRAT.N2E.Coverage.ExecAttOut
1911817058	N.HO.InterRAT.N2E.Coverage.PrepAttOut
1911817054	N.HO.InterRAT.NCell.N2E.PrepAttOut
1911817053	N.HO.InterRAT.NCell.N2E.ExecAttOut
1911817055	N.HO.InterRAT.NCell.N2E.ExecSuccOut
1911817044	N.RRCRedirection.N2E.Coverage
1911822969	N.HO.InterRAT.N2E.Exec.FailOut.ReEst
1911840485	N.RRCRedirection.N2E.Blind.Coverage

Table 5-4 LTE counters related to data service mobility

Counter ID	Counter Name
1526749534	L.IRATHO.N2E.PrepAttIn
1526749535	L.IRATHO.N2E.PrepInSucc
1526749536	L.IRATHO.N2E.ExecSuccIn
1526756789	L.IRATHO.N2E.PrepInFail.HOCancel
1526756788	L.IRATHO.N2E.PrepInFail.TgtNotAllow
1526756787	L.IRATHO.N2E.PrepInFail.AdmitFail

Voice Service Mobility

To evaluate the performance of VoNR-to-VoLTE mobility of voice services, monitor the counters listed in [Table 5-5](#) and [Table 5-6](#).

Table 5-5 NR counters related to voice service mobility

Counter ID	Counter Name
1911822821	N.HO.InterRAT.N2E.PrepAttOut.Voice
1911822820	N.HO.InterRAT.N2E.ExecAttOut.Voice
1911822819	N.HO.InterRAT.N2E.ExecSuccOut.Voice

Counter ID	Counter Name
1911822817	N.HO.InterRAT.N2E.Prep.FailOut.AMF.Voice
1911822816	N.HO.InterRAT.N2E.Prep.FailOut.PrepFailure.Voice
1911822815	N.HO.InterRAT.N2E.Prep.FailOut.NoReply.Voice
1911822818	N.HO.InterRAT.N2E.AMFAbnormRsp.Voice
1911829277	N.HO.InterRAT.NCell.N2E.PrepAttOut.Voice
1911829276	N.HO.InterRAT.NCell.N2E.ExecAttOut.Voice
1911829275	N.HO.InterRAT.NCell.N2E.ExecSuccOut.Voice
1911838095	N.RRCRedirection.N2E.Coverage.Voice
1911840483	N.RRCRedirection.N2E.Blind.Coverage.Voice
1911822968	N.HO.InterRAT.N2E.Exec.FailOut.ReEst.Voice

Table 5-6 LTE counters related to voice service mobility

Counter ID	Counter Name
1526763088	L.IRATHO.N2E.PrepAttIn.Voice
1526763087	L.IRATHO.N2E.ExecSuccIn.Voice
1526763092	L.IRATHO.N2E.PrepInSucc.Voice

5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN

5.3.1 Principles

User experience of UEs is affected when the UEs are in weak coverage areas of NR cells. Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN can now hand over or redirect UEs under weak uplink coverage to LTE cells.

This function is enabled if the **UL_SINR_MOBILITY_TO_EUTRAN_SW** and **MOBILITY_TO_EUTRAN_SW** options of the **NRCellAlgoSwitch.InterRatServiceMobilitySw** parameter are selected. Handover and redirection are supported. They are set for data services and VoNR services separately.

- To enable handover for both data and VoNR services, select the **EUTRAN_HO_SWITCH** option of the gNodeB parameter **NRInterRatHoParam.HoModeSwitch**.

- To enable redirection for both data and VoNR services, select the **EUTRAN_REDIRECT_SWITCH** option of the gNodeB parameter **NRInterRatHoParam.HoModeSwitch**.
- To enable redirection for data services and enable handover for VoNR services, select both the **EUTRAN_REDIRECT_SWITCH** and **VONR_COV_BASED_HO_PREFER_SW** options of the gNodeB parameter **NRInterRatHoParam.HoModeSwitch**.
- To enable redirection to the operating frequency of the strongest neighboring cell indicated in an E-UTRAN measurement report when this cell is not configured as a neighboring cell or encounters PCI confusion, select the **UL_SINR_RED_TO_STGEST_NCELL_SW** option of the gNodeB parameter **NRInterRatHoParam.HoModeSwitch**.

When an NR cell is shared by operators in RAN sharing with common carrier mode, the mobility policy (handover or redirection) for the cell can be set on a per operator basis using the **NRCellOpPolicy** MO. If the **NrCellOpPolicy** MO is not configured, the **NRInterRatHoParam.HoModeSwitch** parameter setting takes effect.

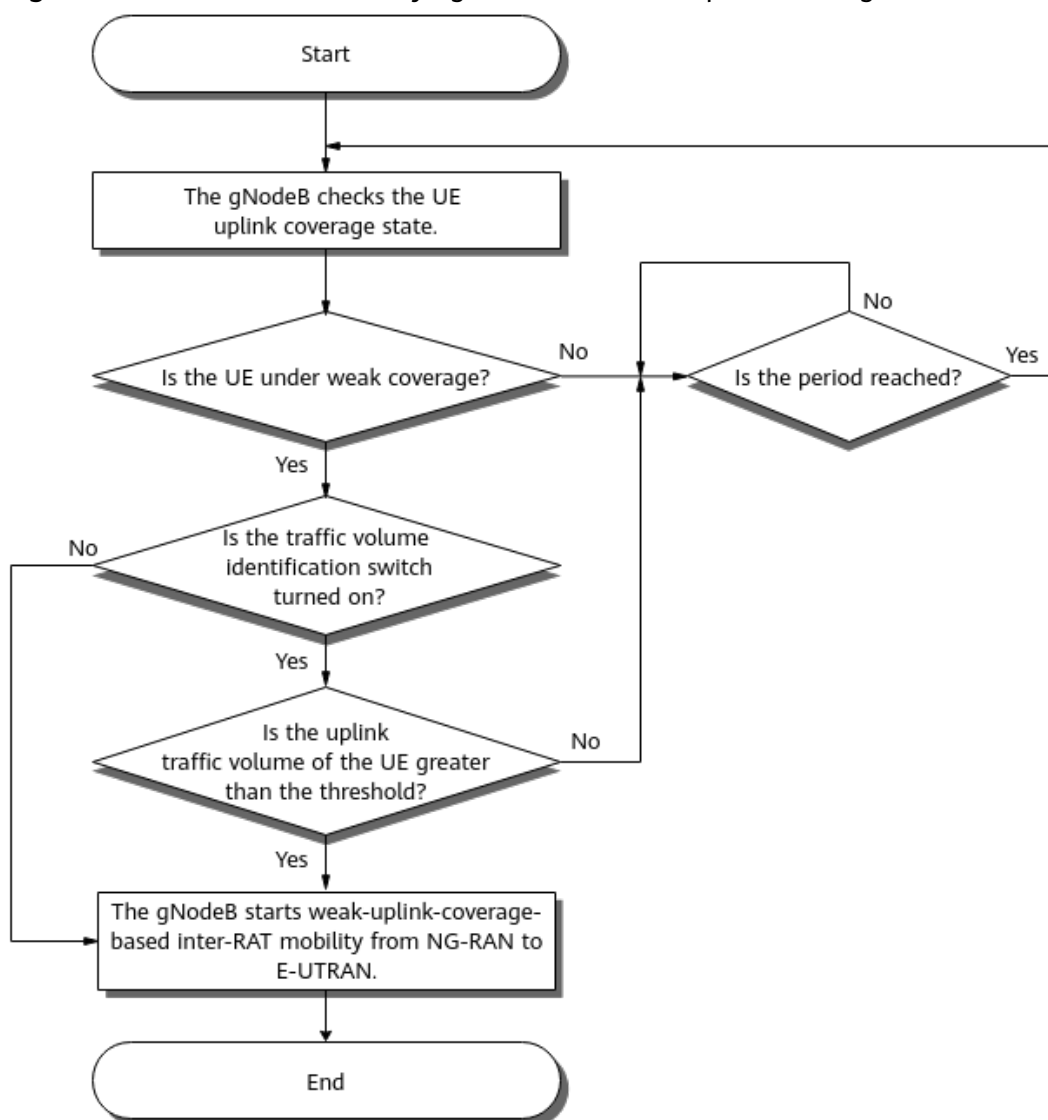
- To enable handover, select both the **EUTRAN_HO_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter and the **EUTRAN_HO_SWITCH** option of the **NRCellOpPolicy.InterRatHoPolicySwitch** parameter.
- Otherwise, redirections will be performed.

This function includes two steps:

1. The gNodeB identifies UEs under weak uplink coverage. For details, see [Identifying UEs Under Weak Uplink Coverage](#).
2. The gNodeB initiates handovers or redirections from NG-RAN to E-UTRAN for the UEs under weak uplink coverage. For details, see [NG-RAN to E-UTRAN Mobility Based on Weak Uplink Coverage](#).

Identifying UEs Under Weak Uplink Coverage

[Figure 5-3](#) shows the procedure for the gNodeB to identify UEs under weak uplink coverage.

Figure 5-3 Procedure for identifying UEs under weak uplink coverage

The procedure is as follows:

1. The gNodeB checks the real-time uplink coverage of all UEs in the cell and determines the status based on five consecutive check results. The check interval is specified by the **NRDUCellSrsMeas.NsaULPathSelSinrTimeToTrig** parameter. All UEs are initially in the strong coverage state.
 - If the **VOICE_UL_COV_BASED_HO_SW** option of the **NRCCellAlgoSwitch.VoiceStrategySwitch** parameter is deselected, or if the **VOICE_UL_COV_BASED_HO_SW** option of the **NRCCellAlgoSwitch.VoiceStrategySwitch** parameter is selected and a UE does not have voice services running (with no 5QI 1 bearer set up), the following rules for evaluating the coverage status of the UE apply:
 - If the SRS SINR of the UE is found to be less than the **NRDUCellSrsMeas.NrToEutranSinrLowThld** parameter value minus 2 dB for five consecutive times, the UE is considered to enter the weak coverage state. The gNodeB proceeds to 2 to evaluate the uplink traffic volume of the UE.

- If the SRS SINR of the UE is found to be greater than the **NRDUCellSrsMeas.NrToEutranSinrLowThld** parameter value plus 2 dB for five consecutive times, the UE is considered to enter the strong coverage state. The procedure ends.
- Otherwise, the UE status remains unchanged.
- If the **VOICE_UL_COV_BASED_HO_SW** option of the **NRCCellAlgoSwitch.VoiceStrategySwitch** parameter is selected and a UE has a voice service running (with a 5QI 1 bearer set up), the following rules for evaluating the coverage status of the UE apply:
 - If the SRS SINR of the UE is found to be less than the **NRCCellMobilityConfig.VoiceULCovBasedHoSinrThld** parameter value minus 2 dB for five consecutive times, the UE is considered to enter the weak coverage state. For UEs considered in the weak coverage state, the gNodeB proceeds to 3 without checking whether they meet the uplink traffic volume conditions.
 - If the SRS SINR of the UE is found to be greater than the **NRCCellMobilityConfig.VoiceULCovBasedHoSinrThld** parameter value plus 2 dB for five consecutive times, the UE is considered to enter the strong coverage state. The procedure ends.
 - Otherwise, the UE status remains unchanged.
- 2. The gNodeB checks whether UEs in the weak uplink coverage state meet the uplink traffic volume condition.
 - If a UE has a voice bearer (that is, a 5QI 1 bearer) but the **VOICE_UL_COV_BASED_HO_SW** option of the **NRCCellAlgoSwitch.VoiceStrategySwitch** parameter is deselected, or if a UE has no voice bearer (that is, no 5QI 1 bearer), the gNodeB determines the traffic volume of the UE based on the following rules:
 - Specifically, the gNodeB checks whether the **VOLUME_IDENTIFY_SW** option of the **NRCCellAlgoSwitch.ServiceFunctionSwitch** parameter is selected.
 - If the option is selected, the gNodeB checks whether the uplink traffic volume to be transmitted is greater than the threshold specified by the **NRDUCellServExp.CommonULLargePktIdentThld** parameter. If the volume is greater than the threshold, the uplink traffic volume condition is met and the gNodeB proceeds to the next step. If the volume is less than or equal to the threshold, the uplink traffic volume condition is not met and the procedure ends.
 - If this option is deselected, the gNodeB proceeds to the next step.
 - If a UE has a voice bearer (that is, a 5QI 1 bearer) and the **VOICE_UL_COV_BASED_HO_SW** option of the **NRCCellAlgoSwitch.VoiceStrategySwitch** parameter is selected, the gNodeB proceeds to the next step.

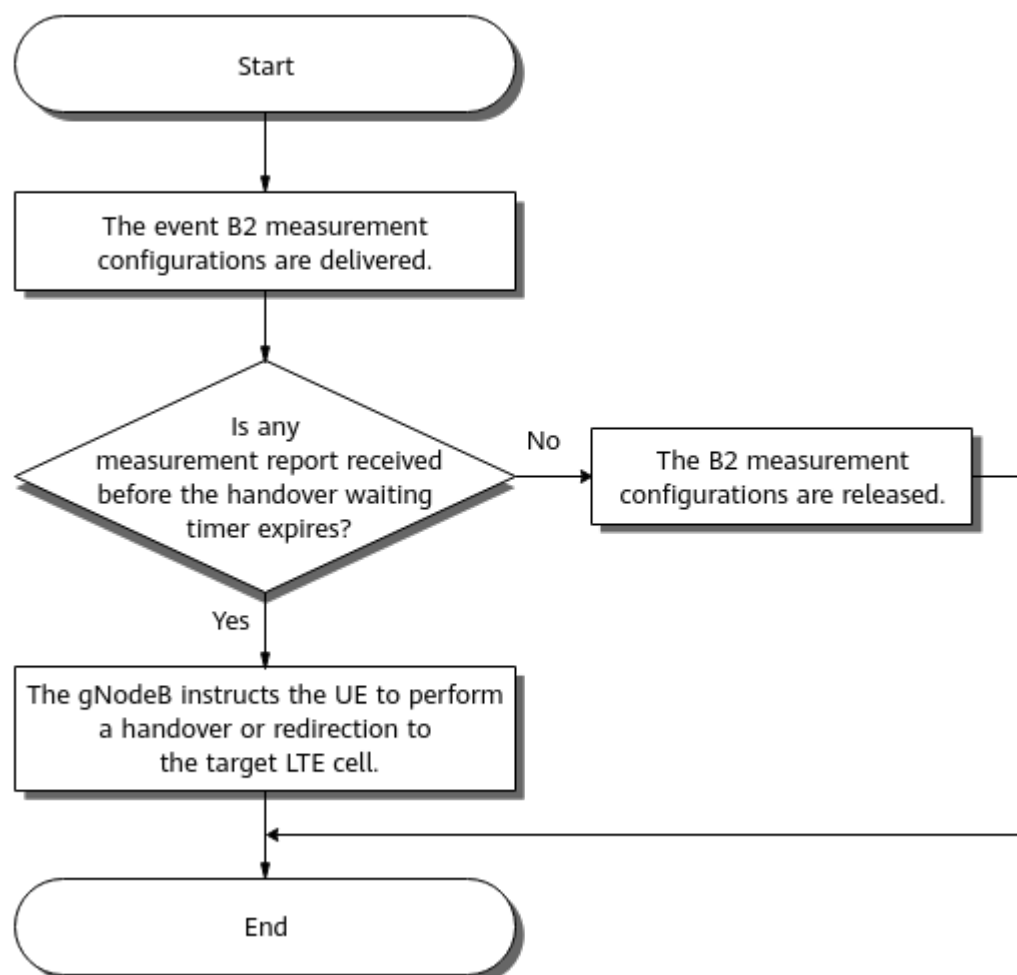
3. The gNodeB starts the procedure for an NG-RAN to E-UTRAN handover or redirection based on weak uplink coverage for the UEs that meet the conditions.

If the handover or redirection fails for a UE, the UE continues camping on the NR cell. The gNodeB periodically identifies UEs that meet the conditions in the cell, and it initiates this procedure for the UEs at the interval specified by the **gNodeBParam.NsaSaSelOptPeriod** parameter.

NG-RAN to E-UTRAN Mobility Based on Weak Uplink Coverage

The gNodeB starts the procedure for NG-RAN to E-UTRAN mobility based on weak uplink coverage for UEs in the weak coverage state, as shown in [Figure 5-4](#).

Figure 5-4 Procedure for NG-RAN to E-UTRAN mobility based on weak uplink coverage



The procedure is as follows:

1. The gNodeB delivers the measurement configurations related to inter-RAT event B2. Threshold 1 for event B2 is specified by the **NRCellInterRHoMeaGrp.UlSinrHoEutranB2RsrpThld1** parameter, and threshold 2 for event B2 is specified by the **NRCellHoEutranMeaGrp.NetworkingOptionOptB1Thld** parameter. Thresh 1

for event B2 must be set to a value larger than the RSRP threshold for inter-RAT handover event A2. For details about the latter, see [4.1.4.2.1 Measurement Events](#).

If the **NRCellEutranNFreq.FreqIntrfFlag** parameter is set to **INTRF** for an LTE frequency on the gNodeB, the gNodeB does not deliver this frequency.

2. The gNodeB checks whether the handover waiting timer (2s) has expired.
 - If the gNodeB has received measurement reports before the handover waiting timer expires, the gNodeB goes to the next step.
 - If the gNodeB does not receive measurement reports for any frequency before the handover waiting timer expires, the gNodeB releases the B2 measurement configurations. The procedure ends.
3. The gNodeB selects the strongest cell as the target LTE cell based on the B2 measurement reports from the UE.
4. The gNodeB performs a handover to the target LTE cell or a redirection to the corresponding frequency.
 - If the PCI of the target LTE cell is included in the neighboring cell list configured on the gNodeB and no PCI confusion occurs:
 - If the UE supports handover, the gNodeB instructs the UE to execute a handover to the target LTE cell.
 - If the UE does not support handover, the gNodeB instructs the UE to redirect to the operating frequency of the target LTE cell.
 - If the PCI of the target LTE cell is not included in the neighboring cell list configured on the gNodeB or PCI confusion occurs:
 - If ANR is enabled on the gNodeB, the UE supports CGI measurement, and handover is supported according to both the gNodeB configuration and the UE capability, then the gNodeB hands over the UE to the neighboring LTE cell with the best signal quality based on the neighboring cell list maintained by ANR.
 - If ANR is enabled on the gNodeB and the UE supports CGI measurement, but handover is not supported according to either the gNodeB configuration or the UE capability, then the gNodeB redirects the UE to the operating frequency of the neighboring LTE cell with the best signal quality based on the neighboring cell list maintained by ANR.
 - If ANR is not enabled on the gNodeB or the UE does not support CGI measurement, the procedure ends.

5.3.2 Network Analysis

5.3.2.1 Benefits

This function increases the average uplink throughput of UEs in an NR cell where UEs are evenly distributed in the cell center, at medium distances from the cell center, and far from the cell center. This throughput is indicated by **User Uplink Average Throughput (DU)**.

5.3.2.2 Impacts

The impacts between this function and other functions are described in [5.3.3.2.3 Function Impacts](#).

NR side:

- The uplink and downlink PRB usages of NR cells decrease. The uplink PRB usage is indicated by **Uplink Resource Block Utilizing Rate (DU)**, and the downlink PRB usage is indicated by **Downlink Resource Block Utilizing Rate (DU)**.
- The uplink and downlink traffic volumes of NR cells decrease. The uplink traffic volume is indicated by **Uplink Traffic Volume (DU)**, and the downlink traffic volume is indicated by **Downlink Traffic Volume (DU)**.
- If the **UL_SINR_MOBILITY_TO_EUTRAN_SW** option of the **NRCCellAlgoSwitch.InterRatServiceMobilitySw** parameter is solely selected, the service drop rate on the NR side may increase, which is indicated by **Service Call Drop Rate (CU, Inactive)**. If the **UL_SINR_MOBILITY_TO_EUTRAN_SW** option of the **NRCCellAlgoSwitch.InterRatServiceMobilitySw** parameter and the **VOLUME_IDENTIFY_SW** option of the **NRCCellAlgoSwitch.ServiceFunctionSwitch** parameter are both selected and the **NRDUCellServExp.CommonULLargePktIdentThld** parameter is appropriately set, the service drop rate decreases.
- When the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is enabled, the **N.NsaDc.SgNB.Mod.Req.Att** counter value decreases.
- If the **VOICE_UL_COV_BASED_HO_SW** option of the **NRCCellAlgoSwitch.VoiceStrategySwitch** parameter is selected, **Service Call Drop Rate (CU)** may increase.

LTE side:

- The uplink and downlink PRB usages of LTE cells increase. The uplink PRB usage is indicated by **Uplink Resource Block Utilizing Rate**, and the downlink PRB usage is indicated by **Downlink Resource Block Utilizing Rate**.
- The uplink and downlink traffic volumes of LTE cells increase. The uplink traffic volume is indicated by **Uplink Traffic Volume**, and the downlink traffic volume is indicated by **Downlink Traffic Volume**.
- When the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is enabled, the **L.NsaDc.SCG.Mod.Req.Att** counter value decreases.

This function may have a negative impact when there is uplink interference to LTE cells or the number of available RBs decreases. An example of such scenarios is that the uplink load is heavy when a large number of UEs are handed over to an LTE cell at the same time.

5.3.3 Requirements

5.3.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
NR	FOFD-021209	Inter-RAT Mobility from NG-RAN to E-UTRAN	NR0SRAT MNE00	Inter-RAT Mobility from NG-RAN to E-UTRAN (NR)	gNodeB	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

5.3.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.2.1 Prerequisite Functions

Uplink SINR-based redirection to the strongest neighboring E-UTRAN cell

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD	Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	UL_SINR_MOBILITY_TO_EUTRAN_SW option of the NRCellAlgoSwitch.InterRatServiceMobilitySw parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>

5.3.3.2.2 Mutually Exclusive Functions

None

5.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Inter-RAT ANR management	NR_EUTRAN_ANR_SW option of the NRCellAlgoSwitch . AnrSwitch parameter	<i>ANR in 5G RAN Feature Documentation</i>	It is recommended that inter-RAT ANR management be enabled so that UEs can be handed over to the best neighboring LTE cells based on the neighboring cell list maintained by ANR when the PCIs of the best cells are not in the neighboring cell list configured on the gNodeB or are involved in PCI confusion.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	User-experience-based coverage expansion	COV_THLD_ADAPT_SW option of the gNBMinimalityMobilityParameters.MobilityAdaptationSwitch parameter	<i>Multi-Frequency Smart Selection in 5G RAN Feature Documentation</i>	Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN uses the same weak coverage evaluation mechanism as user-experience-based coverage expansion does. The weak-uplink-coverage-based handover mechanism is triggered on a first-come, first-served basis. If uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN is enabled before user-experience-based coverage expansion, the number of executions of outgoing handovers from NG-RAN to E-UTRAN (N.HO.InterRAT.N2E.ExecAttOut) decreases. If user-experience-based coverage expansion is enabled before uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, the number of weak-coverage-based inter-frequency handovers (N.HO.InterFreq.UplinkCoverage.ExecAttOut) decreases.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Protection of UEs under weak uplink coverage	UL_WEAK_COV_UE_PROTECT_SW option of the NRCellAlgoSwitch . MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection in 5G RAN Feature Documentation</i>	When protection of UEs under weak uplink coverage is enabled together with uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, the base station performs an inter-frequency or inter-RAT handover based on the measurement report received from the UE. If the base station receives an inter-frequency A5 measurement report first, protection of UEs under weak uplink coverage takes effect. If the base station receives an inter-RAT B2 measurement report first, uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN takes effect.
NR FDD Low-frequency NR TDD	Inter-RAT redirection back to the HPLMN based on roaming UE identification	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, only redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.

5.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all LTE-capable boards support this function.
The LBBPc and LBBPd boards do not support the function of prohibiting large-packet UEs in LTE cells from returning to NR cells.
- For NR, all NR-capable boards support this function.

RF Modules

None

5.3.3.4 Others

- UE requirements
UEs must comply with 3GPP Release 15 or later.
- Core network requirements
 - The core networks must comply with 3GPP Release 15 or later.
 - Handover requires the N26 interface to be configured between the MME and AMF.

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

Table 5-7 describes the parameters related to uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN. For details about basic parameters for interoperability from NG-RAN to E-UTRAN, see **4.4.1.1 Data Preparation** and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-7 NR parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
Inter-RAT Service Mobility Switch	NRCellAlgoSwitch.InterRatServiceMobilitySw	Select the UL_SINR_MOBILITY_TO_EUTRAN_SW and MOBILITY_TO_EUTRAN_SW options.
Handover Mode Switch	NRInterRatHoParam.HoModeSwitch	Set this parameter based on the mobility policies required for data services and VoNR services. For details, see 5.3.1 Principles .
NSA/SA Selection Opt Period	gNodeBParam.NsaSaSelOptPeriod	Set this parameter based on the network plan.
Inter-RAT Handover Trigger Event Type	NRInterRatHoParam.InterRatHoTriggerEventType	The default value is recommended.
NSA UL Path Select SINR Time-to-Trigger	NRDUCellSrsMeasure.NsaULPathSelSnrTimeToTrig	Set this parameter based on the network plan.
NR Move to E-UTRAN UL SINR Low Threshold	NRDUCellSrsMeasure.NrToEutranSnrLowThld	Set this parameter based on the network plan. A larger value of this parameter results in a higher probability of handovers from the NR cell to LTE cells based on the uplink signal quality of the NR cell but also a higher probability of ping-pong handovers. A smaller value of this parameter results in a lower probability of handovers from the NR cell to LTE cells based on the uplink signal quality of the NR cell and a lower probability that UEs enjoy better coverage experience.
Voice Strategy Switch	NRCellAlgoSwitch.VoiceStrategySwitch	If VoNR has been enabled in the NR cell and the thresholds need to be set differently for VoNR and data services, it is recommended that the VOICE_UL_COV_BASED_HO_SW option of this parameter be selected.
Voice UL Coverage-based HO SINR Threshold	NRCellMobilityConfig.VoiceULCoverageBasedHoSnrThld	The default value is recommended.
Service Function Switch	NRCellAlgoSwitch.ServiceFunctionSwitch	It is recommended that the VOLUME_IDENTIFY_SW option be selected.

Parameter Name	Parameter ID	Setting Notes
Common UL Large Pkt Ident Thld	NRDUCellServExp.CommonULLargePktIdentThld	The default value is recommended. A larger value of this parameter results in a lower probability of UE transfer from NG-RAN to E-UTRAN. A smaller value of this parameter results in a higher probability of UE transfer from NG-RAN to E-UTRAN. This may increase the number of abnormal UE context releases in the cell, which can be observed using the N.UECntx.AbnormRel counter.
UL SINR HO to E-UTRAN B2 RSRP Thld1	NRCeIIInterRHoMeaGrp.UlSinnrHoEutranB2RsrpThld1	Set this parameter based on the network plan. The value of this parameter must be set to a value larger than the RSRP threshold for inter-RAT handover event A2. For details about the latter, see 4.1.4.2.1 Measurement Events .
Networking Option Opt RSRP Threshold	NRCeIIHoEutranMeaGrp.NetworkingOptionOptB1Thld	Set this parameter based on the network plan.
Handover Mode switch	NRInterRatHoParam.HoModeSwitch	Select the UL_SINR_RED_TO_STGEST_NCELL_SW option if redirection to the operating frequency of the strongest neighboring cell indicated in an E-UTRAN measurement report is required in the case that this cell is not configured as a neighboring cell or encounters PCI confusion.

5.3.4.1.2 Using MML Commands

The basic configurations for mobility from NG-RAN to E-UTRAN must have been completed before activating this function. For details, see [4.4.1.2 Using MML Commands](#) and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples

Before using MML commands, refer to [5.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Running the following commands for NR cells
//Running the following command if handover and redirection are both required for data and VoNR services
MOD NRINTERRATHOPARAM: NrCellId=0,
HoModeSwitch=EUTRAN_HO_SWITCH-1&EUTRAN_REDIRECT_SWITCH-1,
InterRatHoTriggerEventType=EVENT_B2;
//Running the following command if redirection is required for data services and handover is preferred for
voice services
MOD NRINTERRATHOPARAM: NrCellId=0,
HoModeSwitch=EUTRAN_HO_SWITCH-0&EUTRAN_REDIRECT_SWITCH-1&VONR_COV_BASED_HO_PREFER_S
W-1;
//Turning on the switch for uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=VOICE_UL_COV_BASED_HO_SW-1,
InterRatServiceMobilitySw=MOBILITY_TO_EUTRAN_SW-1&UL_SINR_MOBILITY_TO_EUTRAN_SW-1;
//Changing the settings of NrToEutranSinrLowThld and NsaULPathSelSinrTimeToTrig
MOD NRDUCELLSRSEAS: NrDuCellId=0, NrToEutranSinrLowThld=30, NsaULPathSelSinrTimeToTrig=15;
MOD NRCELLMOBILITYCONFIG: NrCellId=0, VoiceULCovBasedHoSinrThld=-30;
//Changing the setting of NsaSaSelOptPeriod
MOD GNODEBPARAM: NsaSaSelOptPeriod=50;
//Changing the setting of ULSinrHoEutranB2RsrpThld1
MOD NRCELLINTERRHOMEAGRP: NrCellId=0, InterRatHoMeasGroupId=10,
ULSinrHoEutranB2RsrpThld1=-113;
//Changing the setting of NetworkingOptionOptB1Thld
MOD NRCELLHOEUTRANMEAGRP: NrCellId=0, InterRHoToEutranMeasGrpId=10,
NetworkingOptionOptB1Thld=-112;
//((Optional) Enabling the base station to take uplink traffic volume into consideration during evaluation of
handovers or redirections from NG-RAN to E-UTRAN based on weak uplink coverage
MOD NRCELLALGOSWITCH: NrCellId=0, ServiceFunctionSwitch=VOLUME_IDENTIFY_SW-1;
//((Optional) Changing the uplink traffic volume threshold
MOD NRDUCELLSERVEXP: NrDuCellId=0, CommonULLargePktIdentThld=80;
//((Optional) Turning on UL_SINR_RED_TO_STGEST_NCELL_SW if redirection to the operating frequency of
the strongest neighboring cell indicated in an E-UTRAN measurement report is required in the case that this
cell is not configured as a neighboring cell or encounters PCI confusion
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=UL_SINR_RED_TO_STGEST_NCELL_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=UL_SINR_MOBILITY_TO_EUTRAN_SW-0;
```

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

Step 1 Start Uu interface tracing and NG interface tracing for the gNodeB.

Step 2 Observe the signaling messages over the Uu interface. If the Measurement Report message sent from the UE to the gNodeB indicates RSRP threshold 1 for event B2

related to uplink-SINR-based mobility to E-UTRAN, the gNodeB has started the procedure for NG-RAN to E-UTRAN mobility based on weak uplink coverage.

Step 3 Observe the MobilityFromNRCommand message sent by the gNodeB to the UE over the Uu interface. If the value of the *targetRAT-Type* IE in the message is "eutra", the preparation for the handover from NG-RAN to E-UTRAN has succeeded.

Step 4 Observe the signaling messages over the NG interface. If either (i) the gNodeB receives a UE CONTEXT RELEASE COMMAND message with the cause value "Successful Handover" or "Normal Release" from the AMF, or (ii) the gNodeB receives an RRC Reestablishment Complete message from the UE and the RRC connection reestablishment is successful, the handover from NG-RAN to E-UTRAN has been successfully executed.

----End

5.3.4.3 Network Monitoring

Monitor the counters listed in [Table 5-8](#) to evaluate the performance of this function.

Table 5-8 NR counters related to data service mobility

Counter ID	Counter Name
1911831623	N.HO.InterRAT.N2E.ResourceOpt.ULCoverage.ExecSuccOut
1911831624	N.HO.InterRAT.N2E.ResourceOpt.ULCoverage.ExecAttOut
1911831625	N.HO.InterRAT.N2E.ResourceOpt.ULCoverage.PrepAttOut
1911817063	N.HO.InterRAT.N2E.ExecSuccOut
1911817061	N.HO.InterRAT.N2E.ExecAttOut
1911817056	N.HO.InterRAT.N2E.PrepAttOut
1911817057	N.HO.InterRAT.N2E.AMFAbnormRsp
1911817064	N.HO.InterRAT.N2E.Prep.FailOut.AMF
1911817065	N.HO.InterRAT.N2E.Prep.FailOut.PrepFailure
1911817059	N.HO.InterRAT.N2E.Prep.FailOut.NoReply
1911817054	N.HO.InterRAT.NCell.N2E.PrepAttOut
1911817053	N.HO.InterRAT.NCell.N2E.ExecAttOut
1911817055	N.HO.InterRAT.NCell.N2E.ExecSuccOut
1911829365	N.HO.InterRAT.N2E.ResourceOpt.ExecSuccOut
1911829366	N.HO.InterRAT.N2E.ResourceOpt.ExecAttOut

Counter ID	Counter Name
1911829367	N.HO.InterRAT.N2E.ResourceOpt.PrepAttOut
1911847037	N.RRCRedirection.N2E.ULCoverage

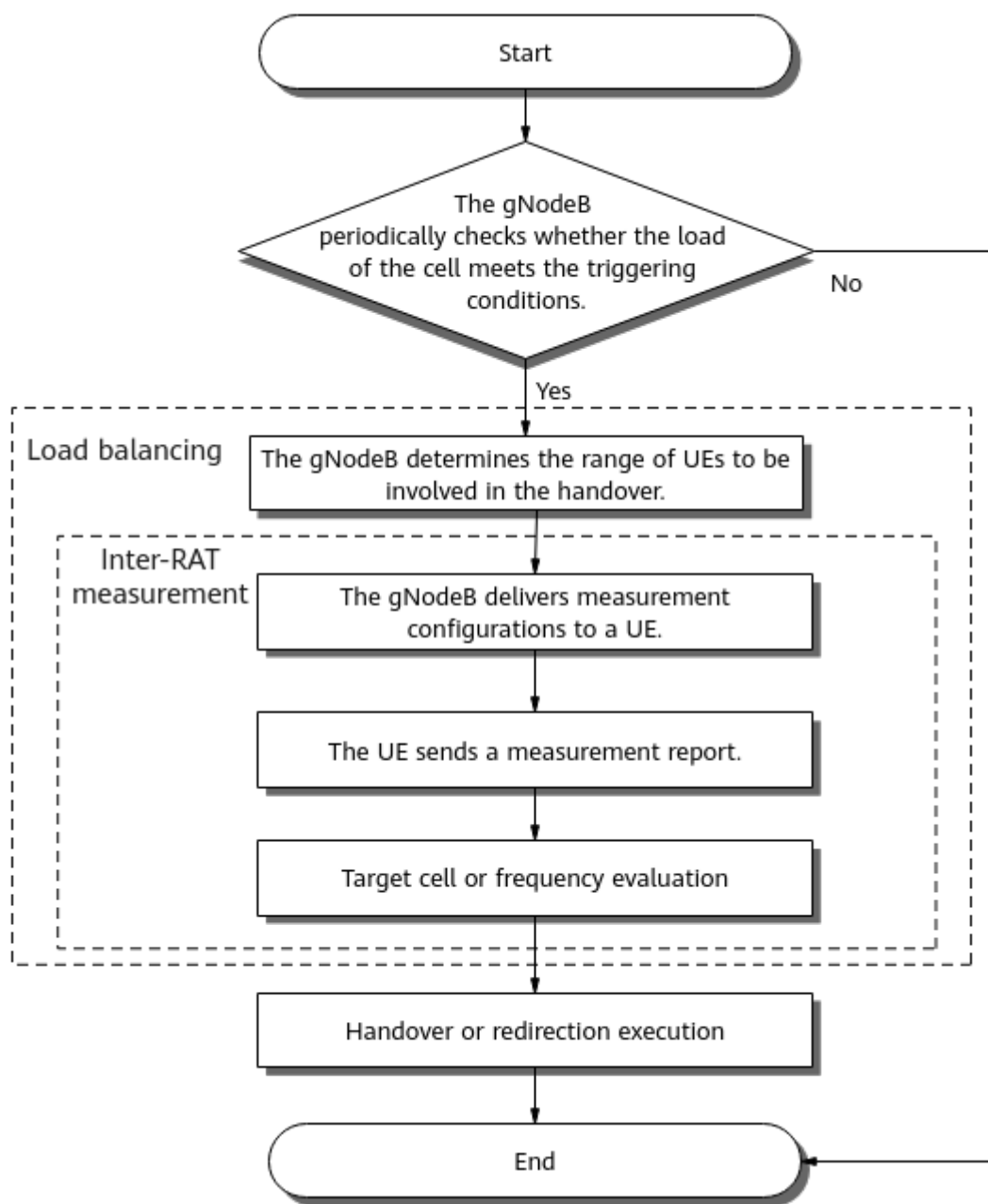
5.4 Load-based Inter-RAT Mobility from NG-RAN to E-UTRAN

5.4.1 Principles

5G capacity is in high demand and may fail to meet user requirements in scenarios with dense UE distribution, potentially causing severe service congestion. To mitigate this issue, the 5G-load-based inter-RAT mobility from NG-RAN to E-UTRAN function is introduced.

This function is controlled by the **LOAD_MOBILITY_TO_EUTRAN_SW** option of the **NRCellMlb.MlbAlgoEnhSwitch** parameter. It supports measurement-based handover and measurement-based redirection from NG-RAN to E-UTRAN.

Figure 5-5 shows the procedure for a 5G-load-based handover from NG-RAN to E-UTRAN.

Figure 5-5 Procedure for a 5G-load-based handover from NG-RAN to E-UTRAN

The procedure is as follows:

1. The gNodeB periodically checks whether the NR cell load meets the triggering conditions. If the triggering conditions are met, the gNodeB proceeds to the next step. If the exiting conditions are met, the procedure ends.

Triggering conditions: The gNodeB proceeds to the next step if either of the following conditions is met throughout an evaluation period specified by the **NRCellMlb.MlbJudgePeriod** parameter in the cell:

- Number of UEs in the downlink in the cell > **NRCellMlb.InterRatMlbUeNumThld** × (1 + Offset (fixed at 5%)), and downlink PRB usage of the cell > **NRCellMlb.InterRatMlbDlPrbUsageThld** × (1 + Offset (fixed at 5%))

- Number of UEs in the uplink in the cell > **NRCellMlb.InterRatMlbUeNumThld** × (1 + Offset (fixed at 5%)), and uplink PRB usage of the cell > **NRCellMlb.InterRatMlbULPrbUsageThld** × (1 + Offset (fixed at 5%))

Exiting conditions: The procedure ends when all of the following conditions are met throughout an evaluation period:

- Total number of UEs in the uplink and downlink in the cell ≤ Inter-RAT MLB UE number threshold
 - Downlink PRB usage of the cell ≤ Inter-RAT MLB downlink PRB usage threshold
 - Uplink PRB usage of the cell ≤ Inter-RAT MLB uplink PRB usage threshold
2. The gNodeB determines the range of UEs to be involved in handover.
 - UE type: The gNodeB excludes the following types of UEs from the UEs to be involved in handover: voice service UEs, VIP UEs (with specific QCI or RFSPs), CA UEs, super uplink UEs, and NSA UEs.
 - UE number: The gNodeB calculates the number of UEs that can be handed over in a period, which is equal to $\min\{\text{NRCellMlb.OffloadMaxUeNum}, \text{Number of UEs in the cell} - \text{Inter-RAT MLB UE number threshold}\}$. If the numbers of UEs in the downlink and uplink in the cell are greater than the inter-RAT MLB UE number threshold, the number of UEs in the cell is equal to number of UEs in the downlink or that in the uplink, whichever is larger.
 3. The gNodeB performs inter-RAT measurement.

Measurement configurations delivery: The gNodeB delivers the measurement configurations related to inter-RAT event B1 to the selected UE. The RSRP threshold for event B1 related to load-based handovers to E-UTRAN is specified by the **NRCellHoEutranMeaGrp.LoadMobToEutranB1RsrpThld** parameter. If the **MOB_INTERRAT_FREQ_FILTER_SW** option of the **NRCellEutranNFreq.FrequencyCtrlSwitch** parameter is selected, the gNodeB does not include the corresponding frequency in the measurement configurations related to inter-RAT event B1. For details about other rules for the gNodeB to filter LTE frequencies, see [4.1.4.1.3 LTE Frequency Selection by the gNodeB](#).

Measurement reporting: The UE performs measurements based on the measurement configurations sent by the gNodeB. When determining that the reporting criteria are met, the UE sends a measurement report on inter-RAT event B1 to the gNodeB.

Target cell or frequency evaluation: The gNodeB receives the measurement report on inter-RAT event B1 (indicating that the signal quality of LTE cells meets certain criteria), generates a candidate cell list based on cell signal quality, and selects the cell with the best signal quality as the target LTE cell.

4. The UE performs a handover or redirection. The gNodeB delivers the target cell or frequency to the UE, instructing the UE to perform a handover or redirection. (The **EUTRAN_HO_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter and the **EUTRAN_REDIRECT_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter must be both selected.) The UE handed over to an LTE cell cannot be handed over back to NG-RAN through "E-UTRAN to NG-RAN traffic steering" or "fast return from E-UTRAN to NG-RAN" within the period of time specified by the

CellHoParaCfg.NrResOptHolnProtectTmr parameter, thereby avoiding ping-pong handovers between E-UTRAN and NG-RAN.

5.4.2 Network Analysis

5.4.2.1 Benefits

When load-based inter-RAT mobility from NG-RAN to E-UTRAN is enabled, the gNodeB transfers NR UEs to LTE networks based on the load of live networks. As a result, the average uplink and downlink throughputs of NR UEs increase.

The definitions of the indicators involved are described as follows:

- Average uplink throughput of NR UEs = **User Uplink Average Throughput (DU)**
- Average downlink throughput of NR UEs = **User Downlink Average Throughput (DU)**

5.4.2.2 Impacts

The impacts between this function and other functions are described in [5.3.3.2.3 Function Impacts](#).

After NR UEs are transferred to the LTE network, the number of UEs on the NR side decreases and that on the LTE side increases, which causes the following impacts:

- The average uplink and downlink UE throughputs in the LTE cell decrease.
- The uplink and downlink PRB usages of the LTE cell increase.
- The uplink and downlink traffic volumes of the LTE cell increase.
- The uplink and downlink PRB usages of NR cells decrease.
- The uplink and downlink traffic volumes of NR cells decrease.
- The service drop rate on the NR side may increase.

The definitions of the indicators involved are described as follows:

- Average uplink throughput of LTE UEs: **User Uplink Average Throughput**
- Average downlink throughput of LTE UEs: **User Downlink Average Throughput**
- Downlink PRB usage: **Downlink Resource Block Utilizing Rate (DU)**
- Uplink PRB usage: **Uplink Resource Block Utilizing Rate (DU)**
- Uplink traffic volume: **Uplink Traffic Volume (DU)**
- Downlink traffic volume: **Downlink Traffic Volume (DU)**
- Service drop rate on the NR side: **Service Call Drop Rate (CU)**

5.4.3 Requirements

5.4.3.1 Licenses

This function is not under license control.

5.4.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Inter-RAT mobility from NG-RAN to E-UTRAN	MOBILITY_T O_EUTRAN_S W option of the NRCellAlgoSwitch.InterRatServiceMobilitySw parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	Load-based inter-RAT mobility from NG-RAN to E-UTRAN requires that inter-RAT mobility from NG-RAN to E-UTRAN be enabled.

5.4.3.2.2 Mutually Exclusive Functions

None

5.4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellMobilitySwitch.MlbAlgorithm parameter NRCellMobilitySwitch.MlbTriggerMode set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode in 5G RAN Feature Documentation</i>	When both load-based inter-RAT mobility from NG-RAN to E-UTRAN and UE-number-based connected mode MLB are enabled, UE-number-based connected mode MLB takes precedence if the conditions for triggering MLB are met for both functions simultaneously. If the conditions are met sequentially, the corresponding MLB policy (intra-RAT MLB or inter-RAT MLB) for each function takes effect in sequence.

5.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all LTE-capable boards support this function.

- For NR, all NR-capable boards support this function.

RF Modules

None

5.4.3.4 Others

None

5.4.4 Operation and Maintenance

5.4.4.1 Data Configuration

5.4.4.1.1 Data Preparation

[Table 5-9](#) describes the parameters related to load-based inter-RAT mobility from NG-RAN to E-UTRAN. For details about basic parameters for interoperability from NG-RAN to E-UTRAN, see [4.4.1.1 Data Preparation](#) and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-9 NR parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
Inter-RAT Service Mobility Switch	NRCeAlgoSwitc h.InterRatService MobilitySw	Select the MOBILITY_TO_EUTRAN_SW option.
MLB Enhancement Algorithm Switch	NRCeMlb.MlbAl goEnhSwitch	Select the LOAD_MOBILITY_TO_EUTRAN_SW option.
Frequency Control Switch	NRCeEutranNFr eq.FrequencyCtrl Switch	To filter out certain frequency from the target frequencies for handovers, select the MOB_INTERRAT_FREQ_FILTER_SW option.
Load Mobility to E-UTRAN B1 RSRP Thld	NRCeHoEutran MeaGrp.LoadMo bToEutranB1Rsrp Thld	Use the recommended value of this parameter.
MLB Judge Period	NRCeMlb.MlbJu dgePeriod	Use the recommended value of this parameter.
Inter-RAT MLB UE Number Threshold	NRCeMlb.InterR atMlbUeNumThl d	Use the recommended value of this parameter.

Parameter Name	Parameter ID	Setting Notes
Inter-RAT MLB DL PRB Usage Threshold	<i>NRCeLLMLb.InterRatMlbDlPrbUsageThld</i>	Set this parameter based on the network plan.
Inter-RAT MLB UL PRB Usage Threshold	<i>NRCeLLMLb.InterRatMlbUlPrbUsageThld</i>	Set this parameter based on the network plan.
Offload Max UE Number	<i>NRCeLLMLb.OffloadMaxUeNum</i>	Use the recommended value of this parameter.

5.4.4.1.2 Using MML Commands

The basic configurations for mobility from NG-RAN to E-UTRAN must have been completed before activating this function. For details, see [4.4.1.2 Using MML Commands](#) and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples

Before using MML commands, refer to [5.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Turning on the switch of mobility to E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=MOBILITY_TO_EUTRAN_SW-1;
//Turning on the load-based mobility to E-UTRAN switch and configuring MLB-related parameters
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=LOAD_MOBILITY_TO_EUTRAN_SW-1,
MlbJudgePeriod=10, InterRatMlbUeNumThld=100, InterRatMlbDlPrbUsageThld=20,
InterRatMlbUlPrbUsageThld=20, OffloadMaxUeNum=10;
//Setting the load mobility to E-UTRAN B1 RSRP threshold (when no parameter group related to
measurements for handovers from NR to E-UTRAN has been added)
ADD NRCELLHOEUTRANMEAGRP: NrCellId=0, InterRHoToEutranMeasGrpId=10,
LoadMobToEutranB1RsrpThld=-111;
//Setting the load mobility to E-UTRAN B1 RSRP threshold (when a parameter group related to
measurements for handovers from NR to E-UTRAN has been added)
MOD NRCELLHOEUTRANMEAGRP: NrCellId=0, InterRHoToEutranMeasGrpId=10,
LoadMobToEutranB1RsrpThld=-111;
//Turning on the mobility inter-RAT frequency filtering switch
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1300,
FrequencyCtrlSwitch=MOB_INTERRAT_FREQ_FILTER_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the load-based mobility to E-UTRAN switch
MOD NRCELLMLB: NrCellId=0, MlbAlgoEnhSwitch=LOAD_MOBILITY_TO_EUTRAN_SW-0;
```

5.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.4.2 Activation Verification

Use the counters listed in the following table to verify whether load-based inter-RAT mobility from NG-RAN to E-UTRAN has taken effect.

If any counter in the table has a non-zero value, load-based inter-RAT mobility from NG-RAN to E-UTRAN has taken effect.

Counter ID	Counter Name
1911849696	N.HO.InterRAT.N2E.Mlb.PrepAttOut
1911849697	N.HO.InterRAT.N2E.Mlb.ExecAttOut
1911849698	N.HO.InterRAT.N2E.Mlb.ExecSuccOut

5.4.4.3 Network Monitoring

To evaluate the performance of data service mobility from NG-RAN to E-UTRAN, monitor the counters listed in [Table 5-10](#).

Table 5-10 NR counters related to data service mobility

Counter ID	Counter Name
1911849696	N.HO.InterRAT.N2E.Mlb.PrepAttOut
1911849697	N.HO.InterRAT.N2E.Mlb.ExecAttOut
1911849698	N.HO.InterRAT.N2E.Mlb.ExecSuccOut

5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN

5.5.1 Principles

When a UE moves from a serving LTE cell to a neighboring NR cell, the signal quality of the LTE cell is poor while that of the NR cell is good. To ensure service

continuity or better user experience, downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN is triggered for the UE.

Downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN is controlled by the **INTER_RAT_MOBILITY_TO_NR_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter. Measurement-based handover and measurement-based redirection are supported.

- To enable handover, select the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter.
- To enable redirection, select the **NrRedirectSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter.
- If the above options are both selected, whether handover can be performed for the UE is preferentially evaluated.

In RAN sharing with common carrier scenarios, the mobility policy (handover or redirection) for a cell can be set on a per operator basis using the **CellOpHoCfg** MO. If no operator-specific mobility policy is set, the mobility policy for the cell is determined by the **CellHoParaCfg.HoModeSwitch** parameter.

- To enable handover, select both the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter and the **HANDOVER_TO_NR_SW** option of the **CellOpHoCfg.HoModeSwitch** parameter.
- To enable redirection, select both the **NrRedirectSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter and the **REDIRECTION_TO_NR_SW** option of the **CellOpHoCfg.HoModeSwitch** parameter.

Downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN supports three types of procedures, as described in [Table 5-11](#).

Table 5-11 Downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN procedures

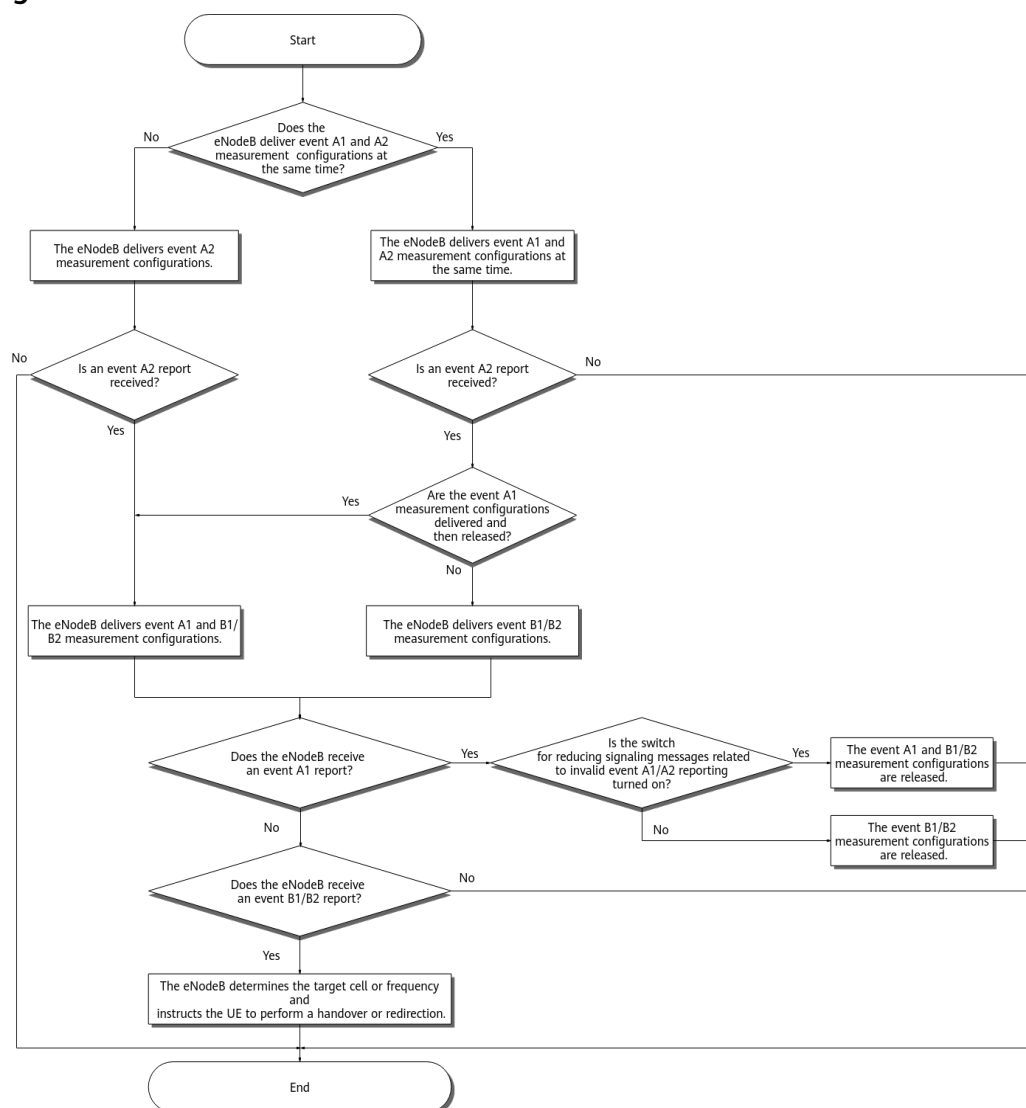
Procedure	Application Scenario	Chapter/Section
Data service handover or redirection	Default scenarios	Procedure with Data Service Handover or Redirection

Procedure	Application Scenario	Chapter/Section
VoLTE-to-VoNR handover	When a UE running a VoLTE service moves from an LTE cell to an NR cell, a VoLTE-to-VoNR handover can be performed if all of the following conditions are met: • At least one bearer with a QCI of 1 has been established for the UE. • The PS_HO option of the InterRatPolicyCfgGroup.NrHoCfg parameter associated with QCI 1 is selected. • The NrHoSwitch option of the CellHoParaCfg.HoModeSwitch parameter is selected for the LTE cell.	Procedure with VoLTE-to-VoNR Handover
NSA-DC-related measurements	In NSA and SA hybrid networking, NSA DC event B1 measurement configurations can be used to replace NR event B1 measurement configurations during handover evaluation for an NSA/SA dual-mode UE.	Procedure with NSA-DC-related Measurements

Procedure with Data Service Handover or Redirection

[Figure 5-6](#) shows the procedure for a coverage-based data service handover or redirection.

Figure 5-6 Data service handover or redirection



The procedure is as follows:

1. The eNodeB delivers the measurement configurations related to inter-RAT events A1 and A2 to the UE. The delivery sequence is controlled by the **ReduceInvalidA1A2RptSigSwitch** option of the **ENodeBAlgoSwitch.HoSignalingOptSwitch** parameter.
 - If the **ReduceInvalidA1A2RptSigSwitch** option of the **ENodeBAlgoSwitch.HoSignalingOptSwitch** parameter is selected, the eNodeB delivers the A2 measurement configurations first. If the eNodeB receives measurement reports on event A2, it delivers the measurement configurations related to events A1 and B1 or events A1 and B2, depending on the setting of the **InterRatHoComm.InterRatHoNrEventType** parameter. If the eNodeB does not receive measurement reports on event A2, the procedure ends.
 - If the **ReduceInvalidA1A2RptSigSwitch** option of the **ENodeBAlgoSwitch.HoSignalingOptSwitch** parameter is deselected, the eNodeB delivers the measurement configurations related to inter-RAT events A1 and A2 at the same time.

- If the eNodeB receives measurement reports on event A2 and the delivered A1 measurement configurations have not been released, the eNodeB delivers the measurement configurations related to event B1 or B2.
- If the eNodeB receives measurement reports on event A2 and the delivered A1 measurement configurations have been released, the eNodeB delivers the measurement configurations related to events A1 and B1 or events A1 and B2.
- If the eNodeB does not receive measurement reports on event A2, the procedure ends.

Event B1 involved in this function is coverage-based inter-RAT event B1.

Thresh for this event is specified by the

InterRatHoNrParamGrp.CovBasedNrB1RsrpThld parameter. For details about this event, see [4.1.4.2.1 Measurement Events](#).

2. The eNodeB waits for measurement reports on events A1 and B1/B2.
 - If the eNodeB receives measurement reports on event A1, it indicates that the signal quality of the serving cell exceeds a specific threshold. If the **ReduceInvalidA1A2RptSigSwitch** option of the **ENodeBAlgoSwitch.HoSignalingOptSwitch** parameter is selected, the eNodeB releases inter-RAT event A1 and B1/B2 measurements. If this option is deselected, the eNodeB releases inter-RAT event B1/B2 measurements. The procedure ends.
 - If the eNodeB receives measurement reports on event B1 or B2, it determines the target cell or frequency based on signal quality. The eNodeB instructs the UE to perform a handover to the target cell or perform a redirection to any cell on the target frequency.
 - If the eNodeB does not receive measurement reports on event A1, B1, or B2, the procedure ends.
3. The UE executes the handover or redirection.

During a handover, handover preparation precedes handover execution.

 - If the preparation for a handover to the target NR cell fails, the eNodeB imposes a penalty on the target NR cell. For details, see [Penalties Upon Failures in the Handover Preparation Phase](#).
 - If the handover execution fails, the procedure ends.

Procedure with VoLTE-to-VoNR Handover

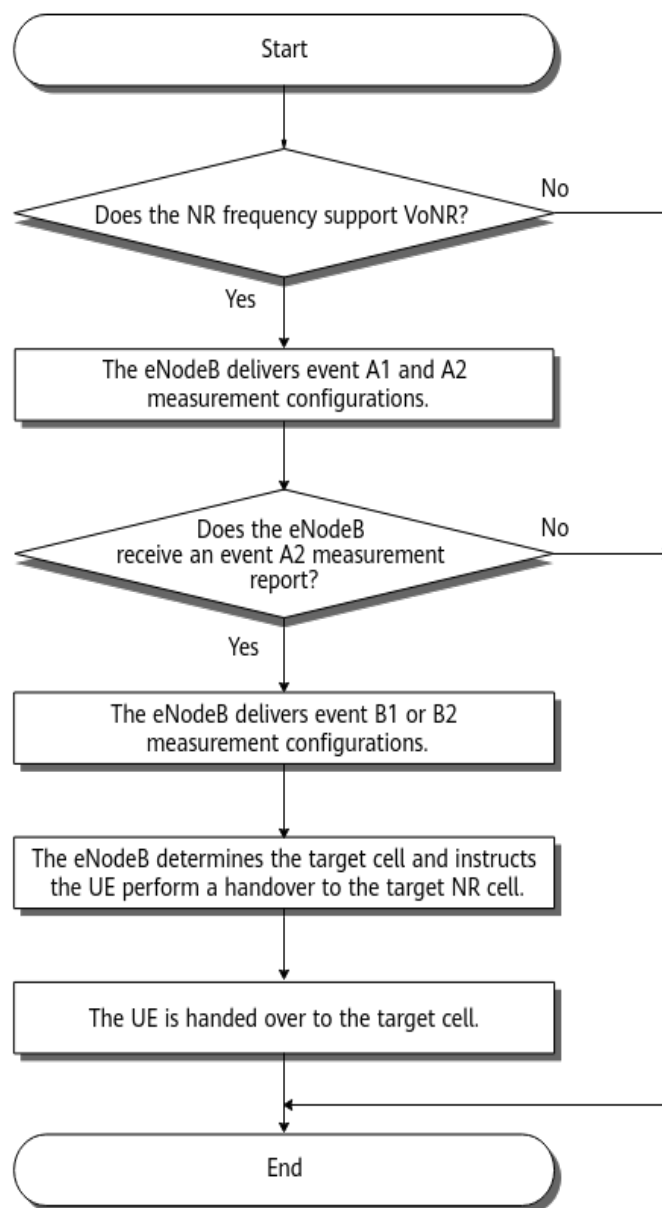
When a UE running a VoLTE service moves from an LTE cell to an NR cell, a VoLTE-to-VoNR handover can be performed if all of the following conditions are met:

- At least one bearer with a QCI of 1 has been established for the UE.
- The **PS_HO** option of the **InterRatPolicyCfgGroup.NrHoCfg** parameter associated with QCI 1 is selected.
- The **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter is selected for the LTE cell.

If the conditions are not met, the eNodeB performs a coverage-based data service handover or redirection.

Figure 5-7 shows the VoLTE-to-VoNR handover procedure.

Figure 5-7 VoLTE-to-VoNR handover



The procedure is as follows:

1. The eNodeB checks the **NrNFreq.VonrPriority** parameter settings to determine whether individual NR frequencies support VoNR.
 - If the **NrNFreq.VonrPriority** parameter is set to **255** for all NR frequencies, none of the NR frequencies supports VoNR. In this case, the eNodeB does not deliver the measurement configurations related to event A1 or A2 to the UE. The procedure ends.
 - If the **NrNFreq.VonrPriority** parameter is set to a value other than **255** for an NR frequency, the NR frequency supports VoNR. In this case, the eNodeB delivers A1 and A2 measurement configurations. The sequence in which the A1 measurement configurations and the A2 measurement

configurations are delivered is determined by the **ReduceInvalidA1A2RptSigSwitch** option of the **ENodeBAlgoSwitch.HoSignalingOptSwitch** parameter.

2. The eNodeB delivers the measurement configurations related to event B1 or B2 based on measurement reports on event A2.
 - If the eNodeB receives measurement reports on event A2, it delivers the measurement configurations related to event B1 or B2, with NR frequencies selected in descending order of priority specified by the **NrNFreq.VonrPriority** parameter.
 - If the number of VoNR-capable frequencies does not exceed the value of the **CellUeMeasControlCfg.MaxNrMeasFreqNum** parameter, the eNodeB selects all of these frequencies.
 - If the number of VoNR-capable frequencies exceeds the value of the **CellUeMeasControlCfg.MaxNrMeasFreqNum** parameter, the eNodeB preferentially selects the NR frequencies with the highest priorities specified by the **NrNFreq.VonrPriority** parameter. If multiple frequencies are assigned the same value of the **NrNFreq.VonrPriority** parameter, the eNodeB preferentially selects the NR frequencies with the highest priorities specified by the **NrNFreq.ConnFreqPriority** parameter. (A larger value indicates a higher priority. The value 0 indicates the lowest priority.) If multiple frequencies are assigned the same combination of the **NrNFreq.VonrPriority** and **NrNFreq.ConnFreqPriority** parameter values, the eNodeB randomly selects NR frequencies.
 - If the eNodeB does not receive measurement reports on event A2, it does not deliver the measurement configurations related to event B1 or B2.
3. If the eNodeB receives measurement reports on event B1 or B2, it determines the target cell based on cell signal quality and instructs the UE to perform a handover to the target NR cell.
4. The UE executes the handover.

Procedure with NSA-DC-related Measurements

With NSA DC-based fast NR handover in effect in NSA and SA hybrid networking, the eNodeB can use NSA DC event B1 measurement configurations instead of NR event B1 measurement configurations during handover evaluation for an NSA/SA dual-mode UE. Upon receiving a measurement report on NSA DC event B1 from a UE, the eNodeB performs the decision-making procedure, which varies depending on the status of the **NR_B1_B2_PLUS_RSRQ_INDICATION** option of the **NrNFreq.AggregationAttribute** parameter for the NR frequencies in the measurement report.

- If this option is deselected, the eNodeB checks whether the RSRP in the measurement report on event B1 meets the leaving condition for event B1. If the leaving condition is not met, the eNodeB instructs the UE to perform a handover or redirection to an NR cell. Otherwise, the procedure with NSA-DC-related measurements ends.
- If this option is selected, the eNodeB checks whether both the RSRP and RSRQ in the measurement report on event B1 meet the leaving condition for event B1. If neither of them meets the leaving condition, the eNodeB instructs the

UE to perform a handover or redirection to an NR cell. Otherwise, the procedure with NSA-DC-related measurements ends.

In the leaving condition for event B1, *Thresh* equals the RSRP threshold or RSRQ threshold for filtering for service-based inter-RAT event B1 plus **CellHoParaCfg.NrB1FastHoThldOffset**. For details about the leaving condition for event B1, see [4.1.4.2.1 Measurement Events](#).

NSA DC-based fast NR handover is controlled by the **NR_B1_FAST_HO_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter, and requires any of the following functions to be enabled beforehand:

- Coverage-based inter-RAT mobility from E-UTRAN to NG-RAN, which is controlled by the **INTER_RAT_MOBILITY_TO_NR_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter
- E-UTRAN to NG-RAN traffic steering, which is controlled by the **ServBasedNrHoSwitch** option of the **CellAlgoSwitch.HoAllowedSwitch** parameter
- Fast return from E-UTRAN to NG-RAN, which is controlled by the **FAST_RETURN_TO_NR_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter

5.5.2 Network Analysis

5.5.2.1 Benefits

With downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN, the coverage in areas with weak E-UTRAN coverage or coverage holes can be supplemented by inter-RAT neighboring cells, decreasing the service drop rate.

When the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter value is greater than the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, NSA-DC-based fast NR handover offers the following benefits:

- The number of SA UEs in the NR cell increases, which can be observed by subtracting the **N.User.NsaDc.PSCell.Avg** counter value from the **N.User.RRCCConn.Avg** counter value.
- The total throughput of SA UEs in the NR cell increases, which can be observed by subtracting the **N.NSA.ThpVol.DL** counter value from the **N.ThpVol.DL** counter value.

When the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter value is greater than the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, a greater difference between them leads to larger benefits. When the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter value is less than or equal to the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, NSA-DC-based fast NR handover does not produce any benefits.

5.5.2.2 Impacts

The impacts between this function and other functions are described in [5.5.3.2.3 Function Impacts](#).

The function has no impact on the network.

5.5.3 Requirements

5.5.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
LTE FDD	LNOFD-151336	Inter-RAT Mobility from E-UTRAN to NG-RAN	LT1SIRME NG00	Inter-RAT Mobility from E-UTRAN to NG-RAN (LTE FDD)	eNodeB	per Cell
LTE TDD	TDLNOFD-151501	Inter-RAT Mobility from E-UTRAN to NG-RAN	LT4SENG MBTDD	Inter-RAT Mobility from E-UTRAN to NG-RAN (LTE TDD)	eNodeB	per Cell

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

5.5.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.5.3.2.1 Prerequisite Functions

None

5.5.3.2.2 Mutually Exclusive Functions

None

5.5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA DC-based fast NR handover	NR_B1_FAST_HO_SW option of the CellAlgoExtSwitch . HoAllowedSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	If (1) NSA DC-based fast NR handover is enabled, (2) NSA DC-related event B1 is reported prior to event A2/B1/B2, and (3) the UE does not meet the leaving conditions of NSA DC-related event B1, the UE will be handed over to an NR cell. For details, see 5.5.1 Principles .

5.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all LTE-capable boards support this function.
- For NR, all NR-capable boards support this function.

RF Modules

No requirements

5.5.3.4 Others

- UE requirements
UEs must comply with 3GPP Release 15 or later. To support inter-RAT service mobility for data services, the corresponding fields in the following table must be carried simultaneously based on the target NR cell situation.

Frequency Band/RAT	TDD	FDD
FR1	• eutra-EPC-HO-ToNR-TDD-FR1-r15 • sa-NR-r15	• eutra-EPC-HO-ToNR-FDD-FR1-r15 • sa-NR-r15
FR2	• eutra-EPC-HO-ToNR-TDD-FR2-r15 • sa-NR-r15	• eutra-EPC-HO-ToNR-FDD-FR2-r15 • sa-NR-r15

To support inter-RAT service mobility for voice services, the corresponding fields in the following table must be carried simultaneously based on the target NR cell situation.

Frequency Band/RAT	TDD	FDD
FR1	• eutra-EPC-HO-ToNR-TDD-FR1-r15 • sa-NR-r15 • ims-VoiceOverNR-FR1-r15	• eutra-EPC-HO-ToNR-FDD-FR1-r15 • sa-NR-r15 • ims-VoiceOverNR-FR1-r15
FR2	• eutra-EPC-HO-ToNR-TDD-FR2-r15 • sa-NR-r15 • ims-VoiceOverNR-FR2-r15	• eutra-EPC-HO-ToNR-FDD-FR2-r15 • sa-NR-r15 • ims-VoiceOverNR-FR2-r15

- Core network requirements
 - The core networks must comply with 3GPP Release 15 or later.
 - Handover requires the N26 interface to be configured between the MME and AMF.

5.5.4 Operation and Maintenance

5.5.4.1 Data Configuration

5.5.4.1.1 Data Preparation

Table 5-12 describes the parameters related to downlink-coverage-based inter-RAT mobility from E-UTRAN to NG-RAN. For details about basic parameters for

interoperability from E-UTRAN to NG-RAN, see [4.5.1.1 Data Preparation](#) and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-12 Parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
Handover Allowed Switch	CellAlgoExtSwitch.HoAllowedSwitch	Select the INTER_RAT_MOBILITY_TO_NR_SW option.
Handover Mode switch	CellHoParaCfg.HoModeSwitch	- To enable handover, select the NrHoSwitch option. - To enable redirection, select the NrRedirectSwitch option.
Handover Algo Switch	CellHoParaCfg.CellHoAlgoSwitch	Select the NrB1TimeToTrigAdjSw option. In this case, the eNodeB adjusts the time-to-trigger for NSA DC event B1 to be the same as that for NR event B1 for NSA/SA dual-mode UEs.
NR Handover Policy Configuration	InterRatPolicyCfgGroup.NrHoCfg	In an inter-RAT policy group to be mapped to QCI 5 or 9, select both the PS_HO and REDIRECTION options. If a UE is running services with different QCIs, the mobility policy for the service with the highest-priority QCI takes effect for the UE.

5.5.4.1.2 Using MML Commands

The basic configurations for mobility from E-UTRAN to NG-RAN must have been completed before activating this function. For details, see [4.5.1.2 Using MML Commands](#) and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples

Before using MML commands, refer to [5.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Activating handover and redirection
MOD CELLHOPARACFG: LocalCellId=0,
```

```
HoModeSwitch=NrHoSwitch-1&NrRedirectSwitch-1,CellHoAlgoSwitch=NrB1TimeToTrigAdjSw-1;
//Turning on the switch of inter-RAT mobility from E-UTRAN to NG-RAN
MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=INTER_RAT_MOBILITY_TO_NR_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch of inter-RAT mobility from E-UTRAN to NG-RAN
MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=INTER_RAT_MOBILITY_TO_NR_SW-0;
```

5.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.5.4.2 Activation Verification

Use either of the methods described in this section to verify function activation.

Observing Performance Counter Values

If handover is set as the mobility policy, observe the counters listed in [Table 5-13](#). If any of those counters returns a non-zero value, the function has taken effect.

Table 5-13 Counters related to handovers from E-UTRAN to NG-RAN

Counter ID	Counter Name
1526758900	L.IRATHO.E2N.Coverage.PrepAttOut
1526784081	L.IRATHO.E2N.Coverage.ExecAttOut
1526784082	L.IRATHO.E2N.Coverage.ExecSuccOut

NOTE

L.IRATHO.E2N.Coverage.ExecAttOut and **L.IRATHO.E2N.Coverage.ExecSuccOut** are not applicable to the scenarios where the UMPTb series main control board works with an EOS baseband processing unit. For details about the EOS hardware list, see the "Last Base Station Version That Supports Board Software Mapping" column in *3900 & 5900 Series Base Station Software and Hardware Compatibility Description of 3900 & 5900 Series Base Station Product Documentation*.

If redirection is set as the mobility policy, observe the counters listed in [Table 5-14](#). If the result of (**L.RRCRedirection.E2N** – **L.RRCRedirection.E2N.Service** – **L.RRCRedirection.E2N.FastReturn**) is not 0, the function has taken effect.

Table 5-14 Counters related to redirections from E-UTRAN to NG-RAN

Counter ID	Counter Name
1526749533	L.RRCRedirection.E2N
1526756893	L.RRCRedirection.E2N.Service
1526756892	L.RRCRedirection.E2N.FastReturn

Tracing Signaling

To verify a handover from E-UTRAN to NG-RAN, perform the following steps:

- Step 1** Start Uu interface tracing and S1 interface tracing for the eNodeB.
- Step 2** Observe the MobilityFromEUTRACommand message sent by the eNodeB to the UE over the Uu interface. If the value of the *targetRAT-Type* IE in the message is "nr", the preparation for the handover from E-UTRAN to NG-RAN has succeeded.
- Step 3** Observe the signaling messages over the S1 interface. If either (i) the eNodeB receives a UE CONTEXT RELEASE COMMAND message with the cause value "Successful Handover" or "Normal Release" from the MME, or (ii) the eNodeB receives an RRC Connection Reestablishment Complete message from the UE and the RRC connection reestablishment is successful, the handover from E-UTRAN to NG-RAN has been successfully executed.

----End

To verify a redirection from E-UTRAN to NG-RAN, perform the following steps:

- Step 1** Start Uu interface tracing for the eNodeB.
- Step 2** Check whether the RRCConnectionRelease message sent by the eNodeB contains the *nr-r15* IE. If it does, the redirection from E-UTRAN to NG-RAN is successful.

----End

5.5.4.3 Network Monitoring

Data Service Mobility

To evaluate the performance of data service mobility from E-UTRAN to NG-RAN, monitor the counters listed in [Table 5-15](#) and [Table 5-16](#).

Table 5-15 LTE counters related to data service mobility

Counter ID	Counter Name
1526749533	L.RRCRedirection.E2N
1526758900	L.IRATHO.E2N.Coverage.PrepAttOut
1526756784	L.IRATHO.E2N.PrepAttOut

Counter ID	Counter Name
1526756777	L.IRATHO.E2N.ExecAttOut
1526756776	L.IRATHO.E2N.ExecSuccOut
1526756781	L.IRATHO.E2N.Prep.FailOut.MME
1526756782	L.IRATHO.E2N.Prep.FailOut.PrepFailure
1526756783	L.IRATHO.E2N.Prep.FailOut.NoReply
1526756779	L.IRATHO.E2N.MMEAbnormRsp
1526756785	L.IRATHO.NCell.E2N.PrepAttOut
1526756780	L.IRATHO.NCell.E2N.ExecAttOut
1526756778	L.IRATHO.NCell.E2N.ExecSuccOut
1526777661	L.RRCRedirection.E2N.RedCap
1526779679	L.IRATHO.E2N.PrepAttOut.RedCap
1526779677	L.IRATHO.E2N.ExecAttOut.RedCap
1526779678	L.IRATHO.E2N.ExecSuccOut.RedCap

Table 5-16 NR counters related to data service mobility

Counter ID	Counter Name
1911820677	N.HO.InterRAT.E2N.PrepAttIn
1911820678	N.HO.InterRAT.E2N.PrepSuccIn
1911820679	N.HO.InterRAT.E2N.ExecSuccIn
1911820683	N.HO.InterRAT.E2N.PrepInFail
1911820682	N.HO.InterRAT.E2N.PrepInFail.TgtNotAllow
1911820681	N.HO.InterRAT.E2N.PrepInFail.RscAdmitFail
1911820680	N.HO.InterRAT.E2N.PrepInFail.HOCancel
1911847073	N.HO.InterRAT.E2N.PrepInFail.RedCap

VoLTE-to-VoNR Handover

To evaluate the performance of VoLTE-to-VoNR handovers, monitor the counters listed in [Table 5-17](#).

Table 5-17 Counters related to VoLTE-to-VoNR handovers

Counter ID	Counter Name
1911822824	N.HO.InterRAT.E2N.PrepAttIn.Voice
1911822823	N.HO.InterRAT.E2N.PrepSucIn.Voice
1911822822	N.HO.InterRAT.E2N.ExecSucIn.Voice
1911822828	N.HO.InterRAT.E2N.PrepInFail.Voice
1911822827	N.HO.InterRAT.E2N.PrepInFail.TgtNotAllow.Voice
1911822826	N.HO.InterRAT.E2N.PrepInFail.AdmitFail.Voice
1911822825	N.HO.InterRAT.E2N.PrepInFail.HOCancel.Voice

5.6 E-UTRAN to NG-RAN Traffic Steering

5.6.1 Principles

E-UTRAN to NG-RAN traffic steering enables a UE to be transferred from an E-UTRAN cell to an NG-RAN cell through a handover or redirection based on the UE service status.

The eNodeB can initiate E-UTRAN to NG-RAN traffic steering for a UE only if the eNodeB determines that the UE supports NR and any of the following occurs:

- The UE has accessed an LTE cell and had its services, including a VoLTE service, set up.
- The UE with services set up in an LTE cell has experienced a service addition or modification.
- The UE has been handed over to an LTE cell from a GSM cell, a UMTS cell, an NR cell, or another LTE cell and had its services set up. If the source cell of that handover is an NR cell and the service-based NR measurement timer (specified by the **CellHoParaCfg.ServBasedNrMeasTimer** parameter) is set to a value other than 0, the E-UTRAN to NG-RAN traffic steering procedure is started after the timer expires.
- The UE has been handed over from an NR cell to an LTE cell for resource optimization reasons, such as user-experience-based switching from SA to NSA. If the protection timer for incoming handovers from NR cells due to resource optimization (specified by the **CellHoParaCfg.NrResOptHolnProtectTmr** parameter) is set to a value other than 0, the E-UTRAN to NG-RAN traffic steering procedure is started after the timer expires.

E-UTRAN to NG-RAN traffic steering (traffic steering for short) is controlled by the **ServBasedNrHoSwitch** option of the **CellAlgoSwitch.HoAllowedSwitch** parameter. Handover and redirection are supported.

- To enable handover, select the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter.

- To enable redirection, select the **NrRedirectSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter.
- If the above options are both selected, whether handover can be performed for the UE is preferentially evaluated. If handover fails, the procedure ends.

In RAN sharing with common carrier scenarios, the mobility policy (handover or redirection) for a cell can be set on a per operator basis using the **CellOpHoCfg** MO. If no operator-specific mobility policy is set, the mobility policy for the cell is determined by the **CellHoParaCfg.HoModeSwitch** parameter.

- To enable handover, select both the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter and the **HANDOVER_TO_NR_SW** option of the **CellOpHoCfg.HoModeSwitch** parameter.
- To enable redirection, select both the **NrRedirectSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter and the **REDIRECTION_TO_NR_SW** option of the **CellOpHoCfg.HoModeSwitch** parameter.

E-UTRAN to NG-RAN traffic steering supports four kinds of procedures, as listed in [Table 5-18](#).

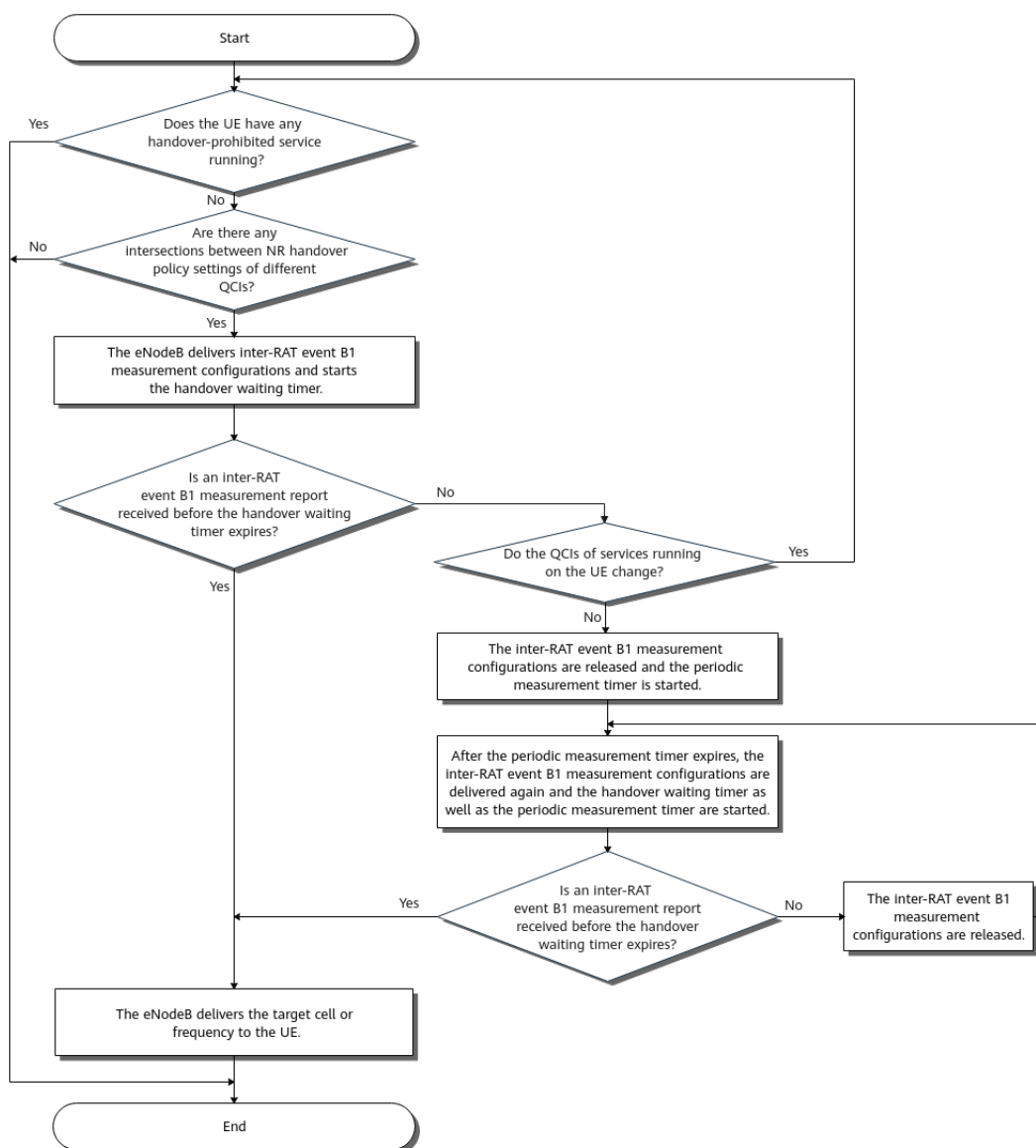
Table 5-18 Traffic steering procedures

Procedure	Application Scenario	Chapter/Section
NR B1 measurements	Default scenarios	Procedure with NR B1 Measurements
NSA DC-based fast NR handover	In NSA and SA hybrid networking, NSA DC event B1 measurement configurations can be used to replace NR event B1 measurement configurations during handover evaluation for an NSA/SA dual-mode UE.	Procedure with NSA DC-based Fast NR Handover
Non-gap-assisted measurements	Scenarios where the NO_GAP_NR_B1_SW option of the CellAlgoExtSwitch.HoAllowedSwitch parameter is selected	Procedure with Non-Gap-assisted Measurements
RSRP prediction based on virtual grids	Scenarios where the NR_HO_WITH_VIRTUAL_GRID_SW option of the LnrCarrierSelection.LnrCarrierSelectionSw parameter is selected	Procedure with RSRP Prediction Based on Virtual Grids

Procedure with NR B1 Measurements

Figure 5-8 shows the procedure for E-UTRAN to NG-RAN traffic steering. By default, the target cell or frequency of a handover or redirection for E-UTRAN to NG-RAN traffic steering is determined based on the obtained reports of gap-assisted NR frequency measurements performed by LTE/NR-dual-mode-capable UEs.

Figure 5-8 Procedure for E-UTRAN to NG-RAN traffic steering



The procedure is as follows:

1. The eNodeB checks whether a UE has handover-prohibited services running. The UE can perform services with different QCLs at the same time, and different handover states can be configured (using the **ServiceIrHoCfGroup.InterRatHoState** parameter) for such services. The UE can be transferred to an NG-RAN cell through a handover or redirection when either of the following conditions is met:

- When the UE is running a service with a QCI for which the handover state is set to **MUST_HO**, the procedure goes to the next step if all of the following conditions are met. Otherwise, the procedure ends.
 - NSA/SA selection based on user experience is enabled as any of the following options is selected:
 - **LTE_FDD_NSA_SA_DL_SEL_OPT_SW** option of the **ENodeBAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter
 - **LTE_TDD_NSA_SA_DL_SEL_OPT_SW** option of the **ENodeBAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter
 - **LTE_FDD_NSA_SA_UL_SEL_OPT_SW** option of the **ENodeBAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter
 - The UE is an NSA UE, that is, the secondary cell group (SCG) has been configured for the UE. Alternatively, NSA carrier combination selection based on user experience is enabled (the **NSA_LTE_FDD_SEL_OPT_SW** and **NSA_LTE_TDD_SEL_OPT_SW** options of the **ENodeBAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter are selected, the license requirements are met, and NSA-related parameters have been configured. For details, see *EPC-based NSA Basics* and *EPC-based NSA Performance Enhancement*.)
 - (Optional) Flexible user steering is enabled, the UE is configured with an SPID, and the **NSA_SA_SEL_OPT_SW** option of the corresponding **SpidCfg.SpidAlgoSwitch** parameter is selected.
- The UE is not running any service with a QCI for which the handover state is set to **NO_HO**.

When the **NrRedirectSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter is selected, reselection to NR is not allowed as long as the UE is running a QCI 1 service.

NOTE

The eNodeB can associate the **CnOperatorQciPara** MO with the **ServiceIrrHoCfgGroup** MO by setting the **CnOperatorQciPara.ServiceHoNrPolicyGroupId** parameter to the parameter group ID of the associated **ServiceIrrHoCfgGroup** MO.

2. The eNodeB checks whether the NR handover policy settings of various QCIs for the UE overlap with each other.

Multiple handover policies can be set for different QCIs. The overall mobility policy set for the UE must be included in policy settings of all QCIs and be consistent with the setting of the **CellHoParaCfg.HoModeSwitch** parameter.

- If the handover policy settings of various QCIs for the UE overlap with each other, the handover policy is determined based on the **InterRatPolicyCfgGroup.NrHoCfg** parameter setting. For example, assume that handover to NR is allowed as the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter is selected. If the UE is running services with QCIs 9 and 5, and the **PS_HO** option of the

InterRatPolicyCfgGroup.NrHoCfg parameter for QCI 9 and that for QCI 5 are both selected, handover is as the overall mobility policy.

- If the NR handover policy settings of various QCIs for the UE do not overlap with each other, the procedure ends. For example, assume that the UE is running services with QCIs 9 and 5. If the **REDIRECTION** and **PS_HO** options of the **InterRatPolicyCfgGroup.NrHoCfg** parameter are selected for QCI 9 and for QCI 5, respectively, there is no overlapped policy settings. E-UTRAN to NG-RAN traffic steering stops.

The relationships between QCIs and inter-RAT policy groups are defined in the **QciPara** MO.

3. The eNodeB delivers the measurement configurations related to inter-RAT event B1 to the UE and starts a handover waiting timer. In the measurement configurations, NR frequencies are arranged in descending order of priority. The handover waiting timer is specified by the **CellHoParaCfg.ServBasedNrB1RptWaitingTmr** parameter.
 - If the UE is not running a QCI 1 service, the NR frequency priorities are determined by the **NrNFreq.ConnFreqPriority** parameter. A larger value indicates a higher priority. The value **0** indicates the lowest priority.
 - If the UE is running a QCI 1 service, the NR frequency priorities are determined by the **NrNFreq.VonrPriority** parameter. If multiple frequencies are assigned the same value of the **NrNFreq.VonrPriority** parameter, NR frequency priorities are determined by the **NrNFreq.ConnFreqPriority** parameter. If multiple NR frequencies are assigned the same value of the **NrNFreq.ConnFreqPriority** parameter, frequencies to deliver are randomly selected.

Event B1 involved in this function is service-based inter-RAT event B1. *Thresh* for this event is specified by the

InterRatHoNrParamGrp.ServBasedNrB1RsrpThld parameter. For a UE performing services with multiple QCIs, *Thresh* takes the maximum value of the thresholds for all these QCIs. For details, see [4.1.4.2.3 QCI-specific Measurement Parameter Configurations](#).

NOTE

Measurements related to event B1 involved in service-based mobility are not terminated by event A1 involved in coverage-based inter-RAT mobility from E-UTRAN to NG-RAN.

When the **SERV_BASED_FR_FORBID_SEL_FLAG** option of the **NrNFreq.AggregationAttribute** parameter is selected for an NR frequency, this NR frequency cannot be selected as a target frequency during service-based measurement, handovers, and redirections from E-UTRAN to NG-RAN.

4. The eNodeB takes different actions, depending on whether it has received inter-RAT event B1 measurement reports before the handover waiting timer expires.
 - If the eNodeB has received inter-RAT event B1 measurement reports on before the handover waiting timer expires, it delivers the target cell or frequency to the UE, instructing the UE to perform a handover or redirection according to the mobility policy.
 - If the eNodeB has not received inter-RAT event B1 measurement reports and the QCIs of services running on the UE have not changed before the handover waiting timer expires:

- i. The eNodeB releases the B1 measurements and starts the measurement timer related to service-based handovers from E-UTRAN to NG-RAN. This timer is referred to as periodic measurement timer for short. It is specified by the **CellHoParaCfg.ServBasedNrMeasTimer** parameter.
- ii. After the periodic measurement timer expires, the eNodeB delivers the measurement configurations related to inter-RAT event B1 again and starts the handover waiting timer and periodic measurement timer.
- iii. The eNodeB takes different actions, depending on whether it has received inter-RAT event B1 measurement reports before the handover waiting timer expires.
 - If the eNodeB has received inter-RAT event B1 measurement reports, it delivers the target cell or frequency to the UE, instructing the UE to perform a handover or redirection according to the mobility policy.
 - If the eNodeB has not received inter-RAT event B1 measurement reports, it releases the B1 measurement configurations and goes back to 4.ii.
- If the eNodeB does not receive inter-RAT event B1 measurement reports and the QCI of services on the UE change before the handover waiting timer expires, the eNodeB checks the service status of different QCIs on the UE again. For details, see 1.
5. The UE performs a handover to the target cell or a redirection to the operating frequency of the target cell.

Procedure with NSA DC-based Fast NR Handover

With NSA DC-based fast NR handover in effect in NSA and SA hybrid networking, the eNodeB can use NSA DC event B1 measurement configurations instead of NR event B1 measurement configurations during handover evaluation for an NSA/SA dual-mode UE. NSA DC-based fast NR handover is controlled by the **NR_B1_FAST_HO_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter. For details about NSA DC-based fast NR handover, see [Procedure with NSA-DC-related Measurements](#).

Procedure with Non-Gap-assisted Measurements

The eNodeB allows non-gap-assisted B1 measurements on NR frequencies that support the non-gap-assisted measurement function, which is controlled by the **NO_GAP_NR_B1_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter.

With the non-gap-assisted B1 measurement function enabled, E-UTRAN to NG-RAN traffic steering works as follows if the eNodeB has not received inter-RAT event B1 measurement reports and the QCIs of services running on the UE have not changed until the handover waiting timer (specified by the **CellHoParaCfg.ServBasedNrB1RptWaitingTmr** parameter) expires as in 4 of [Procedure with NR B1 Measurements](#):

1. The eNodeB releases the B1 measurement configurations and starts the measurement timer related to service-based handovers from E-UTRAN to NG-

RAN. This timer is referred to as periodic measurement timer for short. It is specified by the **CellHoParaCfg.ServBasedNrMeasTimer** parameter.

2. The eNodeB checks whether non-gap-assisted measurement is applicable to any of the configured NR frequencies. It is applicable if the UE complies with Release 16 or an NR frequency meets the conditions described in **4.1.5.2 Non-Gap-assisted B1 Measurement Configuration by the eNodeB**.
 - If non-gap-assisted measurement is applicable to certain NR frequencies, the eNodeB delivers B1 measurement configurations about these frequencies to the UE. These measurement configurations are the same as the measurement configurations related to service-based inter-RAT event B1. The eNodeB releases the delivered non-gap-assisted measurement configurations when any of the following situations occurs:
 - EN-DC band combinations or CA carriers are added or removed. In this situation, the eNodeB needs to query the non-gap-assisted measurement capabilities of NR frequencies again. If a frequency is changed from being capable of non-gap-assisted measurement to being incapable, the eNodeB releases the non-gap-assisted B1 measurement configurations about this frequency. Otherwise, the eNodeB does not release non-gap-assisted B1 measurement configurations.
 - E-UTRAN to NG-RAN traffic steering or fast return is triggered by a UE service change.
 - The eNodeB delivers the measurement configurations related to service-based inter-RAT event B1 as triggered by timeout of the periodic measurement timer.
 - If there are no frequencies that support non-gap-assisted measurement, the eNodeB does not deliver non-gap-assisted measurement configurations to the UE. The procedure with non-gap-assisted measurements ends.
3. The eNodeB takes different actions, depending on whether it has received inter-RAT event B1 measurement reports before releasing non-gap-assisted B1 measurement configurations.
 - If the eNodeB has received inter-RAT event B1 measurement reports, it delivers the target cell or frequency to the UE, instructing the UE to perform a handover or redirection according to the mobility policy. The procedure ends.
 - If the eNodeB has not received inter-RAT event B1 measurement reports, it goes to the next step.
4. After the periodic measurement timer expires, the eNodeB delivers the measurement configurations related to inter-RAT event B1 again and starts the handover waiting timer and periodic measurement timer.
5. The eNodeB takes different actions, depending on whether it has received inter-RAT event B1 measurement reports before the handover waiting timer expires.
 - If the eNodeB has received inter-RAT event B1 measurement reports, it delivers the target cell or frequency to the UE, instructing the UE to perform a handover or redirection according to the mobility policy.

- If the eNodeB has not received inter-RAT event B1 measurement reports, it releases the B1 measurement configurations and goes back to 2.

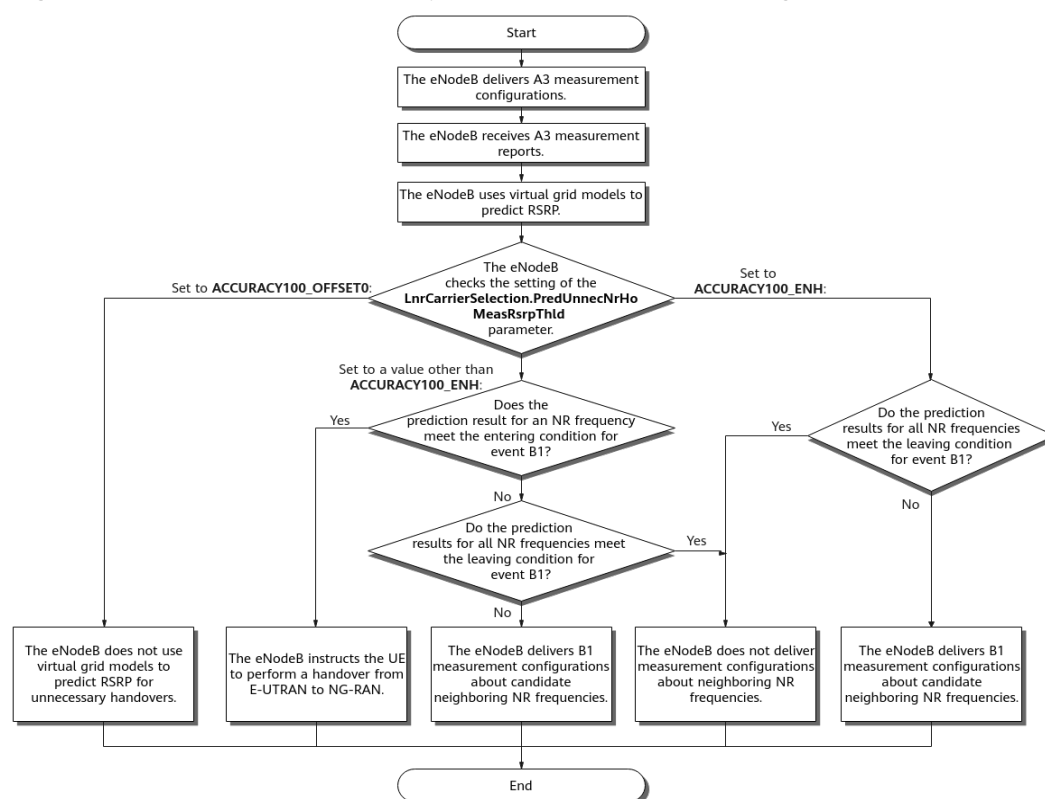
Procedure with RSRP Prediction Based on Virtual Grids

During evaluation of E-UTRAN to NG-RAN traffic steering, use of LN RSRP models can replace UE measurement of RSRP if the **NR_HO_WITH_VIRTUAL_GRID_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected.

This function takes effect only in handover scenarios. It requires the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter to be selected. In RAN sharing with common carrier scenarios, it requires both the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter and the **HANDOVER_TO_NR_SW** option of the **CellOpHoCfg.HoModeSwitch** parameter to be selected.

The procedure for traffic steering with RSRP predicted based on virtual grids is the same as the procedure for traffic steering based on UE measurements, except the following steps (which replace steps 3 and 4 in the latter procedure described in [Procedure with NR B1 Measurements](#)):

Figure 5-9 Procedure with RSRP prediction based on virtual grids



The procedure is as follows:

1. The eNodeB delivers measurement configurations related to event A3.
 - For a UE capable of NR SA (field "sa-NR-r15" reported) access (including initial access in idle mode, incoming handovers, and incoming RRC connection reestablishments): the eNodeB delivers A3 measurement configurations about the serving frequency with a measurement period of

2s to the UE. To prevent ping-pong handovers for the UE if it has been handed over from NG-RAN to E-UTRAN, the eNodeB delivers event A3 measurement configurations only after the timer specified by **CellHoParaCfg.ServBasedNrMeasTimer** expires.

- For other types of UE: the eNodeB directly delivers the A3 measurement configurations.
2. The eNodeB obtains event A3 measurement reports.
- The eNodeB performs this step if the **PS_HO** option of the **InterRatPolicyCfgGroup.NrHoCfg** parameter is selected for all QCI's corresponding to services running on the UE.
 - Otherwise, the eNodeB cannot use virtual grid models to predict the RSRP values.

After the UE sends A3 measurement reports to the eNodeB, the eNodeB uses it (that is, uses the reported RSRP values of the serving cell and intra-frequency neighboring cells) to query coverage information about candidate NR frequencies. The *TimeToTrig* for event A3 is specified by the **LnrCarrierSelection.IntraFreqMeasA3TimeToTrig** parameter.

3. The eNodeB uses virtual grid models to predict RSRP, and determines whether to use the predicted RSRP values to perform service steering based on the setting of the **LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld** parameter.

When LN RSRP models are available on the eNodeB and the model accuracy meets the requirement, the eNodeB uses the models to predict the RSRP for the corresponding neighboring NR frequencies, without the need for UE measurements.

The accuracy requirement is specified by the **LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld** parameter.

- If the **LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld** parameter is set to **ACCURACY100_OFFSET0**, RSRP is not predicted using virtual grid models for unnecessary handovers and the procedure ends.
- If **LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld** is set to a value other than **ACCURACY100_ENH**:
 - During E-UTRAN to NG-RAN traffic steering triggered in one of the following manners, the eNodeB instructs the UE to perform a handover from E-UTRAN to NG-RAN if the prediction result for a neighboring NR frequency meets the entering condition for the corresponding event B1:
 - Event-triggered: A service with a new QCI is to be set up.
 - Timer-triggered: A 2s periodic timer has expired, or the periodic measurement timer (specified by the **CellHoParaCfg.ServBasedNrMeasTimer** parameter) has expired.
 - During E-UTRAN to NG-RAN traffic steering triggered by new-QCI service setup or the periodic measurement timer, if the prediction results for all candidate neighboring NR frequencies meet the leaving condition for the corresponding event B1, the eNodeB does not deliver the measurement configurations about these neighboring NR frequencies. The procedure ends.

- If the preceding conditions are not met, the procedure with NR B1 measurements goes on. After receiving B1 measurement reports from the UE, the eNodeB instructs the UE to perform a handover from E-UTRAN to NG-RAN. For details, see [Procedure with NR B1 Measurements](#).
- If **LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld** is set to **ACCURACY100_ENH**:
 - During E-UTRAN to NG-RAN traffic steering triggered by new-QCI service setup or the periodic measurement timer, if the prediction results for all candidate neighboring NR frequencies meet the leaving condition for the corresponding event B1, the eNodeB does not deliver the B1 measurement configurations about these neighboring NR frequencies.
 - During E-UTRAN to NG-RAN traffic steering triggered by new-QCI service setup or the periodic measurement timer, if the prediction result for any candidate neighboring NR frequency does not meet the leaving condition for the corresponding event B1, the eNodeB delivers the B1 measurement configurations about this frequency. And the procedure with NR B1 measurements goes on. For details, see [Procedure with NR B1 Measurements](#). After the UE sends B1 measurement reports to the eNodeB, the eNodeB instructs the UE to perform a handover from E-UTRAN to NG-RAN.

5.6.2 Network Analysis

5.6.2.1 Benefits

This function provides a means to transfer UEs based on services and allows for flexible networking to meet the traffic steering requirements of operators.

When the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter value is greater than the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, NSA-DC-based fast NR handover offers the following benefits:

- The number of SA UEs in the NR cell increases, which can be observed by subtracting the **N.User.NsaDc.PSCell.Avg** counter value from the **N.User.RRCCConn.Avg** counter value.
- The total throughput of SA UEs in the NR cell increases, which can be observed by subtracting the **N.NSA.ThpVol.DL** counter value from the **N.ThpVol.DL** counter value.

When the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter value is greater than the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, a greater difference between them leads to larger benefits. When the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter value is less than or equal to the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, NSA-DC-based fast NR handover does not produce any benefits.

5.6.2.2 Impacts

The impacts between this function and other functions are described in [5.6.3.2.3 Function Impacts](#).

The function has no impact on the network.

5.6.3 Requirements

5.6.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
LTE FDD	LNOFD-151337	E-UTRAN to NG-RAN Traffic Steering	LT1S0ENGTS00	E-UTRAN to NG-RAN Traffic Steering (LTE FDD)	eNodeB	per Cell
LTE TDD	TDLNOFD-151502	E-UTRAN to NG-RAN Traffic Steering	LT4SENGTSTDD	E-UTRAN to NG-RAN Traffic Steering (LTE TDD)	eNodeB	per Cell

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

5.6.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.6.3.2.1 Prerequisite Functions

None

5.6.3.2.2 Mutually Exclusive Functions

None

5.6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA/SA selection based on downlink traffic volume	LTE_FDD_NSA_SA_DL_SEL_OPT_SW or LTE_TDD_NSA_SA_DL_SEL_OPT_SW option of the EnodeBAlgorithmExtSwitch.MultiNetworkingOptionOptSw parameter	NSA-SA Selection Based on User Experience	After NSA/SA selection based on user experience is enabled, the evaluation of E-UTRAN to NG-RAN traffic steering is changed. (For details, see 5.6.1 Principles.) As a result, the probability that E-UTRAN to NG-RAN traffic steering takes effect decreases, and the value of the L.IRATHO.E2N.ServicePrepAttOut counter decreases.
LTE FDD	NSA/SA selection based on uplink coverage	LTE_FDD_NSA_SA_UL_SEL_OPT_SW option of the EnodeBAlgorithmExtSwitch.MultiNetworkingOptionOptSw parameter	NSA-SA Selection Based on User Experience	

5.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all LTE-capable boards support this function.
- For NR, all NR-capable boards support this function.

RF Modules

No requirements

5.6.3.4 Others

- UE requirements

UEs must comply with 3GPP Release 15 or later. To support inter-RAT traffic steering for data services, the corresponding fields in the following table must be carried simultaneously based on the target NR cell situation.

Frequency Band/RAT	TDD	FDD
FR1	• eutra-EPC-HO-ToNR-TDD-FR1-r15 • sa-NR-r15	• eutra-EPC-HO-ToNR-FDD-FR1-r15 • sa-NR-r15
FR2	• eutra-EPC-HO-ToNR-TDD-FR2-r15 • sa-NR-r15	• eutra-EPC-HO-ToNR-FDD-FR2-r15 • sa-NR-r15

To support inter-RAT traffic steering for voice services, the corresponding fields in the following table must be carried simultaneously based on the target NR cell situation.

Frequency Band/RAT	TDD	FDD
FR1	• eutra-EPC-HO-ToNR-TDD-FR1-r15 • sa-NR-r15 • ims-VoiceOverNR-FR1-r15	• eutra-EPC-HO-ToNR-FDD-FR1-r15 • sa-NR-r15 • ims-VoiceOverNR-FR1-r15
FR2	• eutra-EPC-HO-ToNR-TDD-FR2-r15 • sa-NR-r15 • ims-VoiceOverNR-FR2-r15	• eutra-EPC-HO-ToNR-FDD-FR2-r15 • sa-NR-r15 • ims-VoiceOverNR-FR2-r15

- Core network requirements
 - The core networks must comply with 3GPP Release 15 or later.
 - Handover requires the N26 interface to be configured between the MME and AMF.

5.6.4 Operation and Maintenance

5.6.4.1 Data Configuration

5.6.4.1.1 Data Preparation

Table 5-19 describes the parameters related to E-UTRAN to NG-RAN traffic steering. For details about basic parameters for interoperability from E-UTRAN to NG-RAN, see **4.5.1.1 Data Preparation** and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-19 Parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
Handover Mode switch	CellHoParaCfg.HoModeSwitch	• To enable handover, select the NrHoSwitch option. • To enable redirection, select the NrRedirectSwitch option.
Handover Allowed Switch	CellAlgoSwitch.HoAllowedSwitch	Select the ServBasedNrHoSwitch option.
NR Handover Parameter Group ID	CellQciPara.NrHoParamGroupId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Inter-RAT handover state	ServiceIrHoCfgGroup.InterRatHoState	It is recommended that this parameter be set to NO_HO for QCI 1 services to prevent voice service interruption or user experience deterioration in the following scenarios: • Service-based handovers from LTE cells to NR cells for voice service UEs. In this scenario, voice services may be interrupted due to handover failures. • Service-based redirections from LTE cells to NR cells for voice service UEs. In this scenario, voice service experience may deteriorate due to a long time of user-plane interruption. Other QCIs can be set according to operators' mobility policies. If a UE is running services with different QCIs, the mobility policy for the service with the highest-priority QCI takes effect for the UE.
No Handover Flag	InterRatPolicyCfgGroup.NoHoFlag	The value PERMIT_HO_ENUM is recommended.
NR Handover Policy Configuration	InterRatPolicyCfgGroup.NrHoCfg	• To enable handover, select the PS_HO option. • To enable redirection, select the REDIRECTION option.
Service-based Inter-RAT HO Measure Mode	ServiceIrHoCfgGroup.ServiceIrMeasureMode	Set this parameter based on the network plan.
Service-based NR B1 Report Waiting Timer	CellHoParaCfg.ServBasedNrB1RptWaitingTmr	Set this parameter based on the network plan. If CellHoParaCfg.ServBasedNrMeasureTimer is set to a value other than 0, the value of CellHoParaCfg.ServBasedNrB1RptWaitingTmr must be less than that of CellHoParaCfg.ServBasedNrMeasureTimer .
Service-based NR Measurement Timer	CellHoParaCfg.ServBasedNrMeasureTimer	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Handover Allowed Switch	CellAlgoExtSwitch.HoAllowedSwitch	- To enable mobility based on NSA-DC-related measurement reports, select the NR_B1_FAST_HO_SW option. - To enable mobility based on non-gap-assisted measurement reports, select the NO_GAP_NR_B1_SW option.
NR B1 Fast Handover Threshold Offset	CellHoParaCfg.NrB1FastHoThldOffset	The default value is recommended.
LNR Carrier Selection Switch	LnrCarrierSelection.LnrCarrierSelectionSw	To use the virtual grid technology for E-UTRAN to NG-RAN traffic steering, select the NR_HO_WITH_VIRTUAL_GRID_SW option.
Intra-Freq Measurement A3 Time-to-Trigger	LnrCarrierSelection.IntraFreqMeasA3TimeToTrig	The default value is recommended.
Pred Unnecessary NR Ho Meas RSRP Thld	LnrCarrierSelection.PredUnnecessaryHoMeasRsrpThld	Set this parameter to a value other than ACCURACY100_OFFSET0 .
NR Res Opt Handover-in Protect Timer	CellHoParaCfg.NrResOptHoInProtectTmr	This parameter specifies the length of the protection timer for UEs that have been handed over from NR cells to the LTE cell for resource optimization reasons. E-UTRAN to NG-RAN traffic steering does not take place for these UEs before the timer expires. If this parameter is set to 0 or a value greater than 100, the timer does not take effect.
Aggregation Attribute	NrNFreq.AggregationAttribute	If an NR frequency needs to be filtered out from the target frequencies for E-UTRAN to NG-RAN traffic steering, select the SERV_BASED_FR_FORBID_SEL_FLAG option of this parameter for this frequency.

NOTE

For details about QCI parameter settings, see *QoS Management in eRAN Feature Documentation*.

5.6.4.1.2 Using MML Commands

The basic configurations for mobility from E-UTRAN to NG-RAN must have been completed before activating this function. For details, see [4.5.1.2 Using MML Commands](#) and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples

Before using MML commands, refer to [5.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Enabling handover and redirection
MOD CELLHOPARACFG: LocalCellId=0, HoModeSwitch=NrHoSwitch-1&NrRedirectSwitch-1;
//Turning on the switch for service-based handovers from E-UTRAN to NG-RAN
MOD CELLALGOSWITCH: LocalCellId=0, HoAllowedSwitch=ServBasedNrHoSwitch-1;
//((Optional) Turning on NR_B1_FAST_HO_SW and NO_GAP_NR_B1_SW to replace NR event B1
measurement with NSA DC event B1 measurement for handover decisions and use non-gap-assisted
measurement reports
MOD CELLALGOEXTSWITCH: LocalCellId=0,
HoAllowedSwitch=NR_B1_FAST_HO_SW-1&NO_GAP_NR_B1_SW-1;
//((Optional) Running the following command to use RSRP prediction based on virtual grids
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=NR_HO_WITH_VIRTUAL_GRID_SW-1,
IntraFreqMeasA3TimeToTrig=320MS, PredUnnecNrHoMeasRsrpThld=ACCURACY90_OFFSET0;
//Setting QCI-specific policies
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=9, ServiceHoNrPolicyGroupId=2;
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=5, ServiceHoNrPolicyGroupId=1;
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=1, ServiceHoNrPolicyGroupId=0;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=0, InterRatHoState=NO_HO;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=1, InterRatHoState=PERMIT_HO;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=2, InterRatHoState=MUST_HO;
MOD CELLQCIPARA: LocalCellId=0, Qci=9, NrHoParamGroupId=1;
//Setting an inter-RAT policy group
MOD INTERRATPOLICYCFGGROUP: InterRatPolicyCfgGroupId=0, NoHoFlag=PERMIT_HO_ENUM,
NrHoCfg=PS_HO-1;
//Setting the service-based inter-RAT handover measurement mode (SERVICE_IR_MEAS_NO_ORDER is used
as an example)
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=0,
ServiceIrMeasMode=SERVICE_IR_MEAS_NO_ORDER;
//Binding the group to a QCI
MOD QCIPARA: Qci=1, InterRatPolicyCfgGroupId=0;
//Setting timers
MOD CELLHOPARACFG: LocalCellId=0, ServBasedNrB1RptWaitingTmr=3, ServBasedNrMeasTimer=60,
NrB1FastHoThldOffset=0;
MOD CELLHOPARACFG: LocalCellId=0, NrResOptHoInProtectTmr=10;
//((Optional, when an NR frequency needs to be filtered out for E-UTRAN to NG-RAN traffic steering)
Prohibiting a frequency from being selected for fast return and service-based mobility
MOD NRNFREQ: LocalCellId=0, DIArfcn=428000,
AggregationAttribute=SERV_BASED_FR_FORBID_SEL_FLAG-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for service-based handovers from E-UTRAN to NG-RAN
MOD CELLALGOSWITCH: LocalCellId=0, HoAllowedSwitch=ServBasedNrHoSwitch-0;
//(Optional) Disabling RSRP prediction based on virtual grids
MOD LNRARRIERSELECTION: LnrCarrierSelectionSw=NR_HO_WITH_VIRTUAL_GRID_SW-0;
```

5.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.6.4.2 Activation Verification

To verify handover-based E-UTRAN to NG-RAN traffic steering, perform the following steps:

- Step 1** Start Uu interface tracing and S1 interface tracing for the eNodeB.
- Step 2** Observe the MobilityFromEUTRACommand message sent by the eNodeB to the UE over the Uu interface. If the value of the *targetRAT-Type* IE in the message is "nr", the preparation for the handover from E-UTRAN to NG-RAN has succeeded.
- Step 3** Observe the signaling messages over the S1 interface. If either (i) the eNodeB receives a UE CONTEXT RELEASE COMMAND message with the cause value "Successful Handover" or "Normal Release" from the MME, or (ii) the eNodeB receives an RRC Connection Reestablishment Complete message from the UE and the RRC connection reestablishment is successful, the handover from E-UTRAN to NG-RAN has been successfully executed.

----End

To verify redirection-based E-UTRAN to NG-RAN traffic steering, perform the following steps:

- Step 1** Start Uu interface tracing for the eNodeB.
- Step 2** Check whether the RRCConnectionRelease message sent by the eNodeB contains the *nr-r15* IE. If it does, the redirection from E-UTRAN to NG-RAN is successful.

----End

To verify use of virtual grid models for E-UTRAN to NG-RAN traffic steering, observe the **L.VirtualGrid.Model.NrFreq.InvalidMeasFree.Times** counter. This counter indicates the total number of times the eNodeB did not deliver measurement configurations about neighboring NR frequencies when the virtual-grid-based prediction results for the neighboring NR frequencies meet the leaving condition for event B1. If there are multiple neighboring NR frequencies, the value of this counter is incremented by the number of neighboring NR frequencies. If this counter produces a non-zero value, RSRP prediction using virtual grid models has taken effect.

5.6.4.3 Network Monitoring

Monitor the counters listed in [Table 5-20](#) and [Table 5-21](#) to evaluate the performance of this function.

Table 5-20 LTE counters

Counter ID	Counter Name
1526756893	L.RRCRedirection.E2N.Service
1526758901	L.IRATHO.E2N.Service.PrepAttOut
1526756784	L.IRATHO.E2N.PrepAttOut
1526761060	L.IRATHO.E2N.Service.ExecAttOut
1526761061	L.IRATHO.E2N.Service.ExecSuccOut
1526756781	L.IRATHO.E2N.Prep.FailOut.MME
1526756782	L.IRATHO.E2N.Prep.FailOut.PrepFailure
1526756783	L.IRATHO.E2N.Prep.FailOut.NoReply
1526756779	L.IRATHO.E2N.MMEAbnormRsp
1526756785	L.IRATHO.NCell.E2N.PrepAttOut
1526756780	L.IRATHO.NCell.E2N.ExecAttOut
1526756778	L.IRATHO.NCell.E2N.ExecSuccOut
1526777661	L.RRCRedirection.E2N.RedCap
1526779679	L.IRATHO.E2N.PrepAttOut.RedCap
1526779677	L.IRATHO.E2N.ExecAttOut.RedCap
1526779678	L.IRATHO.E2N.ExecSuccOut.RedCap

Table 5-21 NR counters

Counter ID	Counter Name
1911820677	N.HO.InterRATE2N.PrepAttIn
1911820678	N.HO.InterRATE2N.PrepSuccIn
1911820679	N.HO.InterRATE2N.ExecSuccIn
1911820683	N.HO.InterRATE2N.PrepInFail
1911820682	N.HO.InterRATE2N.PrepInFail.TgtNotAllow
1911820681	N.HO.InterRATE2N.PrepInFail.RscAdmitFail
1911820680	N.HO.InterRATE2N.PrepInFail.HOCancel
1911847073	N.HO.InterRATE2N.PrepInFail.RedCap

5.7 Voice Fallback from NG-RAN to E-UTRAN

5.7.1 Principles

Voice services initiated by UEs in an NG-RAN that does not support VoNR can fall back to E-UTRAN by means of voice fallback, which is also known as EPS fallback. Voice fallback is classified into normal voice fallback, emergency voice fallback, and fast voice fallback. A UE that has fallen back to E-UTRAN is referred to as an EPS fallback UE for short.

In RAN sharing with common carrier scenarios, the mobility policy and blind mode can be set for normal and fast voice fallback on a per operator basis. They take effect based on cell-level and operator-level settings as follows:

- The mobility policy depends on the cell-level setting of **NRInterRatHoParam.EpsFbMode** and the operator-level setting of **NRCellOpPolicy.EpsFbMode**, as described in [Table 5-22](#).

Table 5-22 Mobility policy settings for voice fallback

NRInterRatHoParam.EpsFbMode	NRCellOpPolicy.EpsFbMode	Mobility Policy That Takes Effect
HANDOVER	NOT_CONFIG. Alternatively, the NRCellOpPolicy MO is not configured.	Handover
REDIRECTION	NOT_CONFIG. Alternatively, the NRCellOpPolicy MO is not configured.	Redirection
Any value	HANDOVER	Handover
Any value	REDIRECTION	Redirection

- Whether blind mode takes effect depends on the cell-level setting of **VOICE_BLIND_MODE_SWITCH** of the **NRInterRatHoParam.HoModeSwitch** parameter and the operator-level setting of **VOICE_BLIND_MODE_SWITCH** of the **NRCellOpPolicy.InterRatHoPolicySwitch** parameter, as described in [Table 5-23](#).

Table 5-23 Blind mode settings

VOICE_BLIND_MODE_SWITCH of NRInterRatHoParam.HoModeSwitch	VOICE_BLIND_MODE_SWITCH of NRCellOpPolicy.InterRatHoPolicySwitch	Blind Mode Takes Effect?
On	On	Yes

VOICE_BLIND_MODE_SWITCH of NRInterRatHoParam.HoModeSwitch	VOICE_BLIND_MODE_SWITCH of NRCellOpPolicy.InterRatHoPolicySwitch	Blind Mode Takes Effect?
On	Off	No
Off	On	No
Off	Off	No
On	The NRCellOpPolicy MO is not configured.	Yes
Off	The NRCellOpPolicy MO is not configured.	No

5.7.1.1 Normal Voice Fallback

When a mobile-originated (MO) or mobile-terminated (MT) voice session is triggered, the gNodeB evaluates whether to initiate voice fallback based on factors such as UE capabilities, LTE radio conditions, and the VoNR capability.

The gNodeB determines to perform a normal voice fallback on the UE, and starts the EPS fallback protection timer, which is specified by the **NRInterRatHoParam.EpsFbProtectionTimer** parameter to determine the period of time for event B1 measurement, when one of the following conditions is met:

- The gNodeB sends the core network an INITIAL CONTEXT SETUP RESPONSE message with the cause value "IMS voice EPS fallback or RAT fallback triggered".
- The gNodeB sends the core network a PDU SESSION RESOURCE SETUP RESPONSE message with the cause value "IMS voice EPS fallback or RAT fallback triggered".
- The gNodeB sends the core network a PDU SESSION RESOURCE MODIFY RESPONSE message with the cause value "IMS voice EPS fallback or RAT fallback triggered".

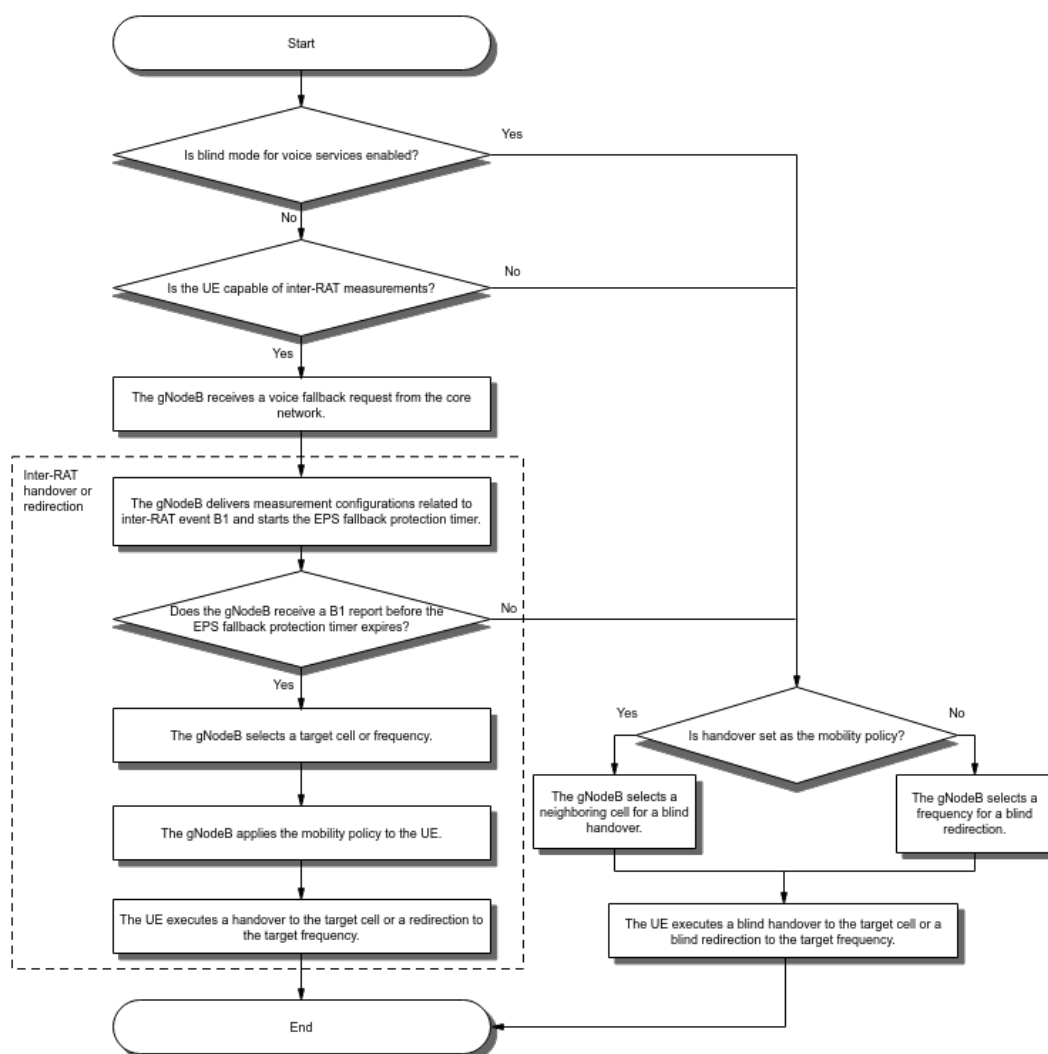
Before the fallback protection timer expires, the voice fallback procedure goes as described in [Procedure with UE Measurements](#) or [Procedure with RSRP Predicted Based on Virtual Grids](#).

When the fallback protection timer expires, the gNodeB takes protection measures as described in [Protection Against Incomplete Voice Fallback Upon Fallback Protection Timer Expiration](#) if the voice fallback procedure has not been completed.

Normal voice fallback is controlled by the **EPS_FB_SWITCH** option of the **NRCellAlgoSwitch.VoiceStrategySwitch** parameter. In addition, this function requires that the **EPS_FB_IN_ADVANCE_MEAS_SWITCH** option of the **NRCellAlgoSwitch.VoiceStrategySwitch** parameter be deselected.

Procedure with UE Measurements

[Figure 5-10](#) shows the procedure for normal voice fallback.

Figure 5-10 Procedure for normal voice fallback

The procedure is as follows:

1. The gNodeB checks whether blind mode for voice fallback has been enabled.
If blind mode for voice services has been enabled by selecting the **VOICE_BLIND_MODE_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter, the gNodeB goes to 5 to perform a blind inter-RAT handover or blind redirection. If it has not, the gNodeB goes to the next step.
2. The gNodeB checks whether the UE is capable of inter-RAT measurements.
 - If the UE reports the *eventB-MeasAndReport* IE, the UE is capable of inter-RAT measurements. In this case, the gNodeB delivers the measurement configurations related to inter-RAT event B1, receives inter-RAT event B1 measurement reports, and then goes to 3.
 - If the UE does not report the *eventB-MeasAndReport* IE, the UE is not capable of inter-RAT measurements. In this case, the gNodeB goes to 5 to perform a blind inter-RAT redirection.
3. The gNodeB receives a voice fallback request from the core network.

4. The gNodeB performs an inter-RAT handover or redirection procedure.
 - a. The gNodeB sorts the filtered frequencies by priority (specified by **NRCellEutranNFreq.VoltePriority**), delivers the measurement configurations related to inter-RAT event B1 to the UE, and starts the EPS fallback protection timer. For details about the frequency filtering rules, see [4.1.4.1.3 LTE Frequency Selection by the gNodeB](#). For details about event B1, see [4.1.4.2.1 Measurement Events](#).
 - b. If the EPS fallback protection timer expires and the gNodeB has not received an inter-RAT event B1 measurement report, the procedure goes to [5](#). In other cases, the procedure goes to [4.c](#).
 - c. The gNodeB determines the target LTE cell for the handover or the target frequency for redirection based on the setting of **NRInterRatHoParam.EpsFbMultiFreqB1RptWaitTmr**.
 - If **NRInterRatHoParam.EpsFbMultiFreqB1RptWaitTmr** takes the default value of 0, the gNodeB selects the cell with the best quality in the measurement reports as the target cell, and instructs the UE to perform a handover to the cell or a redirection to the operating frequency of the cell.
 - If **NRInterRatHoParam.EpsFbMultiFreqB1RptWaitTmr** is set to a non-zero value, selection of the optimal LTE frequency for EPS fallback takes effect.

When selection of the optimal LTE frequency for EPS fallback takes effect:

- If the frequency in the first measurement report received by the gNodeB does not have the highest priority, the gNodeB selects either (i) the neighboring cell with the best signal quality from the report as the target LTE cell for the handover, or (ii) the frequency in the report as the target frequency for the redirection. The procedure goes to [4.4](#).
- If (i) the frequency in the first measurement report received by the gNodeB has the highest priority, and (ii) the RSRP of the neighboring cell with the best signal quality in the report is greater than the value of **NRCellEutranNFreq.EpsFbB1RsrpThld**, then the gNodeB selects either the neighboring cell as the target LTE cell for the handover or the frequency in the report as the target frequency for the redirection. The procedure goes to [4.4](#).
- If (i) the frequency in the first measurement report received by the gNodeB has the highest priority, and (ii) the RSRP of the neighboring cell with the best signal quality in the report is less than or equal to the value of **NRCellEutranNFreq.EpsFbB1RsrpThld**, then the gNodeB buffers the neighboring cell's measurement result and starts the timer specified by **NRInterRatHoParam.EpsFbMultiFreqB1RptWaitTmr**. The gNodeB waits for subsequent measurement reports from the UE:
 - If the RSRP of the neighboring cell with the best signal quality in a subsequent measurement report is greater than the value of **NRCellEutranNFreq.EpsFbB1RsrpThld**, the gNodeB selects either the neighboring cell as the target LTE cell for the handover or the frequency in the report as the target frequency for the redirection. The procedure goes to [4.4](#).

- If (i) the frequency in a subsequent measurement report received by the gNodeB does not have the highest priority, and (ii) the RSRP of the neighboring cell with the best signal quality in the report is less than or equal to the value of **NRCCellEutranNFreq.EpsFbB1RsrpThld**, then the gNodeB discards the report, and waits for the next report from the UE or **executes the timer expiration procedure**.
 - If (i) the frequency in a subsequent measurement report received by the gNodeB has the highest priority, and (ii) the RSRP of the neighboring cell with the best signal quality in the report is less than or equal to the value of **NRCCellEutranNFreq.EpsFbB1RsrpThld**, then the gNodeB updates the buffered B1 measurement results using the neighboring cell's measurement results in the report. After that, the gNodeB waits for the next report from the UE or **executes the timer expiration procedure**.
 - If the timer specified by **NRInterRatHoParam.EpsFbMultiFreqB1RptWaitTmr** expires, the gNodeB selects either the buffered cell as the target LTE cell for the handover or the buffered frequency as the target frequency for the redirection. The procedure goes to 4.4.
- d. The gNodeB applies the mobility policy to the UE as follows:
- If handover takes effect as the mobility policy and the UE supports handovers, the gNodeB instructs the UE to perform a handover to the target LTE cell. However, when the PCI of the target LTE cell is not included in the neighboring cell list configured on the gNodeB or a PCI confusion occurs, the subsequent procedure varies depending on the setting of the **EPS_FB_STRONGEST_NCELL_REDIRECTION_SW** option of the **NRCCellAlgoSwitch.VoiceStrategySwitch** parameter. If this option is selected, the UE performs a redirection to the target frequency. If this option is deselected, the procedure proceeds as follows:
 - In the case that the target LTE cell is not buffered, the gNodeB filters out all neighboring cells in the measurement report that do not meet the handover conditions. If there are remaining neighboring cells for handover after the filtering, the gNodeB attempts to hand over the UE to the cell with the best signal quality. If the handover preparation fails or no neighboring cell remains after the filtering, the gNodeB instructs the UE to perform a redirection to the target frequency.
 - In the case that the target LTE cell is buffered, the gNodeB instructs the UE to perform a redirection to the target frequency.
 - If redirection takes effect as the mobility policy or the UE does not support handovers, the UE performs a redirection to the target frequency.
5. The gNodeB performs a blind handover or blind redirection.
- The gNodeB initiates a blind handover procedure if (i) handover takes effect as the mobility policy, (ii) the UE supports inter-RAT handovers, and (iii) inter-RAT blind-handover neighboring cells have been

configured. In this procedure, the gNodeB selects a target cell for the blind handover as follows:

- The gNodeB selects an inter-RAT neighboring cell to which blind handovers are allowed, as indicated by the setting of the **EPS_FB_BLIND_HO_FLAG** option of the **NRCellEutraNRelation.BlindHoFlag** parameter.
- The gNodeB selects an inter-RAT neighboring cell on the frequency with the highest priority specified by **NRCellEutraNFreq.VolteBlindPriority** among the frequencies whose priority values are neither 0 nor 255.
- If multiple inter-RAT frequencies have the same priority, the gNodeB preferentially selects the frequency with the smallest EARFCN.
- If blind handovers to multiple neighboring cells on an inter-RAT frequency are allowed, the gNodeB preferentially selects the neighboring cell with the smallest PCI.

If blind handovers to none of the neighboring cells are allowed or all blind handover preparations fail, the gNodeB performs a blind redirection procedure.

- The gNodeB initiates a blind redirection procedure if redirection takes effect as the mobility policy. In this procedure, the gNodeB selects the frequency with the highest priority specified by **NRCellEutraNFreq.VolteBlindPriority** among the frequencies whose priority values are neither 0 nor 255. If multiple frequencies have the same priority, as specified by **NRCellEutraNFreq.VolteBlindPriority**, the gNodeB preferentially selects the frequency with the smallest EARFCN.
6. The UE performs a handover to the target cell or a redirection to the target frequency.

For details about the signaling procedure for normal voice fallback, see [7.4 Voice Fallback from NG-RAN to E-UTRAN](#).

Procedure with RSRP Predicted Based on Virtual Grids

During evaluation of voice fallback from NG-RAN to E-UTRAN, use of NL RSRP models can replace UE measurement of RSRP if the **EPS_FB_EUTRAN_RSRP_PRED_SW** option of the **NRCellAlgoSwitch.VoiceStrategySwitch** parameter is selected.

The procedure for normal voice fallback with RSRP predicted based on virtual grids is the same as the procedure for normal voice fallback based on UE measurements, except the following steps (which replace step 2 in the latter procedure described in [Procedure with UE Measurements](#)):

1. For a UE that accesses a cell (at initial access in idle mode, an incoming handover, or an incoming RRC connection reestablishment), the gNodeB delivers A3 measurement configurations about the serving NR frequency with a measurement period of 2s and reporting interval of 240 ms. The time-to-trigger for event A3 is 40 ms.
2. After the UE sends A3 measurement reports to the gNodeB, the gNodeB uses the reported RSRP values of the serving cell and intra-frequency neighboring

cells as an input to virtual grid models, to acquire the RSRP information about LTE frequencies.

- If voice fallback is triggered before the 2s timer expires, the gNodeB goes to the next step.
 - If voice fallback is triggered after the 2s timer expires, the gNodeB delivers A3 measurement configurations again and also delivers B1 measurement configurations.
 - If A3 measurement reports are sent earlier than B1 measurement reports, the procedure goes to the next step.
 - If B1 measurement reports are sent earlier than A3 measurement reports, the B1 measurement reports are used to determine the target cell or frequency.
3. When NL RSRP models are available on the gNodeB and the model accuracy meets the requirement (controlled by the **NRCellSmartMultiCarr.PredAccuForVoiceThld** parameter), the gNodeB uses the models to predict the RSRP for the corresponding neighboring LTE frequencies.
- If the prediction result is positive for the target LTE frequency (that is, the entering condition for the corresponding event B1 is met), the gNodeB initiates a handover procedure without instructing the UE to perform measurements. In case there are multiple LTE frequencies with positive prediction results, the target LTE cell is selected depending on the setting of the **EPS_FB_TARGET_CELL_SEL_SW** option of the **NRCellAlgoSwitch.VoiceStrategySwitch** parameter.
 - When this option is deselected, the LTE frequency with the highest priority specified by the **NRCellEutranNFreq.VoltePriority** parameter is selected. If multiple frequencies have the same priority, the frequency with the smallest absolute radio frequency channel number is preferentially selected. After that, the cell with the highest RSRP on this frequency is selected as the target cell for the handover.
 - If this option is selected, the LTE cell with the highest RSRP is selected as the target cell for the handover.
 - If the prediction result for the target LTE frequency does not meet the entering condition for the corresponding event B1, UE measurements are still required. For details, see [Procedure with UE Measurements](#).

Protection Against Incomplete Voice Fallback Upon Fallback Protection Timer Expiration

The gNodeB takes different protection measures depending on the voice fallback policy used by the gNodeB:

- If the voice fallback policy is blind handover, the gNodeB acts as follows when the fallback protection timer expires:
 - If the outgoing handover preparation phase of the voice fallback procedure is ongoing, the gNodeB sends a HANDOVER CANCEL message to the core network and performs a blind redirection to the operating frequency of the target LTE cell of the handover without waiting for a HANDOVER CANCEL ACKNOWLEDGE message from the core network.

- If the outgoing handover execution phase of the voice fallback procedure is ongoing, the gNodeB continues the handover procedure.
- If the voice fallback policy is measurement-based handover, the gNodeB acts as follows when the fallback protection timer expires:
 - If the gNodeB has received inter-RAT event B1 measurement reports from the UE and:
 - If the outgoing handover preparation phase of the voice fallback procedure is ongoing, the gNodeB sends a HANDOVER CANCEL message to the core network and performs a redirection to the operating frequency of the target LTE cell of the handover without waiting for a HANDOVER CANCEL ACKNOWLEDGE message from the core network.
 - If the outgoing handover execution phase of the voice fallback procedure is ongoing, the gNodeB continues the handover procedure.
 - If the gNodeB has not received inter-RAT event B1 measurement reports from the UE, the gNodeB performs a blind redirection procedure.
- If the voice fallback policy is measurement-based redirection, the gNodeB performs a blind redirection when the fallback protection timer expires.

Frequency Selection During Blind Redirection for Voice Fallback

During blind redirection for voice fallback, a frequency meeting all of the following conditions is selected during target frequency selection:

- The frequency belongs to a band that the UE supports.
- The **NRCellEutranNFreq.MeasActvFlag** parameter is set to **ACTIVE** for the frequency.
- The **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of the **NRCellEutranNFreq.FrequencyCtrlSwitch** parameter is deselected for the frequency.
- The value of the **NRCellEutranNFreq.VolteBlindPriority** parameter corresponding to the frequency is neither **0** nor **255**.
- When the "roaming UEs preferentially back to HPLMN cells" function is enabled (for details, see [4.1.10 Roaming UEs Preferentially Back to HPLMN Cells](#)):
 - The **NRCellEutranNFreq.MeasureType** parameter is set to **ROAMING_MEAS** and the UE is a roaming UE.
 - The **NRCellEutranNFreq.MeasureType** parameter is set to **NORMAL**.

If the **VOICE_BLIND_REDIR_FREQ_SEL_SW** option of the **NRCellAlgoSwitch.VoiceStrategySwitch** parameter is selected and all frequencies have been filtered out, the base station performs frequency filtering again but no longer requires the frequency to meet the following two conditions:

- The **NRCellEutranNFreq.MeasActvFlag** parameter is set to **ACTIVE** for the frequency.
- The **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of the **NRCellEutranNFreq.FrequencyCtrlSwitch** parameter is selected for the frequency.

5.7.1.2 Emergency Voice Fallback

The gNodeB identifies emergency services based on the *Emergency Fallback Indicator* field in the INITIAL CONTEXT SETUP REQUEST or UE CONTEXT MODIFICATION REQUEST message sent from the core network and then initiates emergency voice fallback.

Emergency voice fallback is not under switch control or license control. It takes effect by default. The mobility policy and blind mode evaluation for emergency voice fallback are similar to those for normal voice fallback described in [5.7.1.1 Normal Voice Fallback](#). The differences are as follows:

- The penalty mechanism does not apply to emergency voice fallback.
- In the emergency voice fallback procedure, the following cells are no longer filtered out: cells to which handovers are prohibited according to neighbor relationship settings, cells in the Handover Restriction List, and cells that are not in target PLMNs.
- During emergency voice fallback, only frequencies with the **NRCellEutranNFreq.VolteBlindPriority** parameter set to **0** can be selected for blind redirections or blind handovers; those with the parameter set to **255** are excluded.
- PDU sessions are not established during emergency voice fallback triggered by initial context setup. Therefore, the mobility policy has to be redirection instead of handover.
- During emergency voice fallback, **NRCellEutranNFreq.MeasureType** is not used for frequency filtering.

The procedure is as follows:

- If emergency voice fallback is triggered by initial context setup and the *Target CN IE* in the *Emergency Fallback Indicator* indicates 5GC (which is an abnormal case), the gNodeB performs a blind inter-RAT redirection, adhering to the same target frequency selection policy as in normal voice fallback. In addition, the gNodeB sends an initial context setup failure message to the core network.
- If emergency voice fallback is triggered by initial context setup and the *Target CN IE* in the *Emergency Fallback Indicator* indicates EPC, the gNodeB performs a measurement-based inter-RAT redirection, adhering to the same target frequency selection policy as in normal voice fallback. If the INITIAL CONTEXT SETUP REQUEST message sent from the core network to the gNodeB conveys a PDU session establishment request, the gNodeB responds with a PDU session establishment failure message.
- If emergency voice fallback is triggered by UE context modification, the procedure is the same as that for normal voice fallback.

5.7.1.3 Fast Voice Fallback

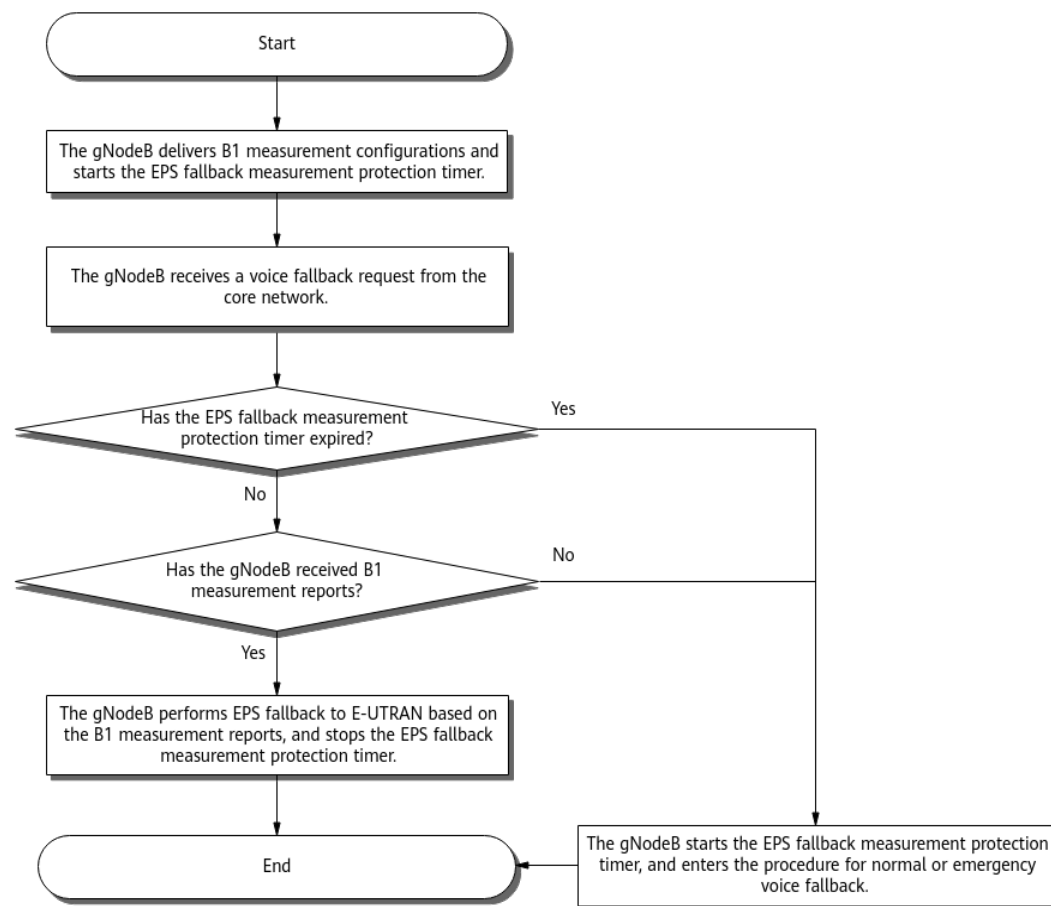
Fast voice fallback allows the gNodeB to deliver the measurement configurations related to inter-RAT event B1 at an earlier time (a time before the UE requests the 5GC to establish a voice session), reducing the time to wait for signaling exchanges between the UE and 5GC. Accordingly, the gNodeB receives measurement reports on event B1 from the UE at an earlier time.

This function takes effect for the following types of UEs when VoNR is disabled or UE capabilities indicate that VoNR is not supported:

- RRC_IDLE or RRC_INACTIVE EPS fallback UEs initiating normal voice services, which are identified as follows:
 - The value of *establishmentCause* in the RRCSetupRequest message is "mo-VoiceCall".
 - The value of *resumeCause* in the RRCResumeRequest or RRCResumeRequest1 message is "mo-VoiceCall".
- RRC_INACTIVE EPS fallback UEs initiating emergency calls, which are identified as follows: The value of *resumeCause* in the RRCResumeRequest or RRCResumeRequest1 message is "emergency".

Figure 5-11 shows the procedure for fast voice fallback.

Figure 5-11 Procedure for fast voice fallback



The procedure is as follows:

1. The gNodeB delivers the measurement configurations related to inter-RAT event B1 and starts the EPS fallback measurement protection timer, which is specified by the **NRInterRatHoParam.EpsFbMeasProtectTimer** parameter.
2. When the gNodeB receives a voice fallback request from the core network:
 - Scenario 1: If the gNodeB receives measurement reports on inter-RAT event B1 before the EPS fallback measurement protection timer expires,

the gNodeB instructs the UE to perform a handover or redirection to E-UTRAN based on the measurement reports and stops the EPS fallback measurement protection timer. For details about the signaling procedure for fast voice fallback, see [7.4 Voice Fallback from NG-RAN to E-UTRAN](#).

- Scenario 2: If the gNodeB does not receive measurement reports on inter-RAT event B1 and the EPS fallback measurement protection timer does not expire, the gNodeB stops the measurement protection timer, delivers EPS fallback measurement configurations again, and starts the EPS fallback measurement protection timer again. The subsequent procedure is the same as that described in [5.7.1.1 Normal Voice Fallback](#) and [5.7.1.2 Emergency Voice Fallback](#).
- Scenario 3: If the EPS fallback measurement protection timer has expired, the gNodeB delivers EPS fallback measurement configurations again and starts the EPS fallback measurement protection timer again. The subsequent procedure is the same as that described in [5.7.1.1 Normal Voice Fallback](#) and [5.7.1.2 Emergency Voice Fallback](#).

In scenarios 1 and 3, if selection of the optimal LTE frequency for EPS fallback is enabled (that is, **NRInterRatHoParam.EpsFbMultiFreqB1RptWaitTmr** is set to a value greater than 0), the gNodeB performs the following operations when it receives event B1 measurement reports:

- If there is no buffered event B1 measurement report, the gNodeB saves the measurement report.
- If there is a buffered event B1 measurement report and the RSRP in the newly received measurement report is greater than the **NRCeUtranNFreq.EpsFbB1RsrpThld** parameter value for the operating frequency:
 - If the RSRP in the buffered measurement report is greater than the **NRCeUtranNFreq.EpsFbB1RsrpThld** parameter value for the operating frequency, the gNodeB saves the measurement results of the frequency with the higher priority as indicated by the **NRCeUtranNFreq.VoltePriority** parameter. If the priorities are the same, the gNodeB saves the newly received measurement report.
 - If the RSRP in the buffered measurement report is less than or equal to the **NRCeUtranNFreq.EpsFbB1RsrpThld** parameter value for the operating frequency, the gNodeB saves the newly received measurement report.
- If there is a buffered B1 measurement report and the RSRP in the newly received measurement report is less than or equal to the **NRCeUtranNFreq.EpsFbB1RsrpThld** parameter value for the operating frequency:
 - If the RSRP in the buffered measurement report is greater than the **NRCeUtranNFreq.EpsFbB1RsrpThld** parameter value for the operating frequency, the gNodeB discards the newly received measurement report.
 - If the RSRP in the buffered measurement report is less than or equal to the **NRCeUtranNFreq.EpsFbB1RsrpThld** parameter value for the operating frequency, the gNodeB saves the measurement results of the frequency with the higher priority as indicated by the

NRCellEutranNFreq.VoltePriority parameter. If the priorities are the same, the gNodeB saves the newly received measurement report.

Fast voice fallback requires both **EPS_FB_SWITCH** of **NRCellAlgoSwitch.VoiceStrategySwitch** and **EPS_FB_IN_ADVANCE_MEAS_SWITCH** of **NRCellAlgoSwitch.VoiceStrategySwitch** to be selected.

NOTE

The **EPS_FB_IN_ADVANCE_MEAS_SWITCH** option of the **NRCellAlgoSwitch.VoiceStrategySwitch** parameter and the **VOICE_BLIND_MODE_SWITCH** option of the **NRInterRatHoParam.HoModeSwitch** parameter cannot be selected at the same time.

5.7.1.4 Collaboration with Other Functions

Handling of Conflicts Between a Handover Procedure and Voice Fallback

During the preparation for an outgoing handover of an RRC_CONNECTED UE from NG-RAN to E-UTRAN, if the core network initiates a procedure for setting up a dedicated voice bearer with a 5QI of 1 or initiates emergency voice fallback, the handover procedure conflicts with the normal or emergency voice fallback procedure. The **gNBMobilityCommParam.HoPduSessConflHandlingSw** parameter is used to control how to handle the conflict.

- If the **VOICE_FB_FIRST_SWITCH** option of this parameter is selected, the gNodeB preferentially handles the normal or emergency voice fallback and sends a handover cancellation request.
- If this option is deselected, the gNodeB preferentially handles the handover procedure and sends a message, notifying the core network that the normal or emergency voice fallback fails because the gNodeB is performing an inter-RAT handover. After the handover is complete, the core network re-initiates a procedure for setting up a dedicated voice bearer with QCI 1 according to 3GPP TS 23.501. If the core network does not support the preceding procedures, the dedicated voice bearer fails to be set up.

Handling of Conflicts Between a QoS Procedure and Voice Fallback

In the case of voice fallback, if the gNodeB receives a PDU session setup or modification message in the outgoing handover preparation phase and the message indicates QoS flow addition or modification, the gNodeB continues the outgoing handover procedure for voice fallback, irrespective of the **QOS_FLOW_FIRST_SW** option setting of the **NRCellQciBearer.HoQosFlowConflProcStrat** parameter.

Handling of Conflicts Between a PDU Session Procedure and Voice Fallback

In the case of voice fallback, if the gNodeB receives a PDU session release message in the outgoing handover preparation phase, the gNodeB continues the outgoing handover procedure for voice fallback, irrespective of the **PDU_SESS_REL_NG_HO_REQ_SEL_SW** option setting of the **gNBMobilityCommParam.HoPduSessConflHandlingSw** parameter.

EPS Fallback UE Identification After Voice Fallback to E-UTRAN

After a UE falls back to an LTE cell, the eNodeB can identify this EPS fallback UE and prevent the EPS fallback failure caused by the collision between the EPS fallback procedure and other interoperability procedures (for example, E-UTRAN to NG-RAN traffic steering).

The method of identifying EPS fallback UEs varies depending on whether the UEs fall back from NR cells to LTE cells by handover or redirection:

- Handover: If the value of the *Handover Type* IE is "5GStoEPS" and the value of *IMS voice EPS fallback from 5G* in the *Source eNB to Target eNB Transparent Container* IE is "true" in the HANDOVER REQUEST message sent from the MME to the eNodeB, the UE is a handover-based EPS fallback UE. A prerequisite for this identification method is that the source gNodeB must comply with 3GPP Release 15 specifications issued in October 2019 or later specifications.
- Redirection: A prerequisite for this identification method is that the **MO_VOICE_CALL_IND_SW** option of the **VolteAlgoConfig.VolteOptSwitch** parameter is selected. With this option selected, if the value of the *establishmentCause* IE in the RRCConnectionRequest message sent from a UE to the eNodeB is *mo-VoiceCall*, the UE is a redirection-based EPS fallback UE.

Within E-UTRAN, the method of identifying EPS fallback UEs varies depending on whether the UEs are transferred from LTE cells to LTE cells by handover or RRC connection reestablishment:

- Handover: If the value of the *Handover Type* IE is "IntraLTE" and the value of *IMS voice EPS fallback from 5G* in the *Source eNB to Target eNB Transparent Container* IE is "true" in the HANDOVER REQUEST message sent from the MME to the eNodeB, the UE is a handover-based EPS fallback UE. Alternatively, if the value of *IMS voice EPS fallback from 5G* is "true" in the HANDOVER REQUEST message sent from the source eNodeB to the target eNodeB, the UE is a handover-based EPS fallback UE. Prerequisites for this identification method are: (1) The **X2S1MsgWithEpsFbIndicationSw** option of the **GlobalProcSwitch.ProtocolSupportSwitch** parameter is selected. (2) The source eNodeB and the target eNodeB must comply with 3GPP Release 16 specifications issued in September 2021 or later specifications.
- RRC connection reestablishment: If the value of *IMS voice EPS fallback from 5G* is "true" in the RETRIEVE UE CONTEXT RESPONSE message sent from the source eNodeB to the target eNodeB, the UE is a reestablishment-based EPS fallback UE. Prerequisites for this identification method are: (1) The **X2S1MsgWithEpsFbIndicationSw** option of the **GlobalProcSwitch.ProtocolSupportSwitch** parameter is selected. (2) The source eNodeB and the target eNodeB must comply with 3GPP Release 16 specifications issued in September 2021 or later specifications.

Delay Optimization After Voice Fallback to E-UTRAN

After an EPS fallback UE falls back to an LTE cell, the following mechanisms take effect for the UE in the LTE cell to shorten delay, if the **FAST_EPS_FALLBACK_SW** option of the **VolteAlgoConfig.VolteOptSwitch** parameter is selected:

- DRX does not take effect for the UE until the eNodeB detects a QCI 1 service in the downlink for the UE.

- The eNodeB performs continuous scheduling on the UE upon the fallback. The scheduling period and data volume are specified by the **VolteAlgoConfig.EpsFbULActiveSchMinPeriod** and **VolteAlgoConfig.EpsFbULActiveSchDataVol** parameters, respectively. The eNodeB does not stop continuous scheduling until 5s elapse or until the eNodeB detects a QCI 1 service in the downlink for the UE.
- If the UE is not of category 0 (CAT0), the eNodeB performs scheduling on the UE based on the TB size for CAT1 by default before querying the UE capability. If the UE is of CAT0, the eNodeB performs scheduling on the UE based on the TB size of CAT0.

Collaboration Between Voice Fallback and ANR

When ANR is enabled, handovers for normal voice fallback or emergency voice fallback can take place regardless of the **NR_EUTRAN_ANR_CTRL_MODE_HO_SW** setting of the **NRCellAnr.AnrModeSwitch** parameter. That is, handovers to unconfirmed neighboring LTE cells (with which neighbor relationships are reported in optimization suggestions but are not confirmed to be added to the neighboring cell list yet) are allowed. For details, see *ANR* in the 5G RAN feature documentation.

If the neighboring LTE cells do not support handovers from NG-RAN to E-UTRAN, the value of the **N.HO.InterRAT.N2E.EPSFB.Prep.FailOut.PrepFailure** counter will increase.

Collaboration Between Normal Voice Fallback with RSRP Predicted Based on Virtual Grids and LTE Cell Deactivation for Energy Saving

After energy conservation and emission reduction features, such as intelligent power-off of carriers in the same coverage, are enabled on the eNodeB, LTE cells may be deactivated at certain moments. In this case, if such a cell is selected as the target cell for a handover based on prediction using virtual grids, the handover preparation will fail. For details about energy conservation and emission reduction, see *Energy Conservation and Emission Reduction* in the eRAN feature documentation.

It is recommended that the **VG_MODEL_ALLOW_BUILD_FLAG** option of the **NRCellEutranNFreq.AggregationAttribute** parameter be deselected for LTE frequencies on which cells are deactivated due to energy conservation and emission reduction. In this case, the gNodeB will not build virtual grid models for these LTE frequencies, and therefore it will not select LTE cells on these frequencies during prediction based on virtual grids for EPS fallback. This prevents handover preparation failures caused by deactivation of the candidate cells selected using prediction based on virtual grids.

5.7.1.5 Protocol Update

This section describes the function changes due to protocol updates. The update in 3GPP Release 16 is as follows:

During an EPS fallback procedure, if *voiceFallbackIndicationEPS-r16* is included in *IMS-Parameters* in the reported UE capability information, the MobilityFromNRCommand or RRCRelease message sent from the gNodeB to the

UE carries the *voiceFallbackIndication-r16* IE. In this case, the UE behaves as follows:

- If the EPS fallback procedure is implemented by handover (including blind handover) and the UE fails to be handed over to an LTE cell, the UE attempts to select another appropriate LTE cell. If no appropriate LTE cell is found, the UE has its RRC connection reestablished with the source NR cell.
- If the EPS fallback procedure is implemented by redirection (including blind redirection) and the *voiceServiceCauseIndication-r12* IE is broadcast in SystemInformationBlockType2 from the eNodeB, the RRCConnectionRequest message sent by the UE in the LTE cell contains the RRC connection setup cause value "mo-VoiceCall".

The preceding functions are controlled by the **VOICE_FB_INDICATION_SWITCH** option of the **NRCellAlgoSwitch.VoiceStrategySwitch** parameter.

If this option is deselected or *voiceFallbackIndicationEPS-r16* is not included in the UE-reported information, the UE behaves as follows:

- If the EPS fallback procedure is implemented by handover (including blind handover) and the UE fails to be handed over to an LTE cell, the UE has its RRC connection reestablished with an NR cell.
- If the EPS fallback procedure is implemented by redirection (including blind redirection), the RRCConnectionRequest message sent by the UE in the LTE cell may or may not contain the RRC connection setup cause value "mo-VoiceCall", depending on UE capabilities.

5.7.2 Network Analysis

5.7.2.1 Benefits

Normal voice fallback and emergency voice fallback offer the following benefits:

- Provide a means for UEs in NR networks to access voice services.
- Help operators reduce costs by reusing their VoLTE networks.

Fast voice fallback shortens the end-to-end delay of normal and emergency voice fallback.

When the **NRInterRatHoParam.EpsFbMultiFreqB1RptWaitTmr** parameter is set to a non-zero value, the gNodeB selects the optimal E-UTRAN frequency for voice fallback, which increases the EPS fallback call completion rate.

5.7.2.2 Impacts

The impacts between this function and other functions are described in [5.7.3.2.3 Function Impacts](#).

UEs in NR networks can access VoLTE services by voice fallback. This function has the following impacts on the access success rate:

- If redirection is used for voice fallback, each fallback procedure is equivalent to an initial access procedure for the initiation of a VoLTE service in E-UTRAN. Therefore, the success rate of access to E-UTRAN by voice fallback is

theoretically the same as the access success rate of VoLTE services in E-UTRAN.

- If handover is used for voice fallback, each fallback procedure involves an inter-RAT handover. The access success rate depends on the success rate of inter-RAT handovers. Handovers have a higher requirement for signal quality than initial access does. Therefore, the success rate of access to E-UTRAN by voice fallback will be slightly lower than the access success rate of VoLTE services in E-UTRAN.

When RSRP prediction based on virtual grids is used for voice fallback and the prediction results are positive for multiple LTE frequencies, the neighboring LTE cell with the strongest RSRP value on the frequencies is selected for a handover. As a result, the number of VoLTE UEs (indicated by the **L.Traffic.User.VoIP.Avg** counter) on the target frequency increases. VoLTE services are sensitive to the delay, jitter, and packet loss rate and take priority over data services. When there are a large number of VoLTE UEs on that frequency, fewer resources are available for data services on the frequency, and therefore the cell traffic volume (indicated by the **L.Thrp.bits.DL** and **L.Thrp.bits.UL** counters) decreases. On other LTE frequencies, the number of VoLTE UEs decreases, more resources can be allocated to data services, and therefore the cell traffic volume increases in certain scenarios.

When a handover procedure conflicts with a normal or emergency voice fallback procedure:

- If the **VOICE_FB_FIRST_SWITCH** option of the **gNB MobilityCommParam.HoPduSessConflHandlingSw** parameter is selected, the values of the counters related to canceled handovers will increase. It is recommended that these counter values be deducted during the calculation of the outgoing handover success rate. The counters are as follows:
 - **N.HO.IntraFreq.Prep.FailOut.HOCancel**
 - **N.HO.IntraFreq.FailOut.HOCancel**
 - **N.HO.InterFreq.Prep.FailOut.HOCancel**
 - **N.HO.InterFreq.FailOut.HOCancel**
 - **N.HO.InterRAT.N2E.Prep.FailOut.HOCancel**
- If the **VOICE_FB_FIRST_SWITCH** option of the **gNB MobilityCommParam.HoPduSessConflHandlingSw** parameter is deselected, the voice fallback preparation success rate (calculated by **N.Emc.FB.PrepSucc/N.Emc.FB.PrepAtt**) decreases.

When the **NRInterRatHoParam.EpsFbMultiFreqB1RptWaitTmr** parameter is set to a non-zero value, the gNodeB selects the optimal E-UTRAN frequency for voice fallback, which causes changes in the values of the following counters:

- The EPS fallback E2E call delay increases.
- The number of EPS fallback redirections increases, which is measured by the **N.RRCRedirection.N2E.EPSFB** counter.
- The number of EPS fallback redirections from NR to E-UTRAN due to timer expiration increases, which is measured by the **N.RRCRedirection.N2E.EPSFB.Timeout** counter.

- The number of NR-to-EUTRAN outgoing EPS fallback preparation attempts decreases, which is measured by the **N.HO.InterRAT.N2E.EPSFB.PrepAttOut** counter.

If the **VOICE_BLIND_REDIR_FREQ_SEL_SW** option of the **NRCellAlgoSwitch.VoiceStrategySwitch** parameter and the **AUTO_EUTRAN_FREQ_RELATION_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter are selected, the **QoS Flow Setup Success Rate (CU)** of 5QI 1 may increase at NR sites with weak LTE coverage.

5.7.3 Requirements

5.7.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
NR	FOFD-021210	Voice Fallback	NR0S00E PSF00	Voice Fallback (NR)	gNodeB	per cell

There are no license requirements for emergency voice fallback.

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

5.7.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD LTE TDD	Support of VoLTE	ENodeBAlgoSwitch.EutranVoipSupportSwitch	<i>VoLTE</i>

5.7.3.2.2 Mutually Exclusive Functions

None

5.7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Inter-RAT redirection back to the HPLMN based on roaming UE identification	VOICE_C APB_MO B_TO_HP LMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In voice fallback from NG-RAN to E-UTRAN, only redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.

5.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all LTE-capable boards support this function.
- For NR, all NR-capable boards support this function.

RF Modules

No requirements

5.7.3.4 Others

- UE requirements
UEs must comply with 3GPP Release 15 or later.
The UE has subscribed to the session transfer number for SRVCC (STN-SR) so that EPS fallback UEs can be handed over to the GSM or UMTS network through SRVCC, ensuring voice service continuity.
- Core network requirements
 - The core networks must comply with 3GPP Release 15 or later.
 - Handover requires the N26 interface to be configured between the MME and AMF.
 - The core network supports SRVCC so that EPS fallback UEs can be handed over to the GSM or UMTS network through SRVCC, ensuring voice service continuity.
- IMS requirements
The IMS must support control-plane functionalities.

5.7.4 Operation and Maintenance

5.7.4.1 Data Configuration

5.7.4.1.1 Data Preparation

[Table 5-24](#) and [Table 5-25](#) describe the parameters related to voice fallback from NG-RAN to E-UTRAN. For details about basic parameters for interoperability from NG-RAN to E-UTRAN, see [4.4.1.1 Data Preparation](#) and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-24 NR parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
Voice Strategy Switch	NRCellAlgoSwitch.VoiceStrategySwitch	- To enable voice fallback, select the EPS_FB_SWITCH option. - To enable fast voice fallback, select the EPS_FB_IN_ADVANCE_MEAS_SWITCH option. - To enable redirection to the strongest cell whose PCI is unknown or is involved in PCI confusion in voice fallback scenarios, select the EPS_FB_STRONGEST_NCELL_REDIRECTION_SWITCH option. - To enable the <i>voiceFallbackIndication-r16</i> IE to be included in the MobilityFromNRCommand and RRCRelease messages, select the VOICE_FB_INDICATION_SWITCH option.
Handover Mode Switch	NRInterRatHoParam.HoModeSwitch	To use blind mode, select the VOICE_BLIND_MODE_SWITCH option. To use measurement-based mode, deselect the VOICE_BLIND_MODE_SWITCH option. When the VOICE_BLIND_MODE_SWITCH option is selected, set NRInterRatHoParam.EpsFbMode to HANDOVER only if the LTE and NR cells are co-sited and cover the same area or if the LTE cell provides better coverage than the NR cell. Otherwise, you are not advised to set NRInterRatHoParam.EpsFbMode to HANDOVER.
EPS FB Mode	NRInterRatHoParam.EpsFbMode	- If this parameter is set to HANDOVER, UEs fall back to E-UTRAN preferentially by handover. - If this parameter is set to REDIRECTION, UEs fall back to E-UTRAN preferentially by redirection.
EPS FB Measurement Protection Timer	NRInterRatHoParam.EpsFbMeasProtectTimer	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
EPS Fallback Mode	NRCellOpPolicy.EpsFbMode	In RAN sharing with common carrier scenarios, set this parameter based on operator policies. If the policies for different operators are the same, the default value NOT_CONFIG is recommended. For details about MML command examples, see 6.4.1.2 Using MML Commands .
EPS FB Protection Timer	NRInterRatHoParam.EpsFbProtectionTimer	The default value is recommended for function activation. During reconfiguration: • A smaller value of this parameter results in a higher probability of fallback to EPS. • A larger value of this parameter results in a lower probability.
EPS FB Multi Frequency B1 Report Wait Timer	NRInterRatHoParam.EpsFbMultiFreqB1RptWaitTmr	Set this parameter to a non-zero value when selection of the optimal LTE frequency for EPS fallback is required. It is recommended that this parameter be set to a value less than the NRInterRatHoParam.EpsFbProtectionTimer parameter value when macro gNodeBs provide indoor coverage.
VoLTE Priority	NRCellEutranNFr eq. VoltePriority	Set this parameter based on the network plan. If this parameter is set to 255 , the neighboring LTE frequency does not support VoLTE.
VOLTE Blind Priority	NRCellEutranNFr eq. VolteBlindPriority	Set this parameter based on the network plan. If this parameter is set to 255 , the neighboring LTE frequency does not support VoLTE.
EPS Fallback B1 RSRP Threshold	NRCellEutranNFr eq. EpsFbB1RsrpThld	It is recommended that this parameter be set to a value greater than the InterFreqHoGroup.InterFreqHoA2ThdRsrp parameter value minus the InterFreqHoGroup.InterFreqHoA1A2Hyst parameter value on the LTE side. The gNodeB uses this parameter solely to filter B1 measurement reports; this parameter is not included in B1 measurement configurations sent to UEs.

Parameter Name	Parameter ID	Setting Notes
Blind Handover Flag	NRCellEutraNRelation.BlindHoFlag	Set this parameter based on the network plan. If an LTE cell and an NR cell cover the same area or the LTE cell provides better coverage than the NR cell, it is recommended that the EPS_FB_BLIND_HO_FLAG option be selected to allow blind handovers for EPS fallback to the LTE cell.
EPS FB B1 Hysteresis	NRCellHoEutranMeaGrp.EpsFbB1Hyst	The default value is recommended for function activation. During reconfiguration: • A smaller value of this parameter results in a higher probability that the entering and leaving conditions for event B1 related to voice fallback are fulfilled. • A larger value of this parameter results in a lower probability.
EPS FB B1 Time to Trigger	NRCellHoEutranMeaGrp.EpsFbB1TimeToTrig	The default value is recommended for function activation. During reconfiguration: • A smaller value of this parameter results in a higher probability that the entering and leaving conditions for event B1 related to voice fallback are fulfilled. • A larger value of this parameter results in a lower probability.

Parameter Name	Parameter ID	Setting Notes
EPS FB B1 RSRP Threshold	NRCellHoEutranMeaGrp.EpsFbB1RsrpThld	The default value is recommended for function activation. During reconfiguration: • A smaller value of this parameter results in a higher probability of triggering event B1 for voice fallback. • A larger value of this parameter results in a lower probability. This parameter is used for the delivery of B1 measurement configurations. NOTE If the NRCellHoEutranMeaGrp.EpsFbB1RsrpThld parameter has been configured for different QCI, the value of this parameter configured for QCI 1 is selected as the threshold for event B1 related to EPS fallback by default to ensure voice service experience.
HO and PDU Session Conflict Handling Sw	gNBMobilityComParam.HoPduSessionConflHandlingSw	Select the VOICE_FB_FIRST_SWITCH option of this parameter if normal or emergency voice fallback needs to be preferentially handled in the event of a conflict between a handover procedure and a normal or emergency voice fallback procedure.
Voice Strategy Switch	NRCellAlgoSwitch.VoiceStrategySwitch	• To use virtual grids for voice fallback from NG-RAN to E-UTRAN, select the EPS_FB_EUTRAN_RSRP_PRED_SW option. • When virtual grids are used for voice fallback from NG-RAN to E-UTRAN and there are multiple LTE frequencies with positive prediction results, the target LTE frequency can be selected based on either VoLTE priorities or RSRP values, depending on the setting of EPS_FB_TARGET_CELL_SEL_SW.
Voice Strategy Switch	NRCellAlgoSwitch.VoiceStrategySwitch	It is recommended that the VOICE_BLIND_REDIR_FREQ_SEL_SW option be selected.

Table 5-25 LTE parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
VoLTE Optimization Switch	VolteAlgoConfig.VolteOptSwitch	- To enable delay optimization after voice fallback, select the FAST_EPS_FALLBACK_SW option. - To enable identification of redirection-based EPS fallback UEs, select the MO_VOICE_CALL_IND_SW option.
UL Active Sch Min Period of EPS FB UE	VolteAlgoConfig.EpsFbULActiveSchMinPeriod	Set this parameter based on the network plan.
UL Active Sch Data Vol of EPS FB UE	VolteAlgoConfig.EpsFbULActiveSchDataVol	Set this parameter based on the network plan.
Protocol Procedure Support Switch	GlobalProcSwitch.ProtocolSupportSwitch	For EPS fallback UEs, to enable the eNodeB to include the <i>IMS voice EPS fallback from 5G</i> IE in the HANDOVER REQUEST and RETRIEVE UE CONTEXT RESPONSE messages over the X2 interface and the HANDOVER REQUEST message over the S1 interface, select the X2S1MsgWithEpsFbIndicationSw option of this parameter.

5.7.4.1.2 Using MML Commands

The basic configurations for interoperability from NG-RAN to E-UTRAN must have been completed before activating this function. For details, see [4.4.1.2 Using MML Commands](#) and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples for Normal and Emergency Voice Fallback

Before using MML commands, refer to [5.7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Using measurement-based mode for voice fallback
//Using handover for voice fallback
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=VOICE_BLIND_MODE_SWITCH-0,
```

```

EpsFbMode=HANDOVER;
//Using redirection for voice fallback
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=VOICE_BLIND_MODE_SWITCH-0,
EpsFbMode=REDIRECTION;
//Turning on the voice policy switch
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=EPS_FB_SWITCH-1;
//((Optional) Selecting the EPS_FB_STRONGEST_NCELL_REDIRECTION option if redirection to the strongest cell
whose PCI in the E-UTRAN measurement report is unknown or is involved in PCI confusion is required in
voice fallback scenarios
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=EPS_FB_STRONGEST_NCELL_REDIRECTION-1;
//((Optional) Selecting the VOICE_FB_INDICATION_SWITCH option if the voiceFallbackIndication-r16 IE
needs to be included in the MobilityFromNRCommand and RRCRelease messages
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=VOICE_FB_INDICATION_SWITCH-1;
//((Optional) Selecting the VOICE_BLIND_REDIRECTION_FREQ_SELECTION option if frequency selection for redirection
is required again based on certain frequency selection conditions when all neighboring LTE frequencies are
filtered out during blind redirection for EPS fallback and emergency voice fallback
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=VOICE_BLIND_REDIRECTION_FREQ_SELECTION-1;
//((Optional) Changing the VoLTE priority of an EARFCN
MOD NRCELLEUTRANRFREQ: NrCellId=0, DLearfcn=1300,VoltePriority=2;
//((Optional) Changing the priority of an EARFCN for voice services in blind mode
MOD NRCELLEUTRANRFREQ: NrCellId=0, DLearfcn=1300,VolteBlindPriority=1;
//((Optional) Running the following command when RSRP prediction based on virtual grids is required
MOD NRCELLALGOSWITCH: NrCellId=0,
VoiceStrategySwitch=EPS_FB_EUTRAN_RSRP_PRED_SWITCH-1&EPS_FB_TARGET_CELL_SELECTION_SWITCH-1;
//((Optional) Turning on VOICE_FB_FIRST_SWITCH of HoPduSessConflHandlingSw so that voice fallback
(including normal and emergency voice fallback) takes precedence over a handover procedure in case of a
conflict between them
MOD GNBMOBILITYCOMMPARAM: HoPduSessConflHandlingSw=VOICE_FB_FIRST_SWITCH-1;
//((If blind handovers are not supported for normal and emergency voice fallback) Adding a neighbor
relationship with the LTE cell
ADD NRCELLEUTRANRELATION: NrCellId=0, Mcc="460", Mnc="20", EnodebId=122, CellId=1;
//((If blind handovers are supported for normal and emergency voice fallback) Adding a neighbor
relationship with the LTE cell with blind handovers to that cell allowed
ADD NRCELLEUTRANRELATION: NrCellId=0, Mcc="460", Mnc="20", EnodebId=122, CellId=1,
BlindHoFlag=EPS_FB_BLIND_HO_FLAG-1;
//((Optional) Running the following commands when selection of the optimal LTE frequency for EPS fallback
is required
MOD NRINTERRATHOPARAM: NrCellId=0, EpsFbMultiFreqB1RptWaitTmr=2;
MOD NRCELLEUTRANRFREQ: NrCellId=0, DLearfcn=1300,EpsFbB1RsrpThld=-103;
//Setting EpsFbProtectionTimer
MOD NRINTERRATHOPARAM: NrCellId=0, EpsFbProtectionTimer=40;
//Setting EpsFbB1TimeToTrig, EpsFbB1RsrpThld, and EpsFbB1Hyst
MOD NRCELLHOEUTRANMEASGRP: NrCellId=0, InterRHoToEutranMeasGrpId=10,
EpsFbB1TimeToTrig=320MS,EpsFbB1RsrpThld=-111, EpsFbB1Hyst=2;

//Using blind mode for voice fallback
//((To use handover for voice fallback) Turning on VOICE_BLIND_MODE_SWITCH and setting the mobility
policy to handover
MOD NRINTERRATHOPARAM: NrCellId=0,
HoModeSwitch=VOICE_BLIND_MODE_SWITCH-1,EpsFbMode=HANDOVER;
//((To use redirection for voice fallback) Turning on VOICE_BLIND_MODE_SWITCH and setting the mobility
policy to redirection
MOD NRINTERRATHOPARAM: NrCellId=0,
HoModeSwitch=VOICE_BLIND_MODE_SWITCH-1,EpsFbMode=REDIRECTION;

```

Activation Command Examples for Fast Voice Fallback

Before using MML commands, refer to [5.7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.


```
//Turning on EPS_FB_IN_ADVANCE_MEAS_SWITCH
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=VOICE_BLIND_MODE_SWITCH-0;
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=EPS_FB_IN_ADVANCE_MEAS_SWITCH-1;
//Setting EpsFbMeasProtectTimer
MOD NRINTERRATHOPARAM: NrCellId=0, EpsFbMeasProtectTimer=50;
```

Optimization Command Examples for Voice Fallback Optimization

Before using MML commands, refer to [5.7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//This example applies to normal voice fallback, emergency voice fallback, and fast voice fallback.
//NR configurations
//Turning on VOICE_FB_FIRST_SWITCH of HoPduSessConflHandlingSw so that voice fallback (including
normal and emergency voice fallback) takes precedence over a handover procedure in case of a conflict
between them
MOD GNBMOBILITYCOMMPARAM: HoPduSessConflHandlingSw=VOICE_FB_FIRST_SWITCH-1;

//LTE configurations
//Enabling EPS fallback UE identification and delay optimization after voice fallback to E-UTRAN
MOD VOLTEALGOCONFIG: LocalCellId=1,
VolteOptSwitch=FAST_EPS_FALLBACK_SW-1&MO_VOICE_CALL_IND_SW-1, EpsFbUIActiveSchMinPeriod=5,
EpsFbUIActiveSchDataVol=80;
//Turning on the protocol procedure support switch if it is required to enable the eNodeB to include the IMS
voice EPS fallback from 5G IE in the HANDOVER REQUEST and RETRIEVE UE CONTEXT RESPONSE
messages over the X2 interface and the HANDOVER REQUEST message over the S1 interface
MOD GLOBALPROCSWITCH: ProtocolSupportSwitch=X2S1MsgWithEpsFbIndicationSw-1;
```

Deactivation Command Examples for Normal and Emergency Voice Fallback

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the voice policy switch
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=EPS_FB_SWITCH-0;
//Turning off VOICE_BLIND_MODE_SWITCH
MOD NRINTERRATHOPARAM: NrCellId=0, HoModeSwitch=VOICE_BLIND_MODE_SWITCH-0;
//((Optional) Disabling RSRP prediction based on virtual grids
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=EPS_FB_EUTRAN_RSRP_PRED_SW-0;
```

Deactivation Command Examples for Fast Voice Fallback

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off EPS_FB_IN_ADVANCE_MEAS_SWITCH
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=EPS_FB_IN_ADVANCE_MEAS_SWITCH-0;
```

5.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.7.4.2 Activation Verification

Observing Performance Counter Values

Observe certain counters. If the relevant counter value is not 0, the function has taken effect. The counters for verifying the activation of EPS fallback differ depending on the mobility policy:

- If handover is set as the mobility policy, observe the **N.HO.InterRAT.N2E.EPSFB.ExecSuccOut** counter.
- If redirection is set as the mobility policy, observe the **N.RRCRedirection.N2E.EPSFB** counter.
- If blind redirection is set as the mobility policy, observe the **N.RRCRedirection.N2E.EPSFB.Timeout** counter.
- If blind handover is set as the mobility policy, observe the **N.BlindHO.InterRAT.N2E.EPSFB.PrepAttOut** counter.

To verify EPS fallback based on RSRP prediction using virtual grid models, observe the **N.HO.InterRAT.N2E.VirtualGrid.EPSFB.PrepAttOut** counter.

Tracing Signaling

To verify the initiation of voice fallback by the gNodeB, perform the following steps:

Step 1 Start NG interface tracing for the gNodeB.

Step 2 Observe the signaling messages over the NG interface. The gNodeB initiates a voice fallback procedure if one of the following conditions is met:

- The core network sends the gNodeB a PDU SESSION RESOURCE MODIFY REQUEST message, instructing the gNodeB to set up a bearer with a 5QI of 1. The gNodeB responds to the core network with a PDU SESSION RESOURCE MODIFY RESPONSE message that contains the cause value of "IMS voice EPS fallback or RAT fallback triggered".
- The core network sends the gNodeB a PDU SESSION RESOURCE SETUP REQUEST message, instructing the gNodeB to set up a bearer with a 5QI of 1. The gNodeB responds to the core network with a PDU SESSION RESOURCE SETUP RESPONSE message that contains the cause value of "IMS voice EPS fallback or RAT fallback triggered".
- The core network sends the gNodeB an INITIAL CONTEXT SETUP REQUEST message, instructing the gNodeB to set up a bearer with a 5QI of 1. The gNodeB responds to the core network with an INITIAL CONTEXT SETUP RESPONSE message that contains the cause value of "IMS voice EPS fallback or RAT fallback triggered".

----End

To verify a measurement-based or blind handover for voice fallback from NG-RAN to E-UTRAN, perform the following steps:

- Step 1** Start Uu interface tracing and NG interface tracing for the gNodeB.
- Step 2** Observe the MobilityFromNRCommand message sent by the gNodeB to the UE over the Uu interface. If the value of the targetRAT-Type IE in the message is "eutra", the preparation for the handover from NG-RAN to E-UTRAN has succeeded.
- Step 3** Observe the signaling messages over the NG interface. If either (i) the gNodeB receives a UE CONTEXT RELEASE COMMAND message with the cause value "Successful Handover" or "Normal Release" from the AMF, or (ii) the gNodeB receives an RRC Reestablishment Complete message from the UE and the RRC connection reestablishment is successful, the handover from NG-RAN to E-UTRAN has been successfully executed.

----End

To verify a measurement-based or blind redirection for voice fallback from NG-RAN to E-UTRAN, perform the following steps:

- Step 1** Start Uu interface tracing for the gNodeB.
- Step 2** Observe the RRCRelease message sent by the gNodeB to the UE over the Uu interface. If the redirectedCarrierInfo IE in the message contains the RedirectedCarrierInfo-EUTRA IE, the redirection or blind redirection from NG-RAN to E-UTRAN has succeeded.

----End

5.7.4.3 Network Monitoring

Monitor the counters listed in [Table 5-26](#) to evaluate the performance of normal voice fallback.

Table 5-26 Counters used to monitor the performance of normal voice fallback

Counter ID	Counter Name
1911817066	N.HO.InterRAT.N2E.EPSFB.ExecSuccOut
1911817067	N.HO.InterRAT.N2E.EPSFB.ExecAttOut
1911817069	N.HO.InterRAT.N2E.EPSFB.PrepAttOut
1911822808	N.HO.InterRAT.N2E.EPSFB.AMFAbnormRsp
1911822811	N.Rcv.VoiceFB.Trig
1911822810	N.VoiceFB.RespSucc
1911817084	N.RRCRedirection.N2E.EPSFB
1911820697	N.HO.InterRAT.NCell.VoiceFB.PrepAttOut
1911820698	N.HO.InterRAT.NCell.VoiceFB.ExecAttOut
1911820699	N.HO.InterRAT.NCell.VoiceFB.ExecSuccOut
1911820702	N.HO.InterRAT.N2E.EPSFB.Prep.FailOut.AMF

Counter ID	Counter Name
1911820701	N.HO.InterRAT.N2E.EPSFB.Prep.FailOut.PrepFailure
1911820700	N.HO.InterRAT.N2E.EPSFB.Prep.FailOut.NoReply
1911820703	N.RRCRedirection.N2E.EPSFB.Timeout
1911827211	N.BlindHO.InterRAT.N2E.EPSFB.PrepAttOut
1911827209	N.BlindHO.InterRAT.N2E.EPSFB.ExecAttOut
1911827210	N.BlindHO.InterRAT.N2E.EPSFB.ExecSuccOut
1911831612	N.CallFB.HO.Duration.Avg
1911831611	N.CallFB.RRCRedirection.Duration.Avg

Monitor the counters listed in [Table 5-27](#) to evaluate the performance of emergency voice fallback.

Table 5-27 Counters used to monitor the performance of emergency voice fallback

Counter ID	Counter Name
1911820687	N.Emc.FB.PrepAtt
1911820688	N.Emc.FB.PrepSucc
1911820689	N.HO.InterRAT.N2E.Emc.EPSFB.PrepAttOut
1911820690	N.HO.InterRAT.N2E.Emc.EPSFB.ExecAttOut
1911820691	N.HO.InterRAT.N2E.Emc.EPSFB.ExecSuccOut
1911820686	N.RRCRedirection.N2E.Emc.EPSFB
1911831612	N.CallFB.HO.Duration.Avg
1911831611	N.CallFB.RRCRedirection.Duration.Avg

Monitor the counters listed in [Table 5-26](#) and [Table 5-27](#) to evaluate the performance of fast voice fallback.

Monitor the counters listed in [Table 5-28](#) to evaluate the performance of EPS fallback with RSRP predicted based on virtual grids.

Table 5-28 Counters used to monitor the performance of EPS fallback with RSRP predicted based on virtual grids

Counter ID	Counter Name
1911831561	N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecSuccOut
1911831562	N.HO.InterRAT.N2E.VirtualGrid.EPSFB.ExecAttOut
1911831563	N.HO.InterRAT.N2E.VirtualGrid.EPSFB.PrepAttOut

5.8 Fast Return from E-UTRAN to NG-RAN

5.8.1 Principles

Fast return from E-UTRAN to NG-RAN enables 5G UEs to quickly return to NG-RAN from E-UTRAN.

Fast return can be implemented through blind redirection, measurement-based handover, and measurement-based redirection. **If blind redirection, measurement-based handover, and measurement-based redirection are all enabled, blind redirection is preferred, followed by measurement-based handover and then measurement-based redirection.**

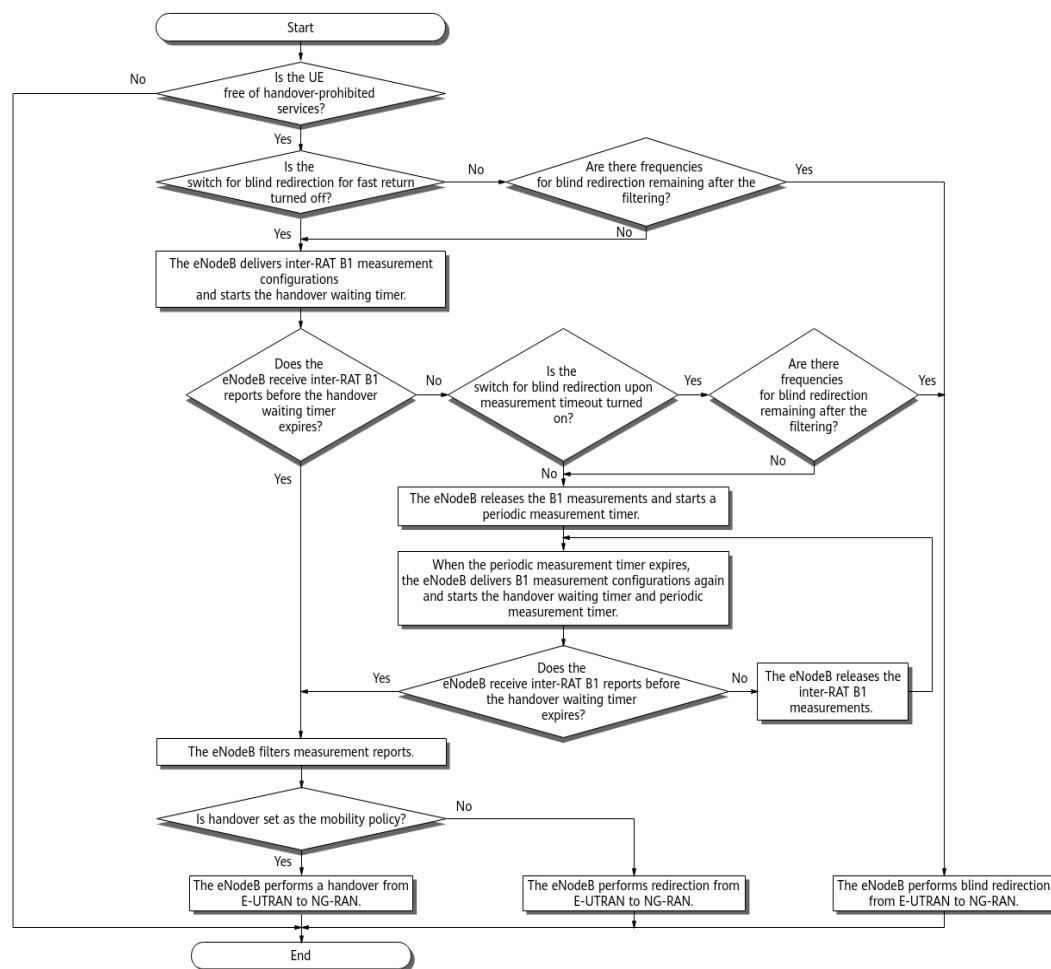
- For details about the overall procedure of fast return, see [Procedure](#).
- If the eNodeB determines to select blind redirection, it performs the procedure as described in [Blind Redirection](#).
- If the eNodeB determines to select measurement-based handover or measurement-based redirection, it performs the procedure as described in [Measurement-based Handover or Redirection](#).
- If non-gap-assisted measurement is used, the eNodeB performs the procedure as described in [Procedure with Non-Gap-assisted Measurements](#).
- If the eNodeB determines to select handover with RSRP predicted based on virtual grids, it performs the procedure as described in [Procedure with RSRP Predicted Based on Virtual Grids](#).
- If the eNodeB receives measurement reports on NSA-DC-related event B1 from the UE, it performs the procedure as described in [NSA DC-based Fast NR Handover](#).
- If the anti-ping-pong mechanism for voice services is enabled, the eNodeB performs the procedure as described in [Anti-Ping-Pong Handling for Voice Services](#).

This function is controlled by the **FAST_RETURN_TO_NR_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter.

Procedure

[Figure 5-12](#) shows the fast return procedure.

Figure 5-12 Procedure for fast return



The procedure is as follows:

1. A UE releases its data or voice services in an LTE network.
If the UE was handed over from an NR cell to the LTE cell for resource optimization reasons and the timer specified by the **CellHoParaCfg.NrResOptHoInProtectTmr** parameter is not set to 0, the procedure varies as follows:
 - Before the timer expires, the voice service (QCI 1) release triggers fast return from E-UTRAN to NG-RAN, but the data service release does not.
 - After the timer expires, both the voice service release and the data service release will trigger fast return from E-UTRAN to NG-RAN.
2. The eNodeB checks the **ServiceIrHoCfgGroup.InterRatHoState** parameter setting for each QCI of services running on the UE to evaluate whether the UE is allowed to return to NG-RAN.
 - When NSA/SA selection based on user experience is enabled (by selecting any of the **LTE_FDD_NSA_SA_DL_SEL_OPT_SW**, **LTE_TDD_NSA_SA_DL_SEL_OPT_SW**, and **LTE_FDD_NSA_SA_UL_SEL_OPT_SW** options of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter):

- If flexible user steering is also enabled, the procedure varies depending on the **NSA_SA_SEL_OPT_SW** setting of the **SpidCfg.SpIdAlgoSwitch** parameter. If this option is selected, the procedure goes to the next step only when the UE has handover-required services (as indicated by the value **MUST_HO**); otherwise, the procedure goes to the next step as long as the UE does not have any handover-prohibited service (as indicated by the value **NO_HO**).
 - If flexible user steering is not enabled, the procedure goes to the next step only when the UE has handover-required services (as indicated by the value **MUST_HO**).
 - When NSA/SA selection based on user experience is disabled, the procedure goes to the next step if the UE does not have any handover-prohibited service (as indicated by the value **NO_HO**). Otherwise, the procedure ends.
3. The eNodeB determines whether to select blind redirection, measurement-based handover, measurement-based redirection, or handover with RSRP predicted based on virtual grids for mobility to NG-RAN for the UE based on the setting of the **CellHoParaCfg.HoModeSwitch** parameter. Blind redirection is preferred.
- If the eNodeB determines to select blind redirection, it performs the procedure as described in **Blind Redirection**.
 - If the eNodeB determines to select measurement-based handover or measurement-based redirection, it performs the procedure as described in **Measurement-based Handover or Redirection**.
 - If the eNodeB determines to select handover with RSRP predicted based on virtual grids, it performs the procedure as described in **Procedure with RSRP Predicted Based on Virtual Grids**.

The mobility policy for the UE is determined by the intersection set of the **PS_HO** and **REDIRECTION** options of the **InterRatPolicyCfgGroup.NrHoCfg** parameter in the inter-RAT policy groups mapped to the QCLs of services on the UE. In addition, this mobility policy must be consistent with the setting of the **CellHoParaCfg.HoModeSwitch** parameter. The relationships between QCLs and inter-RAT policy groups are defined in the **QciPara** MO.

Blind Redirection

During frequency selection, the eNodeB filters out the frequencies whose blind redirection priority (specified by the **NrNFreq.BlindRedirectionPriority** parameter) is **0**.

- If there are frequencies remaining, the eNodeB selects the frequency with the highest blind redirection priority and performs a blind redirection to that frequency. If multiple frequencies have the same priority, the eNodeB randomly selects a frequency for the blind redirection.
- If there are no frequencies remaining, the eNodeB performs a measurement-based handover or redirection.

To enable blind redirection, select both the **FastReturnBlindRedirectSwitch** and **NrRedirectSwitch** options of the **CellHoParaCfg.HoModeSwitch** parameter.

Measurement-based Handover or Redirection

If the eNodeB determines to select measurement-based handover or measurement-based redirection to NG-RAN, it performs the following procedure:

1. The eNodeB delivers inter-RAT measurement configurations and starts a handover waiting timer.

The eNodeB delivers the measurement configurations related to inter-RAT event B1 to the UE. In the measurement configurations, **NR frequencies** are arranged in descending order of **priority**. (A larger value of the NR frequency priority parameter indicates a higher priority. The value **0** indicates the lowest priority.) At the same time, the eNodeB starts a **handover waiting timer**, and waits for and receives NR B1 measurement reports during the period of time specified by this timer.

Inter-RAT event B1 involved in this function is service-based inter-RAT event B1. This event can be triggered only when the signal quality of an inter-RAT neighboring cell exceeds a specific threshold (**Thresh**) throughout a period of time (**TimeToTrig**). For details about event B1, see [4.1.4.2.1 Measurement Events](#).

NOTE

Measurements related to event B1 involved in service-based mobility are not terminated by event A1 involved in coverage-based inter-RAT mobility from E-UTRAN to NG-RAN.

When the **SERV_BASED_FR_FORBID_SEL_FLAG** option of the **NrNFreq.AggregationAttribute** parameter is selected for an NR frequency, this NR frequency will be filtered out from the target frequencies for measurement, handovers, and redirections (including blind redirections) related to fast return from E-UTRAN to NG-RAN.

2. The eNodeB checks whether the handover waiting timer has expired.
 - If the eNodeB has received inter-RAT event B1 measurement reports before the handover waiting timer expires, the eNodeB filters measurement reports. For details about the rules for filtering measurement reports, see [4.1.8 Target Cell or Frequency Evaluation](#). If there are cells remaining after the filtering, the eNodeB sends a handover request to the target cell or sends a redirection request to the operating frequency of the target cell.
 - If the eNodeB has not received inter-RAT event B1 measurement reports when the handover waiting timer expires, it takes the following actions:
 - In the case of a measurement-based handover, the eNodeB releases B1 measurement configurations and starts a periodic measurement timer. After the periodic measurement timer expires, the eNodeB delivers the measurement configurations related to event B1 again and starts the periodic measurement timer and handover waiting timer again. Then, the eNodeB goes to [2](#).
 - In the case of a measurement-based redirection:
 - If **blind redirection upon measurement timeout** is enabled, the eNodeB filters frequencies, selects the one with the highest **blind redirection priority** from the remaining frequencies, and sends a blind redirection request to the UE. Then the eNodeB performs a blind redirection from E-UTRAN to NG-RAN on the UE, and the UE returns to NG-RAN.

- In other cases, the eNodeB releases B1 measurement configurations and starts a service-based NR measurement timer (also named **periodic measurement timer**). After the periodic measurement timer expires, the eNodeB delivers the measurement configurations related to event B1 again and starts the periodic measurement timer and handover waiting timer again. Then, the eNodeB goes to 2.
3. The eNodeB performs a handover or redirection from E-UTRAN to NG-RAN on the UE.
- In case the handover preparation or execution fails and the RRC connection is reestablished with the source eNodeB, the eNodeB selects the operating frequency of the target cell of the handover for a redirection if **switching to a redirection from a handover in case of a handover failure** is enabled and the **handover mode** is set to redirection.

The following table lists the parameters involved in the procedure.

Configuration Item	Parameter
Handover mode	Measurement-based handover: NrHoSwitch option of the CellHoParaCfg.HoModeSwitch parameter Measurement-based redirection: NrRedirectSwitch option of the CellHoParaCfg.HoModeSwitch parameter
NR frequency priority	NrNFreq.ConnFreqPriority
Handover waiting timer	CellHoParaCfg.ServBasedNrB1RptWaitingTmr
Thresh	InterRatHoNrParamGrp.ServBasedNrB1RsrpThld
TimeToTrig	InterRatHoNrParamGrp.FastReturnB1TimeToTrigger
Blind redirection upon measurement timeout	MeasTimeoutToBlindRedirSw option of the GlobalProcSwitch.ProtocolSupportSwitch parameter
Blind redirection priority	NrNFreq.BlindRedirectionPriority
Periodic measurement timer	CellHoParaCfg.ServBasedNrMeasTimer
Switching to a redirection from a handover in case of a handover failure	HoFailureToRedirSw option of the GlobalProcSwitch.ProtocolSupportSwitch parameter

In RAN sharing with common carrier scenarios, the mobility policy (handover or redirection) for a cell can be set on a per operator basis using the **CellOpHoCfg**

MO. If no operator-specific mobility policy is set, the mobility policy for the cell is determined by the **CellHoParaCfg.HoModeSwitch** parameter.

- To enable handover, select both the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter and the **HANDOVER_TO_NR_SW** option of the **CellOpHoCfg.HoModeSwitch** parameter.
- To enable redirection, select both the **NrRedirectSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter and the **REDIRECTION_TO_NR_SW** option of the **CellOpHoCfg.HoModeSwitch** parameter.

Procedure with Non-Gap-assisted Measurements

The eNodeB allows non-gap-assisted B1 measurements on NR frequencies that support the non-gap-assisted measurement function, if the **NO_GAP_NR_B1_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter is selected.

With the non-gap-assisted B1 measurement function enabled, fast return from E-UTRAN to NG-RAN works as follows if the eNodeB has not received inter-RAT event B1 measurement reports and the QCI of services running on the UE have not changed when the handover waiting timer (specified by the **CellHoParaCfg.ServBasedNrB1RptWaitingTmr** parameter) expires as in [2 of Measurement-based Handover or Redirection](#):

1. The eNodeB releases the B1 measurements and starts the measurement timer related to service-based handovers from E-UTRAN to NG-RAN. This timer is referred to as periodic measurement timer for short. It is specified by the **CellHoParaCfg.ServBasedNrMeasTimer** parameter.
2. The eNodeB checks whether non-gap-assisted measurement is applicable to any of the configured NR frequencies. It is applicable if the UE complies with Release 16 or an NR frequency meets the conditions described in [4.1.5.2 Non-Gap-assisted B1 Measurement Configuration by the eNodeB](#).
 - If non-gap-assisted measurement is applicable to certain NR frequencies, the eNodeB delivers B1 measurement configurations about these frequencies to the UE. These measurement configurations are the same as the measurement configurations related to event B1 for service-based inter-RAT mobility. The eNodeB releases the delivered non-gap-assisted measurement configurations when any of the following situations occurs:
 - EN-DC band combinations or CA carriers are added or removed. In this situation, the eNodeB needs to query the non-gap-assisted measurement capabilities of NR frequencies again. If a frequency is changed from being capable of non-gap-assisted measurement to being incapable, the eNodeB releases the non-gap-assisted B1 measurement configurations about this frequency. Otherwise, the eNodeB does not release non-gap-assisted B1 measurement configurations.
 - E-UTRAN to NG-RAN traffic steering or fast return is triggered by a UE service change.
 - The eNodeB delivers the measurement configurations related to event B1 for service-based inter-RAT mobility as triggered by expiration of the periodic measurement timer.

- If non-gap-assisted measurement is not applicable to any frequency, the eNodeB does not deliver non-gap-assisted measurement configurations to the UE. The procedure with non-gap-assisted measurements ends.
- 3. The eNodeB takes different actions, depending on whether it has received inter-RAT event B1 measurement reports before releasing non-gap-assisted B1 measurement configurations.
 - If the eNodeB has received inter-RAT event B1 measurement reports, it delivers the target cell or frequency to the UE, instructing the UE to perform a handover or redirection according to the mobility policy. The procedure ends.
 - If the eNodeB has not received inter-RAT event B1 measurement reports, it goes to the next step.
- 4. After the periodic measurement timer expires, the eNodeB delivers the measurement configurations related to inter-RAT event B1 again and starts the handover waiting timer and periodic measurement timer.
- 5. The eNodeB takes different actions, depending on whether it has received inter-RAT event B1 measurement reports before the handover waiting timer expires.
 - If the eNodeB has received inter-RAT event B1 measurement reports before the handover waiting timer expires, it delivers the target cell or frequency to the UE, instructing the UE to perform a handover or redirection according to the mobility policy.
 - If the eNodeB has not received inter-RAT event B1 measurement reports when the handover waiting timer expires, it releases the B1 measurements and goes back to 2.

Procedure with RSRP Predicted Based on Virtual Grids

During evaluation of fast return from E-UTRAN to NG-RAN, use of LN RSRP models can replace UE measurement of RSRP if the **NR_HO_WITH_VIRTUAL_GRID_SW** option of the **LnrCarrierSelection.LnrCarrierSelectionSw** parameter is selected.

This function takes effect only in handover scenarios. It requires the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter to be selected. In RAN sharing with common carrier scenarios, it requires both the **NrHoSwitch** option of the **CellHoParaCfg.HoModeSwitch** parameter and the **HANDOVER_TO_NR_SW** option of the **CellOpHoCfg.HoModeSwitch** parameter to be selected.

The procedure is as follows:

1. For a UE capable of NR SA access (including initial access in idle mode, incoming handovers, and incoming RRC connection reestablishments), the eNodeB delivers A3 measurement configurations about the serving frequency with a measurement period of 2s to the UE.
2. After the UE sends A3 measurement reports to the eNodeB, the eNodeB uses the reported RSRP values of the serving cell and intra-frequency neighboring cells to query coverage information about candidate NR frequencies. In the A3 measurement configurations, the time-to-trigger for event A3 is specified by the **LnrCarrierSelection.IntraFreqMeasA3TimeToTrig** parameter.

The eNodeB performs this step if it finds that the **PS_HO** option of the **InterRatPolicyCfgGroup.NrHoCfg** parameter is selected for all QCI of

services running on the UE. If **PS_HO** is not selected, RSRP cannot be predicted based on virtual grids.

3. When LN RSRP models are available on the eNodeB and the model accuracy meets the requirement, the eNodeB uses the models to predict the RSRP for the corresponding neighboring NR frequencies, without the need for UE measurements. The accuracy requirement is specified by the **LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld** parameter.

To use virtual grid models, ensure that the

LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld parameter is not set to **ACCURACY100_OFFSET0**. If it is set to this value, RSRP is not predicted using virtual grid models for unnecessary handovers.

- If **LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld** is set to a value other than **ACCURACY100_ENH**:
 - If the prediction result for a candidate neighboring NR frequency meets the entering condition for the corresponding event B1, the eNodeB instructs the UE to perform a handover from E-UTRAN to NG-RAN.
 - If the prediction results for all candidate neighboring NR frequencies meet the leaving condition for the corresponding event B1, the eNodeB does not deliver B1 measurement configurations related to these frequencies.
 - If the preceding conditions are not met, the measurement-based handover procedure described in [Measurement-based Handover or Redirection](#) continues.
- If **LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld** is set to **ACCURACY100_ENH**:
 - If the prediction results for all candidate neighboring NR frequencies meet the leaving condition for the corresponding event B1, the eNodeB does not deliver B1 measurement configurations related to these frequencies.
 - In other scenarios, the eNodeB delivers B1 measurement configurations related to neighboring NR frequencies to the UE. After receiving B1 measurement reports from the UE, the eNodeB instructs the UE to perform a handover from E-UTRAN to NG-RAN.

NSA DC-based Fast NR Handover

With NSA DC-based fast NR handover in effect in NSA and SA hybrid networking, the eNodeB can use NSA DC event B1 measurement configurations instead of NR event B1 measurement configurations during handover evaluation for an NSA/SA dual-mode UE. NSA DC-based fast NR handover is controlled by the **NR_B1_FAST_HO_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter. For details about NSA DC-based fast NR handover, see [Procedure with NSA-DC-related Measurements](#).

Anti-Ping-Pong Handling for Voice Services

When a UE initiates a voice service in E-UTRAN, it may be handed over to NG-RAN before the voice service is set up, and then fall back to E-UTRAN by EPS

fallback. To reduce the probability of this issue, operators can set the **CellHoParaCfg.VolteHoNrDelayTimer** parameter to a non-zero value and select the **FAST_RETURN_HO_OPT_SW** option of the **CellAlgoExtSwitch.HoAllowedSwitch** parameter.

5.8.2 Network Analysis

5.8.2.1 Benefits

With fast return, UEs that meet certain requirements are handed over or redirected to NR cells upon release of their voice bearers. It takes less time for the UEs to return to NR networks than in the case fast return is disabled.

When the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter value is greater than the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, NSA-DC-based fast NR handover offers the following benefits:

- The number of SA UEs in the NR cell increases, which can be observed by subtracting the **N.User.NsaDc.PSCell.Avg** counter value from the **N.User.RRConn.Avg** counter value.
- The total throughput of SA UEs in the NR cell increases, which can be observed by subtracting the **N.NSA.ThpVol.DL** counter value from the **N.ThpVol.DL** counter value.

When the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter value is greater than the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, a greater difference between them leads to larger benefits. When the **InterRatHoNrParamGrp.NrB1B2TimeToTrigger** parameter value is less than or equal to the **NrScgFreqConfig.NrB1TimeToTrigger** parameter value, NSA-DC-based fast NR handover does not produce any benefits.

5.8.2.2 Impacts

The impacts between this function and other functions are described in [5.8.3.2.3 Function Impacts](#).

The function has no impact on the network.

5.8.3 Requirements

5.8.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
LTE FDD	LNOFD-151335	Fast Return from E-UTRAN to NG-RAN	LT1SFREU NG00	Fast Return from E-UTRAN to NG-RAN (LTE FDD)	eNodeB	per Cell
LTE TDD	TDLNOFD-151503	Fast Return from E-UTRAN to NG-RAN	LT4SENFT RTDD	Fast Return from E-UTRAN to NG-RAN (LTE TDD)	eNodeB	per Cell

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation.

5.8.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.8.3.2.1 Prerequisite Functions

None

5.8.3.2.2 Mutually Exclusive Functions

None

5.8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	NSA/SA selection based on downlink traffic volume	LTE_FDD_NSA_SA_DL_SEL_OPT_SW, LTE_TDD_NSA_SA_DL_SEL_OPT_SW, or LTE_FDD_NSA_SA_UL_SEL_OPT_SW option of the EnodeBAlgorithmExtSwitch.MultiNetworkingOptionOptSw parameter	NSA-SA Selection Based on User Experience	After NSA/SA selection based on user experience is enabled, the evaluation of fast return from E-UTRAN to NG-RAN is changed. (For details, see 5.8.1 Principles .) As a result, the probability that fast return from E-UTRAN to NG-RAN takes effect decreases, and the value of the L.IRATHO.E2N.FastReturn.PrepareAttOut counter decreases.
LTE FDD	NSA/SA selection based on uplink coverage	LTE_FDD_NSA_SA_UL_SEL_OPT_SW option of the EnodeBAlgorithmExtSwitch.MultiNetworkingOptionOptSw parameter	NSA-SA Selection Based on User Experience	

5.8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For LTE, all LTE-capable boards support this function.
- For NR, all NR-capable boards support this function.

RF Modules

No requirements

5.8.3.4 Others

- UE requirements
UEs must comply with 3GPP Release 15 or later. To support fast return from E-UTRAN to NG-RAN for data services, the corresponding fields in the following table must be carried simultaneously based on the target NR cell situation.

Frequency Band/RAT	TDD	FDD
FR1	• eutra-EPC-HO-ToNR-TDD-FR1-r15 • sa-NR-r15	• eutra-EPC-HO-ToNR-FDD-FR1-r15 • sa-NR-r15
FR2	• eutra-EPC-HO-ToNR-TDD-FR2-r15 • sa-NR-r15	• eutra-EPC-HO-ToNR-FDD-FR2-r15 • sa-NR-r15

To support fast return from E-UTRAN to NG-RAN for voice services, the corresponding fields in the following table must be carried simultaneously based on the target NR cell situation.

Frequency Band/RAT	TDD	FDD
FR1	• eutra-EPC-HO-ToNR-TDD-FR1-r15 • sa-NR-r15 • ims-VoiceOverNR-FR1-r15	• eutra-EPC-HO-ToNR-FDD-FR1-r15 • sa-NR-r15 • ims-VoiceOverNR-FR1-r15
FR2	• eutra-EPC-HO-ToNR-TDD-FR2-r15 • sa-NR-r15 • ims-VoiceOverNR-FR2-r15	• eutra-EPC-HO-ToNR-FDD-FR2-r15 • sa-NR-r15 • ims-VoiceOverNR-FR2-r15

- Core network requirements
 - The core networks must comply with 3GPP Release 15 or later.
 - Handover requires the N26 interface to be configured between the MME and AMF.

5.8.4 Operation and Maintenance

5.8.4.1 Data Configuration

5.8.4.1.1 Data Preparation

Table 5-29 describes the parameters related to fast return from E-UTRAN to NG-RAN. For details about basic parameters for interoperability from E-UTRAN to NG-RAN, see **4.5.1.1 Data Preparation** and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-29 Parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
Handover Allowed Switch	CellAlgoExtSwitch.HoAllowedSwitch	• To enable fast return from E-UTRAN to NG-RAN, select the FAST_RETURN_TO_NR_SW option. • To enable mobility based on non-gap-assisted measurement reports, select the NO_GAP_NR_B1_SW option.

Parameter Name	Parameter ID	Setting Notes
Handover Mode switch	CellHoParaCfg.HoModeSwitch	- To enable measurement-based handover, select the NrHoSwitch option of the CellHoParaCfg.HoModeSwitch parameter. - To enable measurement-based redirection, select the NrRedirectSwitch option of the CellHoParaCfg.HoModeSwitch parameter. - To enable blind redirection, select both the FastReturnBlindRedirectSwitch and NrRedirectSwitch options of the CellHoParaCfg.HoModeSwitch parameter.
Protocol Procedure Support Switch	GlobalProcSwitch.ProtocolSupportSwitch	Select the following options based on the network plan: - MeasTimeoutToBlindRedirSw - HoFailureToRedirSw
LNR Carrier Selection Switch	LnrCarrierSelection.LnrCarrierSelectionSw	To use the virtual grid technology for fast return from E-UTRAN to NG-RAN, select the NR_HO_WITH_VIRTUAL_GRID_SW option.
Service-based NR B1 Report Waiting Timer	CellHoParaCfg.ServBasedNrB1RptWaitingTmr	The default value is recommended.

Parameter Name	Parameter ID	Setting Notes
Intra-Freq Measurement A3 Time-to-Trigger	LnrCarrierSelection.IntraFreqMeasA3TimeToTrigger	The default value is recommended.
Pred Unnecessary NR Ho Meas RSRP Thld	LnrCarrierSelection.PredUnnecNrHoMeasRsrpThld	Set this parameter to a value other than ACCURACY100_OFFSET0 .
Inter-RAT handover state	ServiceIrrHoCfgGroup.InterRatHoState	It is recommended that this parameter be set to NO_HO for QCI 1 services to prevent voice service interruption or user experience deterioration in the following scenarios: • Service-based handovers from LTE cells to NR cells for voice service UEs. In this scenario, voice services may be interrupted due to handover failures. • Service-based redirections from LTE cells to NR cells for voice service UEs. In this scenario, voice service experience may deteriorate due to a long time of user-plane interruption. The parameter value for other QCIs varies depending on operators' mobility policies. If a UE is running services with different QCIs, the mobility policy for the service with the highest-priority QCI takes effect for the UE.

Parameter Name	Parameter ID	Setting Notes
Fast Return B1 Time-to-Trigger	InterRatHoNrParamGrp.FastReturnB1TimeToTrigger	The default value is recommended.
No Handover Flag	InterRatPolicyCfgGroup.NoHoFlag	The value PERMIT_HO_ENUM is recommended.
NR Handover Policy Configuration	InterRatPolicyCfgGroup.NrHoCfg	- To enable handover, select the PS_HO option. - To enable redirection, select the REDIRECTION option.
NR Res Opt Handover-in Protect Timer	CellHoParaCfg.NrResOptHoInProtectTmr	This parameter specifies the length of the protection timer for UEs that have been handed over from NR cells to the LTE cell for resource optimization reasons. Before the timer expires, the non-QCI-1 service release does not trigger fast return from E-UTRAN to NG-RAN. If this parameter is set to 0 or a value greater than 100, the timer does not take effect.
Aggregation Attribute	NrNFreq.AggregationAttribute	If an NR frequency needs to be filtered out from the target frequencies for fast return from E-UTRAN to NG-RAN, select the SERV_BASED_FR_FORBID_SEL_FLAG option of this parameter for this frequency.

5.8.4.1.2 Using MML Commands

The basic configurations for mobility from E-UTRAN to NG-RAN must have been completed before activating this function. For details, see [4.5.1.2 Using MML](#)

Commands and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples

Before using MML commands, refer to [5.8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Turning on FAST_RETURN_TO_NR_SW and NO_GAP_NR_B1_SW
MOD CELLALGOEXTSWITCH: LocalCellId=0,
HoAllowedSwitch=FAST_RETURN_TO_NR_SW-1&NO_GAP_NR_B1_SW-1;
//Enabling handover and redirection
MOD CELLHOPARACFG: LocalCellId=0, HoModeSwitch=NrHoSwitch-1&NrRedirectSwitch-1;
//Optional) To use RSRP prediction based on virtual grids, run the following command:
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=NR_HO_WITH_VIRTUAL_GRID_SW-1,
IntraFreqMeasA3TimeToTrig=320MS, PredUnnecNrHoMeasRsrpThld=ACCURACY90_OFFSET0;
//Setting QCI-specific policies
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=9, ServiceHoNrPolicyGroupId=2;
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=5, ServiceHoNrPolicyGroupId=1;
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=1, ServiceHoNrPolicyGroupId=0;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=0, InterRatHoState=NO_HO;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=1, InterRatHoState=PERMIT_HO;
MOD SERVICEIRHOCFGGROUP: CnOperatorId=1, ServiceIrHoCfgGroupId=2, InterRatHoState=MUST_HO;
MOD CELLQCIPARA: LocalCellId=0, Qci=9, NrHoParamGroupId=1;
MOD CELLQCIPARA: LocalCellId=0, Qci=5, NrHoParamGroupId=1;
MOD CELLQCIPARA: LocalCellId=0, Qci=1, NrHoParamGroupId=1;
//Changing the setting of ServBasedNrB1RptWaitingTmr
MOD CELLHOPARACFG: LocalCellId=0, ServBasedNrB1RptWaitingTmr=3;
//Setting an inter-RAT policy group
MOD INTERRATPOLICYCFGGROUP: InterRatPolicyCfgGroupId=3, NoHoFlag=PERMIT_HO_ENUM,
NrHoCfg=PS_HO-1&REDIRECTION-1;
//Binding the group to QCIs
MOD QCIPARA: Qci=5, InterRatPolicyCfgGroupId=3;
MOD QCIPARA: Qci=9, InterRatPolicyCfgGroupId=3;
//Optional; to enable blind redirection to NG-RAN for fast return in case of handover failures or
measurement timeout) Turning on HoFailureToRedirSw, MeasTimeoutToBlindRedirSw, or both
MOD GLOBALPROCSWITCH: ProtocolSupportSwitch=HoFailureToRedirSw-1&MeasTimeoutToBlindRedirSw-1;
//Setting NR Res Opt Handover-in Protect Timer
MOD CELLHOPARACFG: LocalCellId=0, NrResOptHoInProtectTmr=10;
//Optional, when an NR frequency needs to be filtered out for fast return from E-UTRAN to NG-RAN)
Prohibiting a frequency from being selected for fast return and service-based mobility
MOD NRNFREQ: LocalCellId=0, DlArfcn=428000,
AggregationAttribute=SERV_BASED_FR_FORBID_SEL_FLAG-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch of fast return from E-UTRAN to NG-RAN
MOD CELLALGOEXTSWITCH: LocalCellId=0, HoAllowedSwitch=FAST_RETURN_TO_NR_SW-0;
//Optional) To disable RSRP prediction based on virtual grids, run the following command:
MOD LNRCARRIERSELECTION: LnrCarrierSelectionSw=NR_HO_WITH_VIRTUAL_GRID_SW-0;
```

5.8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.8.4.2 Activation Verification

Observing Performance Counter Values

- Fast return from E-UTRAN to NG-RAN
Observe the values of the **L.IRATHO.E2N.FastReturn.ExecSuccOut** and **L.RRCRedirection.E2N.FastReturn** counters. If either of the values is not 0, this function has taken effect.
- Virtual-grid-based fast return from E-UTRAN to NG-RAN
When the prediction results for a neighboring NR frequency meet the leaving condition for event B1, the eNodeB does not deliver measurement configurations related to this frequency. This can be observed using the **L.VirtualGrid.Model.NrFreq.InvalidMeasFree.Times** counter. If this counter produces a non-zero value, the virtual grid function has taken effect.

Tracing Signaling

To verify handover-based fast return from E-UTRAN to NG-RAN, perform the following steps:

- Step 1** Start Uu interface tracing and S1 interface tracing for the eNodeB.
- Step 2** Observe the MobilityFromEUTRACommand message sent by the eNodeB to the UE over the Uu interface. If the value of the *targetRAT-Type* IE in the message is "nr", the preparation for the handover from E-UTRAN to NG-RAN has succeeded.
- Step 3** Observe the signaling messages over the S1 interface. If either (i) the eNodeB receives a UE CONTEXT RELEASE COMMAND message with the cause value "Successful Handover" or "Normal Release" from the MME, or (ii) the eNodeB receives an RRC Connection Reestablishment Complete message from the UE and the RRC connection reestablishment is successful, the handover from E-UTRAN to NG-RAN has been successfully executed.

----End

To verify redirection-based fast return from E-UTRAN to NG-RAN, perform the following steps:

- Step 1** Start Uu interface tracing for the eNodeB.
- Step 2** Check whether the RRCConnectionRelease message sent by the eNodeB contains the *nr-r15* IE. If it does, the redirection from E-UTRAN to NG-RAN is successful.

----End

5.8.4.3 Network Monitoring

Monitor the counters listed in [Table 5-30](#) to evaluate the performance of this function.

Table 5-30 LTE counters

Counter ID	Counter Name
1526756890	L.IRATHO.E2N.FastReturn.PrepAttOut
1526756889	L.IRATHO.E2N.FastReturn.ExecAttOut
1526756891	L.IRATHO.E2N.FastReturn.ExecSuccOut
1526756892	L.RRCRedirection.E2N.FastReturn
1526779679	L.IRATHO.E2N.PrepAttOut.RedCap
1526779677	L.IRATHO.E2N.ExecAttOut.RedCap
1526779678	L.IRATHO.E2N.ExecSuccOut.RedCap

5.9 VoNR-to-VoLTE Handover Based on Voice Quality

5.9.1 Principles

If the quality of voice over NR (VoNR) does not meet expectation, the gNodeB can identify such UEs and trigger inter-RAT measurements and handovers for them. This reduces the packet loss rate and improves user experience.

This function requires the following options to be selected:

- **MOBILITY_TO_EUTRAN_SW** option of the **NRCCellAlgoSwitch**.*InterRatServiceMobilitySw* parameter
- **QLTY_BASED_EUTRAN_HO_SW** option of the **gNBQciAlgoParamGrp**.*QciAlgoSwitch* parameter

The procedure is as follows:

1. The gNodeB checks the VoNR quality.
If the uplink or downlink packet loss rate of 5QI 1 services is greater than the value of the **gNBQciAlgoParamGrp**.*ServQltyEutranHoPlrThld* parameter throughout the VoNR packet loss rate evaluation period specified by the **gNBQciAlgoParamGrp**.*ServPlrEvalPeriod* parameter, the gNodeB determines that the VoNR quality is poor. The gNodeB then goes to the next step.
For details about the formulas for calculating the uplink and downlink packet loss rates of 5QI 1 services, see *VoNR* in *5G RAN Feature Documentation*.
2. The gNodeB delivers the measurement configurations related to event B1 used for mobility to E-UTRAN.
 - a. The gNodeB filters out LTE frequencies with the **NRCCellEutranNFreq**.*VoltePriority* parameter set to **255**. If no LTE

frequency is remaining after the filtering, the gNodeB does not deliver the B1 measurement configurations. The procedure ends.

- b. The gNodeB delivers the B1 measurement configurations. In the measurement configurations, the remaining LTE frequencies are arranged based on the VoLTE priority specified by the **NRCCellEutranNFreq.VoltePriority** parameter. For details about the measurement event triggering quantities supported by VoNR-to-VoLTE handover based on voice quality, see [Table 4-18](#). For details about event B1 related to VoNR-to-VoLTE handover based on voice quality, see [Table 4-12](#).

For details about the measurement configuration delivery procedure, see [4.1.4 Measurement Configuration](#).

3. After receiving measurement reports on event B1 from the UE, the gNodeB evaluates a handover based on the target cell evaluation procedure. For details about the target cell evaluation procedure, see [4.1.8 Target Cell or Frequency Evaluation](#).
4. The UE executes the handover.

In the preceding procedure:

- If the gNodeB determines that the VoNR quality has recovered:
 - The gNodeB releases the B1 measurements if the measurement configurations related to event B1 used for mobility to E-UTRAN have been delivered.
 - The gNodeB continues with the handover preparation or execution if the gNodeB has entered the handover preparation or execution phase.

The gNodeB determines that the VoNR quality has recovered when both the following conditions are met:

- Uplink packet loss rate of 5QI 1 services $\leq$ **gNBQciAlgoParamGrp.ServQltyRecoveryPlrThld**
- Downlink packet loss rate of 5QI 1 services $\leq$ **gNBQciAlgoParamGrp.ServQltyRecoveryPlrThld**
- If the gNodeB detects that the 5QI 1 bearer has been deleted:
 - The gNodeB releases the B1 measurements if the measurement configurations related to event B1 used for mobility to E-UTRAN have been delivered.
 - The gNodeB continues with the handover preparation or execution if the gNodeB has entered the handover preparation or execution phase.

When the strongest cell indicated in an E-UTRAN measurement report is not configured as a neighboring cell or encounters PCI confusion, service drops or ping-pong handovers may occur on voice UEs. If the voice-quality-based redirection function is enabled in this case, the gNodeB instructs the voice UE to perform a redirection to the frequency indicated in the measurement report. This function is controlled by the **QLTY_BASED_REDIRECTION** option of the **NRCCellAlgoSwitch.VoiceStrategySwitch** parameter. For details about voice-quality-based redirection, see *VoNR*.

5.9.2 Network Analysis

5.9.2.1 Benefits

VoNR-to-VoLTE handover based on voice quality can be used to improve VoNR experience in strong interference or voice packet loss scenarios. Strong interference may be caused by overlapping coverage or atmospheric duct. Voice packet loss may be caused by a large channel quality difference between uplink and downlink (for example, good downlink quality but poor uplink quality).

5.9.2.2 Impacts

The impacts between this function and other functions are described in [5.9.3.2.3 Function Impacts](#).

The value of the **N.HO.InterRAT.N2E.PrepAttOut** counter increases.

When the function of prohibiting large-packet UEs in LTE cells from returning to NR cells is enabled, the values of **N.NsaDc.SgNB.Mod.Req.Att** and **L.NsaDc.SCG.Mod.Req.Att** decrease.

5.9.3 Requirements

5.9.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
NR	FOFD-021209	Inter-RAT Mobility from NG-RAN to E-UTRAN	NR0SRAT MNE00	Inter-RAT Mobility from NG-RAN to E-UTRAN (NR)	gNodeB	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

5.9.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.9.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD LTE TDD	Support of VoLTE	ENodeBAlgoSwitch.EutranVoipSupportSwitch	<i>VoLTE</i>
NR FDD Low-frequency NR TDD	QoS Class Identifier ^a	NRCellQciBearer.Qci set to a value other than 2, 65, or 66	<i>QoS Management</i>
NR FDD Low-frequency NR TDD	Mobility to E-UTRAN Switch ^b	MOBILITY_TO_EUTRAN_SW option of the NRCellAlgoSwitch.InterRatServiceMobilitySw parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>
Note: a: The QLTY_BASED_EUTRAN_HO_SW option of the gNBQciAlgoParamGrp.QciAlgoSwitch parameter in the parameter group specified by the NRCellQciBearer.QciAlgoParamGroupId parameter can be selected only when the NRCellQciBearer.Qci parameter is set to a value other than 2, 65, or 66. b: Assume that the NRCellQciBearer.Qci parameter is set to 1. If the QLTY_BASED_EUTRAN_HO_SW option of the gNBQciAlgoParamGrp.QciAlgoSwitch parameter in the parameter group specified by the NRCellQciBearer.QciAlgoParamGroupId parameter is selected, the MOBILITY_TO_EUTRAN_SW option of the NRCellAlgoSwitch.InterRatServiceMobilitySw parameter must also be selected.			

5.9.3.2.2 Mutually Exclusive Functions

None

5.9.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Voice-quality-based inter-frequency handover	QLTY_BASSED_INTER_FREQ_HO_SWITCH option of the gNBQciAlgoParamGrp.QciAlgoSwitch parameter	<i>VoNR in 5G RAN Feature Documentation</i>	When VoNR-to-VoLTE handover based on voice quality and voice-quality-based inter-frequency handover are both enabled, the base station selects VoNR UEs with poor voice quality based on the packet loss rate thresholds of the two functions, and delivers inter-RAT or inter-frequency measurement configurations to these UEs. When the base station delivers both inter-RAT and inter-frequency measurement configurations to UEs, it preferentially processes the measurement report that is sent earlier.

5.9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For NR, all NR-capable boards support this function.
- For LTE, all LTE-capable boards support this function.
The LBBPc and LBBPd boards do not support the function of prohibiting large-packet UEs in LTE cells from returning to NR cells.

RF Modules

No requirements

5.9.3.4 Others

- UE requirements
UEs must comply with 3GPP Release 15 or later.
- Core network requirements
 - The core networks must comply with 3GPP Release 15 or later.
 - Handover requires the N26 interface to be configured between the MME and AMF.

5.9.4 Operation and Maintenance

5.9.4.1 Data Configuration

5.9.4.1.1 Data Preparation

Table 5-31 describes the parameters related to VoNR-to-VoLTE handover based on voice quality. For details about basic parameters for interoperability from NG-RAN to E-UTRAN, see **4.4.1.1 Data Preparation** and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-31 Parameters used for configuration

Parameter Name	Parameter ID	Setting Notes
QCI Algorithm Switch	gNBQciAlgoParamGrp.QciAlgoSwitch	Select the QLTY_BASED_EUTRAN_HO_SW option.
Inter-RAT Service Mobility Switch	NRCellAlgoSwitch.InterRatServiceMobilitySw	Select the MOBILITY_TO_EUTRAN_SW option.
Service PLR Evaluation Period	gNBQciAlgoParamGrp.ServPlrEvalPeriod	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Service Quality Eutran HO PLR Thld	gNBQciAlgoParamGrp.ServQltyEutranHoPlrThld	Set this parameter based on the network plan. It is recommended that this parameter be set to a value greater than gNBQciAlgoParamGrp.ServQltyInterFreqHoPlrThld so that inter-frequency VoNR handovers based on voice quality can be preferentially triggered.
Service Quality Recovery PLR Threshold	gNBQciAlgoParamGrp.ServQltyRecoveryPlrThld	Set this parameter based on the network plan.
Voice Strategy Switch	NRCellAlgoSwitch.VoiceStrategySwitch	Select the QLTY_BASED_REDI_SW option to address the issue of service drops or ping-pong handovers of voice UEs when the strongest cell indicated in an E-UTRAN measurement report is not configured as a neighboring cell or encounters PCI confusion.

5.9.4.1.2 Using MML Commands

The basic configurations for mobility from NG-RAN to E-UTRAN must have been completed before activating this function. For details, see [4.4.1.2 Using MML Commands](#) and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples

Before using MML commands, refer to [5.9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Setting parameters related to VoNR-to-VoLTE handover based on voice quality
ADD GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1, QciAlgoSwitch=QLTY_BASED_EUTRAN_HO_SW-1,
ServPlrEvalPeriod=5, ServQltyEutranHoPlrThld=10, ServQltyRecoveryPlrThld=2;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, QciAlgoParamGroupId=1;
```

```
//Turning on the switch of mobility to E-UTRAN
MOD NRCELLALGOSWITCH: NrCellId=0, InterRatServiceMobilitySw=MOBILITY_TO_EUTRAN_SW-1;
//(Optional) Turning on QLTY_BASED_REDI_SW to address the issue of service drops or ping-pong
handovers of voice UEs when the strongest cell indicated in an E-UTRAN measurement report is not
configured as a neighboring cell or encounters PCI confusion
MOD NRCELLALGOSWITCH: NrCellId=0, VoiceStrategySwitch=QLTY_BASED_REDI_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch of VoNR-to-VoLTE handover based on voice quality
MOD GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1, QciAlgoSwitch=QLTY_BASED_EUTRAN_HO_SW-0;
```

5.9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.9.4.2 Activation Verification

Observing Performance Counter Values

If the value of the **N.HO.InterRAT.N2E.VoNRQual.ExecSuccOut.Voice** counter is not 0, this function has taken effect.

Tracing Signaling

To verify VoNR-to-VoLTE handover based on voice quality, perform the following steps:

- Step 1** Start Uu interface tracing and NG interface tracing for the gNodeB.
- Step 2** Observe the signaling messages over the Uu interface. If the Measurement Report message sent from the UE to the gNodeB indicates event B1 related to voice quality, VoNR-to-VoLTE handover based on voice quality has taken effect.
- Step 3** Observe the signaling messages over the NG interface. If the HANDOVER REQUIRED message sent by the gNodeB to the AMF indicates a voice bearer, the gNodeB has triggered a VoNR-to-VoLTE handover.
- Step 4** Observe the signaling messages over the Uu interface. If the value of the *targetRAT-Type* IE in the MobilityFromNRCommand message sent by the gNodeB to the UE is "eutra", the preparation for the VoNR-to-VoLTE handover has succeeded.
- Step 5** Observe the signaling messages over the NG interface. If either (i) the gNodeB receives a UE CONTEXT RELEASE COMMAND message with the cause value "Successful Handover" or "Normal Release" from the AMF, or (ii) the gNodeB receives an RRC Reestablishment Complete message from the UE and the RRC

connection reestablishment is successful, the VoNR-to-VoLTE handover has been successfully executed.

----End

5.9.4.3 Network Monitoring

Table 5-32 lists the counters used to monitor the performance of VoNR-to-VoLTE handover based on voice quality.

Table 5-32 Counters used to monitor the performance of VoNR-to-VoLTE handover based on voice quality

Counter ID	Counter Name
1911829519	N.HO.InterRAT.N2E.VoNRQual.PrepAttOut.Voice
1911829518	N.HO.InterRAT.N2E.VoNRQual.ExecAttOut.Voice
1911829517	N.HO.InterRAT.N2E.VoNRQual.ExecSuccOut.Voice

5.10 Voice-Capability-based Redirection Back to the HPLMN

5.10.1 Inter-RAT Redirection Back to the HPLMN Based on Roaming UE Identification

Inter-RAT redirection/reselection back to the home public land mobile network (HPLMN) based on roaming UE identification applies when the NR coverage area of the serving PLMN of a roaming UE overlaps the LTE coverage area of the HPLMN of the UE. This function can quickly transfer a VoNR-incapable UE back to an LTE cell on a frequency that belongs to the UE's HPLMN through redirection or reselection. It is recommended that this function be enabled when the home network coverage area of a roaming UE overlaps the local network coverage area.

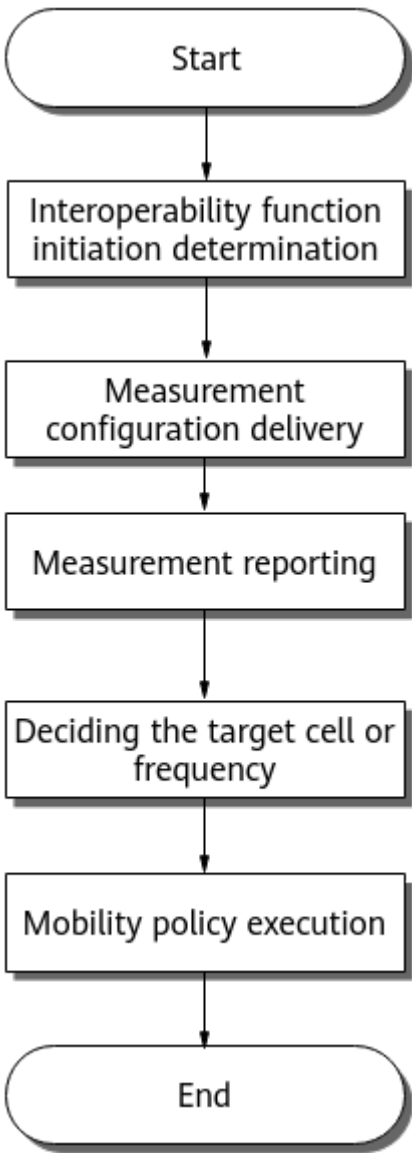
 NOTE

- HPLMN indicates the home operator that a UE subscribes to. The HPLMN of roaming UEs is specified by **gNB***RoamingUeHomePlmn.HomeMcc* and **gNB***RoamingUeHomePlmn.HomeMnc*.
- Serving PLMN indicates an operator that provides services for a UE. If a UE camps on its home network, the serving PLMN is its HPLMN.
- Roaming UE: When a UE leaves the network coverage area of its home operator, the UE may access another network (with which the UE needs to sign a roaming agreement). In this case, the UE on that network is referred to as a roaming UE. A roaming UE is identified based on its PLMN. The base station checks the PLMN and configurations of an SA UE newly accessing the network (through initial access, incoming handover, incoming RRC connection reestablishment, or incoming RRC connection resume). The UE is identified as a PLMN-based roaming UE if the following conditions are met: (1) The serving PLMN of the UE equals the combination of **gNB***Operator.Mcc* and **gNB***Operator.Mnc* of an operator for the base station; (2) **gNB***Operator.RoamingPlmnFlag* is set to **TRUE** for this operator.

5.10.1.1 Principles

Figure 5-13 shows the procedure for inter-RAT redirection back to the HPLMN based on roaming UE identification. This section describes only the differences between this function and the basic cell redirection functions presented in 4 Basic Procedures for Interoperability Between E-UTRAN and NG-RAN in Connected Mode.

Figure 5-13 Basic procedure for interoperability between E-UTRAN and NG-RAN



Redirection Initiation Evaluation

Inter-RAT redirection back to the HPLMN based on roaming UE identification can be performed only in measurement-based mode. Blind mode or RSRP prediction based on virtual grids are not supported. This function can start initiation evaluation if all of the following conditions are met:

- A roaming UE is identified.

- The **VOICE_CAPB_MOB_TO_HPLMN_SW** option of the **gNBOperator.HplmnAlgoSwitch** parameter is selected.
- In the frequency list group bound to the roaming UE, there is a frequency with **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** set to **TRUE**, and this frequency is configured as a neighboring frequency of the serving frequency.
- The roaming UE does not support VoNR.

Measurement Configuration Delivery

After inter-RAT redirection back to the HPLMN based on roaming UE identification is initiated, the gNodeB selects the frequencies with **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** set to **TRUE** from the frequency list group bound to the roaming UE and delivers the measurement configurations related to inter-RAT event B1 to the UE. Specifically:

The threshold of inter-RAT event B1 for inter-RAT redirection back to the HPLMN based on roaming UE identification equals the sum of **NRCellHoEutranMeaGrp.CovBasedHoB1RsrpThld** and **gNBOperatorQciParam.EutranB1B2RsrpThldOffset**.

If no neighboring cell is reported 6s after the inter-RAT event B1 has started, the gNodeB releases the measurement configurations, enables the UE to stop measurements for this event B1, and delivers measurement configurations related to event B1 to the UE every 90s.

Target Frequency or Target Cell Evaluation

After the gNodeB receives the inter-RAT event B1 measurement report for inter-RAT redirection back to the HPLMN based on roaming UE identification, it obtains the frequency information from the measurement report and starts cell evaluation as follows:

If the PLMN of the cell in the measurement report belongs to the HPLMN of the roaming UE, the gNodeB can select the cell as the target cell and performs inter-RAT redirection.

 NOTE

1. Assume that a VoNR-incapable UE of the home network has been transferred from the home 4G cell to the visited 5G cell through cell reselection in idle mode. If the UE initiates a voice call in idle mode, the gNodeB of the visited 5G cell will redirect the VoNR-incapable roaming UE to a 4G cell in the home network. However, no N26 interface exists between the home 4G core network and the visited 5GC. In this case, the voice call fails.
2. If the VoNR capability of a VoNR-capable roaming UE is manually changed from enabled to disabled in connected mode, the visited gNodeB does not perceive the UE capability change. Before querying the UE capability again, the visited gNodeB may misidentify the UE as a VoNR-capable one based on the last query result. In this case, voice call initiated by the UE in the visited 5G cell fails.
3. If the VoNR capability of a VoNR-capable roaming UE is manually changed from disabled to enabled in connected mode, the visited gNodeB does not perceive the UE capability change. Before querying the UE capability again, the visited gNodeB may misidentify the UE as a VoNR-incapable one based on the last query result. In this case, the UE is redirected to a 4G cell in the home network and therefore cannot enjoy the user experience provided by the visited 5G network.
4. Assume that a VoNR-incapable roaming UE of the home network has failed to measure the home 4G coverage during the periodic home 4G frequency measurement. If the UE initiates a voice call on the visited 5G network, the voice call fails.

5.10.1.2 Network Analysis

5.10.1.2.1 Benefits

This function enables a roaming UE to return to its HPLMN as soon as possible if its home network coverage area overlaps the local network coverage area, improving its user experience.

5.10.1.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

If the home network coverage area of a roaming UE overlaps the visited one, enabling this function decreases the number of EPS fallback failures of VoNR-incapable UEs in the visited network.

5.10.1.3 Requirements

5.10.1.3.1 Licenses

None

5.10.1.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

Prerequisite Functions

None

Mutually Exclusive Functions

None

Function Impacts

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	UL_SINR_MOBILITY_TO_EUTRANSW and MOBILITY_TO_EUTRANSW options of the NRCellAIgoSwitch.InterRatServiceMobilitySw parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In uplink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, only redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.
NR FDD Low-frequency NR TDD	Downlink - coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	MOBILITY_TO_EUTRANSW option of the NRCellAIgoSwitch.InterRatServiceMobilitySw parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, only redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Voice fallback from NG-RAN to E-UTRAN	EPS_FB_SWITCH option of the NRCellAlgoSwitch.VoiceStrategySwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In voice fallback from NG-RAN to E-UTRAN, only redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.
NR FDD Low-frequency NR TDD	Timing carrier shutdown	TIMING_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	In timing carrier shutdown, when the gNodeB receives the measurement report of the home 4G network, only redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.
NR FDD Low-frequency NR TDD	LTE and NR intelligent carrier shutdown	LNR_SMART_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	In LTE and NR intelligent carrier shutdown, only blind redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 4G network.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRU.DOROPTFEATSW parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When RF module fast deep dormancy is enabled, only blind redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 5G network.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: In intra-base-station multi-carrier scenarios: INTRA_GNB_MULTI_CARRIER_SD_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingSwitch parameter and the FAST_CARRIER_SHUTDOWN_SWITCH option of the NRDUCellAIgoSwitch.PowerSavingExtensionSwitch parameter	<i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i>	When fast carrier shutdown is enabled, only blind redirection can be performed to transfer VoNR-incapable roaming UEs from the visited 5G network to the home 5G network.

RAT	Function Name	Function Switch	Reference	Description
		In inter-base-station multi-carrier scenarios: INTER_GNB_MULTI_CARR_SD_SWITCH of option the NRDU CellAI goSwitch.PowerSavingSwitch parameter and the FAST_CARRIER_SHUTDOWN_WN_SWITCH option of the NRDU CellAI goSwitch.PowerSavingExtSwitch parameter For basic-layer cells: FAST_CA		

RAT	Function Name	Function Switch	Reference	Description
		RR_SHUT_SERV_AS_SUR_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter		

5.10.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.10.1.3.4 Others

- UE requirements
UEs must support roaming and comply with 3GPP Release 15 or later.
- Core network requirements
The core networks must support roaming and comply with 3GPP Release 15 or later.

5.10.1.4 Operation and Maintenance

5.10.1.4.1 Data Configuration

Data Preparation

Table 5-33 describes parameters related to inter-RAT redirection back to the HPLMN based on roaming UE identification. For details about basic parameters for interoperability from NG-RAN to E-UTRAN, see [4.4.1.1 Data Preparation](#) and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-33 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Roaming UE HPLMN Frequency Flag	gNBFreqPriorityGroup.RoamingUeHplmnFreq-Flag	Set this parameter based on the network plan.
HPLMN Algorithm Switch	gNBOperator.HplmnAlgorithmSwitch	The VOICE_CAPB_MOB_TO_HPLMN_SW option of this parameter specifies whether the gNodeB can transfer VoNR-incapable roaming UEs back to their HPLMNs using inter-RAT redirection or dedicated-priority-based inter-RAT cell reselection. Set this option based on the network plan.
Operator ID	gNBRoamingUeHomePlnm.OperatorId	Set this parameter based on the network plan.
Home Mobile Country Code	gNBRoamingUeHomePlnm.HomeMcc	Set this parameter based on the network plan.
Home Mobile Network Code	gNBRoamingUeHomePlnm.HomeMnc	Set this parameter based on the network plan.

Using MML Commands

The basic configurations for mobility from NG-RAN to E-UTRAN must have been completed before activating this function. For details, see [4.4.1.2 Using MML Commands](#) and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples

Before using MML commands, refer to [5.10.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Adding a frequency priority group on the gNodeB
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0, RatType=EUTRAN,
DEarfcn=1800, RoamingUeHplmnFreqFlag=TRUE;
//Adding operator information with the operator ID set to 0, mobile country code set to 460, mobile
network code set to 00, operator type set to the primary operator, NR networking option set to SA, roaming
PLMN flag set to false, and the switch for voice-capability-based handover back to the HPLMN set to off
ADD GNBOPERATOR: OperatorId=0, OperatorName="Operator A", Mcc="460", Mnc="00",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA, RoamingPlmnFlag=FALSE,
HplmnAlgoSwitch=VOICE_CAPB_MOB_TO_HPLMN_SW-0;
//Adding operator information with the operator ID set to 1, mobile country code set to 460, mobile
network code set to 21, operator type set to the secondary operator, NR networking option set to SA,
roaming PLMN flag set to true, and the switch for voice-capability-based handover back to the HPLMN set
to on
ADD GNBOPERATOR: OperatorId=1, OperatorName="Operator B", Mcc="460", Mnc="21",
OperatorType=SECONDARY_OPERATOR, NrNetworkingOption=SA, RoamingPlmnFlag=TRUE,
HplmnAlgoSwitch=VOICE_CAPB_MOB_TO_HPLMN_SW-1;
//Adding the HPLMN of roaming UEs on the gNodeB
ADD GNBROAMINGUEHOMELMN: OperatorId=1, HomeMcc="460", HomeMnc="11";
//Adding the operator-specific policy of an NR cell
ADD NRCELLOPPOLICY: NrCellId=1, OperatorId=1, gNBFreqPriorityGroupId=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deactivating inter-RAT redirection back to the HPLMN based on roaming UE identification
MOD GNBOPERATOR: OperatorId=1, HplmnAlgoSwitch=VOICE_CAPB_MOB_TO_HPLMN_SW-0;
```

Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.10.1.4.2 Activation Verification

- Step 1** Enable a VoNR-incapable UE to roam on a 5G network.
- Step 2** Enable the UE to run services and trigger an inter-RAT redirection from NR for the UE.
- Step 3** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **Uu Interface Trace**.
- Step 4** Observe the displayed **Uu Interface Trace** dialog box. Check whether the RRCReconfiguration message from the gNodeB, which includes the measurement configurations related to inter-RAT event B1, contains inter-RAT frequencies for

which the roaming UE HPLMN frequency flag is set to **True**. If such frequencies are contained, inter-RAT redirection back to the HPLMN based on roaming UE identification has taken effect.

----End

5.10.1.4.3 Network Monitoring

None

5.10.2 Dedicated-Priority-based Inter-RAT Cell Reselection Back to the HPLMN

Dedicated-priority-based inter-RAT cell reselection back to the HPLMN applies when the NR coverage area of the serving PLMN of a roaming UE overlaps the LTE coverage area of the HPLMN of the UE. This function can quickly perform a reselection for a VoNR-incapable UE, transferring it back to an LTE cell on a frequency that belongs to the UE's HPLMN. It is recommended that this function be enabled when the home network coverage area of a roaming UE overlaps the local network coverage area.

5.10.2.1 Principles

Function Initiation Determination

Dedicated-priority-based inter-RAT cell reselection back to the HPLMN performs reselection for a VoNR-incapable roaming UE, transferring it to an LTE cell working on the HPLMN frequency. This function can start initiation evaluation if all of the following conditions are met:

- A roaming UE is identified.
- The **VOICE_CAPB_MOB_TO_HPLMN_SW** option of the **gNBOperator.HplmnAlgoSwitch** parameter is selected.
- In the frequency list group bound to the roaming UE, there is a frequency with **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** set to **TRUE**, and this frequency is configured as a neighboring frequency of the serving frequency.
- The roaming UE does not support VoNR.
- The roaming UE is ready to enter idle mode from connected mode.

Reselection Procedures

The reselection procedure for VoNR-incapable roaming UEs of the home network is as follows:

1. VoNR-incapable UEs of the home network in roaming scenarios are identified.
2. The gNodeB of the visited network, on which the **VOICE_CAPB_MOB_TO_HPLMN_SW** option of the **gNBOperator.HplmnAlgoSwitch** parameter is selected for the home network, sends an RRCRelease message to those VoNR-incapable UEs. The message contains the LTE frequencies for which the

gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag parameter is set to **TRUE** in the frequency list group bound to the roaming UEs on the gNodeB side.

3. These UEs are transferred to the LTE cells corresponding to their HPLMN frequencies based on the dedicated cell-reselection priorities.

5.10.2.2 Network Analysis

5.10.2.2.1 Benefits

This function enables a roaming UE to return to its HPLMN as soon as possible if its home network coverage area overlaps the local network coverage area, improving its user experience.

5.10.2.2.2 Impacts

The impacts between this function and other functions are described in [Function Impacts](#).

If the home network coverage area of a roaming UE overlaps the visited one, enabling this function decreases the number of EPS fallback failures of VoNR-incapable UEs in the visited network.

5.10.2.3 Requirements

5.10.2.3.1 Licenses

None

5.10.2.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

Prerequisite Functions

None

Mutually Exclusive Functions

None

Function Impacts

None

5.10.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.10.2.3.4 Others

- UE requirements
UEs must support roaming and comply with 3GPP Release 15 or later.
- Core network requirements
The core networks must support roaming and comply with 3GPP Release 15 or later.

5.10.2.4 Operation and Maintenance

5.10.2.4.1 Data Configuration

Data Preparation

Table 5-34 lists the parameters related to dedicated-priority-based inter-RAT cell reselection back to the HPLMN. For details about basic parameters for interoperability from NG-RAN to E-UTRAN, see **4.4.1.1 Data Preparation** and the "Data Preparation" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Table 5-34 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Roaming UE HPLMN Frequency Flag	gNBFreqPriorityGroup.RoamingUeHplmnFreq-Flag	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
HPLMN Algorithm Switch	gNBOperator.HplmnAlgoSwitch	The VOICE_CAPB_MOB_TO_HPLMN_SW option of this parameter specifies whether the gNodeB can transfer VoNR-incapable roaming UEs back to their HPLMNs using dedicated-priority-based inter-RAT cell reselection back to the HPLMN. Set this option based on the network plan.
Operator ID	gNBRoamingUeHplmn.OperatorId	Set this parameter based on the network plan.
Home Mobile Country Code	gNBRoamingUeHplmn.HomeMcc	Set this parameter based on the network plan.
Home Mobile Network Code	gNBRoamingUeHplmn.HomeMnc	Set this parameter based on the network plan.

Using MML Commands

The basic configurations for mobility from NG-RAN to E-UTRAN must have been completed before activating this function. For details, see [4.4.1.2 Using MML Commands](#) and the "Using MML Commands" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

Activation Command Examples

Before using MML commands, refer to [5.10.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

```
//Adding a frequency priority group on the gNodeB
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0, RatType=EUTRAN,
DIEarfcn=1800, RoamingUeHplmnFreqFlag=TRUE;
//Adding operator information with the operator ID set to 0, mobile country code set to 460, mobile
network code set to 00, operator type set to the primary operator, NR networking option set to SA, roaming
PLMN flag set to false, and the switch for voice-capability-based handover back to the HPLMN set to off
ADD GNBOPERATOR: OperatorId=0, OperatorName="Operator A", Mcc="460", Mnc="00",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA, RoamingPlmnFlag=FALSE,
HplmnAlgoSwitch=VOICE_CAPB_MOB_TO_HPLMN_SW-0;
//Adding operator information with the operator ID set to 1, mobile country code set to 460, mobile
network code set to 21, operator type set to the secondary operator, NR networking option set to SA,
roaming PLMN flag set to true, and the switch for voice-capability-based handover back to the HPLMN set
to on
```

```
ADD GNBOPERATOR: OperatorId=1, OperatorName="Operator B", Mcc="460", Mnc="21",
OperatorType=SECONDARY_OPERATOR, NrNetworkingOption=SA, RoamingPlmnFlag=TRUE,
HplmnAlgoSwitch=VOICE_CAPB_MOB_TO_HPLMN_SW-1;
//Adding the HPLMN of roaming UEs on the gNodeB
ADD GNBROAMINGUEHOMELMN: OperatorId=1, HomeMcc="460", HomeMnc="11";
//Adding the operator-specific policy of an NR cell
ADD NRCELLOPPOLICY: NrCellId=1, OperatorId=1, gNBFreqPriorityGroupId=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deactivating dedicated-priority-based inter-RAT cell reselection back to the HPLMN
MOD GNBOPERATOR: OperatorId=1, HplmnAlgoSwitch=VOICE_CAPB_MOB_TO_HPLMN_SW-0;
```

Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.10.2.4.2 Activation Verification

- Step 1** Enable a VoNR-incapable UE to roam on a 5G network.
- Step 2** Enable the UE to run services and trigger an inter-RAT reselection from NR for the UE.
- Step 3** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the left navigation pane of the displayed **Signaling Trace Management** tab page, choose **Uu Interface Trace**.
- Step 4** Observe the displayed **Uu Interface Trace** dialog box. Check whether the RRCRelease message from the gNodeB contains inter-RAT frequencies for which the roaming UE HPLMN frequency flag is set to **True**. If such frequencies are contained, dedicated-priority-based inter-RAT cell reselection back to the HPLMN has taken effect.

----End

5.10.2.4.3 Network Monitoring

None

6 Multi-Operator Sharing Scenarios

6.1 Principles

When LTE or NR cells are shared by multiple operators, the parameters of the following functions can be separately configured for each operator:

- Mobility management for UEs in idle mode: Dedicated reselection priorities can be separately configured for each operator. For details, see the "Dedicated Priority" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- Mobility management for UEs in connected mode:
 - Certain measurement variables can be separately configured for each operator. For details, see [6.1.1.1 Operator-specific Measurement Variables](#).
 - Mobility policies can be separately configured for each operator. For details, see [6.1.1.2 Operator-specific Mobility Policy](#).
 - NR frequency policies can be separately configured for each operator. For details, see [6.1.1.3 Operator-specific NR Frequency Policy](#).
 - LTE frequency policies can be separately configured for each operator. For details, see [6.1.1.4 Operator-specific LTE Frequency Configuration](#).

For details about the configuration requirements in typical multi-operator sharing scenarios, see [6.1.2 Typical Multi-Operator Sharing Scenarios](#).

6.1.1 Function Description

6.1.1.1 Operator-specific Measurement Variables

Certain measurement variables can be separately set for each operator on the gNodeB and eNodeB sides. The measurement variables that are not described in this section are the same as those used in independent operator mode. For details, see [4.1.4.2 Reporting Configurations](#).

On the gNodeB Side

The following measurement variables can be separately set for each operator on the gNodeB side:

- Threshold for inter-RAT event A1 =
NRCellInterRHoMeaGrp.InterRatHoA1RsrpThld + gNBOperatorQciParam.EutranA1RsrpThldOffset + gNBRfspConfig.InterRatA1A2B2RsrpThldOfs + NRCellOpPolicy.EutranA1A2B2RsrpThldOfs
- Threshold for inter-RAT event A2 =
NRCellInterRHoMeaGrp.InterRatHoA2RsrpThld + gNBOperatorQciParam.EutranA2B2RsrpThldOffset + gNBRfspConfig.InterRatA1A2B2RsrpThldOfs + NRCellOpPolicy.EutranA1A2B2RsrpThldOfs
- Threshold for event B1 related to downlink-coverage-based inter-RAT mobility = ***NRCellHoEutranMeaGrp.CovBasedHoB1RsrpThld + gNBOperatorQciParam.EutranB1B2RsrpThldOffset***
- Threshold for event B1 related to EPS fallback =
NRCellHoEutranMeaGrp.EpsFbB1RsrpThld + gNBOperatorQciParam.EpsFbB1RsrpThldOffset
- Threshold 1 for event B2 =
NRCellInterRHoMeaGrp.CovHoToEutranB2RsrpThld1 + gNBOperatorQciParam.EutranA2B2RsrpThldOffset + gNBRfspConfig.InterRatA1A2B2RsrpThldOfs + NRCellOpPolicy.EutranA1A2B2RsrpThldOfs
- Threshold 2 for event B2 = ***NRCellHoEutranMeaGrp.CovBasedHoB1RsrpThld + gNBOperatorQciParam.EutranB1B2RsrpThldOffset***
- Cell individual offset = ***NRCellEutraNRelation.EutranCellIndividualOffset + NRCellOpPolicy.EutranCioOffset***

It is recommended that the parameters be set as required when the coverage areas of LTE cells are inconsistent (for example, when no LTE cell is shared).

On the eNodeB Side

The following threshold parameters for measurement events can be separately set for each operator on the eNodeB side:

- Threshold for inter-RAT event A1 =
InterRatHoCommGroup.InterRatHoNrA1ThldRsrp + CnOperatorQciPara.NrA1RsrpThldOffset
- Threshold for inter-RAT event A2 =
InterRatHoCommGroup.InterRatHoNrA2ThldRsrp + CnOperatorQciPara.NrA2RsrpThldOffset + CnOperatorHoCfg.NrA2ThldRsrpOffset
- Threshold for event B1 related to coverage-based inter-RAT mobility =
InterRatHoNrParamGrp.CovBasedNrB1RsrpThld + CnOpQciEventParam.HandoverThldOffset. The ***CnOpQciEventParam.EventType*** parameter must be set to ***COV_BASED_HO_TO_NR_B1***.
- Threshold for event B1 related to service-based inter-RAT mobility =
InterRatHoNrParamGrp.ServBasedNrB1RsrpThld +

CnOpQciEventParam.HandoverThldOffset. The **CnOpQciEventParam.EventType** parameter must be set to **SERV_BASED_HO_TO_NR_B1**.

It is recommended that the parameters be set as required when the coverage areas of NR cells are inconsistent (for example, when no NR cell is shared).

6.1.1.2 Operator-specific Mobility Policy

Operator-specific mobility policies can be configured for functions involved in interoperability between E-UTRAN and NG-RAN.

E-UTRAN to NG-RAN Mobility Policy

The **CellOpHoCfg.HoModeSwitch** parameter specifies the operator-specific E-UTRAN to NG-RAN mobility policies for the following functions:

- [5.5 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN](#)
- [5.6 E-UTRAN to NG-RAN Traffic Steering](#)
- [5.8 Fast Return from E-UTRAN to NG-RAN](#)

Table 6-1 takes handover as an example to describe the conditions for it to take effect. The conditions for redirection to take effect are similar.

Table 6-1 Conditions for E-UTRAN to NG-RAN handover to take effect

NrHoSwitch in CellHoParaCfg.HoModeSwitch	HANDOVER_TO_NR_SW in CellOpHoCfg.HoModeSwitch	Handover Takes Effect?
On	On	Yes
On	Off	No
Off	On	No
Off	Off	No
On	Not configured	Yes
Off	Not configured	No

NG-RAN to E-UTRAN Mobility Policy

For downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN, the **NRCelOpPolicy.InterRatHoPolicySwitch** parameter specifies operator-specific mobility policies.

- If handover is required for both data and VoNR services, the conditions for handover to take effect are presented in [Table 6-2](#).
- If redirection is required for both data and VoNR services, the conditions for redirection to take effect are presented in [Table 6-3](#).

- If redirection is required for data services and handover is required for VoNR services, the conditions for handover and redirection to take effect are presented in [Table 6-3](#) and [Table 6-4](#).

Table 6-2 Conditions for NG-RAN to E-UTRAN handover to take effect

EUTRAN_HO_SWITCH in NRInterRatHoPar- am.HoModeSwitch	EUTRAN_HO_SWITCH in NRCellOpPolicy.InterRa tHoPolicySwitch	Handover Takes Effect?
On	On	Yes
On	Off	No
Off	On	No
Off	Off	No
On	Not configured	Yes
Off	Not configured	No

Table 6-3 Conditions for NG-RAN to E-UTRAN redirection to take effect

EUTRAN_REDIRECT_SWI TCH in NRInterRatHoPar- am.HoModeSwitch	EUTRAN_REDIRECT_SWI TCH in NRCellOpPolicy.InterRa tHoPolicySwitch	Redirection Takes Effect?
On	On	Yes
On	Off	No
Off	On	No
Off	Off	No
On	Not configured	Yes
Off	Not configured	No

Table 6-4 Conditions for handover to take effect for VoNR when redirection is required for data services and handover is required for VoNR services

VONR_COV_BASED_HO _PREFER_SW in NRInterRatHoPar- am.HoModeSwitch	VONR_COV_BASED_HO _PREFER_SW in NRCellOpPolicy.InterRa tHoPolicySwitch	Handover Takes Effect?
On	On	Yes
On	Off	No

VONR_COV_BASED_HO_PREFER_SW in NRInterRatHoParam.HoModeSwitch	VONR_COV_BASED_HO_PREFER_SW in NRCellOpPolicy.InterRatHoPolicySwitch	Handover Takes Effect?
Off	On	No
Off	Off	No
On	Not configured	Yes
Off	Not configured	No

For voice fallback:

- The **NRCellOpPolicy.EpsFbMode** parameter specifies the operator-specific mobility policies. [Table 6-5](#) presents the conditions for mobility policies to take effect.
- The **VOICE_BLIND_MODE_SWITCH** option of the **NRCellOpPolicy.InterRatHoPolicySwitch** parameter specifies operator-specific blind mode policies. [Table 6-6](#) presents the conditions for blind mode to take effect.

Table 6-5 Mobility policy settings for voice fallback

NRInterRatHoParam.EpsFbMode	NRCellOpPolicy.EpsFbMode	Mobility Policy That Takes Effect
HANDOVER	NOT_CONFIG. Alternatively, the NRCellOpPolicy MO is not configured.	Handover
REDIRECTION	NOT_CONFIG. Alternatively, the NRCellOpPolicy MO is not configured.	Redirection
Any value	HANDOVER	Handover
Any value	REDIRECTION	Redirection

Table 6-6 Blind mode settings

VOICE_BLIND_MODE_SWITCH in NRInterRatHoParam.HoModeSwitch	VOICE_BLIND_MODE_SWITCH in NRCellOpPolicy.InterRatHoPolicySwitch	Blind Mode Takes Effect?
On	On	Yes

VOICE_BLIND_MODE_SWITCH in NRInterRatHoParam.HoModeSwitch	VOICE_BLIND_MODE_SWITCH in NRCellOpPolicy.InterRatHoPolicySwitch	Blind Mode Takes Effect?
On	Off	No
Off	On	No
Off	Off	No
On	The NRCellOpPolicy MO is not configured.	Yes
Off	The NRCellOpPolicy MO is not configured.	No

6.1.1.3 Operator-specific NR Frequency Policy

On the eNodeB side, NR frequency policies can be set on a per operator basis for the following functions:

- NR frequency delivery
- VoNR priority configuration for NR frequencies
- Connected-mode priority configuration for NR frequencies

All the following conditions must be met to enable the preceding functions:

1. The **PlmnBasedNrFreqPolicySw** option of the **CellHoParaCfg.CellHoAlgoSwitch** parameter is selected.
2. NR frequencies are configured for the **RatFreqPriorityGroup.DLEarfcn** parameter associated with the **CellOp.RatFreqPriorityGroupId** parameter of a corresponding operator.
3. The **RatFreqPriorityGroup.RatType** parameter associated with the **CellOp.RatFreqPriorityGroupId** parameter of the corresponding operator is set to **NR**.

If any of the preceding conditions is not met:

- If condition 1 is not met, regardless of whether condition 2 or 3 is met or not, operator-specific NR frequency measurement configurations are not delivered, but common NR frequency measurement configurations are delivered.
- If condition 1 is met but condition 2 or 3 is not met, NR frequency measurement configurations are not delivered.

Operator-specific NR Frequency Delivery

When selecting NR frequencies for a UE in the following functions, the base station filters out NR frequencies that do not belong to the operator serving the UE:

- Mobility management for UEs in idle mode: dedicated reselection priority

- Mobility management for UEs in connected mode:
 - Coverage-based inter-RAT mobility from E-UTRAN to NG-RAN
 - E-UTRAN to NG-RAN traffic steering
 - Fast return from E-UTRAN to NG-RAN

Operator-specific VoNR Priority Configuration for NR Frequencies

If operator-specific VoNR priorities are set for a frequency using the **RatFreqPriorityGroup.VonrPriority** parameter, they override the non-operator-specific VoNR priority of the frequency. The following functions are affected:

- Coverage-based inter-RAT mobility from E-UTRAN to NG-RAN
- E-UTRAN to NG-RAN traffic steering

Operator-specific Connected-mode Priority Configuration for NR Frequencies

If operator-specific connected-mode priorities are set for a frequency using the **RatFreqPriorityGroup.ConnFreqPriority** parameter, they override the non-operator-specific connected-mode priority of the frequency. The following functions are affected:

- Coverage-based inter-RAT mobility from E-UTRAN to NG-RAN
- E-UTRAN to NG-RAN traffic steering

6.1.1.4 Operator-specific LTE Frequency Configuration

The gNodeB allows for operator-specific LTE frequency configuration when all of the following conditions are met:

- The **NRCellOpPolicy** MO is configured for an NR cell with the **NRCellOpPolicy.gNBFreqPriorityGroupId** parameter set to a value in the range of 0 to 254.
- The **gNBFreqPriorityGroup.RatType** parameter in the **gNBFreqPriorityGroup** MO identified by the **NRCellOpPolicy.gNBFreqPriorityGroupId** parameter is set to EUTRAN.

When the **NRCellOpPolicy** MO is configured for an NR cell with the **NRCellOpPolicy.gNBFreqPriorityGroupId** parameter set to a value in the range of 0 to 254:

- If the **OPERATOR_EUTRAN_FREQ_CFG_SW** option of the **NRCellOpPolicy.FreqConfigPolicySwitch** parameter is selected, the LTE frequencies in the intersection of frequencies defined by the **gNBFreqPriorityGroup** and **NRCellEutranNFreq** MOs are treated as candidate frequencies.
 - During a measurement-based handover or redirection, these candidate frequencies are delivered for measurements in descending order of priorities specified by the **gNBFreqPriorityGroup.ConnFreqPriority** parameter (for data services) or by the **gNBFreqPriorityGroup.VoltePriority** parameter (for voice services).
 - During a blind handover or blind redirection, these candidate frequencies are arranged in descending order of priorities as indicated by the

gNBFreqPriorityGroup.ConnFreqPriority parameter (for data services) or by the **gNBFreqPriorityGroup.VolteBlindPriority** parameter (for voice services) for selection as the target cell's frequency of the blind handover or the target frequency of the blind redirection.

If the **gNBFreqPriorityGroup.RatType** parameter in the **gNBFreqPriorityGroup** MO identified by the **NRCelOpPolicy.gNBFreqPriorityGroupId** parameter is set to a value other than **EUTRAN**, the associated frequency is filtered out.

- If the **OPERATOR_EUTRAN_FREQ_CFG_SW** option of the **NRCelOpPolicy.FreqConfigPolicySwitch** parameter is deselected, the LTE frequencies identified by the **NRCelEutranNFreq.DLEarfcn** parameter are treated as candidate frequencies for the operator.
- If the **NRCelEutranNFreq.CovHoA1A2B21FreqOffset** parameter is set to a value other than **0**, thresholds for events A1 and A2, and threshold 1 for event B2 related to coverage-based inter-RAT handovers are affected by the value of this parameter. One of these thresholds plus their relative offset plus the offset specified by this parameter will be used as the threshold for the relevant event.

If the **NRCelOpPolicy** MO is configured for the NR cell but the **NRCelOpPolicy.gNBFreqPriorityGroupId** parameter is set to **255**, the LTE frequencies identified by the **NRCelEutranNFreq.DLEarfcn** parameter are treated as candidate frequencies for the operator.

The following functions are affected:

- [5.2 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN](#)
- [5.3 Uplink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN](#)
- [5.7 Voice Fallback from NG-RAN to E-UTRAN](#)

6.1.2 Typical Multi-Operator Sharing Scenarios

The major multi-operator sharing modes for the eNodeB are as follows:

- RAN sharing with common carrier
- RAN sharing with dedicated carrier
- RAN sharing with special dedicated carrier

The major multi-operator sharing modes for the gNodeB are as follows:

- RAN sharing with common carrier
- RAN sharing with dedicated carrier

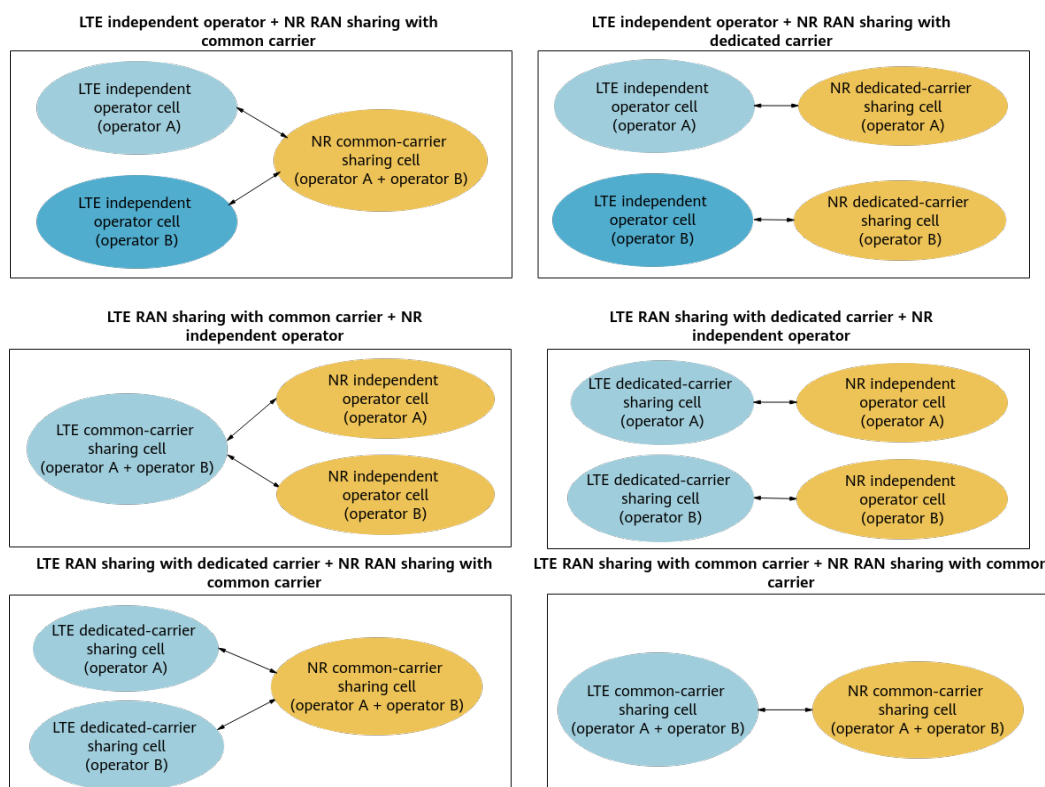
In independent operator mode, the eNodeB or gNodeB is not shared with other operators.

This section describes the following typical eNodeB and gNodeB multi-operator sharing scenarios, as illustrated in [Figure 6-1](#):

- LTE cells in independent operator mode + NR cells in RAN sharing with common carrier mode
- LTE cells in independent operator mode + NR cells in RAN sharing with dedicated carrier mode

- LTE cells in RAN sharing with common carrier mode + NR cells in independent operator mode
- LTE cells in RAN sharing with dedicated carrier mode + NR cells in independent operator mode
- LTE cells in RAN sharing with dedicated carrier mode + NR cells in RAN sharing with common carrier mode
- LTE cells in RAN sharing with common carrier mode + NR cells in RAN sharing with common carrier mode

Figure 6-1 Typical multi-operator sharing scenarios



LTE Cells in Independent Operator Mode + NR Cells in RAN Sharing with Common Carrier Mode

When LTE cells are working in independent operator mode and NR cells are working in RAN sharing with common carrier mode, the following neighboring NR cell data must be configured on the LTE side, in contrast to the data configuration for independent operator mode on both the LTE and NR sides:

- Primary operator on the NR side, involving the **NrNFreq**, **NrExternalCell**, and **NrNRRelationship** MOs
- Secondary operators on the NR side, involving the **NrNFreq**, **NrExternalCell**, **NrNRRelationship**, and **NrExternalCellPlmn** MOs. Specifically:
 - The **NrExternalCell.Mcc** and **NrExternalCell.Mnc** parameters of an external cell are set to the PLMN ID of the primary NR operator.
 - The **NrExternalCell.GnodebId** and **NrExternalCell.CellId** parameters of an external cell must be set to the gNodeB ID and cell ID that are specified for the primary NR operator, respectively.

- The **NrExternalCellPlmn.SharedMcc** and **NrExternalCellPlmn.SharedMnc** parameters in the external-cell PLMN list are set to the PLMN ID of a secondary operator.
 - (Optional; required when operator-specific NR Cell Identity [NCI] configuration is enabled) The following configurations are required:
 - The **NrExternalCellPlmn.SharedPlmnGnodebId** and **NrExternalCellPlmn.SharedPlmnCellId** parameters are set to the gNodeB ID and cell ID that are specified for a secondary NR operator, respectively.
 - The **EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode** or **NbrGnodebPolicy.LteToNrPlmnSelectionMode** parameter is set to **SELECTED_PLMN_ID**. The **EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode** parameter specifies the eNodeB-level PLMN ID filling mode in LTE-to-NR procedures. The **NbrGnodebPolicy.LteToNrPlmnSelectionMode** parameter specifies the target-gNodeB-level PLMN ID filling mode in LTE-to-NR procedures.
- For more details about how to configure operator-specific NCIs, see *Multi-Operator Sharing* in *5G RAN Feature Documentation*.
- When operator-specific TACs are configured on the NR side, **NrExternalCell.Tac** is set to the TAC of the primary operator and **NrExternalCellPlmn.Tac** is set to the TAC of each secondary operator on the LTE side. It is required that the LTE side be upgraded to eRAN16.1. If it is not upgraded, inter-RAT handovers from LTE cells to NR cells may fail.

For details about data configuration for NR RAN sharing with common carrier, see *Multi-Operator Sharing* in *5G RAN Feature Documentation*.

LTE Cells in Independent Operator Mode + NR Cells in RAN Sharing with Dedicated Carrier Mode

When LTE cells are working in independent operator mode and NR cells are working in RAN sharing with dedicated carrier mode, the following neighboring NR cell data must be configured on the LTE side, in contrast to the data configuration for independent operator mode on both the LTE and NR sides:

- Primary operator on the NR side, involving the **NrNFreq**, **NrExternalCell**, and **NrNRRelationship** MOs
- Secondary operators on the NR side, involving the **NrNFreq**, **NrExternalCell**, **NrNRRelationship**, and **NrExternalCellPlmn** MOs. Specifically:
 - If operator-specific NCI configuration is not enabled, the **NrExternalCell.Mcc** and **NrExternalCell.Mnc** parameters are set to the PLMN ID of the primary NR operator, and the **NrExternalCell.MasterPlmnReservedFlag** parameter must be set to **TRUE**.
 - If operator-specific NCI configuration is enabled, the **NrExternalCell.Mcc** and **NrExternalCell.Mnc** parameters must be set to the PLMN ID of a secondary NR operator, and the **NrExternalCell.MasterPlmnReservedFlag** parameter is set to **FALSE**.

For details about data configuration for NR RAN sharing with dedicated carrier, see *Multi-Operator Sharing* in *5G RAN Feature Documentation*.

LTE Cells in RAN Sharing with Common Carrier Mode + NR Cells in Independent Operator Mode

When LTE cells are working in RAN sharing with common carrier mode and NR cells are working in independent operator mode, the following neighboring LTE cell data must be configured on the NR side, in contrast to the data configuration for independent operator mode on both the LTE and NR sides:

- Primary operator on the LTE side, involving the **NRCCellEutranNFreq**, **gNBEutraExternalCell**, and **NRCCellEutraNRelation** MOs
- Secondary operators on the LTE side, involving the **NRCCellEutranNFreq**, **gNBEutraExternalCell**, **NRCCellEutraNRelation**, and **gNBEutraExtCellPlmn** MOs. The **gNBEutraExternalCell.Mcc** and **gNBEutraExternalCell.Mnc** parameters are set to the PLMN ID of the primary LTE operator.

For details about data configuration for LTE RAN sharing with common carrier, see RAN Sharing in *eRAN Feature Documentation*.

LTE Cells in RAN Sharing with Dedicated Carrier Mode + NR Cells in Independent Operator Mode

When LTE cells are working in RAN sharing with dedicated carrier mode and NR cells are working in independent operator mode, the following neighboring LTE cell data must be configured on the NR side, in contrast to the data configuration for independent operator mode on both the LTE and NR sides:

- Primary operator on the LTE side, involving the **NRCCellEutranNFreq**, **gNBEutraExternalCell**, and **NRCCellEutraNRelation** MOs
- Secondary operators on the LTE side, involving the **NRCCellEutranNFreq**, **gNBEutraExternalCell**, and **NRCCellEutraNRelation** MOs. The **gNBEutraExternalCell.Mcc**, **gNBEutraExternalCell.Mnc**, **NRCCellEutraNRelation.Mcc**, and **NRCCellEutraNRelation.Mnc** parameters specify the PLMN IDs of secondary LTE operators.

For details about data configuration for LTE RAN sharing with dedicated carrier, see RAN Sharing in *eRAN Feature Documentation*.

LTE Cells in RAN Sharing with Dedicated Carrier Mode + NR Cells in RAN Sharing with Common Carrier Mode

When LTE cells are working in RAN sharing with dedicated carrier mode and NR cells are working in RAN sharing with common carrier mode, the LTE and NR data configuration requirements described in this section apply, in contrast to the data configuration for independent operator mode on both the LTE and NR sides.

For details about data configuration for NR RAN sharing with common carrier, see *Multi-Operator Sharing* in *5G RAN Feature Documentation*. The following neighboring LTE cell data must be configured on the NR side:

- Primary operator on the LTE side, involving the **NRCCellEutranNFreq**, **gNBEutraExternalCell**, and **NRCCellEutraNRelation** MOs

- Secondary operators on the LTE side, involving the **NRCellEutraNFreq**, **gNBEutraExternalCell**, and **NRCellEutraNRelation** MOs. The **gNBEutraExternalCell.Mcc**, **gNBEutraExternalCell.Mnc**, **NRCellEutraNRelation.Mcc**, and **NRCellEutraNRelation.Mnc** parameters specify the PLMN IDs of secondary LTE operators.

For details about data configuration for LTE RAN sharing with dedicated carrier, see RAN Sharing in *eRAN Feature Documentation*. The following neighboring NR cell data must be configured on the LTE side:

- **NrNFreq**, **NrExternalCell**, **NrNRelationship**, and **gNBEutraExtCellPlmn** MOs
- **NrExternalCellPlmn** MO. The **NrExternalCellPlmn.SharedMcc** and **NrExternalCellPlmn.SharedMnc** parameters are set to the PLMN ID of a secondary operator. If operator-specific NCI configuration is enabled, the following configurations are also required:
 - The **NrExternalCellPlmn.SharedPlmnGnodebId** and **NrExternalCellPlmn.SharedPlmnCellId** parameters are set to the gNodeB ID and cell ID that are specified for a secondary NR operator, respectively.
 - The **EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode** or **NbrGnodebPolicy.LteToNrPlmnSelectionMode** parameter is set to **SELECTED_PLMN_ID**. The **EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode** parameter specifies the eNodeB-level PLMN ID filling mode in LTE-to-NR procedures. The **NbrGnodebPolicy.LteToNrPlmnSelectionMode** parameter specifies the target-gNodeB-level PLMN ID filling mode in LTE-to-NR procedures.
- When operator-specific TACs are configured on the NR side, set **NrExternalCell.Tac** to the TAC of the primary operator and set **NrExternalCellPlmn.Tac** to the TAC of each secondary operator on the LTE side. It is required that the LTE side be upgraded to eRAN16.1. If it is not upgraded, inter-RAT handovers from LTE cells to NR cells may fail.

LTE Cells in RAN Sharing with Common Carrier Mode + NR Cells in RAN Sharing with Common Carrier Mode

When both LTE and NR cells are working in RAN sharing with common carrier mode, the LTE and NR data configuration requirements described in this section apply.

For details about data configuration for LTE RAN sharing with common carrier, see RAN Sharing in *eRAN Feature Documentation*. The following neighboring NR cell data must be configured on the LTE side:

- **NrNFreq**, **NrExternalCell**, **NrNRelationship**, and **NrExternalCellPlmn** MOs
- **NrExternalCellPlmn** MO. The **NrExternalCellPlmn.SharedMcc** and **NrExternalCellPlmn.SharedMnc** parameters are set to the PLMN ID of a secondary operator. If operator-specific NCI configuration is enabled, the following configurations are also required:
 - The **NrExternalCellPlmn.SharedPlmnGnodebId** and **NrExternalCellPlmn.SharedPlmnCellId** parameters are set to the gNodeB ID and cell ID that are specified for a secondary NR operator, respectively.

- The **EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode** or **NbrGnodebPolicy.LteToNrPlmnSelectionMode** parameter is set to **SELECTED_PLMN_ID**. The **EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode** parameter specifies the eNodeB-level PLMN ID filling mode in LTE-to-NR procedures. The **NbrGnodebPolicy.LteToNrPlmnSelectionMode** parameter specifies the target-gNodeB-level PLMN ID filling mode in LTE-to-NR procedures.
- When operator-specific TACs are configured on the NR side, the **NrExternalCell.Tac** parameter is set to the TAC of the primary operator and the **NrExternalCellPlmn.Tac** parameter is set to the TAC of each secondary operator on the LTE side. It is required that the LTE side be upgraded to eRAN16.1. If it is not upgraded, inter-RAT handovers from LTE cells to NR cells may fail.

For details about data configuration for NR RAN sharing with common carrier, see *Multi-Operator Sharing in 5G RAN Feature Documentation*. The following neighboring LTE cell data must be configured on the NR side: **NRCellEutranNFreq**, **gNBEutraExternalCell**, **NRCellEutraNRelation**, and **gNBEutraExtCellPlmn** MOs.

6.2 Network Analysis

6.2.1 Benefits

For details, see *Multi-Operator Sharing in 5G RAN Feature Documentation* and *RAN Sharing in eRAN Feature Documentation*.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

For details about the impacts of the function, see *Multi-Operator Sharing in 5G RAN Feature Documentation* and *RAN Sharing in eRAN Feature Documentation*.

6.3 Requirements

6.3.1 Licenses

Compared with that in independent operator mode, interoperability between E-UTRAN and NG-RAN in multi-operator sharing scenarios has no extra license requirements.

For details about the license requirements for multi-operator sharing on the NR and LTE sides, see *Multi-Operator Sharing in 5G RAN Feature Documentation* and *RAN Sharing in eRAN Feature Documentation*, respectively.

For details about the license requirements for each function of interoperability between E-UTRAN and NG-RAN, see "Licenses" in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#) and the "Mobility Management for

UEs in Idle Mode" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

6.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

Compared with that in independent operator mode, interoperability between E-UTRAN and NG-RAN in multi-operator sharing scenarios has no additional prerequisite functions.

For details about the prerequisite functions for multi-operator sharing on the NR and LTE sides, see *Multi-Operator Sharing* in *5G RAN Feature Documentation* and *RAN Sharing* in *eRAN Feature Documentation*, respectively.

For details about the prerequisite functions for each function of interoperability between E-UTRAN and NG-RAN, see "Prerequisite Functions" in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#) and the "Mobility Management for UEs in Idle Mode" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

6.3.2.2 Mutually Exclusive Functions

Compared with that in independent operator mode, interoperability between E-UTRAN and NG-RAN in multi-operator sharing scenarios has no additional mutually exclusive functions.

For details about the mutually exclusive functions for multi-operator sharing on the NR and LTE sides, see *Multi-Operator Sharing* in *5G RAN Feature Documentation* and *RAN Sharing* in *eRAN Feature Documentation*, respectively.

For details about the mutually exclusive functions for each function of interoperability between E-UTRAN and NG-RAN, see "Mutually Exclusive Functions" in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#) and the "Mobility Management for UEs in Idle Mode" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

6.3.2.3 Function Impacts

For details, see *Multi-Operator Sharing* in *5G RAN Feature Documentation* and *RAN Sharing* in *eRAN Feature Documentation*.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

For LTE, the following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

For NR, the following base stations are compatible with this function:

- 3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For details about the board requirements for multi-operator sharing on the NR and LTE sides, see *Multi-Operator Sharing* in *5G RAN Feature Documentation* and *RAN Sharing* in *eRAN Feature Documentation*, respectively.

For details about the board requirements for each function of interoperability between E-UTRAN and NG-RAN, see "Boards" in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#) and the "Mobility Management for UEs in Idle Mode" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

RF Modules

For details about the RF module requirements for multi-operator sharing on the NR and LTE sides, see *Multi-Operator Sharing* in *5G RAN Feature Documentation* and *RAN Sharing* in *eRAN Feature Documentation*, respectively.

For details about the RF module requirements for each function of interoperability between E-UTRAN and NG-RAN, see "RF Modules" in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#) and the "Mobility Management for UEs in Idle Mode" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

6.3.4 Others

For details about other requirements for multi-operator sharing on the NR and LTE sides, see *Multi-Operator Sharing* in *5G RAN Feature Documentation* and *RAN Sharing* in *eRAN Feature Documentation*, respectively.

For details about other requirements for each function of interoperability between E-UTRAN and NG-RAN, see "Others" in [5 Interoperability Between E-UTRAN and NG-RAN in Connected Mode](#) and the "Mobility Management for UEs in Idle Mode" section in *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Operator-specific Parameters on the NR Side

Table 6-7 describes the operator-specific parameters on the NR side.

Table 6-7 Operator-specific parameters on the NR side

Parameter Name	Parameter ID	Setting Notes
Cell Reselection Priority	gNBFreqPriorityGroup.CellReselPri	Set this parameter based on the network plan.
Cell Reselection Sub-Priority	gNBFreqPriorityGroup.CellReselSubPri	Set this parameter based on the network plan.
VOLTE Blind Priority	gNBFreqPriorityGroup.VolteBlindPriority	Set this parameter based on the network plan. If this parameter is set to 255 , the neighboring LTE frequency does not support VoLTE. If this parameter is set to 65535 , the frequency priority is not obtained from the priority group.
E-UTRAN A2/B2 RSRP Threshold Offset	gNBOperatorQciParam.EutranA2B2RsrpThldOffset	The threshold for event A2 and threshold 1 for event B2 can be set on a per operator basis only in RAN sharing with common carrier scenarios.
E-UTRAN A1 RSRP Threshold Offset	gNBOperatorQciParam.EutranA1RsrpThldOffset	The threshold for event A1 can be set on a per operator basis only in RAN sharing with common carrier scenarios.
E-UTRAN B1/B2 RSRP Threshold Offset	gNBOperatorQciParam.EutranB1B2RsrpThldOffset	The threshold for event B1 and threshold 2 for event B2 related to downlink-coverage-based mobility can be set on a per operator basis only in RAN sharing with common carrier scenarios.
EPS FB B1 RSRP Threshold Offset	gNBOperatorQciParam.EpsFbB1RsrpThldOffset	The threshold for event B1 related to EPS fallback can be set on a per operator basis only in RAN sharing with common carrier scenarios.
E-UTRAN CIO Offset	NRCellOpPolicy.EutranCioOffset	The E-UTRAN CIO can be set on a per operator basis only in RAN sharing with common carrier scenarios.

Parameter Name	Parameter ID	Setting Notes
Inter-RAT Handover Policy Switch	NRCellOpPolicy.<i>InterRatHoPolicy-Switch</i>	In RAN sharing with common carrier scenarios, set this parameter based on operator policies.
EPS Fallback Mode	NRCellOpPolicy.<i>EpsFbMode</i>	In RAN sharing with common carrier scenarios, set this parameter based on operator policies.
gNodeB Frequency Priority Group ID	NRCellOpPolicy.<i>gNBFreqPriorityGroupId</i>	Set this parameter to a value in the range from 0 to 254. The value 255 indicates that the frequency priority group is not configured.
Inter-RAT Handover Policy Switch	NRCellOpPolicy.<i>InterRatHoPolicy-Switch</i>	The VOICE_BLIND_MODE_SWITCH option controls whether the voice services of each operator can be performed in blind mode. In RAN sharing with common carrier scenarios, set this parameter based on operator policies.
Frequency Configuration Policy Switch	NRCellOpPolicy.<i>FreqConfigPolicy-Switch</i>	To enable operator-specific LTE frequency configuration, select the OPERATOR_EUTRAN_FREQ_CFG_SW option.
Inter-RAT A1/A2/B2 RSRP Threshold Offset	gNBRfspConfig.<i>InterRatA1A2B2RsrpThldOfs</i>	The offset to coverage-based inter-RAT handover A1 RSRP threshold, A2 RSRP threshold, and B2 RSRP threshold 1 can be configured on a per operator basis only in RAN sharing with common carrier scenarios.
E-UTRAN A1/A2/B2 RSRP Threshold Offset	NRCellOpPolicy.<i>EutranA1A2B2RsrpThldOfs</i>	The threshold for event A1, threshold for event A2, and threshold 1 for event B2 can be configured on a per operator basis only in RAN sharing with common carrier scenarios.

Operator-specific Parameters on the LTE Side

Table 6-8 describes the operator-specific parameters on the LTE side.

Table 6-8 Operator-specific parameters on the LTE side

Parameter Name	Parameter ID	Setting Notes
NR A2 Threshold RSRP Offset	CnOperatorHoCfg.<i>NrA2ThldRsrpOffset</i>	In RAN sharing with common carrier scenarios, set this parameter to control the operator-specific A2 threshold as required.

Parameter Name	Parameter ID	Setting Notes
NR A2 RSRP Threshold Offset	CnOperatorQciPara.NrA2RsrpThldOffset	Set this parameter when the RSRP threshold for event A2 needs to be flexibly configured on a per operator and QCI basis.
NR A1 RSRP Threshold Offset	CnOperatorQciPara.NrA1RsrpThldOffset	Set this parameter when the RSRP threshold for event A1 needs to be flexibly configured on a per operator and QCI basis.
Event Type	CnOpQciEventParam.EventType	Set this parameter when the threshold for event B1 needs to be flexibly configured on a per operator and QCI basis. - Set this parameter to COV_BASED_HO_TO_NR_B1 when the threshold for coverage-based event B1 needs to be configured. - Set this parameter to SERV_BASED_HO_TO_NR_B1 when the threshold for service-based event B1 needs to be configured.
Handover Threshold Offset	CnOpQciEventParam.HandoverThldOffset	Set this parameter based on operator policies when the threshold for event B1 needs to be flexibly configured on a per operator and QCI basis.
Handover Mode Switch	CellOpHoCfg.HoModeSwitch	In RAN sharing with common carrier scenarios, set this parameter based on operator policies.
VoNR Priority	RatFreqPriorityGroup.VonrPriority	In RAN sharing with common carrier scenarios, set this parameter based on operator policies.
Frequency Priority for Connected Mode	RatFreqPriorityGroup.ConnFreqPriority	In RAN sharing with common carrier scenarios, set this parameter based on operator policies.
VoNR Priority	NrNFreq.VonrPriority	Set this parameter based on the network plan.
Handover Algo Switch	CellHoParaCfg.CellHoAlgoSwitch	In RAN sharing with common carrier scenarios, select the PlmnBasedNrFreqPolicySw option.
Shared PLMN gNodeB ID	NrExternalCellPlmn.SharedPlmnGnodebId	This parameter is mandatory when operator-specific NCI configuration is enabled. Set this parameter to the gNodeB ID of the cell for a secondary NR operator.

Parameter Name	Parameter ID	Setting Notes
Shared PLMN Cell ID	NrExternalCellPlmn.SharedPlmnCellId	This parameter is mandatory when operator-specific NCI configuration is enabled. Set this parameter to the cell ID of the cell for a secondary NR operator.
LTE to NR PLMN Selection Mode	EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode	This parameter specifies the PLMN ID filling mode in LTE-to-NR procedures at the eNodeB level. Set this parameter to SELECTED_PLMN_ID .
LTE to NR PLMN Selection Mode	NbrGnodebPolicy.LteToNrPlmnSelectionMode	This parameter specifies the PLMN ID filling mode in LTE-to-NR procedures at the target gNodeB level. Set this parameter to SELECTED_PLMN_ID . If both the parameters specifying the PLMN ID filling mode are set, the target-gNodeB-level configuration takes effect.
Frequency Priority for Connected Mode	NrNFreq.ConnFreqPriority	In RAN sharing with common carrier scenarios, set this parameter based on operator policies.
Cell Reselection Sub-Priority	NrNFreqSCellOp.CellReselSubPriority	In RAN sharing with common carrier scenarios, set this parameter based on operator policies.
Cell Reselection Sub-Priority	NrNFreqSCellOp.CellReselPriority	In RAN sharing with common carrier scenarios, set this parameter based on operator policies.

6.4.1.2 Using MML Commands

The following provides MML command examples in multi-operator sharing scenarios. In the examples, the PLMN ID of the primary operator is 46000, the PLMN ID of the secondary operator is 46060, and NR cells are FDD cells.

Before running the following MML commands, set NR and LTE cells to work in multi-operator sharing mode. For details, see *Multi-Operator Sharing in 5G RAN Feature Documentation* and *RAN Sharing in eRAN Feature Documentation*. In addition, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

LTE Cells in Independent Operator Mode + NR Cells in RAN Sharing with Common Carrier Mode

```
//Data configuration on the NR side:
//Setting reselection priority values and sub-priority values for different operators
ADD GNBFRECPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=0, RatType=EUTRAN,
DEarfcn=1300, CellReselPri=1, CellReselSubPri=0DOT2;
ADD GNBFRECPRIORITYGROUP: gNBFreqPriorityGroupId=2, FreqIndex=1, RatType=EUTRAN,
DEarfcn=1600, CellReselPri=1, CellReselSubPri=0DOT4;
//Setting mobility policies for different operators
ADD NRCELLOPPOLICY: NrCellId=0, OperatorId=1, gNBFreqPriorityGroupId=1, EpsFbMode=HANDOVER,
InterRatHoPolicySwitch=EUTRAN_REDIRECT_SWITCH-1&EUTRAN_HO_SWITCH-1&VOICE_BLIND_MODE_SWI
TCH-1, EutranCioOffset=1, FreqConfigPolicySwitch=OPERATOR_EUTRAN_FREQ_CFG_SW-1;
ADD NRCELLOPPOLICY: NrCellId=0, OperatorId=2, gNBFreqPriorityGroupId=2, EpsFbMode=REDIRECTION,
InterRatHoPolicySwitch=EUTRAN_REDIRECT_SWITCH-1&EUTRAN_HO_SWITCH-0&VOICE_BLIND_MODE_SWI
TCH-0, EutranCioOffset=2, FreqConfigPolicySwitch=OPERATOR_EUTRAN_FREQ_CFG_SW-1;
//Setting the operator-and-QCI-specific RSRP threshold offset for event A2/B2 related to mobility to E-
UTRAN, RSRP threshold offset for event A1 related to mobility to E-UTRAN, and RSRP threshold offset for
event B1 related to EPS fallback
MOD GNBOPERATORQCIPARAM: OperatorId=1, Qci=1, EutranA2B2RsrpThldOffset=1,
EutranA1RsrpThldOffset=3, EpsFbB1RsrpThldOffset=1;
MOD GNBOPERATORQCIPARAM: OperatorId=2, Qci=1, EutranA2B2RsrpThldOffset=3,
EutranA1RsrpThldOffset=1, EpsFbB1RsrpThldOffset=2;
//Setting the operator-and-QCI-specific RSRP threshold offset for event B1/B2 related to mobility to E-
UTRAN
MOD GNBOPERATORQCIPARAM: OperatorId=1, Qci=9, EutranB1B2RsrpThldOffset=1;
MOD GNBOPERATORQCIPARAM: OperatorId=2, Qci=9, EutranB1B2RsrpThldOffset=2;
//Setting the operator-specific offset relative to the RSRP threshold for event A1/A2/B2 related to mobility
handover
MOD GNBREFSPCONFIG: OperatorId=1, RfspIndex=1, InterRatA1A2B2RsrpThldOfs=1;
MOD GNBREFSPCONFIG: OperatorId=2, RfspIndex=2, InterRatA1A2B2RsrpThldOfs=2;
//Setting the operator-specific offset relative to the RSRP threshold for event A1/A2/B2 related to mobility
to E-UTRAN
MOD NRCELLOPPOLICY: NrCellId=0, OperatorId=1, EutranA1A2B2RsrpThldOfs=1;
MOD NRCELLOPPOLICY: NrCellId=0, OperatorId=2, EutranA1A2B2RsrpThldOfs=2;
//Setting the priority of an EARFCN for voice services in blind mode on a per operator basis
//Running the following commands with an assumption that the following gNBFreqPriorityGroupId and
FreqIndex instances have been configured for the base station
MOD GNBFRECPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=0, RatType=EUTRAN,
DEarfcn=1300, VolteBlindPriority=1;
MOD GNBFRECPRIORITYGROUP: gNBFreqPriorityGroupId=2, FreqIndex=1, RatType=EUTRAN,
DEarfcn=1600, VolteBlindPriority=2;
//Running the following commands with an assumption that the following gNBFreqPriorityGroupId and
FreqIndex instances have not been configured for the base station
ADD GNBFRECPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=0, RatType=EUTRAN,
DEarfcn=1300, VolteBlindPriority=1;
ADD GNBFRECPRIORITYGROUP: gNBFreqPriorityGroupId=2, FreqIndex=1, RatType=EUTRAN,
DEarfcn=1600, VolteBlindPriority=2;

//Data configuration on the LTE side:
//Configuring neighboring NR cell data for LTE cell 0
//Adding a neighboring NR frequency (For NR TDD, set SubcarrierSpacing to 30KHZ.)
ADD NRNFREQ: LocalCellId=0, DEarfcn=428000, ConnFreqPriority=1, VonrPriority=255,
SubcarrierSpacing=15KHZ;
//Setting the frequency band to which the neighboring NR frequency belongs
ADD NRMFBIFREQ: DEarfcn=428000, FrequencyBand=N66;
//Adding PLMN information for neighboring cells
ADD NCELLPLMNLIST: Mcc="460", Mnc="00", RatType=NR, gNBIdLength=32;
//Adding an external NR cell
ADD NREXTERNALCELL: Mcc="460", Mnc="00", GnodebId=123, CellId=1, DEarfcn=428000, PhyCellId=1,
Tac=1;
//Adding a neighbor relationship with the NR cell
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="00", GnodebId=123, CellId=1;
//Adding a RAT/frequency priority group
ADD RATFRECPRIORITYGROUP: RatFreqPriorityGroupId=0, RatType=NR,
DEarfcn=428000, ConnFreqPriority=14, VonrPriority=14;
//Turning on the PLMN-based NR frequency policy switch
MOD CELLHOPARACFG: LocalCellId=0, CellHoAlgoSwitch=PlmnBasedNrFreqPolicySw-1;
```

```
//Configuring neighboring NR cell data for LTE cell 1
//Adding a neighboring NR frequency (For NR TDD, set SubcarrierSpacing to 30KHZ.)
ADD NRNFREQ: LocalCellId=1, DlArfcn=428000, ConnFreqPriority=1, VnrPriority=255,
SubcarrierSpacing=15KHZ;
//Setting the frequency band to which the neighboring NR frequency belongs
ADD NRMFBIFREQ: DlArfcn=428000, FrequencyBand=N66;
//Adding PLMN information for neighboring cells
ADD NCELLPLMNLIST: Mcc="460", Mnc="00", RatType=NR,gNBIdLength=32;
//Adding an external NR cell
ADD NREXTERNALCELL: Mcc="460", Mnc="00", GnodebId=123, CellId=1, DlArfcn=428000, PhyCellId=1,
Tac=1;
//Adding a neighbor relationship with the NR cell
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="00", GnodebId=123, CellId=1;
//((Required when operator-specific NCI configuration is not enabled) Adding the PLMN information of the
secondary NR operator, where GnodebId and CellId are set to the gNodeB ID and cell ID specified for the
primary operator, respectively
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="00", GnodebId=123, CellId=1, SharedMcc="460",
SharedMnc="60", NrNetworkingOption=SA, Tac=2;
//((Required when operator-specific NCI configuration is enabled) Adding the PLMN information of the
secondary NR operator, where SharedPlmnGnodebId and SharedPlmnCellId are set to the gNodeB ID and
cell ID specified for the secondary operator, respectively
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="00", GnodebId=123, CellId=1, SharedMcc="460",
SharedMnc="60", NrNetworkingOption=SA, Tac=2, SharedPlmnGnodebId=456, SharedPlmnCellId=2;
//Setting the eNodeB-level or target-gNodeB-level PLMN ID filling mode in LTE-to-NR procedures when
operator-specific NCI configuration is enabled (If both modes are set, the target-gNodeB-level configuration
takes effect.)
//Setting the eNodeB-level PLMN ID filling mode in LTE-to-NR procedures
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Setting the target-gNodeB-level PLMN ID filling mode in LTE-to-NR procedures
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="00", GnodebId=123,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Adding a RAT/frequency priority group
ADD RATFREQPRIORITYGROUP: RatFreqPriorityGroupId=1, RatType=NR,
DlArfcn=428000,ConnFreqPriority=12,VnrPriority=12;
```

LTE Cells in Independent Operator Mode + NR Cells in RAN Sharing with Dedicated Carrier Mode

```
//Data configuration on the LTE side:
//Configuring neighboring cell data for the primary NR operator
//Adding a neighboring NR frequency (For NR TDD, set SubcarrierSpacing to 30KHZ.)
ADD NRNFREQ: LocalCellId=0, DlArfcn=428000, ConnFreqPriority=1, VnrPriority=255,
SubcarrierSpacing=15KHZ;
//Setting the frequency band to which the neighboring NR frequency belongs
ADD NRMFBIFREQ: DlArfcn=428000, FrequencyBand=N66;
//Adding PLMN information for neighboring cells
ADD NCELLPLMNLIST: Mcc="460", Mnc="00", RatType=NR,gNBIdLength=32;
//Adding an external NR cell
ADD NREXTERNALCELL: Mcc="460", Mnc="00", GnodebId=123, CellId=1, DlArfcn=428000, PhyCellId=1,
Tac=1;
//Adding a neighbor relationship with the NR cell
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="00", GnodebId=123, CellId=1;

//Configuring neighboring cell data for the secondary NR operator
//Adding a neighboring NR frequency (For NR TDD, set SubcarrierSpacing to 30KHZ.)
ADD NRNFREQ: LocalCellId=1, DlArfcn=525000, ConnFreqPriority=1, VnrPriority=255,
SubcarrierSpacing=15KHZ;
//Adding PLMN information for neighboring cells
ADD NCELLPLMNLIST: Mcc="460", Mnc="00", RatType=NR,gNBIdLength=32;
//When operator-specific NCI configuration is not enabled:
//Adding an external NR cell with the MCC and MNC set to those of the primary operator
ADD NREXTERNALCELL: Mcc="460", Mnc="00", GnodebId=123, CellId=2, DlArfcn=525000, PhyCellId=2,
Tac=1, MasterPlmnReservedFlag=TRUE;
//Adding PLMN information about the secondary NR operator
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="00", GnodebId=123, CellId=2, SharedMcc="460",
SharedMnc="60", NrNetworkingOption=SA, Tac=2;
//Adding a neighbor relationship with the NR cell
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="00", GnodebId=123, CellId=2;
//When operator-specific NCI configuration is enabled:
```

```
//Adding an external NR cell with the MCC and MNC set to those of the secondary operator
ADD NREXTERNALCELL: Mcc="460", Mnc="60", GnodebId=456, CellId=2, DlArfcn=525000, PhyCellId=2,
Tac=1, MasterPlmnReservedFlag=FALSE;
//Adding a neighbor relationship with the NR cell
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="60", GnodebId=456, CellId=2;
//Setting the eNodeB-level or target-gNodeB-level PLMN ID filling mode in LTE-to-NR procedures when
operator-specific NCI configuration is enabled (If both modes are set, the target-gNodeB-level configuration
takes effect.)
//Setting the eNodeB-level PLMN ID filling mode in LTE-to-NR procedures
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Setting the target-gNodeB-level PLMN ID filling mode in LTE-to-NR procedures
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="00", GnodebId=123,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
```

LTE Cells in RAN Sharing with Common Carrier Mode + NR Cells in Independent Operator Mode

```
//Data configuration on the LTE side:
//Setting reselection priority values and sub-priority values for different operators
ADD NRNFREQSCCELLOP: LocalCellId=0,
DlArfcn=428000, CnOperatorId=1, CellReselPriorityConfigInd=CFG, CellReselPriority=1, CellReselSubPriority=ZER
O;
ADD NRNFREQSCCELLOP: LocalCellId=0,
DlArfcn=504900, CnOperatorId=2, CellReselPriorityConfigInd=CFG, CellReselPriority=2, CellReselSubPriority=ZER
O;
//Setting mobility policies for different operators
ADD CELLOPHOCFG: LocalCellId=0, CnOperatorId=1,
HoModeSwitch=REDIRECTION_TO_NR_SW-1&HANDOVER_TO_NR_SW-1;
ADD CELLOPHOCFG: LocalCellId=0, CnOperatorId=2,
HoModeSwitch=REDIRECTION_TO_NR_SW-1&HANDOVER_TO_NR_SW-0;
//Setting the operator-specific RSRP threshold offset for event A2 related to mobility from E-UTRAN to NG-
RAN
MOD CNOOPERATORHOCFG: CnOperatorId=1, NrA2ThldRsrpOffset=1;
MOD CNOOPERATORHOCFG: CnOperatorId=2, NrA2ThldRsrpOffset=2;
//Setting the operator-and-QCI-specific RSRP threshold offset parameters for events A1 and A2 related to
mobility from E-UTRAN to NG-RAN
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=9, NrA2RsrpThldOffset=1, NrA1RsrpThldOffset=4;
MOD CNOOPERATORQCIPARA: CnOperatorId=2, Qci=9, NrA2RsrpThldOffset=2, NrA1RsrpThldOffset=2;
//Setting the operator-and-QCI-specific offset for event B1 related to coverage-based mobility
ADD CNOPQCIEVENTPARAM: CnOperatorId=1, Qci=9, EventType=COV_BASED_HO_TO_NR_B1,
HandoverThldOffset=1;
ADD CNOPQCIEVENTPARAM: CnOperatorId=2, Qci=9, EventType=COV_BASED_HO_TO_NR_B1,
HandoverThldOffset=2;
//Setting the operator-and-QCI-specific offset for event B1 related to service-based mobility
ADD CNOPQCIEVENTPARAM: CnOperatorId=1, Qci=9, EventType=SERV_BASED_HO_TO_NR_B1,
HandoverThldOffset=1;
ADD CNOPQCIEVENTPARAM: CnOperatorId=2, Qci=9, EventType=SERV_BASED_HO_TO_NR_B1,
HandoverThldOffset=2;

//Data configuration on the NR side:
//Configuring neighboring cell data for the primary LTE operator
//Adding a neighboring LTE frequency
ADD NRCELLEUTRANNFREQ: NrCellId=0, DlEarfcn=1300, MeasurementBandwidth=MBW100;
//Adding an external LTE cell
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="00", EnodebId=111, CellId=1, DlEarfcn=1300,
PhysicalCellId=183, Tac=4026;
//Adding a neighbor relationship with the LTE cell
ADD NRCELLEUTRANRELATION: NrCellId=0, Mcc="460", Mnc="00", EnodebId=111, CellId=1;

//Configuring neighboring cell data for the secondary LTE operator
//Adding a neighboring LTE frequency
ADD NRCELLEUTRANNFREQ: NrCellId=1, DlEarfcn=1300, MeasurementBandwidth=MBW100;
//Adding an external LTE cell
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="00", EnodebId=111, CellId=1, DlEarfcn=1300,
PhysicalCellId=183, Tac=4026;
//Adding a neighbor relationship with the LTE cell
ADD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="00", EnodebId=111, CellId=1;
//Adding PLMN information about the secondary LTE operator
```

```
ADD GNBEUTRAEXTCELLPLMN: Mcc="460", Mnc="00", EnodebId=111, CellId=1, SharedMcc="460",
SharedMnc="60";
```

LTE Cells in RAN Sharing with Dedicated Carrier Mode + NR Cells in Independent Operator Mode

```
//Data configuration on the NR side:
//Configuring neighboring primary-operator LTE cell data of the NR cell
//Adding a neighboring LTE frequency
ADD NRCELLEUTRANNFREQ: NrCellId=0, DLearfcn=1300, MeasurementBandwidth=MBW100;
//Adding an external LTE cell
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="00", EnodebId=111, CellId=1, DLearfcn=1300,
PhysicalCellId=183, Tac=4026;
//Adding a neighbor relationship with the LTE cell
ADD NRCELLEUTRANRELATION: NrCellId=0, Mcc="460", Mnc="00", EnodebId=111, CellId=1;

//Configuring neighboring secondary-operator LTE cell data of the NR cell
//Adding a neighboring LTE frequency
ADD NRCELLEUTRANNFREQ: NrCellId=1, DLearfcn=1600, MeasurementBandwidth=MBW100;
//Adding an external LTE cell
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="60", EnodebId=111, CellId=2, DLearfcn=1600,
PhysicalCellId=193, Tac=4030;
//Adding a neighbor relationship with the LTE cell
ADD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="60", EnodebId=111, CellId=2;
```

LTE Cells in RAN Sharing with Dedicated Carrier Mode + NR Cells in RAN Sharing with Common Carrier Mode

```
//Data configuration on the LTE side:
//Setting reselection priority values and sub-priority values for different operators
ADD NRNFREQSCCELLOP: LocalCellId=0,
DLearfcn=428000,CnOperatorId=1,CellReselPriorityConfigInd=CFG,CellReselPriority=1,CellReselSubPriority=ZER
O;
ADD NRNFREQSCCELLOP: LocalCellId=1,
DLearfcn=428000,CnOperatorId=2,CellReselPriorityConfigInd=CFG,CellReselPriority=2,CellReselSubPriority=ZER
O;
//Setting mobility policies for different operators
ADD CELLOPHOCFG: LocalCellId=0, CnOperatorId=1,
HoModeSwitch=REDIRECTION_TO_NR_SW-1&HANDOVER_TO_NR_SW-1;
ADD CELLOPHOCFG: LocalCellId=1, CnOperatorId=2,
HoModeSwitch=REDIRECTION_TO_NR_SW-1&HANDOVER_TO_NR_SW-0;
//Setting the operator-specific RSRP threshold offset for event A2 related to mobility from E-UTRAN to NG-
RAN
MOD CNOOPERATORHOCFG: CnOperatorId=1, NrA2ThldRsrpOffset=1;
MOD CNOOPERATORHOCFG: CnOperatorId=2, NrA2ThldRsrpOffset=2;
//Setting the operator-and-QCI-specific RSRP threshold offset parameters for events A1 and A2 related to
mobility from E-UTRAN to NG-RAN
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=9, NrA2RsrpThldOffset=1, NrA1RsrpThldOffset=4;
MOD CNOOPERATORQCIPARA: CnOperatorId=2, Qci=9, NrA2RsrpThldOffset=2, NrA1RsrpThldOffset=2;
//Setting the operator-and-QCI-specific offset for event B1 related to coverage-based mobility
ADD CNOPQCIEVENTPARAM: CnOperatorId=1, Qci=9, EventType=COV_BASED_HO_TO_NR_B1,
HandoverThldOffset=1;
ADD CNOPQCIEVENTPARAM: CnOperatorId=2, Qci=9, EventType=COV_BASED_HO_TO_NR_B1,
HandoverThldOffset=2;
//Setting the operator-and-QCI-specific offset for event B1 related to service-based mobility
ADD CNOPQCIEVENTPARAM: CnOperatorId=1, Qci=9, EventType=SERV_BASED_HO_TO_NR_B1,
HandoverThldOffset=1;
ADD CNOPQCIEVENTPARAM: CnOperatorId=2, Qci=9, EventType=SERV_BASED_HO_TO_NR_B1,
HandoverThldOffset=2;
//Adding RAT/frequency priority groups and associating the groups with operators
ADD RATFREQPRIORITYGROUP: RatFreqPriorityGroupId=0, RatType=NR,
DLearfcn=428000,ConnFreqPriority=14,VonrPriority=14;
ADD RATFREQPRIORITYGROUP: RatFreqPriorityGroupId=1, RatType=NR,
DLearfcn=428000,ConnFreqPriority=12,VonrPriority=12;
MOD CELLOP: LocalCellId=0, TrackingAreald=0, RatFreqPriorityGroupId=0;
MOD CELLOP: LocalCellId=1, TrackingAreald=1, RatFreqPriorityGroupId=1;
//Turning on the PLMN-based NR frequency policy switch
MOD CELLHOPARACFG: LocalCellId=0, CellHoAlgoSwitch=PlmnBasedNrFreqPolicySw-1;
```

```

MOD CELLHOPARACFG: LocalCellId=1, CellHoAlgoSwitch=PlmnBasedNrFreqPolicySw-1;

//Configuring neighboring NR cell data on the LTE side
//Adding a neighboring NR frequency (For NR TDD, set SubcarrierSpacing to 30KHZ.)
ADD NRNFREQ: LocalCellId=0, DlArfcn=428000, ConnFreqPriority=1, VnrPriority=255,
SubcarrierSpacing=15KHZ;
ADD NRNFREQ: LocalCellId=1, DlArfcn=428000, ConnFreqPriority=1, VnrPriority=255,
SubcarrierSpacing=15KHZ;
//Setting the frequency band to which the neighboring NR frequency belongs
ADD NRMFBIFREQ: DlArfcn=428000, FrequencyBand=N66;
//Adding PLMN information for neighboring cells
ADD NCELLPLMNLIST: Mcc="460", Mnc="00", RatType=NR,gNBIdLength=32;
//Adding an external NR cell
ADD NREXTERNALCELL: Mcc="460", Mnc="00", GnodebId=123, CellId=1, DlArfcn=428000, PhyCellId=1,
Tac=1;
//Adding neighbor relationships with the NR cell
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="00", GnodebId=123, CellId=1;
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="00", GnodebId=123, CellId=1;
//((Required when operator-specific NCI configuration is not enabled) Adding the PLMN information of the
secondary NR operator, where GnodebId and CellId are set to the gNodeB ID and cell ID specified for the
primary operator, respectively
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="00", GnodebId=123, CellId=1, SharedMcc="460",
SharedMnc="60", NrNetworkingOption=SA, Tac=2;
//((Required when operator-specific NCI configuration is enabled) Adding the PLMN information of the
secondary NR operator, where SharedPlmnGnodebId and SharedPlmnCellId are set to the gNodeB ID and
cell ID specified for the secondary operator, respectively
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="00", GnodebId=123, CellId=1, SharedMcc="460",
SharedMnc="60", NrNetworkingOption=SA, Tac=2, SharedPlmnGnodebId=456, SharedPlmnCellId=2;
//((Optional) Setting the eNodeB-level PLMN ID filling mode in LTE-to-NR procedures when operator-
specific NR Cell Identity (NCI) configuration is enabled
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//((Optional) Setting the target-gNodeB-level PLMN ID filling mode in LTE-to-NR procedures when operator-
specific NCI configuration is enabled
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="00", GnodebId=123,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;

//Data configuration on the NR side:
//Configuring data related to neighboring LTE cell 0 for the NR cell
//Adding a neighboring LTE frequency
ADD NRCELLEUTRANNFREQ: NrCellId=1, DlEarfcn=1300, MeasurementBandwidth=MBW100;
//Adding an external LTE cell
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="00", EnodebId=111, CellId=0, DlEarfcn=1300,
PhysicalCellId=183, Tac=4026;
//Adding a neighbor relationship with the LTE cell
ADD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="00", EnodebId=111, CellId=0;

//Configuring data related to neighboring LTE cell 1 for the NR cell
//Adding a neighboring LTE frequency
ADD NRCELLEUTRANNFREQ: NrCellId=1, DlEarfcn=1600, MeasurementBandwidth=MBW100;
//Adding an external LTE cell
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="60", EnodebId=111, CellId=1, DlEarfcn=1600,
PhysicalCellId=193, Tac=4030;
//Adding a neighbor relationship with the LTE cell
ADD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="60", EnodebId=111, CellId=1;

```

LTE Cells in RAN Sharing with Common Carrier Mode + NR Cells in RAN Sharing with Common Carrier Mode

```

//Data configuration on the LTE side:
//Setting reselection priority values and sub-priority values for different operators
ADD NRNFREQSCELLOP: LocalCellId=0,
DlArfcn=428000,CnOperatorId=1,CellReselPriorityConfigInd=CFG,CellReselPriority=1,CellReselSubPriority=ZER
O;
ADD NRNFREQSCELLOP: LocalCellId=0,
DlArfcn=428000,CnOperatorId=2,CellReselPriorityConfigInd=CFG,CellReselPriority=2,CellReselSubPriority=ZER
O;
//Setting mobility policies for different operators
ADD CELLOPHOCFG: LocalCellId=0, CnOperatorId=1,
HoModeSwitch=REDIRECTION_TO_NR_SW-1&HANDOVER_TO_NR_SW-1;

```

```

ADD CELLOPHOCFG: LocalCellId=0, CnOperatorId=2,
HoModeSwitch=REDIRECTION_TO_NR_SW-1&HANDOVER_TO_NR_SW-0;
//Setting the operator-specific RSRP threshold offset for event A2 related to mobility from E-UTRAN to NG-
RAN
MOD CNOOPERATORHOCFG: CnOperatorId=1, NrA2ThldRsrpOffset=1;
MOD CNOOPERATORHOCFG: CnOperatorId=2, NrA2ThldRsrpOffset=2;
//Setting the operator-and-QCI-specific RSRP threshold offset parameters for events A1 and A2 related to
mobility from E-UTRAN to NG-RAN
MOD CNOOPERATORQCIPARA: CnOperatorId=1, Qci=9, NrA2RsrpThldOffset=1, NrA1RsrpThldOffset=4;
MOD CNOOPERATORQCIPARA: CnOperatorId=2, Qci=9, NrA2RsrpThldOffset=2, NrA1RsrpThldOffset=2;
//Setting the operator-and-QCI-specific offset for event B1 related to coverage-based mobility
ADD CNOPQCIEVENTPARAM: CnOperatorId=1, Qci=9, EventType=COV_BASED_HO_TO_NR_B1,
HandoverThldOffset=1;
ADD CNOPQCIEVENTPARAM: CnOperatorId=2, Qci=9, EventType=COV_BASED_HO_TO_NR_B1,
HandoverThldOffset=2;
//Setting the operator-and-QCI-specific offset for event B1 related to service-based mobility
ADD CNOPQCIEVENTPARAM: CnOperatorId=1, Qci=9, EventType=SERV_BASED_HO_TO_NR_B1,
HandoverThldOffset=1;
ADD CNOPQCIEVENTPARAM: CnOperatorId=2, Qci=9, EventType=SERV_BASED_HO_TO_NR_B1,
HandoverThldOffset=2;
//Adding RAT/frequency priority groups and associating the groups with operators
ADD RATFREQPRIORITYGROUP: RatFreqPriorityGroupId=0, RatType=NR,
DIEarfcn=428000,ConnFreqPriority=14,VonrPriority=14;
ADD RATFREQPRIORITYGROUP: RatFreqPriorityGroupId=1, RatType=NR,
DIEarfcn=428000,ConnFreqPriority=12,VonrPriority=12;
MOD CELLOP: LocalCellId=0, TrackingAreaid=0, RatFreqPriorityGroupId=0;
MOD CELLOP: LocalCellId=0, TrackingAreaid=1, RatFreqPriorityGroupId=1;
//Turning on the PLMN-based NR frequency policy switch
MOD CELLHOPARACFG: LocalCellId=0, CellHoAlgoSwitch=PlmnBasedNrFreqPolicySw-1;

//Configuring neighboring NR cell data on the LTE side
//Adding a neighboring NR frequency (For NR TDD, set SubcarrierSpacing to 30KHZ.)
ADD NRNFREQ: LocalCellId=0, DIEarfcn=428000, ConnFreqPriority=1, VonrPriority=255,
SubcarrierSpacing=15KHZ;
//Setting the frequency band to which the neighboring NR frequency belongs
ADD NRMFBIFREQ: DIEarfcn=428000, FrequencyBand=N66;
//Adding PLMN information for neighboring cells
ADD NCELLPLMNLIST: Mcc="460", Mnc="00", RatType=NR,gNBIdLength=32;
//Adding an external NR cell
ADD NREXTERNALCELL: Mcc="460", Mnc="00", GnodebId=123, CellId=1, DIEarfcn=428000, PhyCellId=1,
Tac=1;
//Adding a neighbor relationship with the NR cell
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="00", GnodebId=123, CellId=1;
//When operator-specific NCI configuration is not enabled:
//Adding the PLMN information of the secondary NR operator, where GnodebId and CellId are set to the
gNodeB ID and cell ID specified for the primary operator, respectively
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="00", GnodebId=123, CellId=1, SharedMcc="460",
SharedMnc="60", NrNetworkingOption=SA, Tac=2;
//When operator-specific NCI configuration is enabled:
//Adding the PLMN information of the secondary NR operator, where SharedPlmnGnodebId and
SharedPlmnCellId are set to the gNodeB ID and cell ID specified for the secondary operator, respectively
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="00", GnodebId=123, CellId=1, SharedMcc="460",
SharedMnc="60", NrNetworkingOption=SA, Tac=2, SharedPlmnGnodebId=456, SharedPlmnCellId=2;
//Optional Setting the eNodeB-level PLMN ID filling mode in LTE-to-NR procedures when operator-
specific NCI configuration is enabled
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Optional Setting the target-gNodeB-level PLMN ID filling mode in LTE-to-NR procedures when operator-
specific NCI configuration is enabled
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="00", GnodebId=123,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;

//Data configuration on the NR side:
//Setting reselection priority values and sub-priority values for different operators
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=0, RatType=EUTRAN,
DIEarfcn=1300, CellReselPri=1, CellReselSubPri=0DOT2;
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=2, FreqIndex=1, RatType=EUTRAN,
DIEarfcn=1300, CellReselPri=1, CellReselSubPri=0DOT4;
//Setting mobility policies for different operators
ADD NRCELLOPPOLICY: NrCellId=1, OperatorId=1, gNBFreqPriorityGroupId=1, EpsFbMode=HANDOVER,

```



```

InterRatHoPolicySwitch=EUTRAN_REDIRECT_SWITCH-1&EUTRAN_HO_SWITCH-1,
FreqConfigPolicySwitch=OPERATOR_EUTRAN_FREQ_CFG_SW-1;
ADD NRCELLOPPOLICY: NrCellId=1, OperatorId=2, gNBFreqPriorityGroupId=2, EpsFbMode=REDIRECTION,
InterRatHoPolicySwitch=EUTRAN_REDIRECT_SWITCH-1&EUTRAN_HO_SWITCH-0,
FreqConfigPolicySwitch=OPERATOR_EUTRAN_FREQ_CFG_SW-1;
//Setting the operator-and-QCI-specific RSRP threshold offset for event A2/B2 related to mobility to E-
UTRAN, RSRP threshold offset for event A1 related to mobility to E-UTRAN, and RSRP threshold offset for
event B1 related to EPS fallback
MOD GNBOPERATORQCIPARAM: OperatorId=1, Qci=1, EutranA2B2RsrpThldOffset=1,
EutranA1RsrpThldOffset=3, EpsFbB1RsrpThldOffset=1;
MOD GNBOPERATORQCIPARAM: OperatorId=2, Qci=1, EutranA2B2RsrpThldOffset=3,
EutranA1RsrpThldOffset=1, EpsFbB1RsrpThldOffset=2;
//Setting the operator-and-QCI-specific RSRP threshold offset for event B1/B2 related to mobility to E-
UTRAN
MOD GNBOPERATORQCIPARAM: OperatorId=1, Qci=9, EutranB1B2RsrpThldOffset=1;
MOD GNBOPERATORQCIPARAM: OperatorId=2, Qci=9, EutranB1B2RsrpThldOffset=2;
//Setting the operator-and-QCI-specific offset for event A1 related to mobility from NG-RAN to E-UTRAN
ADD NRCELLOPPOLICY: NrCellId=0, OperatorId=1, EutranCioOffset=1;
ADD NRCELLOPPOLICY: NrCellId=0, OperatorId=2, EutranCioOffset=2;
//Setting the operator-specific offset relative to the RSRP threshold for event A1/A2/B2 related to inter-RAT
handover
MOD GNBREFSPCONFIG: OperatorId=1, RfspIndex=1, InterRatA1A2B2RsrpThldOfs=1;
MOD GNBREFSPCONFIG: OperatorId=2, RfspIndex=2, InterRatA1A2B2RsrpThldOfs=2;
//Setting the operator-specific offset relative to the RSRP threshold for event A1/A2/B2 related to mobility
to E-UTRAN
MOD NRCELLOPPOLICY: NrCellId=0, OperatorId=1, EutranA1A2B2RsrpThldOfs=1;
MOD NRCELLOPPOLICY: NrCellId=0, OperatorId=2, EutranA1A2B2RsrpThldOfs=2;
//Setting the VoLTE priority of an LTE frequency for blind handover and blind redirection on a per operator
basis
//Running the following commands with an assumption that the following gNBFreqPriorityGroupId and
FreqIndex instances have been configured for the base station
MOD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=0, RatType=EUTRAN,
DEarfcn=1300, VolteBlindPriority=1;
MOD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=2, FreqIndex=1, RatType=EUTRAN,
DEarfcn=1300, VolteBlindPriority=2;
//Running the following commands with an assumption that the following gNBFreqPriorityGroupId and
FreqIndex instances have not been configured for the base station
ADD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=0, RatType=EUTRAN,
DEarfcn=1300, VolteBlindPriority=1;
ADD GNBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=2, FreqIndex=1, RatType=EUTRAN,
DEarfcn=1300, VolteBlindPriority=2;
//Configuring neighboring LTE cell data of the NR cell
//Adding a neighboring LTE frequency
ADD NRCELLEUTRANNFREQ: NrCellId=1, DEarfcn=1300, MeasurementBandwidth=MBW100;
//Adding an external LTE cell
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="00", EnodebId=111, CellId=0, DEarfcn=1300,
PhysicalCellId=183, Tac=4026;
//Adding a neighbor relationship with the LTE cell
ADD NRCELLEUTRANRELATION: NrCellId=0, Mcc="460", Mnc="00", EnodebId=111, CellId=0;
//Adding PLMN information about the secondary LTE operator
ADD GNBEUTRAEXTCELLPLMN: Mcc="460", Mnc="00", EnodebId=111, CellId=0, SharedMcc="460",
SharedMnc="60";

```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

6.4.2.1 Operator-specific Dedicated Reselection Priority

Use the signaling tracing methods described in this section to check whether operator-specific dedicated reselection priorities have taken effect on the eNodeB and gNodeB sides.

eNodeB Side

Step 1 Start Uu interface tracing for the eNodeB.

Step 2 Check whether the *IdleModeMobilityControlInfo* IE in the RRC Connection Release message sent by the eNodeB to the UE contains the reselection priority specific to a PLMN supported by the UE. If it does, this function has been activated.

----End

gNodeB Side

Step 1 Start Uu interface tracing for the gNodeB.

Step 2 Check whether the RRCRelease message sent from the gNodeB to the UE contains the *CellReselectionPriorities* IE that carries the reselection priority specific to a PLMN supported by the UE. If it does, this function has been activated.

----End

6.4.2.2 Operator-specific Measurement Variables

Use the signaling tracing methods described in this section to check whether operator-specific measurement parameter settings have taken effect on the eNodeB and gNodeB sides.

eNodeB Side

Step 1 Start Uu interface tracing for the eNodeB.

Step 2 Check whether the threshold for the related event in the *ReportConfigEUTRA* IE in the RRCConnectionReconfiguration message delivered by the eNodeB is consistent with the configured value. For example, if the value of *a2-Threshold* of *eventA2* in the *ReportConfigEUTRA* IE in the RRCConnectionReconfiguration message delivered by the eNodeB is the same as the configured value, this function has taken effect.

----End

gNodeB Side

Step 1 Start Uu interface tracing for the gNodeB.

Step 2 Check whether the threshold for the related event in the *ReportConfigNR* IE in the RRCReconfiguration message delivered by the gNodeB is consistent with the configured value. For example, if the value of *a2-Threshold* of *eventA2* in the *ReportConfigNR* IE in the RRCReconfiguration message delivered by the gNodeB is the same as the configured value, this function has taken effect.

----End

6.4.2.3 Operator-specific Mobility Policy

Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN

- Observing performance counter values
In NR RAN sharing with common carrier mode, if the value of **N.HO.InterRAT.N2E.ExecSuccOut.PLMN** is not 0, this function has taken effect.
In NR RAN sharing with dedicated carrier mode, if the value of **N.HO.InterRAT.N2E.Coverage.ExecSuccOut** or **N.RRCRedirection.N2E.Coverage** is not 0, this function has taken effect.
- Tracing signaling
Check whether the result complies with the operator's mobility policy. The verification method is the same as that for independent operator mode. For details, see [5.2.4.2 Activation Verification](#).

Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN

- Observing performance counter values
In LTE RAN sharing with common carrier mode, if the value of **L.IRATHO.E2N.ExecSuccOut.PLMN** or **L.RRCRedirection.E2N.PLMN** is not 0, this function has taken effect.
In LTE RAN sharing with dedicated carrier mode, if the value of **L.IRATHO.E2N.ExecSuccOut** or **L.RRCRedirection.E2N** is not 0, this function has taken effect.
- Tracing signaling
Check whether the result complies with the operator's mobility policy. The verification method is the same as that for independent operator mode. For details, see [5.2.4.2 Activation Verification](#).

E-UTRAN to NG-RAN Traffic Steering

Check whether the result complies with the operator's mobility policy. The verification method is the same as that for independent operator mode. For details, see [5.2.4.2 Activation Verification](#).

Voice Fallback from NG-RAN to E-UTRAN

- Observing performance counter values
In NR RAN sharing with common carrier mode, if the value of **N.HO.InterRAT.N2E.EPSFB.ExecSuccOut.PLMN** or **N.RRCRedirection.N2E.EPSFB.PLMN** is not 0, this function has taken effect.
In NR RAN sharing with dedicated carrier mode, if the value of **N.HO.InterRAT.N2E.EPSFB.ExecSuccOut** or **N.RRCRedirection.N2E.EPSFB** is not 0, this function has taken effect.
- Tracing signaling
Check whether the result complies with the operator's mobility policy. The verification method is the same as that for independent operator mode. For details, see [5.2.4.2 Activation Verification](#).

Fast Return from E-UTRAN to NG-RAN

- Observing performance counter values
Observe the values of the **L.IRATHO.E2N.FastReturn.ExecSuccOut** and **L.RRCRedirection.E2N.FastReturn** counters. If either of the values is not 0, this function has taken effect.
- Tracing signaling
Check whether the result complies with the operator's mobility policy. The verification method is the same as that for independent operator mode. For details, see [5.2.4.2 Activation Verification](#).

6.4.2.4 Operator-specific NR Frequency Delivery

Observe signaling tracing results to check whether operator-specific NR frequency filtering has taken effect on the eNodeB side.

Step 1 Start Uu interface tracing for the eNodeB.

Step 2 Check whether the value of *carrierFreq-r15* of *measObjectNR-r15* in the *MeasObjectToAddMod* IE in the RRCConnectionReconfiguration message delivered by the eNodeB is the same as the configured value. If it is, this function has taken effect.

----End

6.4.3 Network Monitoring

In RAN sharing with common carrier scenarios, certain cell-level counters can be measured on a per operator basis. For details, see [Table 6-9](#) and [Table 6-10](#).

Table 6-9 NR counters related to data service mobility to be monitored in RAN sharing with common carrier scenarios

Counter ID	Counter Name
1911822955	N.HO.InterRAT.N2E.PrepAttOut.PLMN
1911822954	N.HO.InterRAT.N2E.ExecAttOut.PLMN
1911822953	N.HO.InterRAT.N2E.ExecSuccOut.PLMN
1911823080	N.HO.InterRAT.N2E.ExecSuccOut.Voice.PLMN
1911823081	N.HO.InterRAT.N2E.ExecAttOut.Voice.PLMN
1911827245	N.HO.InterRAT.E2N.PrepAttIn.PLMN
1911827244	N.HO.InterRAT.E2N.PrepSuccIn.PLMN
1911827243	N.HO.InterRAT.E2N.ExecSuccIn.PLMN
1911838092	N.HO.InterRAT.N2E.ResourceOpt.ULCoverage.PrepAttOut.PLMN
1911838093	N.HO.InterRAT.N2E.ResourceOpt.ULCoverage.ExecAttOut.PLMN

Counter ID	Counter Name
1911838094	N.HO.InterRAT.N2E.ResourceOpt.ULCoverage.ExecSuccOut.PLMN
1911840486	N.RRCRedirection.N2E.Blind.Coverage.PLMN
1911844996	N.HO.InterRAT.NCell.N2E.PrepAttOut.PLMN
1911844997	N.HO.InterRAT.NCell.N2E.ExecAttOut.PLMN
1911844998	N.HO.InterRAT.NCell.N2E.ExecSuccOut.PLMN

Table 6-10 LTE counters related to data service mobility to be monitored in RAN sharing with common carrier scenarios

Counter ID	Counter Name
1526759003	L.IRATHO.E2N.PrepAttOut.PLMN
1526759001	L.IRATHO.E2N.ExecAttOut.PLMN
1526759002	L.IRATHO.E2N.ExecSuccOut.PLMN
1526759023	L.RRCRedirection.E2N.PLMN

In RAN sharing with common carrier scenarios, certain cell-level counters related to EPS fallback can be measured on a per operator basis. For details, see [Table 6-11](#).

Table 6-11 NR counters related to EPS fallback to be monitored in RAN sharing with common carrier scenarios

Counter ID	Counter Name
1911822949	N.HO.InterRAT.N2E.EPSFB.PrepAttOut.PLMN
1911822948	N.HO.InterRAT.N2E.EPSFB.ExecAttOut.PLMN
1911822947	N.HO.InterRAT.N2E.EPSFB.ExecSuccOut.PLMN
1911822950	N.RRCRedirection.N2E.EPSFB.PLMN
1911822952	N.Rcv.VoiceFB.Trig.PLMN
1911822951	N.VoiceFB.RespSucc.PLMN
1911838086	N.HO.InterRAT.N2E.Emc.EPSFB.PrepAttOut.PLMN
1911838087	N.HO.InterRAT.N2E.Emc.EPSFB.ExecAttOut.PLMN
1911838088	N.HO.InterRAT.N2E.Emc.EPSFB.ExecSuccOut.PLMN

In RAN sharing with common carrier scenarios, the counters listed in [Table 6-12](#) can be used to monitor the performance of VoNR-to-VoLTE handover based on voice quality.

Table 6-12 Counters related to VoNR-to-VoLTE handover based on voice quality to be monitored in RAN sharing with common carrier scenarios

Counter ID	Counter Name
1911836045	N.HO.InterRAT.N2E.VoNRQual.PrepareOut.Voice.PLMN
1911836046	N.HO.InterRAT.N2E.VoNRQual.ExecOut.Voice.PLMN
1911836047	N.HO.InterRAT.N2E.VoNRQual.ExecSuccOut.Voice.PLMN

7 Appendix

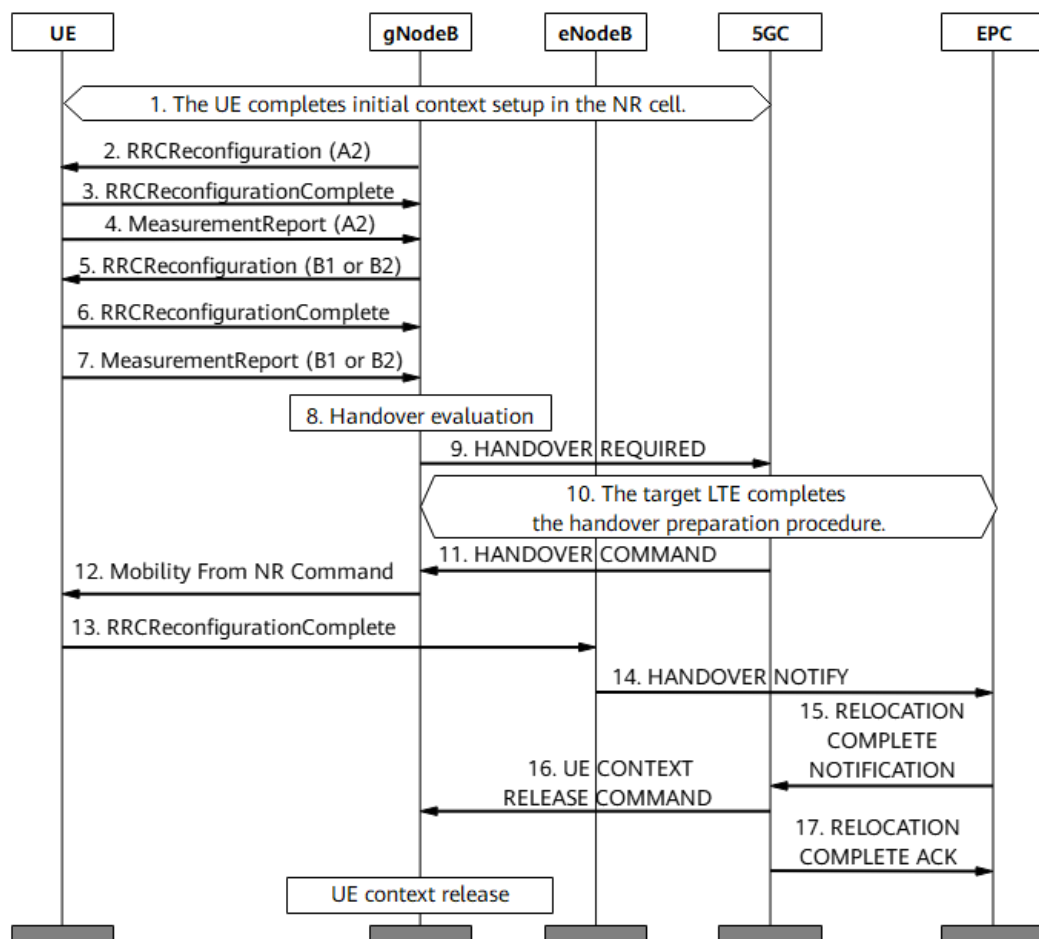
7.1 Downlink-Coverage-based Inter-RAT Mobility from NG-RAN to E-UTRAN

More details about the signaling procedures described in this section can be found in 3GPP TS 23.502 "Procedures for the 5G System; Stage 2" and 3GPP TS 38.331 "NR; Radio Resource Control (RRC) protocol specification".

Handover

Figure 7-1 shows the signaling procedure for a downlink-coverage-based inter-RAT handover from NG-RAN to E-UTRAN.

Figure 7-1 Signaling procedure for a downlink-coverage-based inter-RAT handover from NG-RAN to E-UTRAN



The details are as follows:

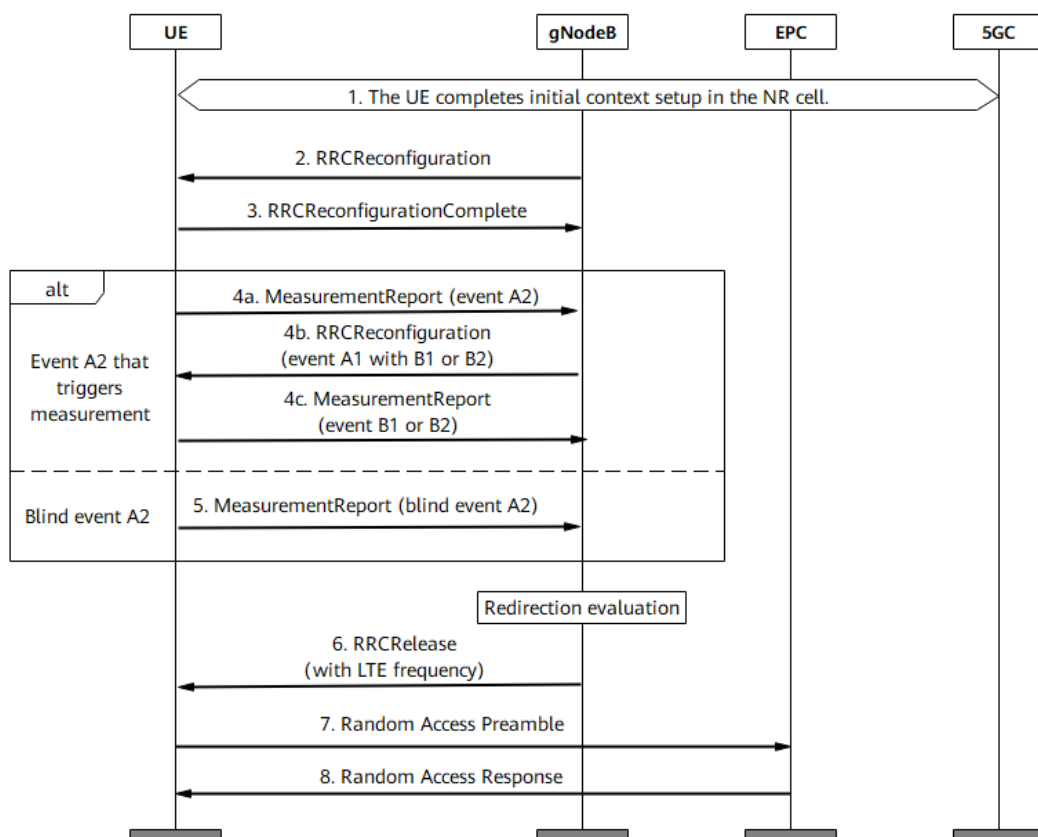
1. The UE completes initial context setup in the NR cell. For details, see *5G Networking and Signaling* in the 5G RAN feature documentation.
2. The gNodeB delivers the measurement configurations related to inter-RAT event A2.
3. The UE responds that the measurement configuration is complete.
4. The UE sends measurement reports on event A2.
5. The gNodeB receives the A2 measurement reports and then delivers the measurement configurations related to inter-RAT event A1 and event B1 or B2, depending on the type of event used to trigger inter-RAT handovers.
6. The UE responds that the measurement configuration is complete.
7. The UE sends measurement reports on event B1 or B2.
8. After receiving the B1 or B2 measurement reports, the gNodeB generates a candidate cell list based on the reported cell signal quality and selects the cell with the best signal quality as the target LTE cell.
9. The gNodeB sends a handover request to the 5GC.
10. The target LTE cell completes the handover preparation procedure, which involves the following messages:

- a. AMF to MME: RELOCATION REQUEST
 - b. MME to eNodeB: HANDOVER REQUEST
 - c. eNodeB to MME: HANDOVER REQUEST ACK
 - d. MME to AMF: RELOCATION RESPONSE
11. The 5GC notifies the gNodeB that the handover preparation is complete.
 12. The gNodeB initiates handover control towards the UE.
 13. The UE sends an RRC connection reconfiguration complete message.
 14. The eNodeB informs the EPC that the handover is complete.
 15. The EPC notifies the 5GC that the data forwarding tunnel has been established.
 16. The gNodeB receives a UE context release message and then releases the UE context.
 17. The 5GC responds to the EPC that data forwarding is complete.

Redirection

Figure 7-2 shows the signaling procedure for a downlink-coverage-based inter-RAT redirection from NG-RAN to E-UTRAN.

Figure 7-2 Signaling procedure for a downlink-coverage-based inter-RAT redirection from NG-RAN to E-UTRAN



The details are as follows:

1. The UE completes initial context setup in the NR cell. For details, see *5G Networking and Signaling* in the 5G RAN feature documentation.
2. The gNodeB delivers the measurement configurations related to inter-RAT event A2 and blind event A2 to the UE.
3. The UE responds that the RRC connection reconfiguration is complete.
4. For inter-RAT event A2, which triggers inter-RAT measurements:
 - a. The UE sends measurement reports on event A2.
 - b. The gNodeB receives the A2 measurement reports and then delivers the measurement configurations related to inter-RAT event A1 and event B1 or B2, depending on the type of event used to trigger inter-RAT handovers.
 - i. If the gNodeB receives measurement reports on event A1, it releases the measurement configurations related to event A1 and event B1 or B2.
 - ii. If the gNodeB receives measurement reports on event B1 or B2, it selects the operating frequency of the cell with the best signal quality among the reported cells for a redirection.
5. For blind event A2, the UE sends measurement reports on blind event A2.
6. The gNodeB sends the UE an RRCRelease message that carries the target frequency information.
7. The UE performs random access to an LTE cell based on the received frequency information.
8. The UE receives a random access response from the target LTE cell.

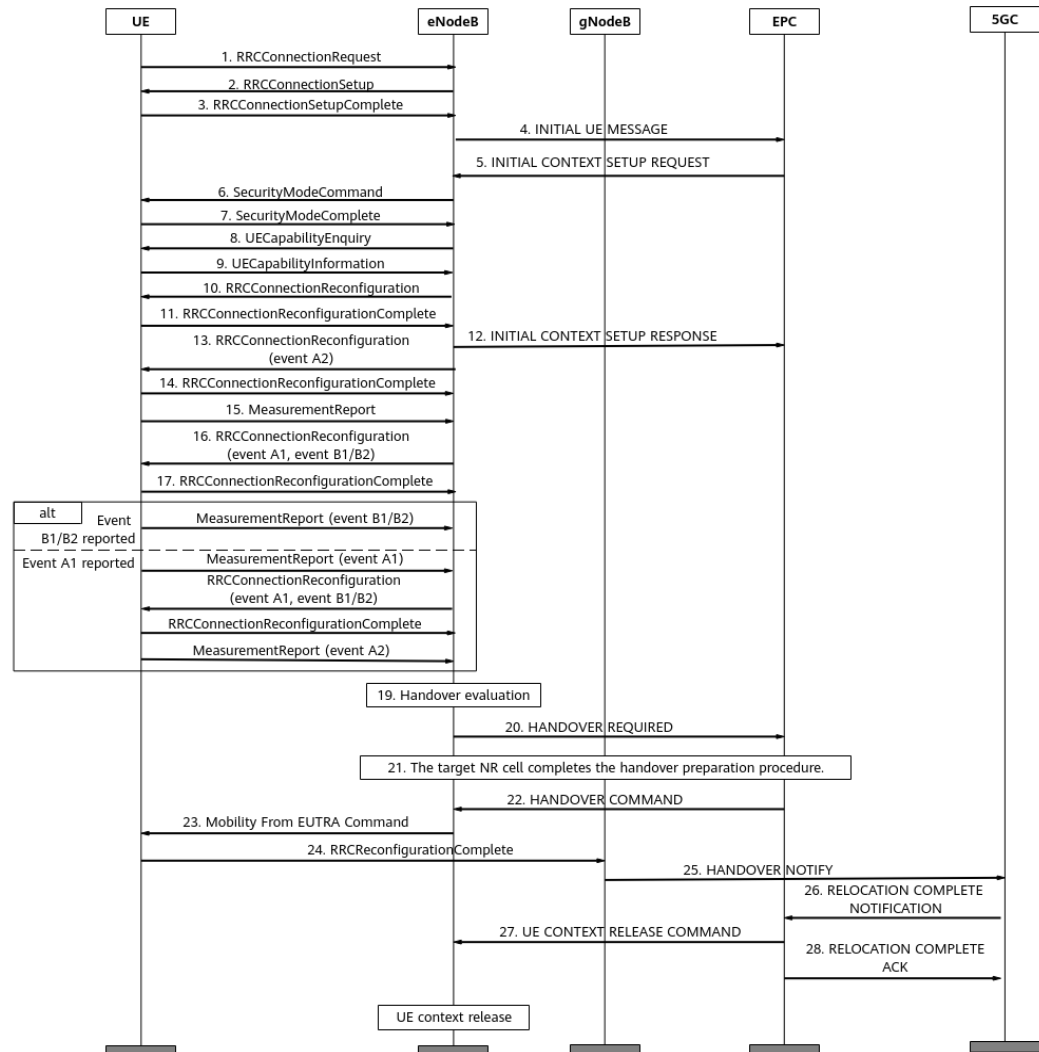
7.2 Downlink-Coverage-based Inter-RAT Mobility from E-UTRAN to NG-RAN

More details about the signaling procedures described in this section can be found in 3GPP TS 23.502 "Procedures for the 5G System; Stage 2" and 3GPP TS 36.331 "RRC Protocol Specification".

Handover

Figure 7-3 shows the signaling procedure for a coverage-based inter-RAT handover from E-UTRAN to NG-RAN.

Figure 7-3 Signaling procedure for a coverage-based inter-RAT handover from E-UTRAN to NG-RAN



The details are as follows:

1. The UE sends an RRCConnectionRequest message that carries a NAS PDU of REGISTRATION REQUEST.
2. The eNodeB sends an RRCConnectionSetup message to set up SRB1.
3. The UE responds that the RRC connection setup is complete.
4. The eNodeB sends the EPC an INITIAL UE MESSAGE to transparently transmit the NAS PDU of REGISTRATION REQUEST.
5. The EPC sends the eNodeB an INITIAL CONTEXT SETUP REQUEST message that contains a NAS PDU of REGISTRATION ACCEPT. After receiving the message, the eNodeB sets up the UE context.
6. The eNodeB sends a security mode command.

 NOTE

If the **UeCapbEnquirySecurityEnhSwitch** option of the **GlobalProcSwitch.ProtocolMsgOptSwitch** parameter is selected, the eNodeB can initiate UE capability transfer only after sending a security mode command. If this option is deselected, the eNodeB initiates UE capability transfer before sending a security mode command.

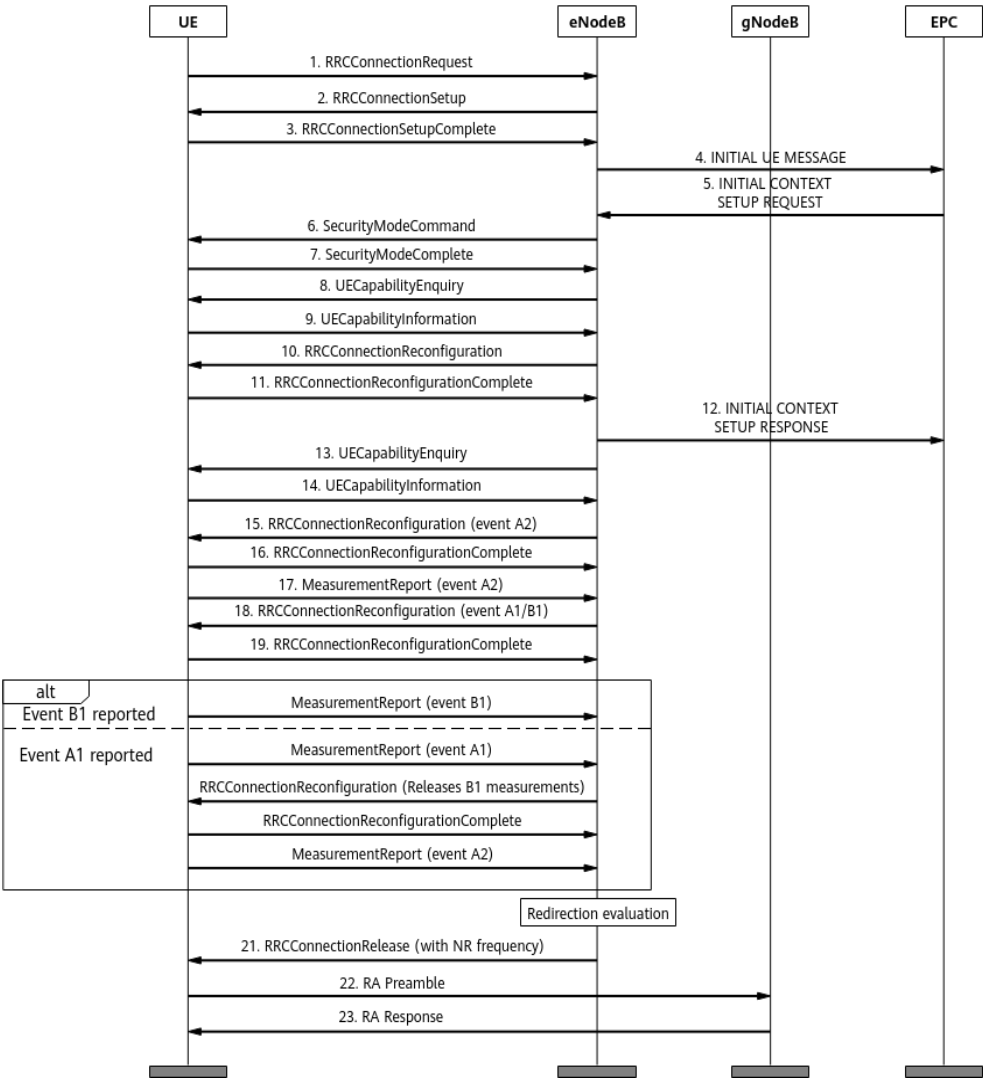
7. The UE responds with a SecurityModeComplete message.
8. The eNodeB initiates UE capability transfer.
9. The UE reports its capability information to the eNodeB. If UE-EUTRA-Capability contains the sa-Nr-r15 IE, the UE supports NR. The eNodeB saves the UE capability.
10. The eNodeB sends an RRCConnectionReconfiguration message to set up SRB2 and transparently transmit the NAS PDU of REGISTRATION ACCEPT to the UE.
11. The UE responds that the RRC connection reconfiguration is complete.
12. The eNodeB responds to the EPC that the UE context setup is complete.
13. The eNodeB delivers measurement configurations related to inter-RAT event A2.
14. The UE responds that the measurement configuration is complete.
15. The UE sends event A2 measurement reports.
16. The eNodeB receives the event A2 measurement reports and then delivers measurement configurations related to inter-RAT events A1 and B1/B2 to the UE. In the measurement configurations related to event B1/B2, NR frequencies are arranged in descending order of priority.
17. The UE responds that the RRC connection reconfiguration is complete.
18. The UE sends event B1/B2 or A1 measurement reports.
 - If the UE sends event B1/B2 measurement reports, the eNodeB goes to the next step.
 - If the UE sends event A1 measurement reports, the signal quality of the serving cell is favorable, in which case an inter-RAT handover is not required. The eNodeB releases the measurement configurations related to events A1 and B1. The UE continues A2 measurements and goes back to [17](#).
19. The eNodeB evaluates whether to initiate a handover.
20. The eNodeB sends a handover request to the EPC.
21. The target NR cell completes the handover preparation procedure, which involves the following messages:
 - a. RELOCATION REQUEST
 - b. HANDOVER REQUEST
 - c. HANDOVER REQUEST ACK
 - d. RELOCATION RESPONSE
 - e. HANDOVER COMMAND
22. The EPC sends a handover command to the eNodeB.
23. The eNodeB initiates handover control towards the UE.
24. The UE informs the gNodeB that the RRC connection reconfiguration is complete.

- 25. The gNodeB informs the 5GC that the handover is complete.
- 26. The 5GC notifies the EPC that the data forwarding tunnel has been established.
- 27. The eNodeB receives a UE context release message and then releases the UE context.
- 28. The EPC responds to the 5GC that data forwarding is complete.

Redirection

Figure 7-4 shows the signaling procedure for a coverage-based inter-RAT redirection from E-UTRAN to NG-RAN.

Figure 7-4 Signaling procedure for a coverage-based inter-RAT redirection from E-UTRAN to NG-RAN



The details are as follows:

- 1. The UE sends an RRCConnectionRequest message that carries a NAS PDU of ATTACH REQUEST.

2. The eNodeB sends an RRCConnectionSetup message to set up SRB1.
3. The UE responds that the RRC connection setup is complete.
4. The eNodeB sends the EPC an INITIAL UE MESSAGE to transparently transmit the NAS PDU of ATTACH REQUEST.
5. The EPC sends the eNodeB an INITIAL CONTEXT SETUP REQUEST message that contains a NAS PDU of ATTACH ACCEPT. After receiving the message, the eNodeB sets up the UE context.
6. The eNodeB sends a security mode command.

 NOTE

If the **UeCapbEnquirySecurityEnhSwitch** option of the **GlobalProcSwitch.ProtocolMsgOptSwitch** parameter is selected, the eNodeB can initiate UE capability transfer only after sending a security mode command. If this option is deselected, the eNodeB initiates UE capability transfer before sending a security mode command.

7. The UE responds with a SecurityModeComplete message.
8. The eNodeB initiates UE capability transfer.
9. The UE reports its NR capability. If UE-EUTRA-Capability contains the sa-Nr-r15 IE, the UE supports NR. The eNodeB saves the UE capability.
10. The eNodeB sends an RRCConnectionReconfiguration message to set up SRB2 and transparently transmit the NAS PDU of ATTACH ACCEPT to the UE.
11. The UE responds that the RRC connection reconfiguration is complete.
12. The eNodeB responds to the EPC that the UE context setup is complete.
13. If both of the following conditions are met, the eNodeB initiates NR capability query towards the UE by sending a UECapabilityEnquiry message in which RAT-TYPE is NR. If either of the following conditions is not met, the procedure ends.
 - The UE has reported the SupportedBandListNR-SA IE.
 - The switch of inter-RAT mobility from E-UTRAN to NG-RAN and the switch of redirection to NG-RAN are both turned on.
14. The UE reports its NR capability information to the eNodeB.
15. The eNodeB delivers measurement configurations related to event A2.
16. The UE responds that the RRC connection reconfiguration is complete.
17. The UE sends event A2 measurement reports.
18. The eNodeB receives the event A2 measurement reports and then delivers measurement configurations related to inter-RAT events A1 and B1 to the UE. In the measurement configurations related to event B1, NR frequencies are arranged in descending order of priority.
19. The UE responds that the RRC connection reconfiguration is complete.
20. The UE sends event B1 or A1 measurement reports.
 - If the UE sends event B1 measurement reports, the eNodeB goes to the next step.
 - If the UE sends event A1 measurement reports, the signal quality of the serving cell is favorable, in which case an inter-RAT redirection is not required. The eNodeB releases the measurement configurations related to events A1 and B1. The UE continues A2 measurements and goes back to [17](#).

21. Based on the redirection evaluation result, the eNodeB sends the UE an RRCConnectionRelease message that carries target NR frequency information.
22. The UE performs random access to an NR cell based on the received frequency information.
23. The UE receives a random access response from the target NR cell.

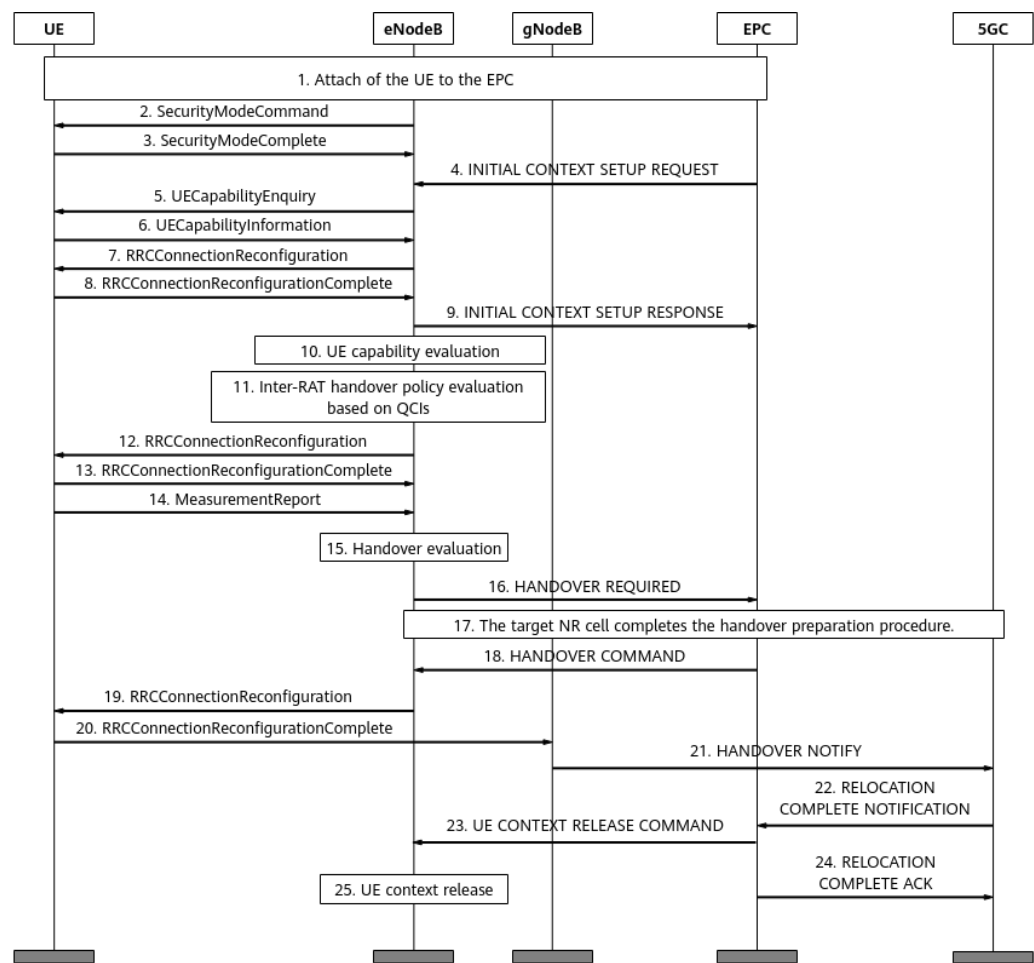
7.3 E-UTRAN to NG-RAN Traffic Steering

More details about the signaling procedures described in this section can be found in 3GPP TS 36.331 "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC) Protocol specification" and 3GPP TS 38.331 "NR; Radio Resource Control (RRC) protocol specification".

Handover

Figure 7-5 shows the signaling procedure for handover-based E-UTRAN to NG-RAN traffic steering.

Figure 7-5 Signaling procedure for handover-based E-UTRAN to NG-RAN traffic steering



The details are as follows:

1. The UE attaches to the EPC.
2. The eNodeB sends a security mode command.

 NOTE

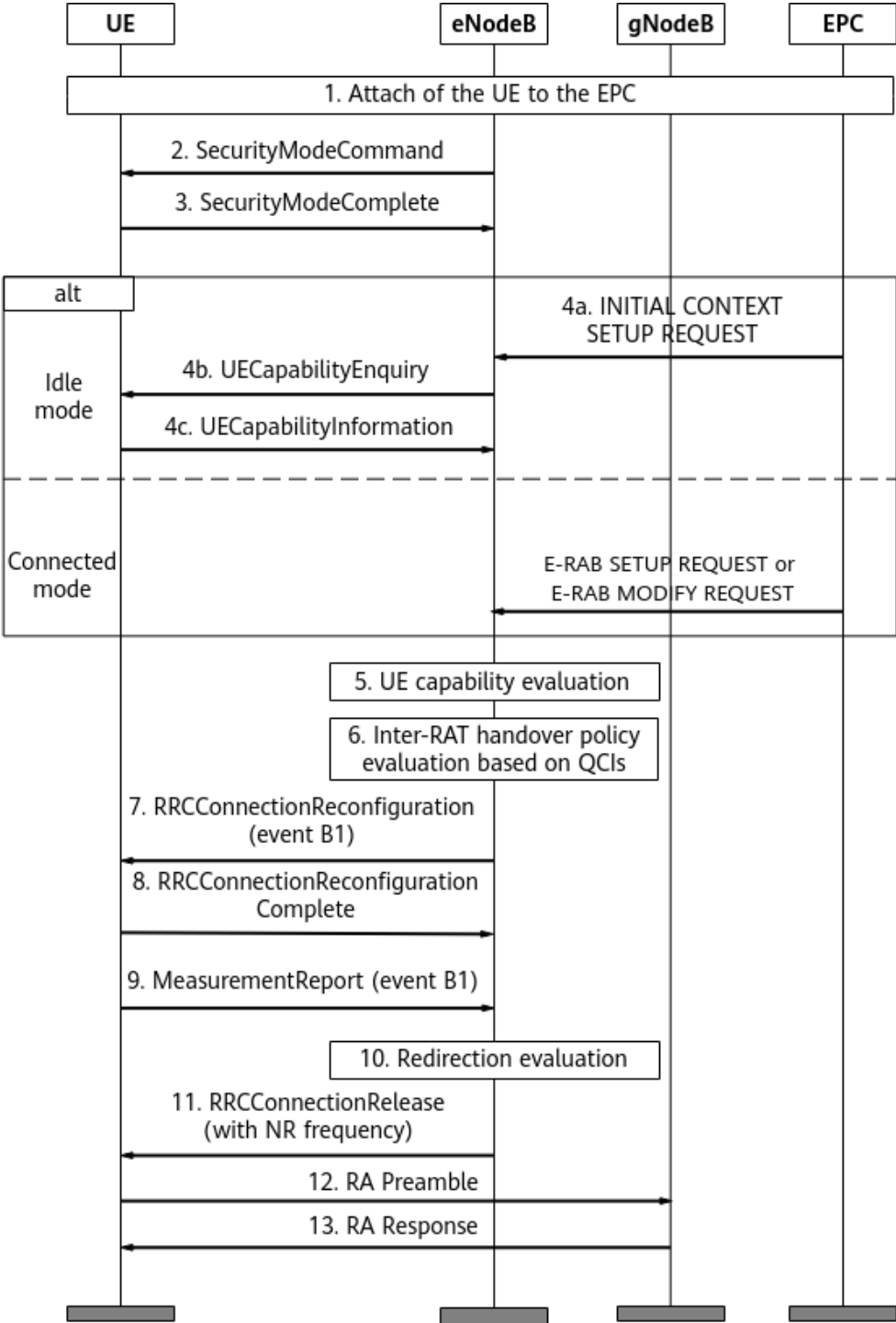
If the **UeCapbEnquirySecurityEnhSwitch** option of the **GlobalProcSwitch.ProtocolMsgOptSwitch** parameter is selected, the eNodeB can initiate UE capability transfer only after sending a security mode command. If this option is deselected, the eNodeB initiates UE capability transfer before sending a security mode command.

3. The UE responds with a SecurityModeComplete message.
4. The EPC sends an INITIAL CONTEXT SETUP REQUEST message to the eNodeB.
5. The eNodeB initiates UE capability transfer.
6. The UE reports its capability information to the eNodeB.
7. The eNodeB initiates RRC connection reconfiguration.
8. The UE responds that the RRC connection reconfiguration is complete.
9. The eNodeB sends an INITIAL CONTEXT SETUP RESPONSE message to the EPC.
10. The eNodeB evaluates UE capabilities.
11. The eNodeB evaluates whether to initiate a service-based handover based on the policies mapped to the QCLs of services running on the UE.
12. The eNodeB delivers the measurement configurations related to event B1.
13. The UE responds that the RRC connection reconfiguration is complete.
14. The UE sends measurement reports on event B1.
15. The eNodeB receives the B1 measurement reports and then evaluates whether to initiate a handover.
16. The eNodeB sends a HANDOVER REQUIRED message to the EPC.
17. The target NR cell completes the handover preparation procedure.
18. The EPC notifies the eNodeB that the handover preparation is complete.
19. The eNodeB sends the UE an RRCConnectionReconfiguration message that carries the target NR cell.
20. The UE sends an RRCConnectionReconfigurationComplete message to the gNodeB.
21. The gNodeB sends a HANDOVER NOTIFY message to the 5GC.
22. The 5GC sends a RELOCATION COMPLETE NOTIFICATION message to the EPC.
23. The EPC sends a UE CONTEXT RELEASE COMMAND message to the eNodeB.
24. The EPC sends a RELOCATION COMPLETE ACK message to the 5GC.
25. The eNodeB receives the UE CONTEXT RELEASE COMMAND message and then releases the UE context.

Redirection

Figure 7-6 shows the signaling procedure for redirection-based E-UTRAN to NG-RAN traffic steering.

Figure 7-6 Signaling procedure for redirection-based E-UTRAN to NG-RAN traffic steering



- The details are as follows:
- 1. The UE attaches to the EPC.
 - 2. The eNodeB sends a security mode command.

 NOTE

If the **UeCapbEnquirySecurityEnhSwitch** option of the **GlobalProcSwitch.ProtocolMsgOptSwitch** parameter is selected, the eNodeB can initiate UE capability transfer only after sending a security mode command. If this option is deselected, the eNodeB initiates UE capability transfer before sending a security mode command.

3. The UE responds with a SecurityModeComplete message.
4. A service is initiated for the UE. The procedure varies depending on UE states:
 - Idle mode:
 - 4a. The eNodeB sets up the UE context and an E-RAB according to the INITIAL CONTEXT SETUP REQUEST message sent by the EPC.
 - 4b. If the INITIAL CONTEXT SETUP REQUEST message does not contain the *UE Radio Capability* IE, the eNodeB sends a UECapabilityEnquiry message to query UE capabilities.
 - 4c. The UE sends a UECapabilityInformation message to report its capabilities.
 - Connected mode: The eNodeB sets up an E-RAB according to the E-RAB SETUP REQUEST or E-RAB MODIFY REQUEST message sent by the EPC. In this case, the eNodeB has stored UE capability information.
5. The eNodeB checks whether the UE capability information includes the *sa-Nr-r15* IE. If the UE capability information includes this IE, the UE supports NR and the procedure continues. Otherwise, the procedure ends.
6. The eNodeB checks the mobility policies for the services on the UE to evaluate whether an inter-RAT handover is allowed. If an inter-RAT handover is allowed, the procedure continues. Otherwise, the procedure ends.
7. The eNodeB delivers the measurement configurations related to event B1.
8. The UE responds that the RRC connection reconfiguration is complete.
9. The UE sends measurement reports on event B1.
10. The eNodeB receives the B1 measurement reports and then evaluates whether to initiate a redirection.
11. The eNodeB sends the UE an RRCConnectionRelease message that carries the target NR frequency.
12. The UE performs random access to an NR cell on the target frequency.
13. The UE receives a random access response from the target NR cell.

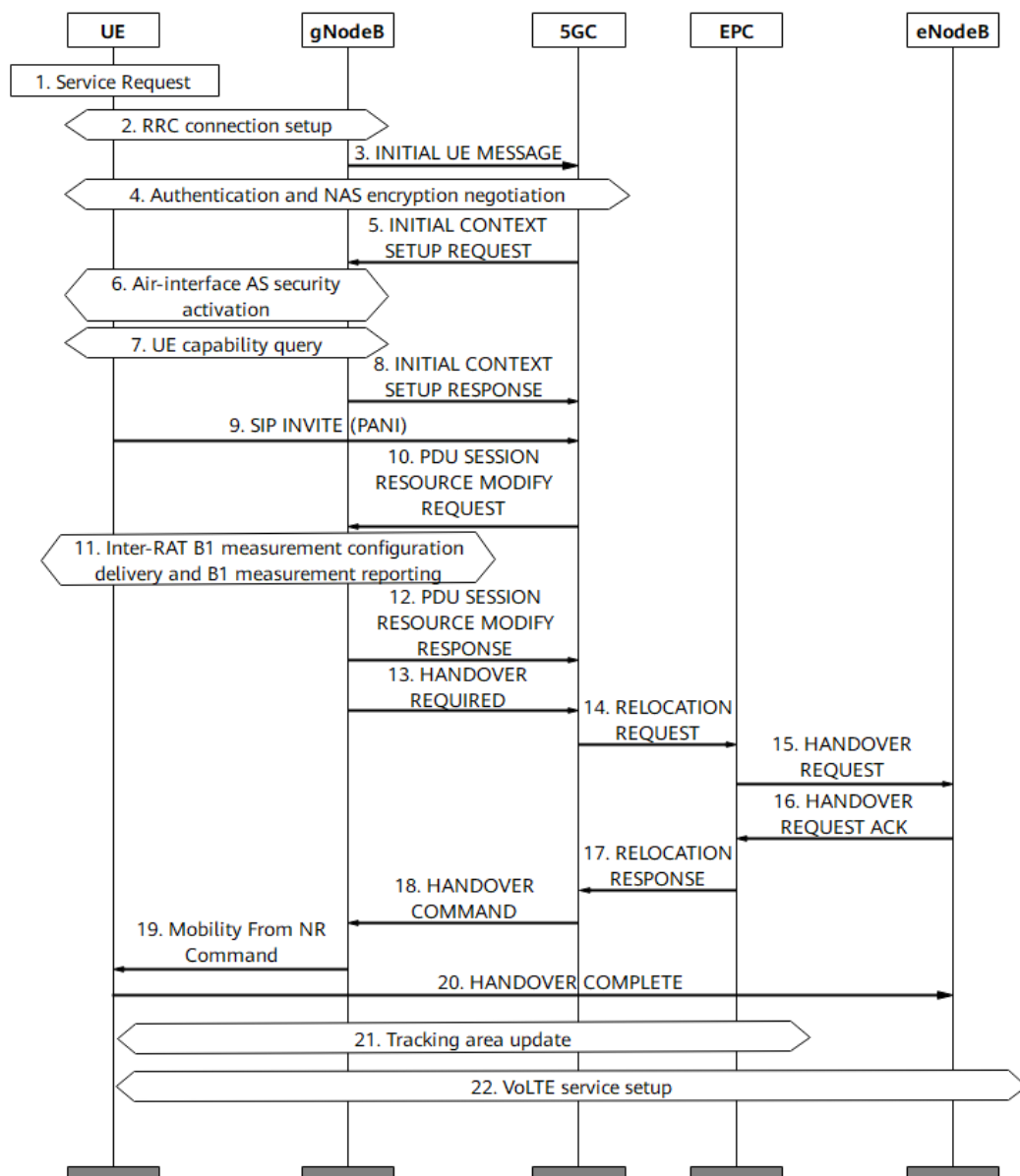
7.4 Voice Fallback from NG-RAN to E-UTRAN

More details about the signaling procedures described in this section can be found in 3GPP TS 23.502 "Procedures for the 5G System; Stage 2" and 3GPP TS 38.331 "NR; Radio Resource Control (RRC) protocol specification".

Normal Handover-based Voice Fallback

Figure 7-7 shows the signaling procedure for common handover-based voice fallback.

Figure 7-7 Signaling procedure for common handover-based voice fallback



The details are as follows:

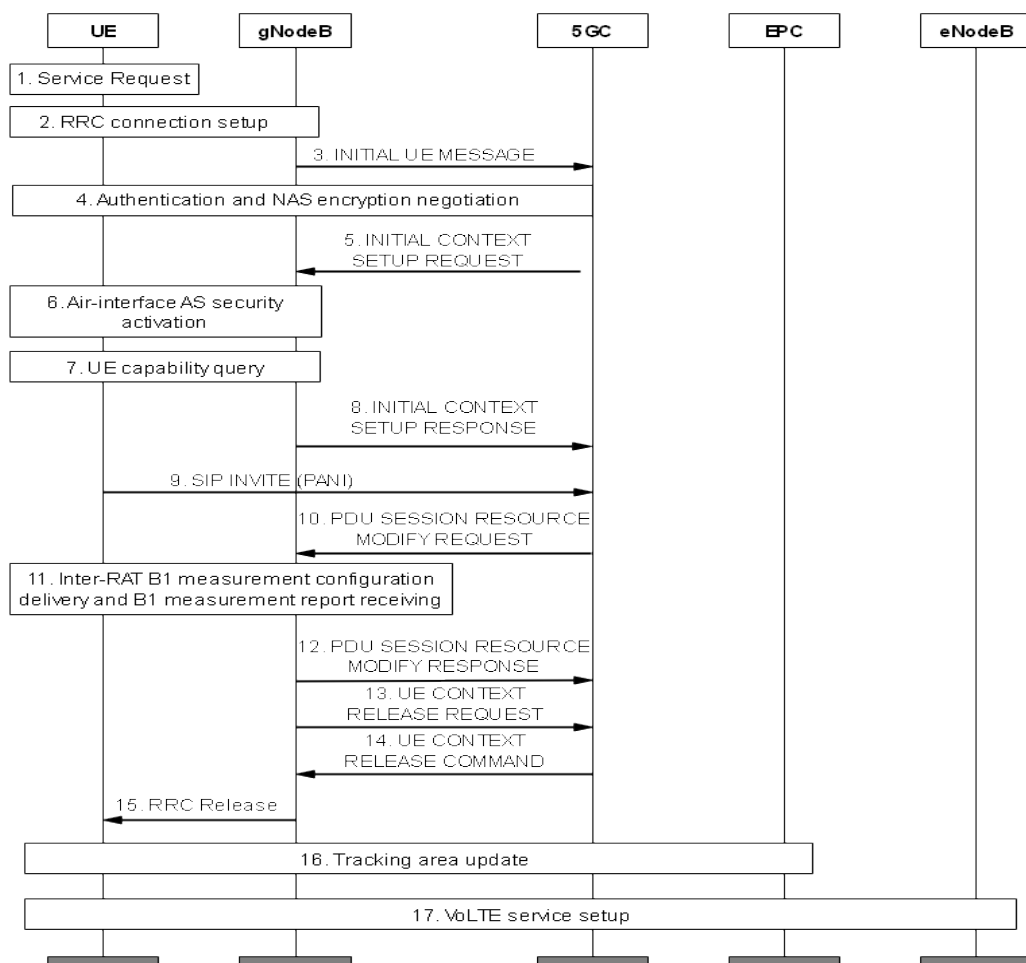
1. The UE initiates a voice service request.
2. The UE and gNodeB perform RRC connection setup, involving the following interactions:
 - a. The UE sends an RRCSetupRequest message and the gNodeB responds with an RRCSetup message to set up SRB1.
 - b. The UE sends an RRCSetupComplete message, informing the gNodeB that the RRC connection setup is complete. This message also carries the service request.
3. The gNodeB sends the 5GC an INITIAL UE MESSAGE to transparently transmit the service request.
4. The UE and 5GC perform authentication and NAS encryption negotiation, involving the following interactions:

- a. 5GC to UE: AUTHENTICATION REQUEST
 - b. UE to 5GC: AUTHENTICATION RESPONSE
 - c. 5GC to UE: SECURITY MODE COMMAND
 - d. UE to 5GC: SECURITY MODE COMPLETE
5. The gNodeB receives an INITIAL CONTEXT SETUP REQUEST message and sets up the UE context and an IMS signaling bearer.
6. The gNodeB conducts air-interface AS security activation, involving the following interactions:
 - a. gNodeB to UE: SecurityModeCommand
 - b. UE to gNodeB: SecurityModeComplete
7. The gNodeB initiates UE capability transfer and the UE reports its capabilities, involving the following interactions:
 - a. gNodeB to UE: UECapabilityEnquiry
 - b. UE to gNodeB: UECapabilityInformation
8. The gNodeB sends an INITIAL CONTEXT SETUP RESPONSE message, indicating that the PDU session has been set up.
9. The UE sends the 5GC a SIP INVITE message, requesting the 5GC to establish a voice session.
10. The gNodeB receives a PDU SESSION RESOURCE MODIFY REQUEST message, instructing the gNodeB to set up a dedicated voice bearer with a 5G QoS Identifier (5QI) of 1.
11. The gNodeB delivers measurement configurations related to inter-RAT event B1 and then receives event B1 measurement reports, involving the following interactions:
 - a. gNodeB to UE: RRCReconfiguration
 - b. UE to gNodeB: RRCReconfigurationComplete
 - c. UE to gNodeB: MeasurementReport
12. The gNodeB responds to the 5GC that it rejects PDU session modification, and indicates IMS voice fallback in progress.
13. The gNodeB sends a handover request to the 5GC based on the switch parameter settings and UE capability.
14. The 5GC forwards the UE context to the EPC.
15. The EPC sends a handover request to the eNodeB.
16. The EPC receives a handover response from the eNodeB.
17. The EPC forwards the handover success response message from the eNodeB to the 5GC.
18. The 5GC sends a handover command to the gNodeB.
19. The gNodeB sends a handover command to the UE.
20. The UE performs a handover to the target LTE cell.
21. A TAU procedure is initiated between the UE and the EPC.
22. The EPC initiates setup of a dedicated voice bearer with a QCI of 1. The subsequent procedure is the same as the VoLTE signaling procedure. For details, see VoLTE in *eRAN Feature Documentation*.

Normal Redirection-based Voice Fallback

Figure 7-8 shows the signaling procedure for common redirection-based voice fallback.

Figure 7-8 Signaling procedure for common redirection-based voice fallback



The details are as follows:

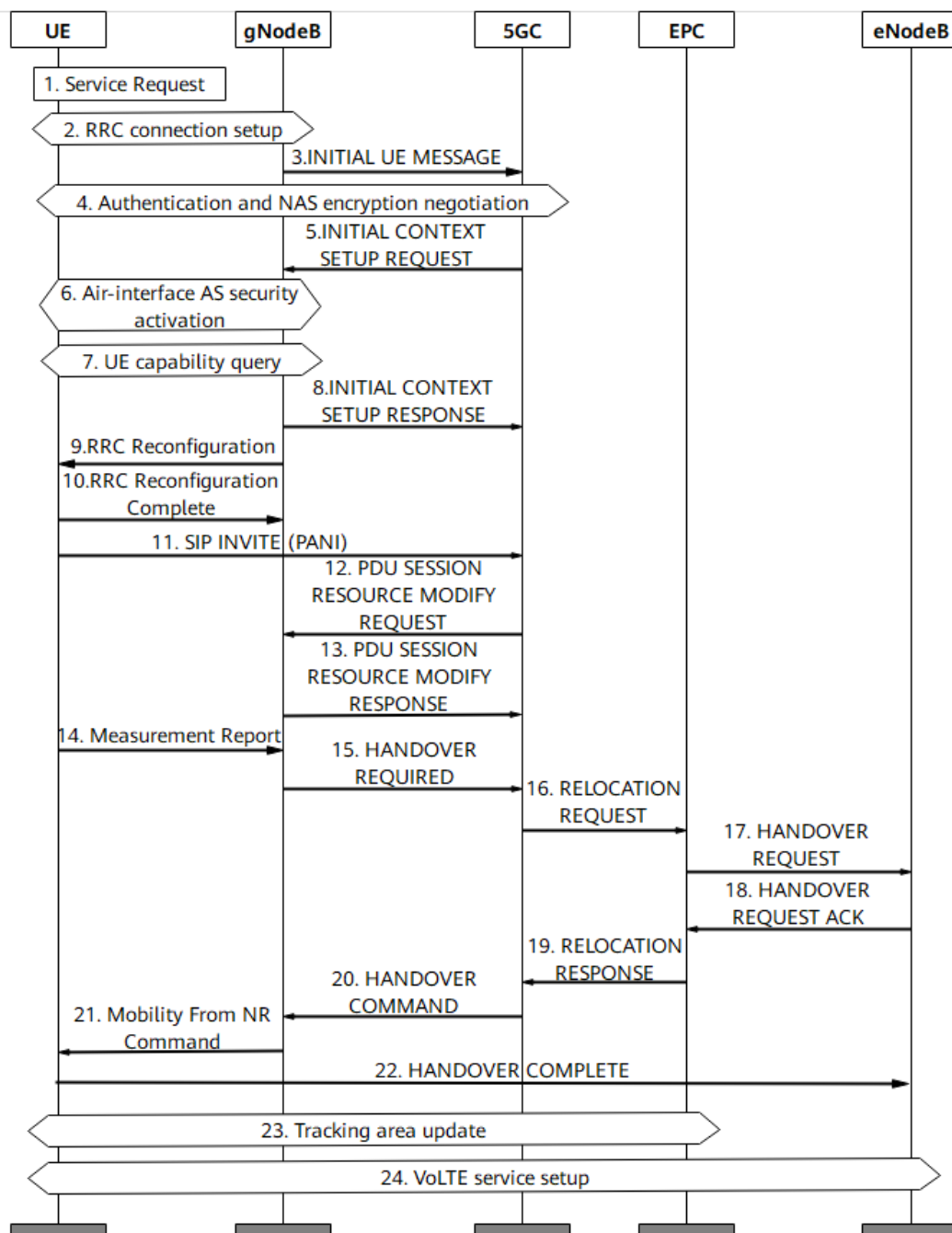
1. The UE initiates a voice service request.
2. The UE and gNodeB perform RRC connection setup, involving the following interactions:
 - a. The UE sends an RRCSetupRequest message and the gNodeB responds with an RRCSetup message to set up SRB1.
 - b. The UE sends an RRCSetupComplete message, informing the gNodeB that the RRC connection setup is complete. This message also carries the service request.
3. The gNodeB sends a service request to the 5GC through an INITIAL UE MESSAGE.
4. The UE and 5GC perform authentication and NAS encryption negotiation, involving the following interactions:

- a. 5GC to UE: AUTHENTICATION REQUEST
- b. UE to 5GC: AUTHENTICATION RESPONSE
- c. 5GC to UE: SECURITY MODE COMMAND
- d. UE to 5GC: SECURITY MODE COMPLETE
5. The gNodeB receives an INITIAL CONTEXT SETUP REQUEST message and sets up the UE context and an IMS signaling bearer.
6. The gNodeB conducts air-interface AS security activation, involving the following interactions:
 - a. gNodeB to UE: SecurityModeCommand
 - b. UE to gNodeB: SecurityModeComplete
7. The gNodeB initiates UE capability transfer and the UE reports its capabilities, involving the following interactions:
 - a. gNodeB to UE: UECapabilityEnquiry
 - b. UE to gNodeB: UECapabilityInformation
8. The gNodeB sends an INITIAL CONTEXT SETUP RESPONSE message, indicating that the PDU session has been set up.
9. The UE sends the 5GC a SIP INVITE message, requesting the 5GC to establish a voice session.
10. The gNodeB receives a PDU SESSION RESOURCE MODIFY REQUEST message, instructing the gNodeB to set up a dedicated voice bearer with a 5G QoS Identifier (5QI) of 1.
11. The gNodeB delivers measurement configurations related to inter-RAT event B1 and then receives event B1 measurement reports, involving the following interactions:
 - a. gNodeB to UE: RRCReconfiguration
 - b. UE to gNodeB: RRCReconfigurationComplete
 - c. UE to gNodeB: MeasurementReport
12. The gNodeB responds to the 5GC that it rejects PDU session modification, and indicates IMS voice fallback in progress.
13. The gNodeB sends the 5GC a UE CONTEXT RELEASE REQUEST message, requesting the 5GC to release the UE context.
14. The gNodeB receives a UE CONTEXT RELEASE COMMAND message from the 5GC, instructing the gNodeB to release all bearers.
15. The gNodeB sends the UE an RRCRelease message that carries the target LTE frequency.
16. The UE initiates a TAU procedure towards the EPC.
17. The EPC initiates setup of a dedicated voice bearer with a QCI of 1. The subsequent procedure is the same as the VoLTE signaling procedure. For details, see VoLTE in *eRAN Feature Documentation*.

Handover for Fast Voice Fallback

Figure 7-9 shows the signaling procedure for a handover-based fast voice fallback.

Figure 7-9 Signaling procedure for a handover-based fast voice fallback



The details are as follows:

1. The UE initiates a voice service request.
2. The UE and gNodeB perform RRC connection setup, involving the following interactions:
 - a. The UE sends an RRCSetupRequest message and the gNodeB responds with an RRCSetup message to set up SRB1.
 - b. The UE sends an RRCSetupComplete message, informing the gNodeB that the RRC connection setup is complete. This message also carries the service request.

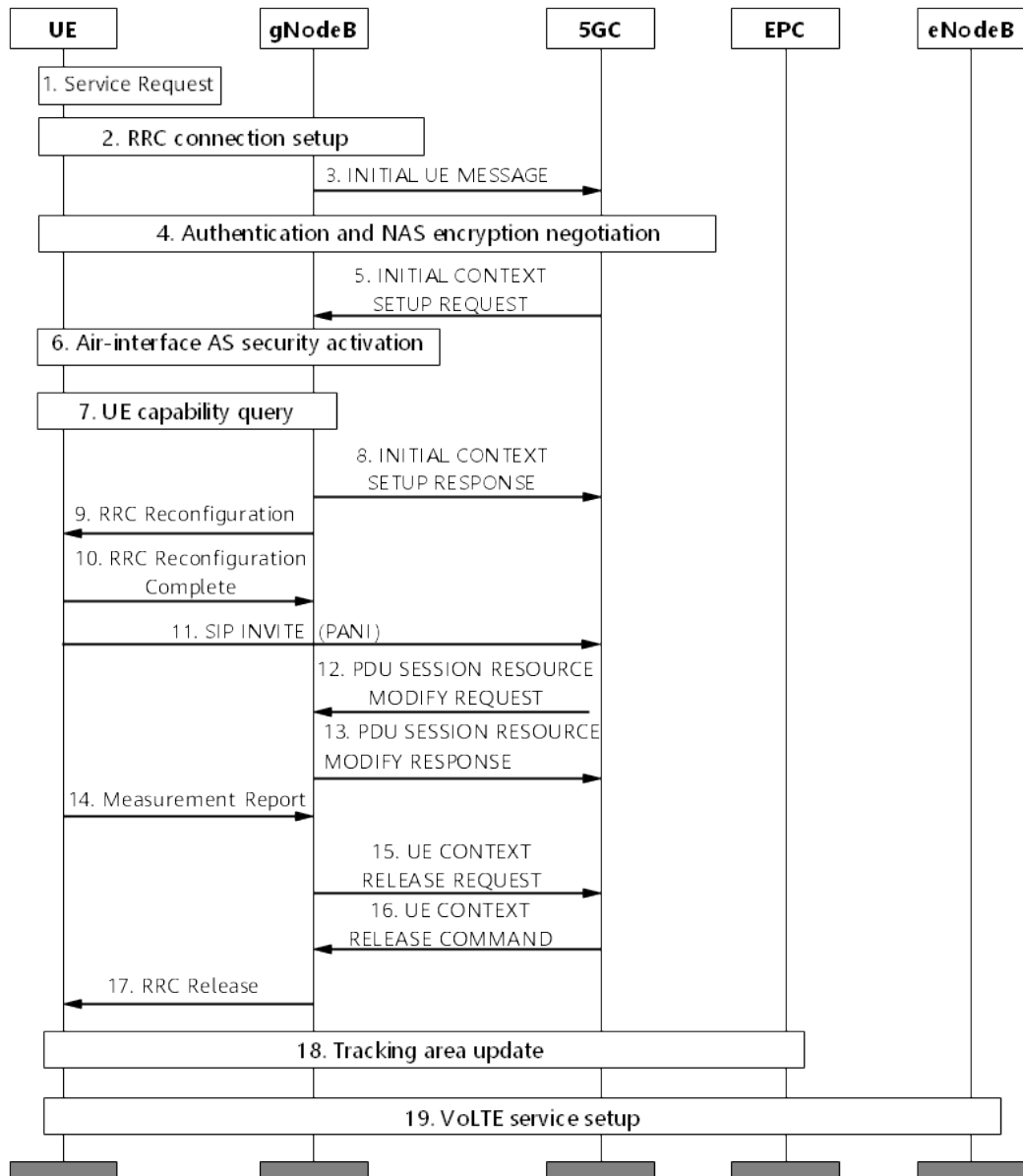
3. The gNodeB sends the 5GC an INITIAL UE MESSAGE to transparently transmit the service request.
4. The UE and 5GC perform authentication and NAS encryption negotiation, involving the following interactions:
 - a. 5GC to UE: AUTHENTICATION REQUEST
 - b. UE to 5GC: AUTHENTICATION RESPONSE
 - c. 5GC to UE: SECURITY MODE COMMAND
 - d. UE to 5GC: SECURITY MODE COMPLETE
5. The gNodeB receives an INITIAL CONTEXT SETUP REQUEST message and sets up the UE context and an IMS signaling bearer.
6. The gNodeB conducts air-interface AS security activation, involving the following interactions:
 - a. gNodeB to UE: SecurityModeCommand
 - b. UE to gNodeB: SecurityModeComplete
7. The gNodeB initiates UE capability transfer and the UE reports its capabilities, involving the following interactions:
 - a. gNodeB to UE: UECapabilityEnquiry
 - b. UE to gNodeB: UECapabilityInformation
8. The gNodeB sends an INITIAL CONTEXT SETUP RESPONSE message, indicating that the PDU session has been set up.
9. The gNodeB delivers measurement configurations related to inter-RAT event B1.
10. The UE reports that the inter-RAT event B1 measurement configuration is complete.
11. The UE sends the 5GC a SIP INVITE message, requesting the 5GC to establish a voice session.
12. The gNodeB receives a PDU SESSION RESOURCE MODIFY REQUEST message, instructing the gNodeB to set up a dedicated voice bearer with a 5G QoS Identifier (5QI) of 1.
13. The gNodeB responds to the 5GC that it rejects PDU session modification, and indicates IMS voice fallback in progress.
14. The UE sends a measurement report to the gNodeB.
15. The gNodeB sends a handover request to the 5GC based on the switch parameter settings and UE capabilities.
16. The 5GC forwards the UE context to the EPC.
17. The EPC sends a handover request to the eNodeB.
18. The eNodeB sends a handover request acknowledgment to the EPC.
19. The EPC forwards the handover request acknowledgment to the 5GC.
20. The 5GC sends a handover command to the gNodeB.
21. The gNodeB sends a handover command to the UE.
22. The UE performs a handover to the target LTE cell.
23. A tracking area update procedure is initiated between the UE and the EPC.

24. The EPC initiates setup of a dedicated voice bearer with a QoS class identifier (QCI) of 1. The subsequent procedure is the same as the VoLTE signaling procedure. For details, see VoLTE in *eRAN Feature Documentation*.

Redirection for Fast Voice Fallback

Figure 7-10 shows the signaling procedure for redirection-based fast voice fallback.

Figure 7-10 Signaling procedure for redirection-based fast voice fallback



The details are as follows:

1. The UE initiates a voice service request.
2. The UE and gNodeB perform RRC connection setup, involving the following interactions:

- a. The UE sends an RRCSetupRequest message and the gNodeB responds with an RRCSetup message to set up SRB1.
 - b. The UE sends an RRCSetupComplete message, informing the gNodeB that the RRC connection setup is complete. This message also carries the service request information.
3. The gNodeB sends the service request to the 5GC through an INITIAL UE MESSAGE.
4. The UE and 5GC perform authentication and NAS encryption negotiation, involving the following interactions:
 - a. 5GC to UE: AUTHENTICATION REQUEST
 - b. UE to 5GC: AUTHENTICATION RESPONSE
 - c. 5GC to UE: SECURITY MODE COMMAND
 - d. UE to 5GC: SECURITY MODE COMPLETE
5. The gNodeB receives an INITIAL CONTEXT SETUP REQUEST message and sets up the UE context and an IMS signaling bearer.
6. The gNodeB conducts air-interface AS security activation, involving the following interactions:
 - a. gNodeB to UE: SecurityModeCommand
 - b. UE to gNodeB: SecurityModeComplete
7. The gNodeB initiates UE capability transfer and the UE reports its capabilities, involving the following interactions:
 - a. gNodeB to UE: UECapabilityEnquiry
 - b. UE to gNodeB: UECapabilityInformation
8. The gNodeB sends an INITIAL CONTEXT SETUP RESPONSE message, indicating that the PDU session has been set up.
9. The gNodeB delivers measurement configurations related to inter-RAT event B1.
10. The UE reports that the inter-RAT event B1 measurement configuration is complete.
11. The UE sends the 5GC a SIP INVITE message, requesting the 5GC to establish a voice session.
12. The gNodeB receives a PDU SESSION RESOURCE MODIFY REQUEST message, instructing the gNodeB to set up a dedicated voice bearer with a 5QI of 1.
13. The gNodeB responds to the 5GC that it rejects PDU session modification, and indicates IMS voice fallback in progress.
14. The UE sends a measurement report to the gNodeB.
15. The gNodeB sends the 5GC a UE CONTEXT RELEASE REQUEST message, requesting the 5GC to release the UE context.
16. The gNodeB receives a UE CONTEXT RELEASE COMMAND message from the 5GC, instructing the gNodeB to release all bearers.
17. The gNodeB sends the UE an RRCRelease message that carries the target LTE frequency.
18. The UE initiates a tracking area update procedure towards the EPC.

19. The EPC initiates setup of a dedicated voice bearer with a QCI of 1. The subsequent procedure is the same as the VoLTE signaling procedure. For details, see VoLTE in *eRAN Feature Documentation*.

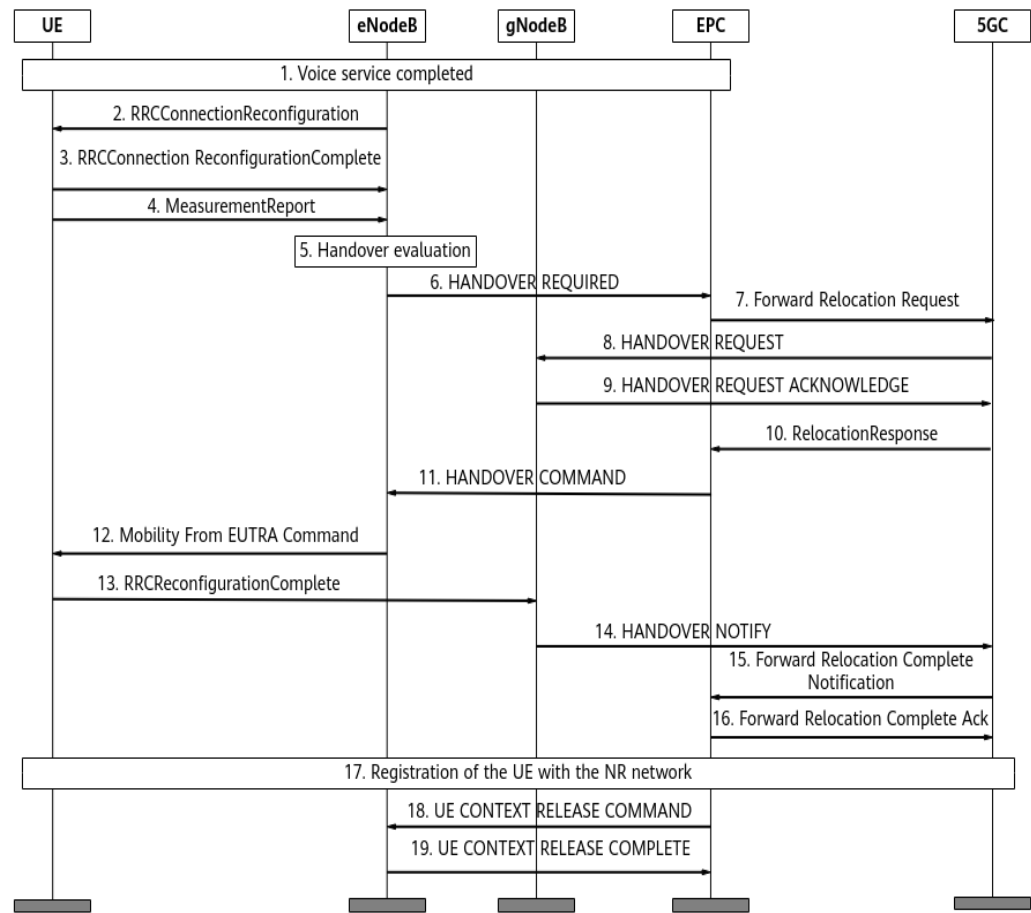
7.5 Fast Return from E-UTRAN to NG-RAN

More details about the signaling procedures described in this section can be found in 3GPP TS 36.331 "RRC Protocol Specification".

Handover

Figure 7-11 shows the signaling procedure for handover-based fast return from E-UTRAN to NG-RAN.

Figure 7-11 Signaling procedure for handover-based fast return from E-UTRAN to NG-RAN



The details are as follows:

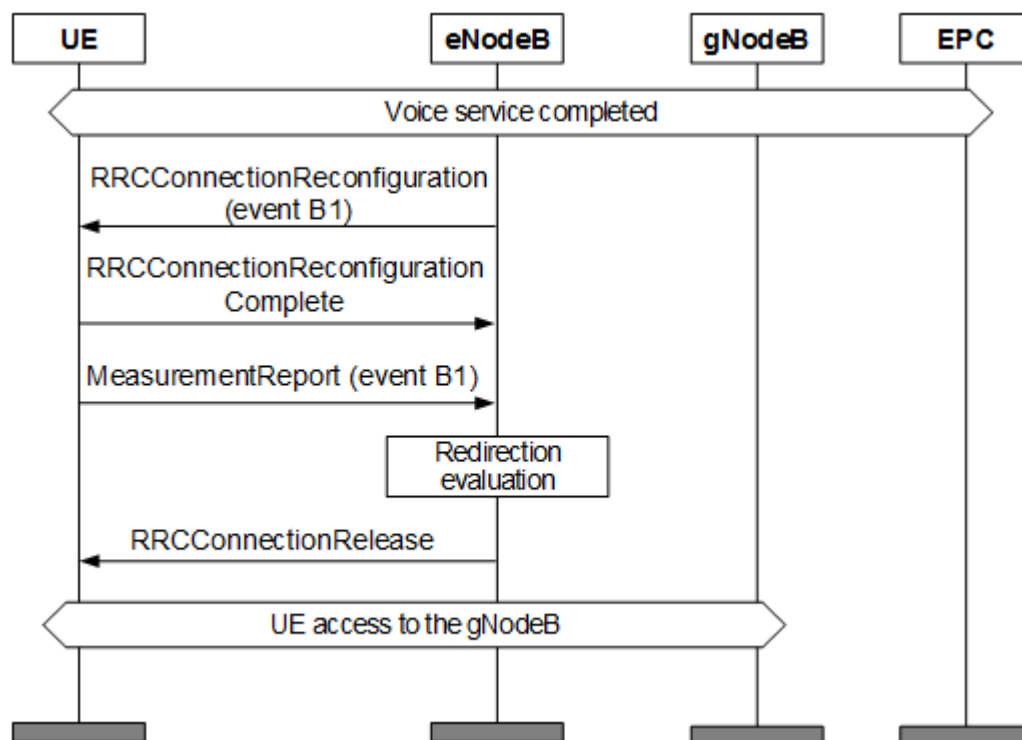
1. The voice service is complete, and the QCI 1 bearer is released.
2. The eNodeB sends the UE an RRCConnectionReconfiguration message, which contains measurement configurations related to inter-RAT event B1.
3. The UE sends the eNodeB an RRCConnectionReconfigurationComplete message, notifying the eNodeB that the RRC connection reconfiguration is complete.

4. The UE sends the eNodeB a MeasurementReport message, which contains event B1 measurement results.
5. The eNodeB performs handover evaluation and determines the target gNodeB.
6. The eNodeB sends the EPC a HANDOVER REQUIRED message, requesting a handover.
7. The EPC sends the 5GC a Forward Relocation Request message, which contains the UE context.
8. The 5GC sends the gNodeB a HANDOVER REQUEST message, instructing the gNodeB to prepare for the handover.
9. The gNodeB sends the 5GC a HANDOVER REQUEST ACKNOWLEDGE message, indicating that the handover preparation is complete.
10. The 5GC sends the EPC a Relocation Response message, indicating that the handover preparation is complete.
11. The EPC sends a HANDOVER COMMAND message to the eNodeB.
12. The eNodeB sends the UE a Mobility From EUTRA Command message, instructing the UE to perform a handover.
13. The UE sends the target gNodeB an RRCReconfigurationComplete message, notifying the gNodeB that the handover is complete.
14. The gNodeB notifies the 5GC of the handover.
15. The 5GC sends a Forward Relocation Complete Notification message, notifying the UE that the UE has arrived in the target area.
16. The EPC responds to the 5GC with a Forward Relocation Complete Ack message.
17. The UE registers with the NR network.
18. The EPC instructs the eNodeB to release the UE context.
19. The eNodeB sends a UE CONTEXT RELEASE COMPLETE message to the EPC.

Redirection

Figure 7-12 shows the signaling procedure for redirection-based fast return from E-UTRAN to NG-RAN.

Figure 7-12 Signaling procedure for redirection-based fast return from E-UTRAN to NG-RAN



The details are as follows:

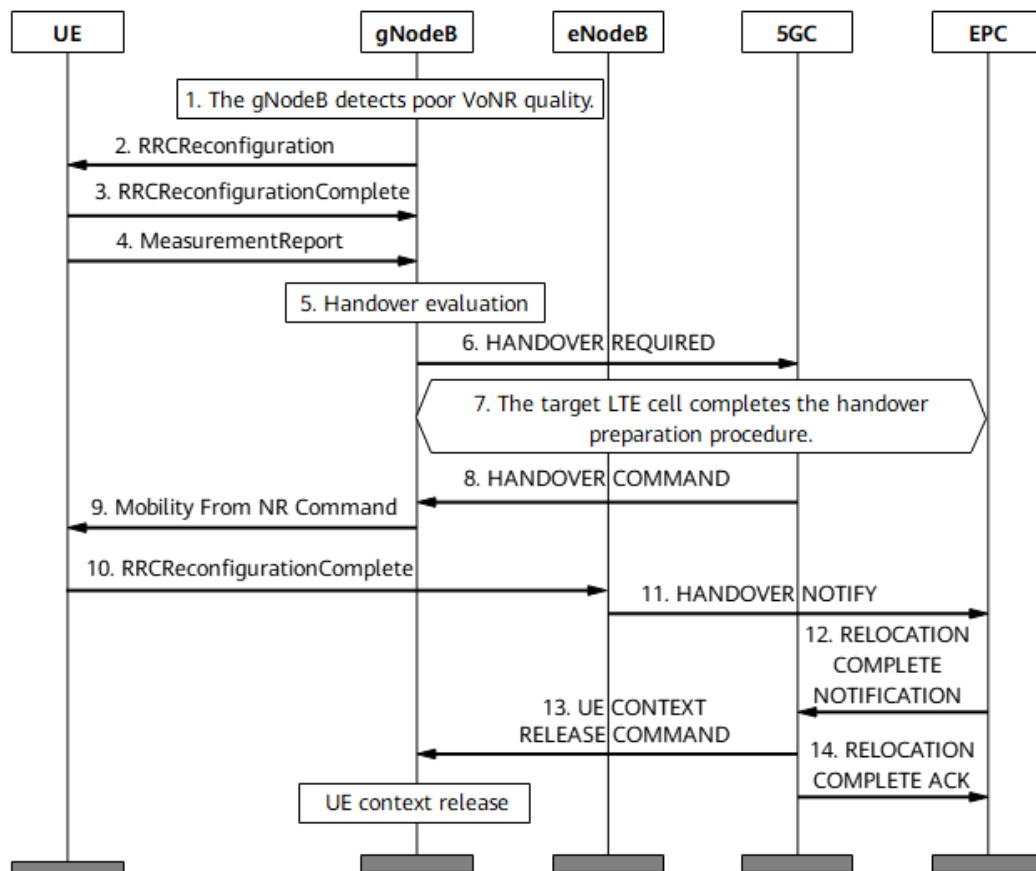
1. The voice service is complete.
2. The eNodeB sends the UE an RRConnectionReconfiguration message, which contains measurement configurations related to inter-RAT event B1.
3. The UE sends the eNodeB an RRConnectionReconfigurationComplete message, notifying the eNodeB that the RRC connection reconfiguration is complete.
4. The UE sends the eNodeB a MeasurementReport message, which contains event B1 measurement results.
5. The eNodeB selects the operating frequency of the cell with the best signal quality among the reported cells.
6. The eNodeB sends the UE an RRConnectionRelease message, informing the UE of the target NR frequency for redirection.
7. The UE is redirected to an NR cell based on the NR frequency information in the RRConnectionRelease message.

7.6 VoNR-to-VoLTE Handover Based on Voice Quality

More details about the signaling procedures described in this section can be found in 3GPP TS 23.502 "Procedures for the 5G System; Stage 2" and 3GPP TS 38.331 "NR; Radio Resource Control (RRC) protocol specification".

Figure 7-13 shows the signaling procedure for a VoNR-to-VoLTE handover based on voice quality.

Figure 7-13 Signaling procedure for a VoNR-to-VoLTE handover based on voice quality



The details are as follows:

1. The gNodeB finds that the VoNR quality is poor.
2. The gNodeB delivers the measurement configurations related to inter-RAT event B1.
3. The UE responds that the measurement configuration is complete.
4. The UE sends measurement reports on event B1.
5. After receiving the B1 measurement reports, the gNodeB generates a candidate cell list based on the reported cell signal quality and selects the cell with the best signal quality as the target LTE cell.
6. The gNodeB sends a handover request to the 5GC.
7. The target LTE cell completes the handover preparation procedure, which involves the following messages:
 - a. AMF to MME: RELOCATION REQUEST
 - b. MME to eNodeB: HANDOVER REQUEST
 - c. eNodeB to MME: HANDOVER REQUEST ACK
 - d. MME to AMF: RELOCATION RESPONSE
8. The 5GC notifies the gNodeB that the handover preparation is complete.
9. The gNodeB initiates handover control towards the UE.

10. The UE sends an RRC connection reconfiguration complete message.
11. The eNodeB informs the EPC that the handover is complete.
12. The EPC notifies the 5GC that the data forwarding tunnel has been established.
13. The gNodeB receives a UE context release message and then releases the UE context.
14. The 5GC responds to the EPC that data forwarding is complete.

8 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

11 Reference Documents

- *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- *NSA-SA Selection Based on User Experience*
- *EPC-based NSA Basics*
- *EPC-based NSA Performance Enhancement*
- *Multi-RAT Coordinated Energy Saving*
- *LTE FDD and NR Dynamic Spectrum Sharing*
- *VoLTE in eRAN Feature Documentation*
- *Flexible User Steering in eRAN Feature Documentation*
- *RAN Sharing in eRAN Feature Documentation*
- *Emergency Call in eRAN Feature Documentation*
- *Energy Conservation and Emission Reduction in eRAN Feature Documentation*
- *ANR Management in eRAN Feature Documentation*
- *SA Mobility Management in Connected Mode in 5G RAN Feature Documentation*
- *Multi-Operator Sharing in 5G RAN Feature Documentation*
- *5G Networking and Signaling in 5G RAN Feature Documentation*
- *VoNR in 5G RAN Feature Documentation*
- *Flexible User Steering in 5G RAN Feature Documentation*
- *ANR in 5G RAN Feature Documentation*
- *Virtual Grid-based Multi-Frequency Coordination in 5G RAN Feature Documentation*
- *Multi-Frequency Smart Selection in 5G RAN Feature Documentation*
- *Energy Conservation and Emission Reduction in 5G RAN Feature Documentation*
- *QoS Management in eRAN Feature Documentation*
- *Super Uplink*
- *License Control Item Lists (FDD), License Control Item Lists (CIoT), and License Control Item Lists (TDD) in eRAN Feature Documentation*
- *License Control Item Lists in 5G RAN Feature Documentation*

- *RedCap in 5G RAN Feature Documentation*
- *Terminal Awareness Differentiation in eRAN Feature Documentation*
- *Mobility Load Balancing in Connected Mode in 5G RAN Feature Documentation*
- 3900 & 5900 Series Base Station Product Documentation
- 3GPP TS 23.501: "System Architecture for the 5G System"
- 3GPP TS 23.502: "Procedures for the 5G System"
- 3GPP TS 36.304: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) procedures in idle mode"
- 3GPP TS 36.331: "RRC Protocol Specification"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"